UM-2 field equations

The coupled field equations of Einstein-Lovelock gravity, the dirac16complex field, three generations of quarks and leptons, the gluons, the photon and the W^+, W^- and Z bosons (no Higgs particle), the tip sigma field and the block stabiliser, with every interaction

Underlying theory: Patrick L. Nash Derived and verified by Claude (Anthropic)

Field equations, version 4, 2026-10-18 (versions 1 to 3: 2026-10-17 and 2026-10-18)

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1 How to read this document

1.1 What this document contains

This document states the complete set of coupled field equations of the universe model UM-2, derives them from one action, and explains every term. It is self-contained: every equation that it uses is written in it, and every symbol of every equation is defined right after the equation. The sections are:

- **The action** (IN-1 to IN-18): every term of the action, written out and explained: what it couples, and why it is consistent with QCD and with the electroweak theory.
- **Gravity** (FE-1 to FE-11): the Einstein-Lovelock equations, the energy-momentum tensor of every field and its conservation.
- **The matter fields** (FE-12 to FE-30): the field equations of dirac16complex, of the three generations of quarks and leptons, of the gauge fields (gluons, $W^{\pm}$, Z, photon), of the tip sigma field, and their boundary

conditions.

- **The background** (FE-31 to FE-49): the exact order-0 equations of the metric, the static block and its stabiliser, the radion, the response of the block to matter and four-dimensional gravity.
- **The masses from the hidden coordinate** (FE-50 to FE-62): the x_0 eigenvalue problems that give every particle its mass (the gauge bosons, the graviton tower, the twelve fermions with their mixing, dirac16complex).
- **dirac16complex coupled to all fields** (FE-63 to FE-86): the Kohn-Sham equations of dirac16complex with the mean fields of every other field, the portals on the zero modes, the condensates of the Standard-Model plasma, and the block response that closes the loop.
- **The linear perturbations** (FE-87 to FE-99): the metric as a power series, the Einstein-Boltzmann equations of the CMB, the CNB and the BAO.
- **The three neutrino flavours** (FE-100 to FE-120): the neutrino masses, the four-fermion interactions, the flavour evolution in vacuum, in matter, in the early Universe and in dense neutrino gases.
- **The thermal history** (FE-121 to FE-136): the degrees of freedom, the Boltzmann equations of the dark matter, of the fold bands of dirac16complex and of the right-handed neutrinos.
- **The coupled system**: which fields appear in which equations, and how the system is solved, step by step.
- **What was verified**: the 287 checks of the computations and the residuals of the solution in 26 of the equations.

1.2 The status labels

Every equation and every result carries one of the labels

- **AUTHOR**: from the author's notebooks (the metric, the coordinates, the Lovelock weights, the field dirac16complex);
- **RECORD**: from the Revision record of this repository (the dirac16complex Lagrangian and its Kohn-Sham treatment);
- **STANDARD**: textbook physics;
- **ASSUMED**: a choice made in UM-2 without derivation;
- **INVENTED**: a new interaction or construction of UM-2 (the tip sigma field, the portals I11, I12, I13);
- **DERIVED**: computed and verified in UM-2 (the verification is stated);
- **ESTIMATE**: an order-of-magnitude evaluation;
- **OPEN**: not settled.

1.3 Every equation is followed by its symbols

Every display equation is followed by the list "Symbols in (FE-n)" (or "Symbols in this equation" for a display without a tag) of EVERY symbol that appears in it: every quantity, field, coupling, constant and operator, every index letter with the values it takes, every mark (prime, dagger, transpose, star), the imaginary unit, Euler's number and the differential. Each item gives the meaning, the unit (natural units $\hbar = c = 1$), the value where one exists, and the status. A field equation carries the tag FE-1 to FE-136 and an interaction term of the action the tag IN-1 to IN-18; the numbering is that of the inventory of UM-2 and is the same in every UM-2 document. Displays without a tag are steps of a derivation.

1.4 Coordinates, signature and units

The author's coordinates are $X = \{x_0, x_1, \ldots, x_7\}$, always with the hidden direction first: x_0 is the hidden space-like coordinate, on which the metric depends periodically with the period $\pi/(6H)$ through the angle $z = 6Hx_0$; x_1, x_2, x_3 are ordinary space, x_4 is the time, and x_5, x_6, x_7 are three extra time-like coordinates. The signature is $(+,+,+,+,-,-,-,-)$ in this order; the four observable coordinates $x_1, \ldots, x_4$ have the signature $(+,+,+,-)$. $H = 14.296$ TeV is the author's constant (an inverse length). Units are natural, $\hbar = c = 1$; in eight dimensions a fermion field has the dimension $\mathrm{GeV}^{7/2}$ and the gravitational coupling $\kappa = 8\pi G_8$ the dimension GeV^{-6}. A dot is the derivative with respect to the time x_4; a prime is used only in the cosmological perturbation equations, for the derivative with respect to the conformal time.

1.5 The fields of UM-2

The author orders a model universe with the following coupled fields; UM-2 adds the tip sigma field (which replaces the Higgs field) and the stabiliser of the block.

field	symbol	what it is	it couples to	field equations
Einstein-Lovelock gravity	g_{MN}	the metric of eight dimensions	every field	FE-1 to FE-11, FE-31 to FE-49, FE-87 to FE-99
dirac16complex	Ψ	the author's 16-component spinor field; its odd level is the dark matter	gravity, itself, every fermion (I11, I13), every gauge field (I12)	FE-5, FE-12, FE-13, FE-28, FE-62 to FE-73
left-handed neutrinos (three flavours)	ν_e, ν_μ, ν_τ in ℓ_i	the upper members of the lepton doublets	gravity, $W^\pm$, Z, Σ (masses with ν_R), Ψ (I11)	FE-17, FE-19, FE-100 to FE-120
left-handed electrons and the right-handed electron singlet, with the muon and tau generations	ℓ_i, e_{Ri}	the lower members of the lepton doublets and the charged-lepton singlets	gravity, $W^\pm$, Z, photon, Σ, Ψ (I11)	FE-17, FE-18
three quark generations	Q_i, u_{Ri}, d_{Ri}	the quark doublets and singlets	gravity, gluons, $W^\pm$, Z, photon, Σ, Ψ (I11, I13)	FE-14 to FE-16
gluons	G^A_M	the gauge field of SU(3)	quarks, Ψ (I12), gravity	FE-7, FE-22, FE-74, FE-75
photon	A_M	the massless combination of W^3 and B	charged fermions, gravity	FE-24, FE-25, FE-50 to FE-52
W^+, W^-, Z	$W^\pm_M$, Z_M	the massive weak bosons	fermions, Σ, Ψ (I12), gravity	FE-23 to FE-25, FE-51, FE-55
no Higgs particle				
tip sigma field	Σ	a unitary 2×2 matrix field on the tip: the longitudinal $W^\pm$, Z	W, B, every fermion at the tip	FE-8, FE-26, FE-30
block stabiliser	ρ_{stab}	a vacuum energy of the block	gravity	FE-4, FE-43

The gauge fields of the action are the SU(3) field G, the SU(2) field W (components W^1, W^2, W^3) and the hypercharge field B; W^+, W^-, Z and the photon A are their physical combinations (IN-10). The right-handed neutrinos ν_R are gauge singlets: they give the neutrinos Dirac masses at the tip and are never thermalised.

1.6 Conventions of version 4

Version 4 changes the presentation of version 3, not its physics. Every symbol has one meaning in the whole document (the symbols of version 3 that had several meanings are replaced); Euler's number is e, the imaginary unit i, the differential d; a word or a particle name in a subscript is upright (M_{Pl}, m_{W}, n_{e}), an italic letter in a subscript is an index. Δ denotes a change (a variation, a first-order perturbation, a level shift), δ the Kronecker delta and δ_{D} the Dirac delta function. The gauge fields G, W, B are in the geometric normalisation (the couplings stand in front of their kinetic terms), in eight and in four dimensions; $W^\pm$, Z, A and $F^V_{\mu\nu}$ are canonically normalised. W^+ carries the electric charge $+e$ and W^- the charge $-e$. Two equations of version 3 are written more explicitly: the block components of the background equations (FE-40, FE-42, FE-43) show the pressure p_0 of the hidden direction and the pressure p_5 of the extra times separately (version 3 wrote only their average $\bar{p}$, and in one place p_0; section

"The background", subsection "The block pressures"), and the overlap integrals of the portals carry their factor $1/H$ explicitly (FE-70, FE-135). No number of version 3 changes.

2 The action

All field equations of UM-2 follow from one action by varying it with respect to every field. This section writes out every term of the action (IN-1 to IN-18) and says in plain words what it couples. The Standard-Model part has exactly the gauge group, the fermion content (three generations, each with a right-handed neutrino) and the gauge couplings of the Standard Model, but no Higgs field: the electroweak symmetry is broken at the tip of the hidden direction by a non-linear sigma field, and every mass comes from the dependence of the wave functions on x_0. dirac16complex is a singlet of the Standard-Model gauge group and couples to the Standard Model only through three gauge-invariant portals.

2.1 The sum of all terms

$$S = S_{\mathrm{EL}} + S_{\mathrm{stab}} + S_{\Psi} + S_{\mathrm{gauge}} + S_{\mathrm{f}} + S_{\mathrm{tip}} + S_{\mathrm{I11}} + S_{\mathrm{I12}} + S_{\mathrm{I13}} \tag{IN-1}$$

Symbols in (IN-1).

- S: The action of UM-2 (the integral of the Lagrangian density over spacetime); its variation with respect to every field gives the field equations. Unit: dimensionless ($\hbar = 1$). Status: ASSUMED.
- S_{EL}: The Einstein-Lovelock action of gravity. Unit: dimensionless. Status: AUTHOR.
- S_{stab}: The action of the block stabiliser (a vacuum energy of the block). Unit: dimensionless. Status: ASSUMED.
- S_{Ψ}: The action of the dirac16complex field (kinetic term with the canonical spin connection, mass term and contact self-interaction). Unit: dimensionless. Status: RECORD.
- S_{gauge}: The action of the gauge fields of SU(3), SU(2) and $\mathrm{U}(1)_Y$ in eight dimensions. Unit: dimensionless. Status: ASSUMED.
- S_{f}: The action of the Standard-Model fermions of three generations (kinetic and bulk-mass terms). Unit: dimensionless. Status: ASSUMED.
- S_{tip}: The action on the tip $y = -L$: the sigma field and the tip Yukawa couplings. Unit: dimensionless. Status: INVENTED.
- S_{I11}: The action of the portal I11 (dirac16complex vector current times the $B - L$ current of the fermions). Unit: dimensionless. Status: INVENTED.
- S_{I12}: The action of the portal I12 (scalar density of dirac16complex times the gauge kinetic invariants). Unit: dimensionless. Status: INVENTED.
- S_{I13}: The action of the portal I13 (the x_0-chiral vector current of dirac16complex times the baryon-number current). Unit: dimensionless. Status: INVENTED.

The action is the sum of nine terms: Einstein-Lovelock gravity (S_{EL}), the block stabiliser (S_{stab}), the field dirac16complex (S_{Ψ}), the gauge fields of SU(3), SU(2) and $\mathrm{U}(1)_Y$ (S_{gauge}), the Standard-Model fermions of three generations (S_{f}), the tip terms (the sigma field and the tip Yukawa couplings, S_{tip}) and the three portals of dirac16complex (S_{I11}, S_{I12}, S_{I13}). Varying S with respect to the metric gives the gravitational equations FE-1; with respect to $\bar{\Psi}$ the dirac16complex equation FE-12; with respect to every fermion its Dirac equation FE-14 to FE-19; with respect to the gauge fields the Yang-Mills equations FE-22 to FE-25; the boundary terms of the variation give the boundary conditions FE-27 to FE-30. There is no Higgs field. Status: ASSUMED (the choice of the terms).

2.2 Gravity

$$\begin{aligned} S_{\mathrm{EL}} &= \frac{1}{2\kappa}\int \mathrm{d}^8X\,\sqrt{|g|}\,\Big(\sum_{p=1}^{3}\alpha_{(p)}\,\mathcal{L}_{(p)} - 2\Lambda\Big), \qquad \kappa = 8\pi G_8, \\ \alpha_{(p)} &= w_{(p)}\,H^{2-2p}, \qquad w_{(1)} = 1, \quad w_{(2)} = \frac{1}{32}, \quad w_{(3)} = \frac{1}{1152}, \qquad \Lambda = -10H^2 \end{aligned} \tag{IN-2}$$

Symbols in (IN-2).

- S_{EL}: The Einstein-Lovelock action of gravity. Unit: dimensionless. Status: AUTHOR.

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_\mathrm{N}\mathcal{G}(K_*) = 2.328 \times 10^{-36}\ \mathrm{GeV}^{-2}$. Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)}H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- $\mathcal{L}_{(p)}$: The Lovelock density of order p of the metric (a scalar built from p Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -10H^2$. Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_8: The eight-dimensional Newton constant ($\kappa = 8\pi G_8$). Unit: GeV^{-6}. Status: AUTHOR.
- $w_{(p)}$: The author's Lovelock weights, $w_{(1)} = 1$, $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$. Unit: dimensionless. Value: $(1, 1/32, 1/1152)$. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.

Gravity is the Einstein-Lovelock theory in eight dimensions with the author's weights. $\mathcal{L}_{(1)}$ is the Einstein-Hilbert density (the Ricci scalar), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density and $\mathcal{L}_{(3)}$ the cubic Lovelock density; in eight dimensions the series ends with $p = 3$ (the density of order 4 is a total derivative in eight dimensions). The couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ carry the author's dimensionless weights $w_{(1)} = 1$, $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$, and the cosmological constant is $\Lambda = -10H^2$. Gravity couples to every field through the metric that appears in every other term of the action (in $\sqrt{|g|}$, in the contractions of indices and in the spin connection). Status: AUTHOR.

$$\begin{aligned} \mathcal{L}_{(p)} &= \frac{1}{2^p}\,\delta^{M_1N_1\cdots M_pN_p}_{P_1Q_1\cdots P_pQ_p}\,R^{P_1Q_1}{}_{M_1N_1}\cdots R^{P_pQ_p}{}_{M_pN_p}, \\ \mathcal{L}_{(1)} &= R, \qquad \mathcal{L}_{(2)} = R^2 - 4R_{MN}R^{MN} + R_{MNPQ}R^{MNPQ} \end{aligned} \tag{IN-3}$$

Symbols in (IN-3).

- $\mathcal{L}_{(p)}$: The Lovelock density of order p of the metric (a scalar built from p Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic). Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{\alpha e}$, δ_{l0}, δ^{I3}); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1\cdots M_n}_{N_1\cdots N_n}$, the determinant of the $n \times n$ matrix of Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Q: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor, with two indices the Ricci tensor $R_{MN} = R^{P}{}_{MPN}$, without indices the Ricci (scalar) curvature $R = g^{MN}R_{MN}$; $R^{ab}{}_{ab}$ (no sum) is the sectional curvature of the coordinate plane (a, b). Unit: 1/length^2. Status: STANDARD.

The Lovelock densities are built with the generalised Kronecker delta from p Riemann tensors. Their key property (Lovelock's theorem, STANDARD) is that their metric variation contains no derivatives of the metric higher than the second, so that the field equations FE-1 remain second-order differential equations, as in general relativity. For $p = 1$ the formula gives the Ricci scalar, for $p = 2$ the Gauss-Bonnet combination written in the second line. Status: STANDARD.

2.3 The block stabiliser

$$S_{\text{stab}} = -\int \mathrm{d}^8X\,\sqrt{|g|}\,\rho_{\text{stab}}(f), \qquad \kappa\rho_{\text{stab}}(f) = \kappa\rho_{\text{stab}}(f_*)\Big(\frac{f_*}{f}\Big)^8, \qquad f = \mathrm{e}^{-a_4}\,b_4^{1/2} \tag{IN-4}$$

Symbols in (IN-4).

- S_{stab}: The action of the block stabiliser (a vacuum energy of the block). Unit: dimensionless. Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- ρ_{stab}: The energy density of the block stabiliser, a function of the block scale, $\rho_{\text{stab}} \propto f^{-8}$; its value at the static block is $\rho_{\text{stab}}(f_*)$. Unit: GeV^8. Value: $\kappa\rho_{\text{stab}}(f_*) = (26 + 4\sqrt{51})H^2 = 54.566\,H^2$. Status: ASSUMED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_{\text{N}}\mathcal{G}(K_*) = 2.328 \times 10^{-36}$ GeV^{-2}. Status: AUTHOR.
- f_*: The block scale of the stabilised (static) block, $f_* = K_*^{-1/2}$. Unit: dimensionless. Value: 0.215556. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.

The four directions x_0, x_5, x_6, x_7 form the "block". The order-0 metric (FE-31) scales all four of them with one block scale $f = \mathrm{e}^{-a_4}b_4^{1/2}$. With the author's weights and Λ alone there is no static block with attractive four-dimensional gravity (FE-42); UM-2 therefore assumes a vacuum energy of the block, ρ_{stab}, that falls with the block scale as f^{-8} (the law of a Casimir-type energy of a block of size f). Its normalisation $\kappa\rho_{\text{stab}}(f_*)$ is fixed by the requirement that the block be static (FE-43). It couples only to gravity: its energy-momentum tensor is FE-4. Status: ASSUMED; its microscopic origin is OPEN.

2.4 The field dirac16complex

$$\begin{aligned} S_\Psi &= \int \mathrm{d}^8X\,\sqrt{|g|}\,\Big[\tfrac{1}{2}\big(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi\big) - m_\Psi\,\bar\Psi\Psi - \frac{\lambda}{2}\big(\bar\Psi\Psi\big)^2\Big], \\ m_\Psi &= 3H\mathcal{M} = H \qquad (\mathcal{M} = 1/3) \end{aligned} \tag{IN-5}$$

Symbols in (IN-5).

- S_Ψ: The action of the dirac16complex field (kinetic term with the canonical spin connection, mass term and contact self-interaction). Unit: dimensionless. Status: RECORD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- λ: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ of the record). Unit: GeV^{-6}. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm 7 \times 10^{-5} H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4 \times 10^{-10} H^{-6}$ (for $\Delta N_{\mathrm{eff}} \leq 0.001$). Status: RECORD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $\mathcal{M}$: The record's dimensionless mass parameter of dirac16complex, $\mathcal{M} = 1/3$ ($m_\Psi = 3H\mathcal{M}$). Unit: dimensionless. Value: 1/3. Status: RECORD.

dirac16complex Ψ is the author's 16-component complex spinor field of eight dimensions. Its Lagrangian is the record's: the symmetrised Dirac kinetic term (both Ψ and $\bar\Psi$ are differentiated, so the action is real), the mass term with the mass parameter $m_\Psi = 3H\mathcal{M} = H$, and the contact self-interaction $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$. The covariant derivative D_M contains the canonical spin connection (IN-7), through which Ψ couples to gravity. Ψ carries no colour, weak isospin or hypercharge. Status: RECORD.

$$\begin{aligned}
&\gamma^a\gamma^b + \gamma^b\gamma^a = 2\eta^{ab}\,\mathbf{1}, \qquad \eta_{ab} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1), \qquad \gamma^M = \mathbf{e}_a{}^M\gamma^a,\\
&\bar\Psi = \Psi^\dagger C, \qquad C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}, \qquad \Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}, \qquad \mathcal{B} = -\mathrm{i}C\gamma^{x_4},\\
&\Psi = \Psi_A + \Psi_B, \qquad \mathcal{B}\Psi_A = -\Psi_A, \qquad \mathcal{B}\Psi_B = +\Psi_B
\end{aligned} \tag{IN-6}$$

Symbols in (IN-6).

- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$, aligned with $x_0, \ldots, x_7$; raised and lowered with η_{ab}; a repeated index is summed (except in a sectional curvature $R^{ab}{}_{ab}$). Status: STANDARD.
- b: A frame index running over $0, \ldots, 7$ (as a). Status: STANDARD.
- η_{ab}: The flat metric of the frame, $\eta_{ab} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$ (η^{ab} the same matrix). Unit: dimensionless. Status: AUTHOR.

- **1**: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mathbf{e}^a{}_M$, $\mathbf{e}_a{}^M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M \mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is the inverse matrix. Unit: dimensionless. Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- C: The matrix of the Dirac adjoint, $C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}$ ($\bar{\Psi} = \Psi^\dagger C$); in each block of the reduction $C = \varsigma_2\sigma_2$. Unit: dimensionless. Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space (also the symmetry axis of the fast-flavour analysis). Unit: length. Status: AUTHOR.
- Γ_4: The four-dimensional chirality matrix, $\Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$; a doublet zero mode has $\Gamma_4 = +1$ (left-handed), a singlet zero mode $\Gamma_4 = -1$ (right-handed). Unit: dimensionless. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{B}$: The author's operator $\mathcal{B} = -\mathrm{i}C\gamma^{x_4}$; its eigenvalues ∓ 1 define the halves Ψ_A and Ψ_B, and it commutes with every interaction of the good sector. Unit: dimensionless. Status: AUTHOR.
- Ψ_A, Ψ_B: The two halves of dirac16complex, $\Psi = \Psi_A + \Psi_B$, the eigenspaces of $\mathcal{B}$ with the eigenvalues -1 and $+1$; each half carries one dark-matter species. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.

The Dirac matrices satisfy the Clifford algebra of the signature $(4,4)$; the author's matrices are real 16×16 matrices, and the algebra holds exactly (31 checks of `code/um2_algebra.py`). The Dirac adjoint uses the matrix $C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}$ (in four-four signature the usual γ^0 is replaced by the product of the four space-like matrices). Γ_4 is the four-dimensional chirality. The author's operator $\mathcal{B} = -\mathrm{i}C\gamma^{x_4}$ has the eigenvalues ± 1 and splits Ψ into two halves, Ψ_A and Ψ_B; it commutes with every interaction of the good sector, so the two halves never mix (each carries one dark-matter species). The same algebra and adjoint serve every Standard-Model fermion. Status: AUTHOR.

$$
\begin{aligned}
&D_M\Psi = \partial_M\Psi + \Omega_M\Psi, \qquad D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M, \qquad \Omega_M = \tfrac{1}{2}\,\omega_{Mab}\,\mathcal{S}^{ab}, \qquad \mathcal{S}^{ab} = \tfrac{1}{4}\big[\gamma^a,\gamma^b\big],\\
&\omega_M{}^a{}_b = \mathbf{e}^a{}_N\big(\partial_M \mathbf{e}_b{}^N + \Gamma^N{}_{MP}\,\mathbf{e}_b{}^P\big), \qquad \omega_{Mab} = \eta_{ac}\,\omega_M{}^c{}_b, \qquad g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\,\mathbf{e}^b{}_N,\\
&\Gamma^N{}_{MP} = \tfrac{1}{2}\,g^{NQ}\big(\partial_M g_{QP} + \partial_P g_{QM} - \partial_Q g_{MP}\big)
\end{aligned}
\tag{IN-7}
$$

Symbols in (IN-7).

- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.

- Ω_M: The spin connection acting on spinors, the matrix $\Omega_M = \frac{1}{2}\omega_{Mab}\mathcal{S}^{ab}$. Unit: 1/length. Status: STANDARD.
- ${\omega_M}^a{}_b$: The coefficients of the spin connection of the vielbein (the Levi-Civita connection in the frame basis); $\omega_{Mab} = \eta_{ac}\,{\omega_M}^c{}_b$. Unit: 1/length. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$, aligned with $x_0, \ldots, x_7$; raised and lowered with η_{ab}; a repeated index is summed (except in a sectional curvature $R^{ab}{}_{ab}$). Status: STANDARD.
- b: A frame index running over $0, \ldots, 7$ (as a). Status: STANDARD.
- $\mathcal{S}^{ab}$: The generators of frame rotations on spinors, $\mathcal{S}^{ab} = \frac{1}{4}[\gamma^a, \gamma^b]$ (the commutator). Unit: dimensionless. Status: STANDARD.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4, 4)$; $\gamma^M = {\mathbf{e}_a}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- ${\mathbf{e}^a}_M$, ${\mathbf{e}_a}^M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,{\mathbf{e}^a}_M{\mathbf{e}^b}_N$; ${\mathbf{e}_a}^M$ is the inverse matrix. Unit: dimensionless. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- ${\Gamma^N}_{MP}$: The Christoffel symbols (the Levi-Civita connection) of the metric, ${\Gamma^N}_{MP} = \frac{1}{2}g^{NQ}(\partial_M g_{QP} + \partial_P g_{QM} - \partial_Q g_{MP})$. Unit: 1/length. Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- η_{ab}: The flat metric of the frame, $\eta_{ab} = \mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$ (η^{ab} the same matrix). Unit: dimensionless. Status: AUTHOR.
- c: A frame index running over $0, \ldots, 7$ (as a). Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+, +, +, +, -, -, -, -$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, ${g^M}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- Q: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

The spin connection couples every spinor to gravity: Ω_M is built from the vielbein ${\mathbf{e}^a}_M$ of the metric and the Christoffel symbols, and it makes the Dirac matrices covariantly constant (verified symbolically). On the order-0 metric it is the non-zero term FE-13. Status: STANDARD.

2.5 The gauge fields

$$S_{\text{gauge}} = -\frac{1}{4}\int \mathrm{d}^8X\,\sqrt{|g|}\,\Big(\frac{1}{\hat{g}_3^2}\,G^A_{MN}G^{AMN} + \frac{1}{\hat{g}_2^2}\,W^I_{MN}W^{IMN} + \frac{1}{\hat{g}_1^2}\,B_{MN}B^{MN}\Big) \tag{IN-8}$$

Symbols in (IN-8).

- S_{gauge}: The action of the gauge fields of SU(3), SU(2) and $\mathrm{U}(1)_Y$ in eight dimensions. Unit: dimensionless. Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+, +, +, +, -, -, -, -$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, ${g^M}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $\hat{g}_3$: The eight-dimensional SU(3) coupling (it multiplies the gluon kinetic term as $1/\hat{g}_3^2$); $g_3^2 = \hat{g}_3^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- G^A_M, G_M, G_{MN}, G^A_{MN}: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components G^A_M ($A = 1, \ldots, 8$), the matrix $G_M = G^A_M t^A$, the field strength

G_{MN} (matrix) and its components G^A_{MN}; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in x_0). Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.

- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- W^I_M, W_M, W_{MN}, W^I_{MN}: The SU(2) gauge field (geometric normalisation): the components W^I_M ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength W_{MN} and its components W^I_{MN}; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- $\hat{g}_1$: The eight-dimensional $\mathrm{U}(1)_Y$ coupling; $g_1^2 = \hat{g}_1^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- B_M, B_{MN}: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.

The gauge group is that of the Standard Model, $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y$, in eight dimensions. The fields are in the geometric normalisation: the eight-dimensional couplings $\hat{g}_3$, $\hat{g}_2$, $\hat{g}_1$ stand in front of the kinetic terms, and the four-dimensional couplings are $g_i^2 = \hat{g}_i^2/V_\perp$ (FE-49). The gauge zero modes are flat in x_0 (FE-52), so the four-dimensional theory has exactly the gauge interactions of QCD and of the electroweak theory. Status: ASSUMED (the eight-dimensional form).

$$
\begin{aligned}
&G_M = G^A_M\, t^A, \qquad G_{MN} = \partial_M G_N - \partial_N G_M - \mathrm{i}\big[G_M, G_N\big], \qquad G^A_{MN} = 2\,\mathrm{tr}\big(t^A G_{MN}\big),\\
&W_M = W^I_M\,\frac{\tau^I}{2}, \qquad W_{MN} = \partial_M W_N - \partial_N W_M - \mathrm{i}\big[W_M, W_N\big], \qquad W^I_{MN} = \mathrm{tr}\big(\tau^I W_{MN}\big),\\
&B_{MN} = \partial_M B_N - \partial_N B_M
\end{aligned}
\tag{IN-9}
$$

Symbols in (IN-9).

- G^A_M, G_M, G_{MN}, G^A_{MN}: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components G^A_M ($A = 1, \ldots, 8$), the matrix $G_M = G^A_M t^A$, the field strength G_{MN} (matrix) and its components G^A_{MN}; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in x_0). Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- t^A: The generators of SU(3) in the fundamental representation (the Gell-Mann matrices divided by 2), $\mathrm{tr}(t^A t^B) = \frac{1}{2}\delta^{AB}$. Unit: dimensionless. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- W^I_M, W_M, W_{MN}, W^I_{MN}: The SU(2) gauge field (geometric normalisation): the components W^I_M ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength W_{MN} and its components W^I_{MN}; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- τ^I: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- B_M, B_{MN}: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.

The field strengths are the standard non-abelian ones (STANDARD): t^A are the generators of SU(3) and $\tau^I/2$ those of SU(2); the commutator terms make the gluons and the W bosons self-interacting, as in QCD and in the electroweak theory.

$$W_M^{\pm} = \frac{W_M^1 \mp \mathrm{i} W_M^2}{\sqrt{2}\,\hat{g}_2}, \qquad Z_M = \frac{W_M^3 - B_M}{\hat{g}_Z}, \qquad A_M = \frac{\hat{g}_1^2\, W_M^3 + \hat{g}_2^2\, B_M}{\hat{g}_1 \hat{g}_2 \hat{g}_Z},$$
$$\hat{g}_Z = \sqrt{\hat{g}_2^2 + \hat{g}_1^2}, \qquad \cos\theta_{\mathrm{W}} = \frac{\hat{g}_2}{\hat{g}_Z}, \qquad \sin\theta_{\mathrm{W}} = \frac{\hat{g}_1}{\hat{g}_Z} \tag{IN-10}$$

Symbols in (IN-10).

- $W_M^{\pm}$, W^+, W^-: The charged weak bosons, canonically normalised: $W_M^{\pm} = (W_M^1 \mp \mathrm{i} W_M^2)/(\sqrt{2}\,\hat{g}_2)$; W^+ carries the electric charge $+e$ and W^- the charge $-e$ (they are each other's antiparticles and have one mass). Unit: GeV^3. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- W_M^I, W_M, W_{MN}, W_{MN}^I: The SU(2) gauge field (geometric normalisation): the components W_M^I ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W_M^I \tau^I/2$, the field strength W_{MN} and its components W_{MN}^I; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- Z_M: The Z boson field, canonically normalised, $Z_M = (W_M^3 - B_M)/\hat{g}_Z$. Unit: GeV^3. Status: STANDARD.
- B_M, B_{MN}: The hypercharge gauge field U(1)$_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- $\hat{g}_Z$: The eight-dimensional coupling of the Z, $\hat{g}_Z = (\hat{g}_2^2 + \hat{g}_1^2)^{1/2}$. Unit: GeV^{-2}. Status: STANDARD.
- A_M: The photon field, canonically normalised, $A_M = (\hat{g}_1^2 W_M^3 + \hat{g}_2^2 B_M)/(\hat{g}_1 \hat{g}_2 \hat{g}_Z)$. Unit: GeV^3. Status: STANDARD.
- $\hat{g}_1$: The eight-dimensional U(1)$_Y$ coupling; $g_1^2 = \hat{g}_1^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- θ_{W}: The weak mixing angle of the couplings, $\cos\theta_{\mathrm{W}} = \hat{g}_2/\hat{g}_Z$, $\sin\theta_{\mathrm{W}} = \hat{g}_1/\hat{g}_Z$; fixed by the measured W and Z masses with $\rho_{\mathrm{EW},0}$. Unit: dimensionless. Value: $\cos^2\theta_{\mathrm{W}} = 0.776368$. Status: DERIVED.

The physical electroweak fields are the canonically normalised combinations of the SU(2) and hypercharge fields: W^+ and W^- from W^1 and W^2, the Z and the photon A from W^3 and B, rotated by the weak mixing angle θ_{W} of the couplings. W^+ carries the electric charge $+e$ and W^- the charge $-e$ (DERIVED from the charge operator of IN-12); they are each other's antiparticles. Status: STANDARD.

2.6 The Standard-Model fermions

$$S_{\mathrm{f}} = \sum_{i=1}^{3} \sum_{\psi} \int \mathrm{d}^8 X\, \sqrt{|g|}\, \Big[\tfrac{1}{2} \big(\bar{\psi}_i \gamma^M D_M \psi_i - (D_M \bar{\psi}_i) \gamma^M \psi_i \big) - \mu_{\psi i}\, \bar{\psi}_i \psi_i \Big],$$
$$\psi \in \{Q,\ u_R,\ d_R,\ \ell,\ e_R,\ \nu_R\}, \qquad \mu_{\psi i} = \mu_Q\ (\psi = Q), \qquad \mu_{\psi i} = \mu_\ell\ (\psi = \ell) \tag{IN-11}$$

Symbols in (IN-11).

- S_{f}: The action of the Standard-Model fermions of three generations (kinetic and bulk-mass terms). Unit: dimensionless. Status: ASSUMED.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- ψ: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over ψ runs over the six. Status: STANDARD.
- d: The differential: dX is an infinitesimal change of X, dX/dY the derivative of X with respect to Y, d$^n X$ the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8 X = \mathrm{d}x_0\, \mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate

volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- ψ_i, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: ψ_i is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation i, ψ_f one fermion component f (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- $\mu_{\psi i}$: The bulk mass of a Standard-Model fermion: $\mu_{\psi i}$ of the multiplet ψ of the generation i; with the hidden direction it fixes the localisation $c = f\mu L$ of the zero mode. Unit: GeV. Value: $\mu_{\psi i} = c/(fL)$ with the localisation c of the zero mode (doublets: $\mu_Q = 5.2304\,H$, $\mu_\ell = 9.2055\,H$). Status: ASSUMED.
- Q_i, $\bar{Q}_i$: The quark doublet of the generation i (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- u_R, u_{Ri}, $\bar{u}_{Ri}$: The up-type quark singlet: u_R the multiplet, u_{Ri} that of the generation i (u_R, c_R, t_R), $\bar{u}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- d_R, d_{Ri}, $\bar{d}_{Ri}$: The down-type quark singlet: d_R the multiplet, d_{Ri} that of the generation i (d_R, s_R, b_R), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- ℓ_i, $\bar{\ell}_i$: The lepton doublet of the generation i (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- e_R, e_{Ri}: The right-handed charged-lepton singlet: e_R the multiplet, e_{Ri} that of the generation i (e_R, μ_R, τ_R). Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- ν_R, ν_{Ri}: The right-handed neutrino (a gauge singlet): ν_R the multiplet, ν_{Ri} that of the generation i; its zero modes give Dirac neutrino masses. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- μ_Q: The common bulk mass of the three quark doublets. Unit: GeV. Value: $5.2304\,H$ ($c_Q = 3.38230$). Status: DERIVED.
- μ_ℓ: The common bulk mass of the three lepton doublets. Unit: GeV. Value: $9.2055\,H$ ($c_\ell = 5.95288$). Status: DERIVED.

Each generation $i = 1, 2, 3$ contains the quark doublet Q_i (left-handed up and down quarks, three colours, hypercharge $\frac{1}{6}$), the up-type singlet u_{Ri} ($\frac{2}{3}$), the down-type singlet d_{Ri} ($-\frac{1}{3}$), the lepton doublet ℓ_i (left-handed neutrino and charged lepton, $-\frac{1}{2}$), the charged-lepton singlet e_{Ri} (-1) and the right-handed neutrino ν_{Ri} (a gauge singlet). These are eight-dimensional spinor fields with bulk masses $\mu_{\psi i}$; the doublets of a sector share one bulk mass (μ_Q, μ_ℓ). The bulk mass fixes the localisation of the four-dimensional zero mode along x_0, and with the tip mixing it fixes the four-dimensional mass (FE-56 to FE-60). With this content the gauge anomalies cancel generation by generation, exactly as in the Standard Model (DERIVED, exact fractions; a lepton generation without its quark generation would leave the anomalies $\mathcal{A}_{SU(2)^2U(1)} = -1$ and $\mathcal{A}_{U(1)^3} = 3/2$): the three quark generations are required together with the three lepton generations. Status: ASSUMED (the eight-dimensional form; the boundary conditions FE-29 are INVENTED).

$$D_M = \partial_M + \Omega_M - \mathrm{i}G^A_M\, t^A - \mathrm{i}W^I_M\, \frac{\tau^I}{2} - \mathrm{i}\, Y B_M, \qquad Q_{\mathrm{em}} = T_3 + Y \tag{IN-12}$$

Symbols in (IN-12).

- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.

- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- Ω_M: The spin connection acting on spinors, the matrix $\Omega_M = \frac{1}{2}\omega_{Mab}\mathcal{S}^{ab}$. Unit: 1/length. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- G^A_M, G_M, G_{MN}, G^A_{MN}: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components G^A_M ($A = 1, \ldots, 8$), the matrix $G_M = G^A_M t^A$, the field strength G_{MN} (matrix) and its components G^A_{MN}; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in x_0). Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- t^A: The generators of SU(3) in the fundamental representation (the Gell-Mann matrices divided by 2), $\mathrm{tr}(t^A t^B) = \frac{1}{2}\delta^{AB}$. Unit: dimensionless. Status: STANDARD.
- W^I_M, W_M, W_{MN}, W^I_{MN}: The SU(2) gauge field (geometric normalisation): the components W^I_M ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength W_{MN} and its components W^I_{MN}; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- τ^I: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- Y: The hypercharge (the generator of $\mathrm{U}(1)_Y$); Y_ψ the hypercharge of the multiplet ψ (Q: $\frac{1}{6}$, u_R: $\frac{2}{3}$, d_R: $-\frac{1}{3}$, ℓ: $-\frac{1}{2}$, e_R: -1, ν_R: 0). Unit: dimensionless. Status: STANDARD.
- B_M, B_{MN}: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- Q_{em}: The electric charge in units of the elementary charge, $Q_{\mathrm{em}} = T_3 + Y$; Q^f_{em} that of the fermion f. Unit: dimensionless. Status: STANDARD.
- T_3: The third component of the weak isospin ($+\frac{1}{2}$ for the upper and $-\frac{1}{2}$ for the lower member of a doublet, 0 for a singlet); T^f_3 that of the fermion f. Unit: dimensionless. Status: STANDARD.

Every fermion couples to gravity through the spin connection and to the gauge fields through its covariant derivative, with the generators of its representation (t^A for quarks, $\tau^I/2$ for doublets, the hypercharge Y). The electric charge is $Q_{\mathrm{em}} = T_3 + Y$: $+\frac{2}{3}$ and $-\frac{1}{3}$ for the quarks, 0 and -1 for the leptons. Status: STANDARD.

2.7 The tip: the sigma field and the Yukawa couplings (no Higgs particle)

$$
\begin{aligned}
S_{\mathrm{tip}} &= \int_{y=-L} \mathrm{d}^7 X_{\mathrm{tip}} \sqrt{|g_{\mathrm{tip}}|}\, \Big[-\frac{v_\Sigma^2}{4}\, \mathrm{tr}\big((D_\mu \Sigma)^\dagger D^\mu \Sigma\big) - \mathcal{L}_{\mathrm{Y}} \Big], \\
\mathcal{L}_{\mathrm{Y}} &= \sum_{i,j=1}^{3} \Big[\bar{\ell}_i\, \Sigma \begin{pmatrix} (\mathcal{Y}_\nu)_{ij}\, \nu_{Rj} \\ (\mathcal{Y}_e)_{ij}\, e_{Rj} \end{pmatrix} + \bar{Q}_i\, \Sigma \begin{pmatrix} (\mathcal{Y}_u)_{ij}\, u_{Rj} \\ (\mathcal{Y}_d)_{ij}\, d_{Rj} \end{pmatrix} \Big] + \mathrm{h.c.}
\end{aligned}
\qquad \text{(IN-13)}
$$

Symbols in (IN-13).

- S_{tip}: The action on the tip $y = -L$: the sigma field and the tip Yukawa couplings. Unit: dimensionless. Status: INVENTED.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- X_{tip}: The seven coordinates $x_1, \ldots, x_7$ of the tip $y = -L$ of the hidden direction; $\mathrm{d}^7 X_{\mathrm{tip}}$ is their coordinate volume element. Unit: length (each coordinate). Status: DERIVED.
- g_{tip}: The metric induced on the seven-dimensional tip $y = -L$ (the metric g_{MN} restricted to the tip directions $x_1, \ldots, x_7$); $|g_{\mathrm{tip}}|$ is the absolute value of its determinant. Unit: dimensionless. Status: DERIVED.

- v_Σ: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$); on the static block it fixes the Robin parameter β_{W}. Unit: $\mathrm{GeV}^{5/2}$. Value: fixed through $\beta_{\mathrm{W}} = 0.011912$. Status: INVENTED.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- Σ: The tip sigma field: a unitary 2×2 matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal W^+, W^- and Z, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathcal{L}_{\mathrm{Y}}$: The tip Yukawa Lagrangian: the couplings of the doublets to the singlets through the sigma field and the tip matrices $\mathcal{Y}$ (plus the Hermitian conjugate, h.c.). Unit: GeV^7. Status: INVENTED.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- j: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- ℓ_i, $\bar{\ell}_i$: The lepton doublet of the generation i (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\mathcal{Y}_\nu$, $\mathcal{Y}_e$, $\mathcal{Y}_u$, $\mathcal{Y}_d$: The complex 3×3 tip Yukawa matrices of the neutrino, charged-lepton, up-quark and down-quark sectors (generation indices i, j); on the static block they become the tip mixing matrices $\mathcal{K}$ (the normalisation of that relation is not fixed: OPEN). Unit: dimensionless. Status: INVENTED.
- ν_R, ν_{Ri}: The right-handed neutrino (a gauge singlet): ν_R the multiplet, ν_{Ri} that of the generation i; its zero modes give Dirac neutrino masses. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- e_R, e_{Ri}: The right-handed charged-lepton singlet: e_R the multiplet, e_{Ri} that of the generation i (e_R, μ_R, τ_R). Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- Q_i, $\bar{Q}_i$: The quark doublet of the generation i (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- u_R, u_{Ri}, $\bar{u}_{Ri}$: The up-type quark singlet: u_R the multiplet, u_{Ri} that of the generation i (u_R, c_R, t_R), $\bar{u}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- d_R, d_{Ri}, $\bar{d}_{Ri}$: The down-type quark singlet: d_R the multiplet, d_{Ri} that of the generation i (d_R, s_R, b_R), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.

At the regular tip $y = -L$ of the hidden direction, a non-linear sigma field Σ (a unitary 2×2 matrix) breaks the electroweak symmetry. Its kinetic term is the gauged non-linear sigma model of the electroweak chiral Lagrangian, located on the seven-dimensional tip: it gives masses to W^+, W^- and Z and none to the photon (IN-15). Σ has no radial mode, so there is no Higgs particle; its three modes become the longitudinal W^+, W^- and Z. The Yukawa terms $\mathcal{L}_{\mathrm{Y}}$ couple the doublets to the singlets through Σ with complex 3×3 matrices $\mathcal{Y}_\nu$, $\mathcal{Y}_e$, $\mathcal{Y}_u$, $\mathcal{Y}_d$ (generation indices i, j); they mix the doublet and singlet zero modes at the tip and give the fermions their masses and their mixing (FE-30, FE-59, FE-60). Every term is invariant under $\mathrm{SU}(2)\times\mathrm{U}(1)_Y$ (IN-14), so the construction is consistent with the electroweak theory at tree level; it is the Higgsless limit of the Standard Model, which conflicts with the observed Higgs boson at 125 GeV and lets the longitudinal W^+W^- scattering violate perturbative unitarity near 1.2 TeV (a conflict with data inherent in the order to omit the Higgs particle). Status: INVENTED.

$$D_\mu\Sigma = \partial_\mu\Sigma - \mathrm{i}W_\mu^I\frac{\tau^I}{2}\Sigma + \mathrm{i}B_\mu\Sigma\frac{\tau^3}{2}, \qquad \Sigma^\dagger\Sigma = \mathbf{1}, \qquad \Sigma \to U_L\,\Sigma\,U_Y^\dagger, \qquad U_Y = \mathrm{e}^{\mathrm{i}\vartheta\tau^3/2} \tag{IN-14}$$

Symbols in (IN-14).

- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.

- Σ: The tip sigma field: a unitary 2×2 matrix field on the tip ($\Sigma^\dagger \Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal W^+, W^- and Z, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- W^I_M, W_M, W_{MN}, W^I_{MN}: The SU(2) gauge field (geometric normalisation): the components W^I_M ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength W_{MN} and its components W^I_{MN}; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- τ^I: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- B_M, B_{MN}: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- U_L: An SU(2) gauge transformation (a unitary 2×2 matrix acting on doublets from the left). Unit: dimensionless. Status: STANDARD.
- U_Y: A $\mathrm{U}(1)_Y$ gauge transformation of the sigma field, $U_Y = \mathrm{e}^{\mathrm{i}\vartheta\tau^3/2}$. Unit: dimensionless. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- ϑ: The angle (gauge parameter) of a $\mathrm{U}(1)_Y$ transformation. Unit: dimensionless. Status: STANDARD.

The covariant derivative of Σ contains the SU(2) field from the left and the hypercharge field from the right, so that Σ transforms as $\Sigma \to U_L \, \Sigma \, U_Y^\dagger$. The tip terms are invariant under this transformation for arbitrary complex tip matrices (DERIVED with random group elements, deviation 1.1×10^{-14}). Status: INVENTED.

$$-\frac{v_\Sigma^2}{4} \operatorname{tr}\big((D_\mu \Sigma)^\dagger D^\mu \Sigma\big)\Big|_{\Sigma=\mathbf{1}} = -\frac{v_\Sigma^2}{8}\Big[2\hat{g}_2^2 \, W^+_\mu W^{-\mu} + \hat{g}_Z^2 \, Z_\mu Z^\mu\Big] \tag{IN-15}$$

Symbols in (IN-15).

- v_Σ: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$); on the static block it fixes the Robin parameter β_{W}. Unit: $\mathrm{GeV}^{5/2}$. Value: fixed through $\beta_{\mathrm{W}} = 0.011912$. Status: INVENTED.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+, +, +, -$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- Σ: The tip sigma field: a unitary 2×2 matrix field on the tip ($\Sigma^\dagger \Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal W^+, W^- and Z, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- $W^\pm_M$, W^+, W^-: The charged weak bosons, canonically normalised: $W^\pm_M = (W^1_M \mp \mathrm{i}W^2_M)/(\sqrt{2}\,\hat{g}_2)$; W^+ carries the electric charge $+e$ and W^- the charge $-e$ (they are each other's antiparticles and have one mass). Unit: GeV^3. Status: STANDARD.
- $\hat{g}_Z$: The eight-dimensional coupling of the Z, $\hat{g}_Z = (\hat{g}_2^2 + \hat{g}_1^2)^{1/2}$. Unit: GeV^{-2}. Status: STANDARD.

- Z_M: The Z boson field, canonically normalised, $Z_M = (W^3_M - B_M)/\hat{g}_Z$. Unit: GeV^3. Status: STANDARD.

In the unitary gauge $\Sigma = \mathbf{1}$ the sigma-field kinetic term becomes a mass term of W^+, W^- and Z on the tip. W^+ and W^- share one mass (particle and antiparticle); the Z term has the coefficient $\hat{g}_Z^2 = \hat{g}_2^2 + \hat{g}_1^2$ instead of $\hat{g}_2^2$ (at leading order the Z is heavier by $1/\cos\theta_{\mathrm{W}}$, the custodial relation); the photon receives no term and stays massless. On the static block this term becomes the Robin condition of the $W^\pm$ and Z profiles (FE-51). Status: DERIVED.

2.8 The portals of dirac16complex

Ψ is a singlet of the Standard-Model gauge group, so it can couple to the Standard Model only through gauge-invariant composite operators of the Standard-Model fields. UM-2 contains three of them; they respect every symmetry of QCD and of the electroweak theory, and none of them breaks a gauge symmetry or gives a gauge boson a mass. Their strengths are fixed by the cosmology (the relic density of the dark matter, the freeze-in of the fold bands, partial-wave unitarity).

$$S_{\mathrm{I11}} = \int \mathrm{d}^8X\,\sqrt{|g|}\,\frac{c_{\mathrm{V}}}{\Lambda_\star^6}\left(\mathrm{i}\bar{\Psi}\gamma^M\Psi\right)J^{B-L}_M, \qquad J^{B-L}_M = \sum_{i=1}^{3}\sum_{\psi} q_\psi\left(\mathrm{i}\bar{\psi}_i\gamma_M\psi_i\right),$$
$$q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}, \qquad q_\ell = q_{e_R} = q_{\nu_R} = -1 \tag{IN-16}$$

Symbols in (IN-16).

- S_{I11}: The action of the portal I11 (dirac16complex vector current times the $B-L$ current of the fermions). Unit: dimensionless. Status: INVENTED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, ${g^M}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- c_{V}: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_{\mathrm{V}} = 4.38\times10^{-17}$ GeV^{-2}. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: GeV$^{7/2}$. Status: RECORD.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = {\mathbf{e}_a}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- J^{B-L}_M: The $B-L$ current of the Standard-Model fermions (baryon minus lepton number), summed over the generations. Unit: GeV^7. Status: STANDARD.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- ψ: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over ψ runs over the six. Status: STANDARD.
- q_ψ, q_f: The $B-L$ charge: q_ψ of the multiplet ψ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), q_f of the fermion component f. Unit: dimensionless. Status: STANDARD.

- ψ_i, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: ψ_i is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation i, ψ_f one fermion component f (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.

I11 couples the vector current of Ψ (both halves, including the massless fold bands) to the $B - L$ current of the Standard-Model fermions of three generations: baryon number minus lepton number, with the charges q_ψ. $B - L$ is conserved in the Standard Model and, with the right-handed neutrinos, free of gauge anomalies, so I11 is consistent with QCD and the electroweak theory; every term is separately gauge invariant and generation blind. c_V is dimensionless and $\Lambda_\star$ is the common scale of the portals. On the zero modes it becomes the contact interaction FE-70; its benchmark strength keeps the fold bands out of equilibrium (FE-129). Status: INVENTED.

$$S_{\mathrm{I12}} = -\int \mathrm{d}^8X\,\sqrt{|g|}\,\frac{\bar{\Psi}\Psi}{\Lambda_\star^3}\,\mathcal{F}, \qquad \mathcal{F} = c_\mathrm{G}\,G^A_{MN}G^{AMN} + c_\mathrm{W}\,W^I_{MN}W^{IMN} + c_\mathrm{B}\,B_{MN}B^{MN} \tag{IN-17}$$

Symbols in (IN-17).

- S_{I12}: The action of the portal I12 (scalar density of dirac16complex times the gauge kinetic invariants). Unit: dimensionless. Status: INVENTED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- Ψ, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $\mathcal{F}$: The gauge-kinetic operator of the portal I12, $\mathcal{F} = c_\mathrm{G}G^A_{MN}G^{AMN} + c_\mathrm{W}W^I_{MN}W^{IMN} + c_\mathrm{B}B_{MN}B^{MN}$; $\langle\mathcal{F}\rangle$ its mean value in an environment. Unit: GeV^4. Status: INVENTED.
- c_G: The coefficient of the gluon term of the portal I12. Unit: dimensionless. Value: $c_\mathrm{G}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- G^A_M, G_M, G_{MN}, G^A_{MN}: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components G^A_M ($A = 1, \ldots, 8$), the matrix $G_M = G^A_M t^A$, the field strength G_{MN} (matrix) and its components G^A_{MN}; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in x_0). Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- c_W: The coefficient of the SU(2) term of the portal I12. Unit: dimensionless. Value: $c_\mathrm{W}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- W^I_M, W_M, W_{MN}, W^I_{MN}: The SU(2) gauge field (geometric normalisation): the components W^I_M ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W^I_M\tau^I/2$, the field strength W_{MN} and its components W^I_{MN}; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c_B: The coefficient of the hypercharge term of the portal I12. Unit: dimensionless. Value: $c_\mathrm{B}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- B_M, B_{MN}: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.

I12 couples the scalar density $\bar{\Psi}\Psi$ of dirac16complex to the gauge-kinetic invariants of the three gauge groups. Each invariant is gauge invariant and a Lorentz scalar; the coupling acts as a field-dependent gauge coupling (FE-21) and does not break any symmetry of QCD or of the electroweak theory. On the dark matter it is FE-71; it is set to the unitarity-safe benchmark. Status: INVENTED.

$$S_{\mathrm{I13}} = \int \mathrm{d}^8X\,\sqrt{|g|}\,\frac{c_{\mathrm{A}}}{\Lambda_\star^6}\sum_{M\neq x_0}\left(\mathrm{i}\bar{\Psi}\gamma^M P_b\Psi\right)J^B_M, \qquad P_b=\frac{1-\gamma^{x_0}}{2}, \qquad P_a=\frac{1+\gamma^{x_0}}{2},$$
$$J^B_M=\frac{1}{3}\sum_{i=1}^{3}\left(\mathrm{i}\bar{Q}_i\gamma_M Q_i+\mathrm{i}\bar{u}_{Ri}\gamma_M u_{Ri}+\mathrm{i}\bar{d}_{Ri}\gamma_M d_{Ri}\right) \tag{IN-18}$$

Symbols in (IN-18).

- S_{I13}: The action of the portal I13 (the x_0-chiral vector current of dirac16complex times the baryon-number current). Unit: dimensionless. Status: INVENTED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X=\{x_0,\ldots,x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X=\{x_0,x_1,\ldots,x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X=\mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1,\ldots,x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, ${g^M}_N=\delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- c_{A}: The coefficient of the x_0-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_{\mathrm{A}}=1.278\times10^{-9}$ GeV^{-2} (dark-matter relic). Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2,\ldots$): it runs over $x_0,x_1,\ldots,x_7$; a repeated index is summed. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z=6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi}=\Psi^\dagger C$ is its Dirac adjoint. Unit: GeV$^{7/2}$. Status: RECORD.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M={\mathbf{e}_a}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M=g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- P_a, P_b: The projectors on the two x_0-chiralities, $P_a=(1+\gamma^{x_0})/2$ and $P_b=(1-\gamma^{x_0})/2$ (in a block: the upper and the lower component). Unit: dimensionless. Status: INVENTED.
- J^B_M: The baryon-number current of the quarks (one third per quark), summed over the generations. Unit: GeV^7. Status: STANDARD.
- i: The generation index ($i=1,2,3$), which also labels the neutrino mass states ν_1,ν_2,ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- Q_i, $\bar{Q}_i$: The quark doublet of the generation i (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: GeV$^{7/2}$. Status: ASSUMED.
- u_R, u_{Ri}, $\bar{u}_{Ri}$: The up-type quark singlet: u_R the multiplet, u_{Ri} that of the generation i (u_R, c_R, t_R), $\bar{u}_{Ri}$ its Dirac adjoint. Unit: GeV$^{7/2}$. Status: ASSUMED.
- d_R, d_{Ri}, $\bar{d}_{Ri}$: The down-type quark singlet: d_R the multiplet, d_{Ri} that of the generation i (d_R, s_R, b_R), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: GeV$^{7/2}$. Status: ASSUMED.

I13 couples the x_0-chiral vector current of Ψ (the projector P_b selects one x_0-chirality) to the baryon-number current of the quarks (one third per quark, three generations). The sum leaves out the direction x_0, for which the bilinear is not Hermitian (DERIVED with the author's matrices: $\mathrm{i}\bar{\Psi}\gamma^M P_b\Psi$ is Hermitian for $M\neq x_0$ only).

Baryon number is conserved in QCD and in the perturbative electroweak theory, and J^B_M is gauge invariant, so I13 is consistent with both. On the zero modes it is the contact interaction FE-70, whose strength is fixed by the relic density of the dark matter (FE-126); it vanishes on the fold bands (to 7×10^{-14}), which have no lower component. Status: INVENTED.

3 Gravity

3.1 The Einstein-Lovelock equations

Varying the gravitational action IN-2 with respect to the inverse metric gives, up to a total derivative,

$$\Delta S_{\mathrm{EL}} = \frac{1}{2\kappa}\int \mathrm{d}^8X\,\sqrt{|g|}\,\Big(\sum_{p=1}^{3}\alpha_{(p)}\,\mathcal{E}_{(p)MN}+\Lambda\,g_{MN}\Big)\Delta g^{MN},$$

Symbols in this equation.

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21}=m^2_2-m^2_1$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- S_{EL}: The Einstein-Lovelock action of gravity. Unit: dimensionless. Status: AUTHOR.
- κ: The eight-dimensional gravitational coupling, $\kappa=8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp=8\pi G_{\mathrm{N}}\mathcal{G}(K_*)=2.328\times10^{-36}\ \mathrm{GeV}^{-2}$. Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X=\{x_0,\dots,x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X=\{x_0,x_1,\dots,x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X=\mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1,\dots,x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\dots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, ${g^M}_N=\delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- p: The Lovelock order $p=1,2,3$ (Einstein, Gauss-Bonnet, cubic). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)}=w_{(p)}H^{2-2p}$ ($\alpha_{(1)}=1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- ${\mathcal{E}_{(p)}}^M{}_N$: The Lovelock tensor of order p, the metric variation of $\sqrt{|g|}\mathcal{L}_{(p)}$: $\mathcal{E}_{(1)}$ is the Einstein tensor ${R^M}_N-\frac12\delta^M_N R$; every $\mathcal{E}_{(p)}$ is divergence free. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1,M_2,\dots$): it runs over $x_0,x_1,\dots,x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1,N_2,\dots$), running over $x_0,\dots,x_7$; a repeated index is summed. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda=-10H^2$. Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.

because the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ is $\sqrt{|g|}\,\mathcal{E}_{(p)MN}\Delta g^{MN}$ (the defining property of the Lovelock tensor $\mathcal{E}_{(p)}$, STANDARD) and the variation of $\sqrt{|g|}$ is $-\frac12\sqrt{|g|}\,g_{MN}\Delta g^{MN}$. The variation of the rest of the action defines the energy-momentum tensor (FE-2), $\Delta S_{\mathrm{m}}=-\frac12\int\mathrm{d}^8X\sqrt{|g|}\,\mathcal{T}_{MN}\Delta g^{MN}$; the sum of the two vanishes for every Δg^{MN}:

$$\begin{aligned}&\sum_{p=1}^{3}\alpha_{(p)}\,{\mathcal{E}_{(p)}}^M{}_N+\Lambda\,\delta^M_N=\kappa\,{\mathcal{T}}^M{}_N,\\ &{\mathcal{E}_{(p)}}^M{}_N=-\frac{1}{2^{p+1}}\,\delta^{MM_1N_1\cdots M_pN_p}_{NP_1Q_1\cdots P_pQ_p}\,{R^{P_1Q_1}}_{M_1N_1}\cdots{R^{P_pQ_p}}_{M_pN_p}\end{aligned}\qquad\text{(FE-1)}$$

Symbols in (FE-1).

- p: The Lovelock order $p=1,2,3$ (Einstein, Gauss-Bonnet, cubic). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)}=w_{(p)}H^{2-2p}$ ($\alpha_{(1)}=1$). Unit: GeV^{2-2p}. Status: AUTHOR.

- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order p, the metric variation of $\sqrt{|g|}\mathcal{L}_{(p)}$: $\mathcal{E}_{(1)}$ is the Einstein tensor $R^{M}{}_{N} - \frac{1}{2}\delta^{M}_{N}R$; every $\mathcal{E}_{(p)}$ is divergence free. Unit: $1/\text{length}^{2p}$. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -10H^2$. Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.
- δ: The Kronecker delta: with two indices $\delta^{M}_{N} = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{\alpha e}$, δ_{l0}, δ^{I3}); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1\cdots M_n}_{N_1\cdots N_n}$, the determinant of the $n \times n$ matrix of Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_{\text{N}}\mathcal{G}(K_*) = 2.328 \times 10^{-36}\ \text{GeV}^{-2}$. Status: AUTHOR.
- $\mathcal{T}^{M}{}_{N}$: The energy-momentum tensor of all matter (every field except gravity, the stabiliser included), the source of gravity; defined by the change of the matter action under a change of the metric. Unit: GeV^8. Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Q: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor, with two indices the Ricci tensor $R_{MN} = R^{P}{}_{MPN}$, without indices the Ricci (scalar) curvature $R = g^{MN}R_{MN}$; $R^{ab}{}_{ab}$ (no sum) is the sectional curvature of the coordinate plane (a, b). Unit: $1/\text{length}^2$. Status: STANDARD.

The terms. $\mathcal{E}_{(1)}$ is the Einstein tensor $R^{M}{}_{N} - \frac{1}{2}\delta^{M}_{N}R$ of general relativity; $\mathcal{E}_{(2)}$ (Gauss-Bonnet) and $\mathcal{E}_{(3)}$ (cubic Lovelock) are the higher-order Lovelock tensors, products of two and three curvature tensors contracted with the generalised Kronecker delta, weighted by $\alpha_{(2)} = H^{-2}/32$ and $\alpha_{(3)} = H^{-4}/1152$. $\Lambda\delta^{M}_{N}$ is the eight-dimensional cosmological constant. The right side is the energy-momentum tensor of every field (FE-3), multiplied by the gravitational coupling $\kappa = 8\pi G_8$. These equations couple the metric to every field of UM-2. Each $\mathcal{E}_{(p)}$ contains at most second derivatives of the metric, so the system is of second order. Status: AUTHOR (the weights and Λ), the equations DERIVED from the action. They are solved exactly at order 0 (FE-38 to FE-43: the residual of the stored static block in the printed equations is 3.9×10^{-13} of the largest term) and at the first order through the four-dimensional reduction (FE-48, FE-49).

3.2 The energy-momentum tensors

$$\begin{aligned}
&\Delta S_{\text{m}} = -\frac{1}{2}\int \mathrm{d}^8X\,\sqrt{|g|}\,\mathcal{T}_{MN}\,\Delta g^{MN}, \qquad \rho = -\mathcal{T}^{x_4}{}_{x_4}, \qquad p_M = \mathcal{T}^{M}{}_{M} \quad (M \neq x_4, \text{ no sum}),\\
&\mathcal{T}^{M}{}_{N} = \operatorname{diag}\big(p_0,\ p_3,\ p_3,\ p_3,\ -\rho,\ p_5,\ p_5,\ p_5\big) \quad (\text{a diagonal source}), \qquad \bar{p} = \frac{p_0 + 3p_5}{4}
\end{aligned} \tag{FE-2}$$

Symbols in (FE-2).

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21} = m^2_2 - m^2_1$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- S_{m}: The matter action: all terms of the action except S_{EL} (the stabiliser included). Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^{M}{}_{N} = \delta^{M}_{N}$, and $|g|$ the absolute value

of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{T}^M{}_N$: The energy-momentum tensor of all matter (every field except gravity, the stabiliser included), the source of gravity; defined by the change of the matter action under a change of the metric. Unit: GeV^8. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- p: A pressure: $p_M = \mathcal{T}^M{}_M$ (no sum) for the direction $M \neq x_4$, p_s the pressure of the species s; Δp the pressure perturbation in the cosmological perturbation equations; in the Feynman-Hellmann relation the pressure of a gauge plasma. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- p_0: The pressure along the hidden direction x_0, $p_0 = \mathcal{T}^{x_0}{}_{x_0}$. Unit: GeV^8. Status: STANDARD.
- p_3: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: GeV^8. Status: STANDARD.
- p_5: The pressure along the extra times, $p_5 = \mathcal{T}^{x_5}{}_{x_5} = \mathcal{T}^{x_6}{}_{x_6} = \mathcal{T}^{x_7}{}_{x_7}$. Unit: GeV^8. Status: STANDARD.
- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial \ln f$. Unit: GeV^8 (or GeV^4 for a four-dimensional density). Status: DERIVED.

The terms. $\mathcal{T}_{MN}$ is the response of the matter action S_m (every term of IN-1 except S_EL) to a change Δg^{MN} of the metric (for spinors the change is made in the vielbein, which carries the metric). Its time-time component gives the energy density ρ and its diagonal space-like components the pressures p_M along each direction: p_0 along the hidden direction x_0, p_3 along ordinary space, p_5 along the extra times. Every source of UM-2 is diagonal on the order-0 metric, with one pressure for x_1, x_2, x_3 and one for x_5, x_6, x_7; $\bar{p}$ is the pressure averaged over the four block directions, the hidden pressure. Status: STANDARD.

$$\mathcal{T}^M{}_N = \mathcal{T}_\mathrm{stab}{}^M{}_N + \mathcal{T}_\Psi{}^M{}_N + \mathcal{T}_\mathrm{gauge}{}^M{}_N + \mathcal{T}_\mathrm{f}{}^M{}_N + \mathcal{T}_\mathrm{tip}{}^M{}_N + \mathcal{T}_\mathrm{I11}{}^M{}_N + \mathcal{T}_\mathrm{I12}{}^M{}_N + \mathcal{T}_\mathrm{I13}{}^M{}_N \qquad \text{(FE-3)}$$

Symbols in (FE-3).

- $\mathcal{T}^M{}_N$: The energy-momentum tensor of all matter (every field except gravity, the stabiliser included), the source of gravity; defined by the change of the matter action under a change of the metric. Unit: GeV^8. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mathcal{T}_\mathrm{stab}$: The energy-momentum tensor of the stabiliser. Unit: GeV^8. Status: DERIVED.
- $\mathcal{T}_\Psi$: The energy-momentum tensor of the dirac16complex field. Unit: GeV^8. Status: DERIVED.
- $\mathcal{T}_\mathrm{gauge}$: The energy-momentum tensor of the gauge fields. Unit: GeV^8. Status: STANDARD.
- $\mathcal{T}_\mathrm{f}$: The energy-momentum tensor of the Standard-Model fermions. Unit: GeV^8. Status: DERIVED.
- $\mathcal{T}_\mathrm{tip}$: The energy-momentum tensor of the tip terms (non-zero only for the tip directions). Unit: GeV^8. Status: DERIVED.
- $\mathcal{T}_\mathrm{I11}$: The energy-momentum tensor of the portal I11. Unit: GeV^8. Status: DERIVED.
- $\mathcal{T}_\mathrm{I12}$: The energy-momentum tensor of the portal I12. Unit: GeV^8. Status: DERIVED.
- $\mathcal{T}_\mathrm{I13}$: The energy-momentum tensor of the portal I13. Unit: GeV^8. Status: DERIVED.

The terms. The total source is the sum of the energy-momentum tensors of the stabiliser (FE-4), of dirac16complex (FE-5), of the Standard-Model fermions (FE-6), of the gauge fields (FE-7), of the tip terms (FE-8) and of the three portals (FE-9, FE-10). At order 0 the stabiliser dominates; the matter terms enter the block through their block

averages (FE-45 to FE-47): today their size relative to the stabiliser is 2.4×10^{-93}, at the dark-matter freeze-out 3.0×10^{-32} (DERIVED from the stored sources). Status: DERIVED.

The stabiliser density depends on the metric only through the block scale f, the fourth root of the proper volume of the block directions. A change of the metric changes f by

$$\Delta \ln f = -\frac{1}{8} \sum_{M \in \{x_0,\, x_5,\, x_6,\, x_7\}} g_{MM}\, \Delta g^{MM}, \qquad \Delta\big(\sqrt{|g|}\,\rho_{\mathrm{stab}}\big) = \sqrt{|g|}\,\Big(-\frac{1}{2}\,\rho_{\mathrm{stab}}\, g_{MN} \Delta g^{MN} + f\,\frac{\mathrm{d}\rho_{\mathrm{stab}}}{\mathrm{d}f}\,\Delta \ln f \Big),$$

Symbols in this equation.

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21} = m^2_2 - m^2_1$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- ρ_{stab}: The energy density of the block stabiliser, a function of the block scale, $\rho_{\mathrm{stab}} \propto f^{-8}$; its value at the static block is $\rho_{\mathrm{stab}}(f_*)$. Unit: GeV^8. Value: $\kappa\rho_{\mathrm{stab}}(f_*) = (26 + 4\sqrt{51})H^2 = 54.566\,H^2$. Status: ASSUMED.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.

and for $\rho_{\mathrm{stab}} \propto f^{-8}$ (IN-4), $f\,\mathrm{d}\rho_{\mathrm{stab}}/\mathrm{d}f = -8\rho_{\mathrm{stab}}$: the second term turns the coefficient of the block components from $-\frac{1}{2}$ into $+\frac{1}{2}$, so that $\mathcal{T}_{MN} = -g_{MN}\rho_{\mathrm{stab}}$ for the observable directions $x_1, \ldots, x_4$ and $\mathcal{T}_{MN} = +g_{MN}\rho_{\mathrm{stab}}$ for the block directions:

$$\mathcal{T}_{\mathrm{stab}}{}^M{}_N = \rho_{\mathrm{stab}}\,\mathrm{diag}(1,\ -1,\ -1,\ -1,\ -1,\ 1,\ 1,\ 1) : \qquad \rho = \rho_{\mathrm{stab}}, \qquad p_3 = -\rho_{\mathrm{stab}}, \qquad p_0 = p_5 = \rho_{\mathrm{stab}} \ \text{(FE-4)}$$

Symbols in (FE-4).

- $\mathcal{T}_{\mathrm{stab}}$: The energy-momentum tensor of the stabiliser. Unit: GeV^8. Status: DERIVED.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- ρ_{stab}: The energy density of the block stabiliser, a function of the block scale, $\rho_{\mathrm{stab}} \propto f^{-8}$; its value at the static block is $\rho_{\mathrm{stab}}(f_*)$. Unit: GeV^8. Value: $\kappa\rho_{\mathrm{stab}}(f_*) = (26 + 4\sqrt{51})H^2 = 54.566\,H^2$. Status: ASSUMED.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.

- p_3: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: GeV^8. Status: STANDARD.
- p_0: The pressure along the hidden direction x_0, $p_0 = \mathcal{T}^{x_0}{}_{x_0}$. Unit: GeV^8. Status: STANDARD.
- p_5: The pressure along the extra times, $p_5 = \mathcal{T}^{x_5}{}_{x_5} = \mathcal{T}^{x_6}{}_{x_6} = \mathcal{T}^{x_7}{}_{x_7}$. Unit: GeV^8. Status: STANDARD.

The terms. The stabiliser is a vacuum energy of the block: in ordinary space it acts like a cosmological constant (energy density ρ_{stab}, pressure $p_3 = -\rho_{\text{stab}}$), along the block directions it pushes outwards with the pressures $p_0 = p_5 = \rho_{\text{stab}}$. It is block-isotropic ($p_0 = p_5$). It is conserved (FE-41 with $\mathcal{H}_f$: $\dot\rho = -4\mathcal{H}_f(\rho + \bar p) = -8\mathcal{H}_f\rho$ gives $\rho \propto f^{-8}$, consistent with IN-4). Its value at the static block is fixed in FE-43. Status: DERIVED (from the ASSUMED IN-4).

$$
\begin{aligned}
(\mathcal{T}_\Psi)_{MN} &= -\frac{1}{4}\Big[\bar\Psi\gamma_M D_N\Psi + \bar\Psi\gamma_N D_M\Psi - (D_M\bar\Psi)\gamma_N\Psi - (D_N\bar\Psi)\gamma_M\Psi\Big] + g_{MN}\,\mathcal{L}_\Psi,\\
\mathcal{L}_\Psi &= \tfrac{1}{2}\big(\bar\Psi\gamma^P D_P\Psi - (D_P\bar\Psi)\gamma^P\Psi\big) - m_\Psi\,\bar\Psi\Psi - \frac{\lambda}{2}\big(\bar\Psi\Psi\big)^2
\end{aligned}
\qquad \text{(FE-5)}
$$

Symbols in (FE-5).

- $\mathcal{T}_\Psi$: The energy-momentum tensor of the dirac16complex field. Unit: GeV^8. Status: DERIVED.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: GeV$^{7/2}$. Status: RECORD.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{L}_\Psi$: The Lagrangian density of dirac16complex (the bracket of S_Ψ). Unit: GeV^8. Status: RECORD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- λ: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ of the record). Unit: GeV^{-6}. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm7\times10^{-5}H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4\times10^{-10}H^{-6}$ (for $\Delta N_{\text{eff}} \le 0.001$). Status: RECORD.

The terms. The first bracket is the symmetrised kinetic stress of the spinor (each of $\gamma_M D_N$ and $\gamma_N D_M$, acting on Ψ and on $\bar\Psi$, with the spin connection inside D); $g_{MN}\mathcal{L}_\Psi$ comes from the variation of $\sqrt{|g|}$. $\mathcal{L}_\Psi$ contains the mass m_Ψ and the contact self-interaction λ. It couples dirac16complex to gravity. For the Kohn-Sham states of dirac16complex it takes the form FE-67 (verified by the energy identity and the y-conservation law FE-68). Status: DERIVED.

$$
\begin{aligned}
(\mathcal{T}_{\text{f}})_{MN} &= \sum_{i=1}^{3}\sum_{\psi}\Big\{-\frac{1}{4}\Big[\bar\psi_i\gamma_M D_N\psi_i + \bar\psi_i\gamma_N D_M\psi_i - (D_M\bar\psi_i)\gamma_N\psi_i - (D_N\bar\psi_i)\gamma_M\psi_i\Big] + g_{MN}\,\mathcal{L}_{\psi i}\Big\},\\
\mathcal{L}_{\psi i} &= \tfrac{1}{2}\big(\bar\psi_i\gamma^P D_P\psi_i - (D_P\bar\psi_i)\gamma^P\psi_i\big) - \mu_{\psi i}\,\bar\psi_i\psi_i
\end{aligned}
\qquad \text{(FE-6)}
$$

Symbols in (FE-6).

- $\mathcal{T}_{\mathrm{f}}$: The energy-momentum tensor of the Standard-Model fermions. Unit: GeV^8. Status: DERIVED.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- ψ: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over ψ runs over the six. Status: STANDARD.
- ψ_i, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: ψ_i is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation i, ψ_f one fermion component f (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4, 4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+, +, +, +, -, -, -, -$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{L}_{\psi i}$: The Lagrangian density of the fermion multiplet ψ of the generation i (the bracket of S_{f}). Unit: GeV^8. Status: ASSUMED.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mu_{\psi i}$: The bulk mass of a Standard-Model fermion: $\mu_{\psi i}$ of the multiplet ψ of the generation i; with the hidden direction it fixes the localisation $c = f\mu L$ of the zero mode. Unit: GeV. Value: $\mu_{\psi i} = c/(fL)$ with the localisation c of the zero mode (doublets: $\mu_Q = 5.2304\,H$, $\mu_\ell = 9.2055\,H$). Status: ASSUMED.

The terms. The same form for every Standard-Model fermion multiplet ψ of every generation i, with the bulk mass $\mu_{\psi i}$ in place of m_Ψ, no self-interaction, and the gauge fields in D. It enters the block through the block averages of the plasma and of cold matter (ideal and degenerate gases of particles with their x_0 masses) in the block response FE-45 and in the hidden pressure FE-81. Status: DERIVED.

$$
\begin{aligned}
(\mathcal{T}_{\mathrm{gauge}})_{MN} = {} & \frac{1}{\hat{g}_3^2}\Big(G^A_{MP}G^A{}_N{}^P - \frac{1}{4}g_{MN}G^A_{PQ}G^{APQ}\Big) + \frac{1}{\hat{g}_2^2}\Big(W^I_{MP}W^I{}_N{}^P - \frac{1}{4}g_{MN}W^I_{PQ}W^{IPQ}\Big) \\
& + \frac{1}{\hat{g}_1^2}\Big(B_{MP}B_N{}^P - \frac{1}{4}g_{MN}B_{PQ}B^{PQ}\Big)
\end{aligned}
\tag{FE-7}
$$

Symbols in (FE-7).

- $\mathcal{T}_{\mathrm{gauge}}$: The energy-momentum tensor of the gauge fields. Unit: GeV^8. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\hat{g}_3$: The eight-dimensional SU(3) coupling (it multiplies the gluon kinetic term as $1/\hat{g}_3^2$); $g_3^2 = \hat{g}_3^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- G^A_M, G_M, G_{MN}, G^A_{MN}: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components G^A_M ($A = 1, \ldots, 8$), the matrix $G_M = G^A_M t^A$, the field strength G_{MN} (matrix) and its components G^A_{MN}; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in x_0). Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.

- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- Q: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- W^I_M, W_M, W_{MN}, W^I_{MN}: The SU(2) gauge field (geometric normalisation): the components W^I_M ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength W_{MN} and its components W^I_{MN}; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- $\hat{g}_1$: The eight-dimensional $\mathrm{U}(1)_Y$ coupling; $g_1^2 = \hat{g}_1^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- B_M, B_{MN}: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.

The terms. The standard energy-momentum tensor of the gauge fields (geometric normalisation, so each group carries its inverse squared coupling): the stress $F_{MP}F_N{}^P$ minus the trace part. It is traceless in four dimensions, so radiation drops out of the block response (FE-47: $c_3 = -3c_\rho$); what remains is the trace anomaly of the interacting plasma (FE-75, FE-82). Status: STANDARD.

$$
\begin{aligned}
(\mathcal{T}_{\mathrm{tip}})_{\mu\nu} &= \delta_{\mathrm{tip}}\Big[\frac{v_\Sigma^2}{4}\,\mathrm{tr}\big((D_\mu\Sigma)^\dagger D_\nu\Sigma + (D_\nu\Sigma)^\dagger D_\mu\Sigma\big) + g_{\mu\nu}\,\mathcal{L}_{\mathrm{tip}}\Big], \qquad \delta_{\mathrm{tip}} = \frac{\delta_{\mathrm{D}}(y+L)}{f}, \\
\mathcal{L}_{\mathrm{tip}} &= -\frac{v_\Sigma^2}{4}\,\mathrm{tr}\big((D_\mu\Sigma)^\dagger D^\mu\Sigma\big) - \mathcal{L}_{\mathrm{Y}}
\end{aligned}
\tag{FE-8}
$$

Symbols in (FE-8).

- $\mathcal{T}_{\mathrm{tip}}$: The energy-momentum tensor of the tip terms (non-zero only for the tip directions). Unit: GeV^8. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index running over $x_1, \ldots, x_4$ (as μ). Status: STANDARD.
- δ_{tip}: The density that places a tip term in the field equations of eight dimensions, $\delta_{\mathrm{tip}} = \delta_{\mathrm{D}}(y+L)/f$ (the Dirac delta function at the tip with the proper length of the hidden direction). Unit: 1/length. Status: DERIVED.
- v_Σ: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$); on the static block it fixes the Robin parameter β_{W}. Unit: $\mathrm{GeV}^{5/2}$. Value: fixed through $\beta_{\mathrm{W}} = 0.011912$. Status: INVENTED.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- Σ: The tip sigma field: a unitary 2×2 matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal W^+, W^- and Z, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{L}_{\mathrm{tip}}$: The Lagrangian density of the tip action (the bracket of S_{tip}). Unit: GeV^7. Status: INVENTED.
- δ_{D}: The Dirac delta function: $\delta_{\mathrm{D}}(y+L)$ is zero except at $y = -L$ and has the unit integral over y. Unit: 1/length. Status: STANDARD.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.

- L: The depth of the regular tip: each half of a period of x_0 ends at $y=-L$; $HL=3$. Unit: length. Value: $L=3/H$ ($fL=8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f=\mathrm{e}^{-a_4}b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\mathcal{L}_\mathrm{Y}$: The tip Yukawa Lagrangian: the couplings of the doublets to the singlets through the sigma field and the tip matrices $\mathcal{Y}$ (plus the Hermitian conjugate, h.c.). Unit: GeV^7. Status: INVENTED.

The terms. The tip terms act only on the tip $y=-L$: $\delta_\mathrm{tip}=\delta_\mathrm{D}(y+L)/f$ is the Dirac delta function in proper length. The first part is the stress of the sigma field, the second the tip Lagrangian (sigma field and Yukawa terms) times the metric; the indices μ,ν run over the tip directions, and the components along y vanish. Its gravitational effect is part of the rest energies of $W^\pm$, Z and the fermions, which are counted in the plasma. Status: DERIVED.

$$(\mathcal{T}_\mathrm{I11})_{MN}=g_{MN}\,\frac{c_\mathrm{V}}{\Lambda_\star^6}\big(\mathrm{i}\bar\Psi\gamma^P\Psi\big)J^{B-L}_P,\qquad (\mathcal{T}_\mathrm{I13})_{MN}=g_{MN}\,\frac{c_\mathrm{A}}{\Lambda_\star^6}\sum_{P\neq x_0}\big(\mathrm{i}\bar\Psi\gamma^P P_b\Psi\big)J^{B}_P \tag{FE-9}$$

Symbols in (FE-9).

- $\mathcal{T}_\mathrm{I11}$: The energy-momentum tensor of the portal I11. Unit: GeV^8. Status: DERIVED.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, ${g^M}_N=\delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- c_V: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_\mathrm{V}=4.38\times10^{-17}$ GeV^{-2}. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi=\Psi^\dagger C$ is its Dirac adjoint. Unit: GeV$^{7/2}$. Status: RECORD.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M={\mathbf{e}_a}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M=g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- J^{B-L}_M: The $B-L$ current of the Standard-Model fermions (baryon minus lepton number), summed over the generations. Unit: GeV^7. Status: STANDARD.
- $\mathcal{T}_\mathrm{I13}$: The energy-momentum tensor of the portal I13. Unit: GeV^8. Status: DERIVED.
- c_A: The coefficient of the x_0-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_\mathrm{A}=1.278\times10^{-9}$ GeV^{-2} (dark-matter relic). Status: INVENTED.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z=6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- P_a, P_b: The projectors on the two x_0-chiralities, $P_a=(1+\gamma^{x_0})/2$ and $P_b=(1-\gamma^{x_0})/2$ (in a block: the upper and the lower component). Unit: dimensionless. Status: INVENTED.
- J^B_M: The baryon-number current of the quarks (one third per quark), summed over the generations. Unit: GeV^7. Status: STANDARD.

The terms. The portal currents are frame currents, $\mathrm{i}\bar\Psi\gamma^a\Psi$ and $\mathrm{i}\bar\psi\gamma^a\psi$, which do not depend on the vielbein; only $\sqrt{|g|}$ and the explicit metric in the contraction vary, so the energy-momentum tensor is the metric times the interaction density. They couple dirac16complex and the fermions to gravity; their effect on the block is negligible in every environment (the level shifts FE-72 are below 10^{-9} GeV). Status: DERIVED.

$$(\mathcal{T}_{\mathrm{I12}})_{MN} = \frac{4\bar\Psi\Psi}{\Lambda_\star^3}\Big[c_{\mathrm{G}}\Big(G^A_{MP}G^A{}_N{}^P - \frac{1}{4}g_{MN}G^A_{PQ}G^{APQ}\Big) + c_{\mathrm{W}}\Big(W^I_{MP}W^I{}_N{}^P - \frac{1}{4}g_{MN}W^I_{PQ}W^{IPQ}\Big) + c_{\mathrm{B}}\Big(B_{MP}B_N{}^P - \frac{1}{4}g_{MN}B_{PQ}B^{PQ}\Big)\Big] \tag{FE-10}$$

Symbols in (FE-10).

- $\mathcal{T}_{\mathrm{I12}}$: The energy-momentum tensor of the portal I12. Unit: GeV^8. Status: DERIVED.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: GeV$^{7/2}$. Status: RECORD.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- c_{G}: The coefficient of the gluon term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{G}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- G^A_M, G_M, G_{MN}, G^A_{MN}: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components G^A_M ($A = 1, \ldots, 8$), the matrix $G_M = G^A_M t^A$, the field strength G_{MN} (matrix) and its components G^A_{MN}; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in x_0). Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- Q: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- c_{W}: The coefficient of the SU(2) term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{W}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- W^I_M, W_M, W_{MN}, W^I_{MN}: The SU(2) gauge field (geometric normalisation): the components W^I_M ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength W_{MN} and its components W^I_{MN}; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c_{B}: The coefficient of the hypercharge term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{B}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- B_M, B_{MN}: The hypercharge gauge field U(1)$_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.

The terms. I12 multiplies the gauge kinetic invariants by $\bar\Psi\Psi$, so its energy-momentum tensor is that of the gauge fields (FE-7) with $1/\hat g_i^2$ replaced by $4c\,\bar\Psi\Psi/\Lambda_\star^3$: I12 acts as a field-dependent gauge coupling (FE-21). Its effect in the coupled solution is at most 1.7×10^{-16}. Status: DERIVED.

$$\nabla_M\,\mathcal{T}^M{}_N = 0, \qquad \text{because} \qquad \nabla_M\,\mathcal{E}_{(p)}{}^M{}_N = 0 \quad (p = 1, 2, 3) \tag{FE-11}$$

Symbols in (FE-11).

- ∇: The covariant derivative of a tensor with the Levi-Civita connection of the metric g_{MN} (Christoffel symbols $\Gamma^N{}_{MP}$); ∇_M acts on the index M, $\nabla^M = g^{MN}\nabla_N$. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

- $\mathcal{T}^{M}{}_{N}$: The energy-momentum tensor of all matter (every field except gravity, the stabiliser included), the source of gravity; defined by the change of the matter action under a change of the metric. Unit: GeV^8. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order p, the metric variation of $\sqrt{|g|}\mathcal{L}_{(p)}$: $\mathcal{E}_{(1)}$ is the Einstein tensor $R^{M}{}_{N} - \frac{1}{2}\delta^{M}_{N}R$; every $\mathcal{E}_{(p)}$ is divergence free. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic). Status: STANDARD.

The terms. Every Lovelock tensor is divergence-free identically (the Bianchi identities of the Lovelock tensors, STANDARD), so the field equations FE-1 imply that the total source is conserved; the conservation holds when the matter fields obey their own field equations. Status: DERIVED (the Bianchi identities verified for the order-0 metric for $p = 1, 2, 3$, sympy, exact). At order 0 the conservation is FE-41.

4 The matter fields

4.1 dirac16complex

The terms of the action that contain $\bar{\Psi}$ are S_Ψ (IN-5) and the three portals (IN-16 to IN-18). Varying them with respect to $\bar{\Psi}$ (the kinetic term after an integration by parts, which turns $-(D_M\bar{\Psi})\gamma^M\Psi$ into $\bar{\Psi}\gamma^M D_M\Psi$) gives

$$\frac{\Delta S}{\Delta\bar{\Psi}} = \sqrt{|g|}\left[\gamma^M D_M\Psi - \Big(m_\Psi + \lambda\,\bar{\Psi}\Psi + \frac{\mathcal{F}}{\Lambda_\star^3}\Big)\Psi + \mathrm{i}\,\frac{c_\mathrm{V}}{\Lambda_\star^6}\,J^{B-L}_M\gamma^M\Psi + \mathrm{i}\,\frac{c_\mathrm{A}}{\Lambda_\star^6}\sum_{M\neq x_0} J^B_M\gamma^M P_b\Psi\right] = 0,$$

Symbols in this equation.

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21} = m_2^2 - m_1^2$ a difference of squared masses, ΔN_eff an addition to N_eff, ΔS_m the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- S: The action of UM-2 (the integral of the Lagrangian density over spacetime); its variation with respect to every field gives the field equations. Unit: dimensionless ($\hbar = 1$). Status: ASSUMED.
- Ψ, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^{M}{}_{N} = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- λ: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar{\Psi}\Psi)^2$ of the record). Unit: GeV^{-6}. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm7\times10^{-5}H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4\times10^{-10}H^{-6}$ (for $\Delta N_\mathrm{eff} \leq 0.001$). Status: RECORD.
- $\mathcal{F}$: The gauge-kinetic operator of the portal I12, $\mathcal{F} = c_\mathrm{G}G^A_{MN}G^{AMN} + c_\mathrm{W}W^I_{MN}W^{IMN} + c_\mathrm{B}B_{MN}B^{MN}$; $\langle\mathcal{F}\rangle$ its mean value in an environment. Unit: GeV^4. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda^n_\star$ enter. Status: INVENTED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.

- c_{V}: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_{\mathrm{V}} = 4.38\times10^{-17}$ GeV^{-2}. Status: INVENTED.
- J^{B-L}_M: The $B-L$ current of the Standard-Model fermions (baryon minus lepton number), summed over the generations. Unit: GeV^7. Status: STANDARD.
- c_{A}: The coefficient of the x_0-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_{\mathrm{A}} = 1.278\times10^{-9}$ GeV^{-2} (dark-matter relic). Status: INVENTED.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- J^B_M: The baryon-number current of the quarks (one third per quark), summed over the generations. Unit: GeV^7. Status: STANDARD.
- P_a, P_b: The projectors on the two x_0-chiralities, $P_a = (1+\gamma^{x_0})/2$ and $P_b = (1-\gamma^{x_0})/2$ (in a block: the upper and the lower component). Unit: dimensionless. Status: INVENTED.

which is the field equation of dirac16complex:

$$\gamma^M D_M \Psi = \Big[m_\Psi + \lambda\,\bar\Psi\Psi + \frac{\mathcal{F}}{\Lambda_\star^3}\Big]\Psi - \mathrm{i}\,\frac{c_{\mathrm{V}}}{\Lambda_\star^6}\,J^{B-L}_M \gamma^M \Psi - \mathrm{i}\,\frac{c_{\mathrm{A}}}{\Lambda_\star^6}\sum_{M\neq x_0} J^B_M \gamma^M P_b \Psi \qquad \text{(FE-12)}$$

Symbols in (FE-12).

- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- m_Ψ: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- λ: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ of the record). Unit: GeV^{-6}. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm7\times10^{-5}H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4\times10^{-10}H^{-6}$ (for $\Delta N_{\mathrm{eff}} \le 0.001$). Status: RECORD.
- $\mathcal{F}$: The gauge-kinetic operator of the portal I12, $\mathcal{F} = c_{\mathrm{G}} G^A_{MN} G^{AMN} + c_{\mathrm{W}} W^I_{MN} W^{IMN} + c_{\mathrm{B}} B_{MN} B^{MN}$; $\langle\mathcal{F}\rangle$ its mean value in an environment. Unit: GeV^4. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- c_{V}: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_{\mathrm{V}} = 4.38\times10^{-17}$ GeV^{-2}. Status: INVENTED.
- J^{B-L}_M: The $B-L$ current of the Standard-Model fermions (baryon minus lepton number), summed over the generations. Unit: GeV^7. Status: STANDARD.
- c_{A}: The coefficient of the x_0-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_{\mathrm{A}} = 1.278\times10^{-9}$ GeV^{-2} (dark-matter relic). Status: INVENTED.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- J^B_M: The baryon-number current of the quarks (one third per quark), summed over the generations. Unit: GeV^7. Status: STANDARD.
- P_a, P_b: The projectors on the two x_0-chiralities, $P_a = (1+\gamma^{x_0})/2$ and $P_b = (1-\gamma^{x_0})/2$ (in a block: the upper and the lower component). Unit: dimensionless. Status: INVENTED.

The terms. $\gamma^M D_M\Psi$ is the eight-dimensional Dirac operator with the canonical spin connection (gravity). On the right, m_Ψ is the mass parameter, $\lambda\bar\Psi\Psi$ the mean field of the contact self-interaction, and $\mathcal{F}/\Lambda_\star^3$ the I12 term:

the gauge-field invariants act on Ψ as an extra, field-dependent mass. The c_{V} term is I11: the $B-L$ current of the Standard-Model fermions acts on Ψ as a vector potential. The c_{A} term is I13: the baryon-number current acts on the x_0-chirality selected by P_b only. So dirac16complex couples to gravity, to itself, to every fermion (through J^{B-L} and J^B) and to every gauge field (through $\mathcal{F}$). Status: DERIVED. It is reduced exactly to a two-component problem in every block (FE-56), solved exactly for the free spectrum (FE-62) and by Kohn-Sham theory with the mean fields of every other field (FE-63 to FE-73).

$$\gamma^M \Omega_M = \frac{3}{2}\,\frac{H}{f}\,\gamma^{x_0} + \Big(\frac{\dot b_4}{b_4} - \frac{\dot a_4}{2}\Big)\gamma^{x_4} \tag{FE-13}$$

Symbols in (FE-13).

- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Ω_M: The spin connection acting on spinors, the matrix $\Omega_M = \frac{1}{2}\omega_{Mab}\mathcal{S}^{ab}$. Unit: 1/length. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $\dot b_4$: The time derivative $\mathrm{d}b_4/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- b_4: The author's metric function of the time x_4 in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.
- $\dot a_4$: The time derivative $\mathrm{d}a_4/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.

The terms. On the order-0 metric (FE-31) the spin connection contracted with the Dirac matrices has two parts: a term along γ^{x_0} of strength $\frac{3}{2}H/f$, from the warping of the extra times along the hidden direction (e^{2Hy} in FE-32), and a term along γ^{x_4} from the time dependence of the metric functions a_4 and b_4. It vanishes only in flat four-plus-four space ($H=0$, constant a_4, b_4), so the coupling of every spinor to gravity is non-trivial. The prefactors of the reduction ansatz of FE-56 remove it exactly. Status: DERIVED (exact for every a_4, b_4 and on both halves of a period).

4.2 The Standard-Model fermions

Varying S_{f} (IN-11), the tip action (IN-13) and the portals I11, I13 with respect to the adjoint of every multiplet gives its Dirac equation. The quark doublets:

$$\begin{aligned}\gamma^M D_M Q_i = \mu_Q\, Q_i - \mathrm{i}\,\frac{c_{\mathrm{V}}}{\Lambda_\star^6}\,q_Q\,\big(\mathrm{i}\bar\Psi\gamma_M\Psi\big)\gamma^M Q_i - \frac{\mathrm{i}}{3}\,\frac{c_{\mathrm{A}}}{\Lambda_\star^6}\sum_{M\neq x_0}\big(\mathrm{i}\bar\Psi\gamma_M P_b\Psi\big)\gamma^M Q_i \\ + \delta_{\mathrm{tip}}\sum_{j=1}^{3}\Sigma\begin{pmatrix}(\mathcal{Y}_u)_{ij}\,u_{Rj}\\ (\mathcal{Y}_d)_{ij}\,d_{Rj}\end{pmatrix}\end{aligned} \tag{FE-14}$$

Symbols in (FE-14).

- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4,

$\dots$), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.

- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \dots$): it runs over $x_0, x_1, \dots, x_7$; a repeated index is summed. Status: STANDARD.
- $D_M,\ D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- $Q_i,\ \bar{Q}_i$: The quark doublet of the generation i (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- μ_Q: The common bulk mass of the three quark doublets. Unit: GeV. Value: $5.2304\,H$ ($c_Q = 3.38230$). Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- c_V: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_\mathrm{V} = 4.38\times10^{-17}$ GeV^{-2}. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $q_\psi,\ q_f$: The $B - L$ charge: q_ψ of the multiplet ψ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), q_f of the fermion component f. Unit: dimensionless. Status: STANDARD.
- $\Psi,\ \bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- c_A: The coefficient of the x_0-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_\mathrm{A} = 1.278 \times 10^{-9}$ GeV^{-2} (dark-matter relic). Status: INVENTED.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $P_a,\ P_b$: The projectors on the two x_0-chiralities, $P_a = (1 + \gamma^{x_0})/2$ and $P_b = (1 - \gamma^{x_0})/2$ (in a block: the upper and the lower component). Unit: dimensionless. Status: INVENTED.
- δ_tip: The density that places a tip term in the field equations of eight dimensions, $\delta_\mathrm{tip} = \delta_\mathrm{D}(y + L)/f$ (the Dirac delta function at the tip with the proper length of the hidden direction). Unit: 1/length. Status: DERIVED.
- j: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- Σ: The tip sigma field: a unitary 2×2 matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal W^+, W^- and Z, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $\mathcal{Y}_\nu,\ \mathcal{Y}_e,\ \mathcal{Y}_u,\ \mathcal{Y}_d$: The complex 3×3 tip Yukawa matrices of the neutrino, charged-lepton, up-quark and down-quark sectors (generation indices i, j); on the static block they become the tip mixing matrices $\mathcal{K}$ (the normalisation of that relation is not fixed: OPEN). Unit: dimensionless. Status: INVENTED.
- $u_R,\ u_{Ri},\ \bar{u}_{Ri}$: The up-type quark singlet: u_R the multiplet, u_{Ri} that of the generation i (u_R, c_R, t_R), $\bar{u}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $d_R,\ d_{Ri},\ \bar{d}_{Ri}$: The down-type quark singlet: d_R the multiplet, d_{Ri} that of the generation i (d_R, s_R, b_R), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.

The terms. $\gamma^M D_M Q_i$ contains gravity (the spin connection) and the gauge fields: gluons, W and B (IN-12). μ_Q is the common bulk mass of the three quark doublets, which localises their zero modes at the fold. The c_V term is I11 (the vector current of dirac16complex acting with the $B - L$ charge $q_Q = \frac{1}{3}$), the c_A term I13 (the x_0-chiral current of dirac16complex acting with the baryon number $\frac{1}{3}$). The last term is the tip Yukawa coupling: at $y = -L$ the doublet couples, through Σ and the tip matrices $\mathcal{Y}_u$, $\mathcal{Y}_d$, to the singlets u_{Rj}, d_{Rj} of every generation j. Status: DERIVED. Solved on the static block by the zero modes and the tip mixing (FE-56 to FE-60; exact roots). The portal terms act on the quarks as mean fields of the vector densities of dirac16complex; they are neglected, because in every environment the density of the dark matter is far below that of the Standard-Model fermions, so they are far smaller than the shifts of dirac16complex by the quarks (FE-72) (ESTIMATE; not computed).

$$\gamma^M D_M u_{Ri} = \mu_{\psi i}\, u_{Ri} - \mathrm{i}\,\frac{c_\mathrm{V}}{\Lambda_\star^6}\, q_{u_R}\,\big(\mathrm{i}\bar\Psi\gamma_M\Psi\big)\gamma^M u_{Ri} - \frac{\mathrm{i}}{3}\,\frac{c_\mathrm{A}}{\Lambda_\star^6}\sum_{M\neq x_0}\big(\mathrm{i}\bar\Psi\gamma_M P_b\Psi\big)\gamma^M u_{Ri} + \delta_\mathrm{tip}\sum_{j=1}^{3}(\mathcal{Y}_u)^*_{ji}\,\big(\Sigma^\dagger Q_j\big)_1 \qquad (\psi = u_R) \tag{FE-15}$$

Symbols in (FE-15).

- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- u_R, u_{Ri}, $\bar u_{Ri}$: The up-type quark singlet: u_R the multiplet, u_{Ri} that of the generation i (u_R, c_R, t_R), $\bar u_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\mu_{\psi i}$: The bulk mass of a Standard-Model fermion: $\mu_{\psi i}$ of the multiplet ψ of the generation i; with the hidden direction it fixes the localisation $c = f\mu L$ of the zero mode. Unit: GeV. Value: $\mu_{\psi i} = c/(fL)$ with the localisation c of the zero mode (doublets: $\mu_Q = 5.2304\,H$, $\mu_\ell = 9.2055\,H$). Status: ASSUMED.
- ψ: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over ψ runs over the six. Status: STANDARD.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- c_V: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_\mathrm{V} = 4.38\times10^{-17}$ GeV^{-2}. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- q_ψ, q_f: The $B-L$ charge: q_ψ of the multiplet ψ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), q_f of the fermion component f. Unit: dimensionless. Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- c_A: The coefficient of the x_0-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_\mathrm{A} = 1.278\times10^{-9}$ GeV^{-2} (dark-matter relic). Status: INVENTED.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- P_a, P_b: The projectors on the two x_0-chiralities, $P_a = (1+\gamma^{x_0})/2$ and $P_b = (1-\gamma^{x_0})/2$ (in a block: the upper and the lower component). Unit: dimensionless. Status: INVENTED.
- δ_tip: The density that places a tip term in the field equations of eight dimensions, $\delta_\mathrm{tip} = \delta_\mathrm{D}(y+L)/f$ (the Dirac delta function at the tip with the proper length of the hidden direction). Unit: 1/length. Status: DERIVED.
- j: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- $\mathcal{Y}_\nu$, $\mathcal{Y}_e$, $\mathcal{Y}_u$, $\mathcal{Y}_d$: The complex 3×3 tip Yukawa matrices of the neutrino, charged-lepton, up-quark and down-quark sectors (generation indices i, j); on the static block they become the tip mixing matrices $\mathcal{K}$ (the normalisation of that relation is not fixed: OPEN). Unit: dimensionless. Status: INVENTED.
- *: Complex conjugation of a number or of each entry of a matrix. Status: STANDARD.
- Σ: The tip sigma field: a unitary 2×2 matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal W^+, W^- and Z, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators

the adjoint operator). Status: STANDARD.

- Q_i, $\bar{Q}_i$: The quark doublet of the generation i (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- ψ_i, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: ψ_i is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation i, ψ_f one fermion component f (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.

The terms. The up-type singlets of the three generations (the right-handed up, charm and top quarks) couple to gravity, gluons and B (they carry no weak isospin), have the bulk mass $\mu_{\psi i}$ of their generation, feel I11 with $q_{u_R} = \frac{1}{3}$ and I13 with the baryon number $\frac{1}{3}$, and couple at the tip to the upper component of $\Sigma^\dagger Q_j$ through the conjugate tip matrix. Status: DERIVED; exact roots for the up, charm and top quarks.

$$\begin{aligned}\gamma^M D_M d_{Ri} = \mu_{\psi i}\, d_{Ri} - \mathrm{i}\,\frac{c_\mathrm{V}}{\Lambda_\star^6}\, q_{d_R}\left(\mathrm{i}\bar{\Psi}\gamma_M\Psi\right)\gamma^M d_{Ri} - \frac{\mathrm{i}}{3}\,\frac{c_\mathrm{A}}{\Lambda_\star^6}\sum_{M\neq x_0}\left(\mathrm{i}\bar{\Psi}\gamma_M P_b\Psi\right)\gamma^M d_{Ri} \\ + \delta_\mathrm{tip}\sum_{j=1}^{3}(\mathcal{Y}_d)^*_{ji}\left(\Sigma^\dagger Q_j\right)_2 \qquad (\psi = d_R)\end{aligned} \tag{FE-16}$$

Symbols in (FE-16).

- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- d_R, d_{Ri}, $\bar{d}_{Ri}$: The down-type quark singlet: d_R the multiplet, d_{Ri} that of the generation i (d_R, s_R, b_R), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\mu_{\psi i}$: The bulk mass of a Standard-Model fermion: $\mu_{\psi i}$ of the multiplet ψ of the generation i; with the hidden direction it fixes the localisation $c = f\mu L$ of the zero mode. Unit: GeV. Value: $\mu_{\psi i} = c/(fL)$ with the localisation c of the zero mode (doublets: $\mu_Q = 5.2304\,H$, $\mu_\ell = 9.2055\,H$). Status: ASSUMED.
- ψ: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over ψ runs over the six. Status: STANDARD.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- c_V: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_\mathrm{V} = 4.38\times10^{-17}$ GeV^{-2}. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- q_ψ, q_f: The $B-L$ charge: q_ψ of the multiplet ψ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), q_f of the fermion component f. Unit: dimensionless. Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- c_A: The coefficient of the x_0-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_\mathrm{A} = 1.278\times10^{-9}$ GeV^{-2} (dark-matter relic). Status: INVENTED.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- P_a, P_b: The projectors on the two x_0-chiralities, $P_a = (1+\gamma^{x_0})/2$ and $P_b = (1-\gamma^{x_0})/2$ (in a block: the upper and the lower component). Unit: dimensionless. Status: INVENTED.

- δ_{tip}: The density that places a tip term in the field equations of eight dimensions, $\delta_{\mathrm{tip}} = \delta_{\mathrm{D}}(y+L)/f$ (the Dirac delta function at the tip with the proper length of the hidden direction). Unit: 1/length. Status: DERIVED.
- j: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- $\mathcal{Y}_\nu$, $\mathcal{Y}_e$, $\mathcal{Y}_u$, $\mathcal{Y}_d$: The complex 3×3 tip Yukawa matrices of the neutrino, charged-lepton, up-quark and down-quark sectors (generation indices i, j); on the static block they become the tip mixing matrices $\mathcal{K}$ (the normalisation of that relation is not fixed: OPEN). Unit: dimensionless. Status: INVENTED.
- *: Complex conjugation of a number or of each entry of a matrix. Status: STANDARD.
- Σ: The tip sigma field: a unitary 2×2 matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal W^+, W^- and Z, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- Q_i, $\bar{Q}_i$: The quark doublet of the generation i (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- ψ_i, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: ψ_i is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation i, ψ_f one fermion component f (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.

The terms. As FE-15 for the down-type singlets (the right-handed down, strange and bottom quarks) with the lower component of $\Sigma^\dagger Q_j$ and the tip matrix $\mathcal{Y}_d$; in UM-2 the tip mixing matrix of this sector is the CKM matrix (FE-30). Status: DERIVED.

$$\gamma^M D_M \ell_i = \mu_\ell\, \ell_i - \mathrm{i}\,\frac{c_{\mathrm{V}}}{\Lambda_\star^6}\, q_\ell \left(\mathrm{i}\bar{\Psi}\gamma_M \Psi\right)\gamma^M \ell_i + \delta_{\mathrm{tip}} \sum_{j=1}^{3} \Sigma \begin{pmatrix} (\mathcal{Y}_\nu)_{ij}\, \nu_{Rj} \\ (\mathcal{Y}_e)_{ij}\, e_{Rj} \end{pmatrix} \tag{FE-17}$$

Symbols in (FE-17).

- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4, 4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- ℓ_i, $\bar{\ell}_i$: The lepton doublet of the generation i (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- μ_ℓ: The common bulk mass of the three lepton doublets. Unit: GeV. Value: $9.2055\,H$ ($c_\ell = 5.95288$). Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- c_{V}: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_{\mathrm{V}} = 4.38 \times 10^{-17}$ GeV^{-2}. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- q_ψ, q_f: The $B - L$ charge: q_ψ of the multiplet ψ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), q_f of the fermion component f. Unit: dimensionless. Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- δ_{tip}: The density that places a tip term in the field equations of eight dimensions, $\delta_{\mathrm{tip}} = \delta_{\mathrm{D}}(y+L)/f$ (the Dirac delta function at the tip with the proper length of the hidden direction). Unit: 1/length. Status: DERIVED.

- j: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- Σ: The tip sigma field: a unitary 2×2 matrix field on the tip ($\Sigma^\dagger \Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal W^+, W^- and Z, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $\mathcal{Y}_\nu$, $\mathcal{Y}_e$, $\mathcal{Y}_u$, $\mathcal{Y}_d$: The complex 3×3 tip Yukawa matrices of the neutrino, charged-lepton, up-quark and down-quark sectors (generation indices i, j); on the static block they become the tip mixing matrices $\mathcal{K}$ (the normalisation of that relation is not fixed: OPEN). Unit: dimensionless. Status: INVENTED.
- ν_R, ν_{Ri}: The right-handed neutrino (a gauge singlet): ν_R the multiplet, ν_{Ri} that of the generation i; its zero modes give Dirac neutrino masses. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- e_R, e_{Ri}: The right-handed charged-lepton singlet: e_R the multiplet, e_{Ri} that of the generation i (e_R, μ_R, τ_R). Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.

The terms. The lepton doublets contain the three flavours of left-handed neutrinos and the left-handed charged leptons (electron, muon, tau). They couple to gravity, W and B (no colour), have the common bulk mass μ_ℓ, feel I11 with $q_\ell = -1$ and no I13 term (leptons carry no baryon number), and couple at the tip to the right-handed neutrinos (through $\mathcal{Y}_\nu$) and to the right-handed charged leptons (through $\mathcal{Y}_e$). One doublet zero mode serves the three generations. Status: DERIVED; exact roots for the three neutrinos and the three charged leptons.

$$\gamma^M D_M e_{Ri} = \mu_{\psi i}\, e_{Ri} - \mathrm{i}\,\frac{c_\mathrm{V}}{\Lambda_\star^6}\, q_{e_R} \left(\mathrm{i}\bar{\Psi}\gamma_M \Psi\right)\gamma^M e_{Ri} + \delta_\mathrm{tip} \sum_{j=1}^{3} (\mathcal{Y}_e)^*_{ji} \left(\Sigma^\dagger \ell_j\right)_2 \qquad (\psi = e_R) \tag{FE-18}$$

Symbols in (FE-18).

- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4, 4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu \Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- e_R, e_{Ri}: The right-handed charged-lepton singlet: e_R the multiplet, e_{Ri} that of the generation i (e_R, μ_R, τ_R). Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\mu_{\psi i}$: The bulk mass of a Standard-Model fermion: $\mu_{\psi i}$ of the multiplet ψ of the generation i; with the hidden direction it fixes the localisation $c = f\mu L$ of the zero mode. Unit: GeV. Value: $\mu_{\psi i} = c/(fL)$ with the localisation c of the zero mode (doublets: $\mu_Q = 5.2304\,H$, $\mu_\ell = 9.2055\,H$). Status: ASSUMED.
- ψ: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over ψ runs over the six. Status: STANDARD.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- c_V: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_\mathrm{V} = 4.38 \times 10^{-17}$ GeV^{-2}. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- q_ψ, q_f: The $B - L$ charge: q_ψ of the multiplet ψ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), q_f of the fermion component f. Unit: dimensionless. Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- δ_tip: The density that places a tip term in the field equations of eight dimensions, $\delta_\mathrm{tip} = \delta_\mathrm{D}(y + L)/f$ (the Dirac delta function at the tip with the proper length of the hidden direction). Unit: 1/length. Status: DERIVED.
- j: A second generation index ($j = 1, 2, 3$). Status: STANDARD.

- $\mathcal{Y}_\nu$, $\mathcal{Y}_e$, $\mathcal{Y}_u$, $\mathcal{Y}_d$: The complex 3×3 tip Yukawa matrices of the neutrino, charged-lepton, up-quark and down-quark sectors (generation indices i, j); on the static block they become the tip mixing matrices $\mathcal{K}$ (the normalisation of that relation is not fixed: OPEN). Unit: dimensionless. Status: INVENTED.
- *: Complex conjugation of a number or of each entry of a matrix. Status: STANDARD.
- Σ: The tip sigma field: a unitary 2×2 matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal W^+, W^- and Z, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- ℓ_i, $\bar{\ell}_i$: The lepton doublet of the generation i (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- ψ_i, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: ψ_i is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation i, ψ_f one fermion component f (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.

The terms. The right-handed charged leptons of the three generations (electron, muon and tau singlets) couple to gravity and to B, have their bulk masses, feel I11 with $q_{e_R} = -1$, and couple at the tip to the lower component of $\Sigma^\dagger \ell_j$. Status: DERIVED; exact roots for the electron, muon and tau with the tip matrix $\mathbf{1}$.

$$\gamma^M D_M \nu_{Ri} = \mu_{\psi i}\, \nu_{Ri} - \mathrm{i}\, \frac{c_\mathrm{V}}{\Lambda_\star^6}\, q_{\nu_R} \left(\mathrm{i}\bar{\Psi}\gamma_M \Psi\right)\gamma^M \nu_{Ri} + \delta_\mathrm{tip} \sum_{j=1}^{3} (\mathcal{Y}_\nu)^*_{ji} \left(\Sigma^\dagger \ell_j\right)_1 \qquad (\psi = \nu_R) \tag{FE-19}$$

Symbols in (FE-19).

- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4, 4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- ν_R, ν_{Ri}: The right-handed neutrino (a gauge singlet): ν_R the multiplet, ν_{Ri} that of the generation i; its zero modes give Dirac neutrino masses. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\mu_{\psi i}$: The bulk mass of a Standard-Model fermion: $\mu_{\psi i}$ of the multiplet ψ of the generation i; with the hidden direction it fixes the localisation $c = f\mu L$ of the zero mode. Unit: GeV. Value: $\mu_{\psi i} = c/(fL)$ with the localisation c of the zero mode (doublets: $\mu_Q = 5.2304\, H$, $\mu_\ell = 9.2055\, H$). Status: ASSUMED.
- ψ: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over ψ runs over the six. Status: STANDARD.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- c_V: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_\mathrm{V} = 4.38 \times 10^{-17}$ GeV^{-2}. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- q_ψ, q_f: The $B - L$ charge: q_ψ of the multiplet ψ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), q_f of the fermion component f. Unit: dimensionless. Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- δ_tip: The density that places a tip term in the field equations of eight dimensions, $\delta_\mathrm{tip} = \delta_\mathrm{D}(y + L)/f$ (the Dirac delta function at the tip with the proper length of the hidden direction). Unit: 1/length. Status: DERIVED.

- j: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- $\mathcal{Y}_\nu$, $\mathcal{Y}_e$, $\mathcal{Y}_u$, $\mathcal{Y}_d$: The complex 3×3 tip Yukawa matrices of the neutrino, charged-lepton, up-quark and down-quark sectors (generation indices i, j); on the static block they become the tip mixing matrices $\mathcal{K}$ (the normalisation of that relation is not fixed: OPEN). Unit: dimensionless. Status: INVENTED.
- *: Complex conjugation of a number or of each entry of a matrix. Status: STANDARD.
- Σ: The tip sigma field: a unitary 2×2 matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal W^+, W^- and Z, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- ℓ_i, $\bar{\ell}_i$: The lepton doublet of the generation i (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- ψ_i, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: ψ_i is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation i, ψ_f one fermion component f (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.

The terms. The right-handed neutrinos are gauge singlets: no gauge field acts on them; they couple to gravity, to dirac16complex through I11 ($q_{\nu_R} = -1$), and at the tip to the upper component of $\Sigma^\dagger\ell_j$. Their zero modes are sharply localised at the fold ($c_R = -31$ to -35), so their values at the tip are tiny; mixed there with the doublet zero mode they give the small Dirac masses of the three neutrinos (FE-100). They are never thermalised (FE-134). Status: DERIVED.

4.3 The gauge fields

$$J_3^{AN} = \sum_{i=1}^{3} \big(\mathrm{i}\bar{Q}_i\gamma^N t^A Q_i + \mathrm{i}\bar{u}_{Ri}\gamma^N t^A u_{Ri} + \mathrm{i}\bar{d}_{Ri}\gamma^N t^A d_{Ri}\big), \qquad J_2^{IN} = \sum_{i=1}^{3} \Big(\mathrm{i}\bar{Q}_i\gamma^N \frac{\tau^I}{2} Q_i + \mathrm{i}\bar{\ell}_i\gamma^N \frac{\tau^I}{2}\ell_i\Big),$$
$$J_1^N = \sum_{i=1}^{3}\sum_{\psi} Y_\psi\, \mathrm{i}\bar{\psi}_i\gamma^N\psi_i, \qquad J_{\mathrm{em}}^N = J_2^{3N} + J_1^N, \qquad J_2^{\pm N} = \frac{J_2^{1N} \mp \mathrm{i}J_2^{2N}}{\sqrt{2}} \tag{FE-20}$$

Symbols in (FE-20).

- J_3^{AN}: The colour current of the quarks (the source of the gluons), summed over the generations. Unit: GeV^7. Status: STANDARD.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- Q_i, $\bar{Q}_i$: The quark doublet of the generation i (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4, 4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- t^A: The generators of SU(3) in the fundamental representation (the Gell-Mann matrices divided by 2), $\mathrm{tr}(t^A t^B) = \frac{1}{2}\delta^{AB}$. Unit: dimensionless. Status: STANDARD.
- u_R, u_{Ri}, $\bar{u}_{Ri}$: The up-type quark singlet: u_R the multiplet, u_{Ri} that of the generation i (u_R, c_R, t_R), $\bar{u}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- d_R, d_{Ri}, $\bar{d}_{Ri}$: The down-type quark singlet: d_R the multiplet, d_{Ri} that of the generation i (d_R, s_R, b_R), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- J_2^{IN}: The weak-isospin current of the doublets Q_i, ℓ_i (the source of W^I). Unit: GeV^7. Status: STANDARD.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- τ^I: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.

- ℓ_i, $\bar{\ell}_i$: The lepton doublet of the generation i (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- J_1^N: The hypercharge current of all fermions (the source of B). Unit: GeV^7. Status: STANDARD.
- ψ: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over ψ runs over the six. Status: STANDARD.
- Y: The hypercharge (the generator of $\mathrm{U}(1)_Y$); Y_ψ the hypercharge of the multiplet ψ (Q: $\frac{1}{6}$, u_R: $\frac{2}{3}$, d_R: $-\frac{1}{3}$, ℓ: $-\frac{1}{2}$, e_R: -1, ν_R: 0). Unit: dimensionless. Status: STANDARD.
- ψ_i, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: ψ_i is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation i, ψ_f one fermion component f (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.
- J_{em}^N: The electromagnetic current, $J_{\mathrm{em}} = J_2^3 + J_1$ (in units of the elementary charge). Unit: GeV^7. Status: STANDARD.
- $J_2^{\pm N}$: The charged combinations of the isospin current, $J_2^\pm = (J_2^1 \mp \mathrm{i}J_2^2)/\sqrt{2}$ (the sources of $W^\pm$). Unit: GeV^7. Status: STANDARD.

The terms. The currents are the sources of the gauge fields, summed over the three generations: J_3^{AN} (colour) from the quarks, J_2^{IN} (weak isospin) from the doublets, J_1^N (hypercharge) from every multiplet with its hypercharge Y_ψ; the electromagnetic current J_{em} and the charged currents $J_2^\pm$, the sources of the photon and of $W^\pm$, are combinations of them. Status: STANDARD.

The gauge kinetic term IN-8 and the portal I12 (IN-17) contain the same invariants; for the gluons together

$$-\frac{1}{4}\Big(\frac{1}{\hat{g}_3^2} + \frac{4c_{\mathrm{G}}\,\bar{\Psi}\Psi}{\Lambda_\star^3}\Big)G^A_{MN}G^{AMN},$$

Symbols in this equation.

- $\hat{g}_3$: The eight-dimensional SU(3) coupling (it multiplies the gluon kinetic term as $1/\hat{g}_3^2$); $g_3^2 = \hat{g}_3^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- c_{G}: The coefficient of the gluon term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{G}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- Ψ, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- G_M^A, G_M, G_{MN}, G_{MN}^A: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components G_M^A ($A = 1, \ldots, 8$), the matrix $G_M = G_M^A t^A$, the field strength G_{MN} (matrix) and its components G_{MN}^A; $G_{\mu\nu}^A$ are its components along the observable directions (the zero mode, flat in x_0). Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

and likewise for W and B. The scalar density of dirac16complex therefore changes the effective couplings:

$$\frac{1}{\hat{g}_{3,\mathrm{eff}}^2} = \frac{1}{\hat{g}_3^2} + \frac{4c_{\mathrm{G}}\bar{\Psi}\Psi}{\Lambda_\star^3}, \qquad \frac{1}{\hat{g}_{2,\mathrm{eff}}^2} = \frac{1}{\hat{g}_2^2} + \frac{4c_{\mathrm{W}}\bar{\Psi}\Psi}{\Lambda_\star^3}, \qquad \frac{1}{\hat{g}_{1,\mathrm{eff}}^2} = \frac{1}{\hat{g}_1^2} + \frac{4c_{\mathrm{B}}\bar{\Psi}\Psi}{\Lambda_\star^3} \tag{FE-21}$$

Symbols in (FE-21).

- $\hat{g}_{3,\mathrm{eff}}$, $\hat{g}_{2,\mathrm{eff}}$, $\hat{g}_{1,\mathrm{eff}}$: The field-dependent eight-dimensional couplings produced by I12: $1/\hat{g}_{i,\mathrm{eff}}^2 = 1/\hat{g}_i^2 + 4c\,\bar{\Psi}\Psi/\Lambda_\star^3$ with $c = c_{\mathrm{G}}, c_{\mathrm{W}}, c_{\mathrm{B}}$. Unit: GeV^{-2}. Status: DERIVED.

- $\hat{g}_3$: The eight-dimensional SU(3) coupling (it multiplies the gluon kinetic term as $1/\hat{g}_3^2$); $g_3^2 = \hat{g}_3^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- c_G: The coefficient of the gluon term of the portal I12. Unit: dimensionless. Value: $c_\mathrm{G}/\Lambda_\star^3 = c_{\mathrm{I}12}/\Lambda_\star^3$. Status: INVENTED.
- Ψ, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- c_W: The coefficient of the SU(2) term of the portal I12. Unit: dimensionless. Value: $c_\mathrm{W}/\Lambda_\star^3 = c_{\mathrm{I}12}/\Lambda_\star^3$. Status: INVENTED.
- $\hat{g}_1$: The eight-dimensional $\mathrm{U}(1)_Y$ coupling; $g_1^2 = \hat{g}_1^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- c_B: The coefficient of the hypercharge term of the portal I12. Unit: dimensionless. Value: $c_\mathrm{B}/\Lambda_\star^3 = c_{\mathrm{I}12}/\Lambda_\star^3$. Status: INVENTED.

The terms. Where dirac16complex has a scalar density $\bar{\Psi}\Psi$, every gauge coupling is shifted by the I12 coefficient of its group. In four dimensions this is the coupling shift of FE-79 (at most 1.7×10^{-16} at the dark-matter freeze-out). Status: DERIVED.

Varying the gauge action with these couplings, the fermion action (through the covariant derivatives) and the tip action with respect to the gauge fields gives the Yang-Mills equations:

$$\nabla_M\Big(\frac{G^{MN}}{\hat{g}_{3,\mathrm{eff}}^2}\Big) - \mathrm{i}\Big[G_M,\ \frac{G^{MN}}{\hat{g}_{3,\mathrm{eff}}^2}\Big] = J_3^{AN}\, t^A \tag{FE-22}$$

Symbols in (FE-22).

- ∇: The covariant derivative of a tensor with the Levi-Civita connection of the metric g_{MN} (Christoffel symbols $\Gamma^N{}_{MP}$); ∇_M acts on the index M, $\nabla^M = g^{MN}\nabla_N$. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- G_M^A, G_M, G_{MN}, G_{MN}^A: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components G_M^A ($A = 1, \ldots, 8$), the matrix $G_M = G_M^A t^A$, the field strength G_{MN} (matrix) and its components G_{MN}^A; $G_{\mu\nu}^A$ are its components along the observable directions (the zero mode, flat in x_0). Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\hat{g}_{3,\mathrm{eff}}$, $\hat{g}_{2,\mathrm{eff}}$, $\hat{g}_{1,\mathrm{eff}}$: The field-dependent eight-dimensional couplings produced by I12: $1/\hat{g}_{i,\mathrm{eff}}^2 = 1/\hat{g}_i^2 + 4c\,\bar{\Psi}\Psi/\Lambda_\star^3$ with $c = c_\mathrm{G}, c_\mathrm{W}, c_\mathrm{B}$. Unit: GeV^{-2}. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- J_3^{AN}: The colour current of the quarks (the source of the gluons), summed over the generations. Unit: GeV^7. Status: STANDARD.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- t^A: The generators of SU(3) in the fundamental representation (the Gell-Mann matrices divided by 2), $\mathrm{tr}(t^A t^B) = \frac{1}{2}\delta^{AB}$. Unit: dimensionless. Status: STANDARD.

The terms. The covariant divergence of the gluon field strength (with the field-dependent coupling of FE-21) plus the commutator term (the self-interaction of the gluons) equals the colour current of the quarks. There is no tip term: the gluons do not couple to Σ. Status: DERIVED. On the static block the gluon zero mode is exactly massless (flat x_0 profile, FE-50 to FE-52); below the confinement scale QCD enters through the nucleon masses and the condensates (FE-74, FE-75).

$$\nabla_M\Big(\frac{W^{MN}}{\hat{g}_{2,\mathrm{eff}}^2}\Big) - \mathrm{i}\Big[W_M,\ \frac{W^{MN}}{\hat{g}_{2,\mathrm{eff}}^2}\Big] = J_2^{IN}\,\frac{\tau^I}{2} + \delta_\mathrm{tip}\,\frac{v_\Sigma^2}{4}\Big(W^N - B^N\,\frac{\tau^3}{2}\Big) \qquad (\Sigma = \mathbf{1}) \tag{FE-23}$$

Symbols in (FE-23).

- ∇: The covariant derivative of a tensor with the Levi-Civita connection of the metric g_{MN} (Christoffel symbols $\Gamma^N{}_{MP}$); ∇_M acts on the index M, $\nabla^M = g^{MN}\nabla_N$. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- W^I_M, W_M, W_{MN}, W^I_{MN}: The SU(2) gauge field (geometric normalisation): the components W^I_M ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength W_{MN} and its components W^I_{MN}; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\hat{g}_{3,\mathrm{eff}}$, $\hat{g}_{2,\mathrm{eff}}$, $\hat{g}_{1,\mathrm{eff}}$: The field-dependent eight-dimensional couplings produced by I12: $1/\hat{g}^2_{i,\mathrm{eff}} = 1/\hat{g}^2_i + 4c\,\bar{\Psi}\Psi/\Lambda^3_\star$ with $c = c_\mathrm{G}, c_\mathrm{W}, c_\mathrm{B}$. Unit: GeV^{-2}. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- J^{IN}_2: The weak-isospin current of the doublets Q_i, ℓ_i (the source of W^I). Unit: GeV^7. Status: STANDARD.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- τ^I: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- δ_tip: The density that places a tip term in the field equations of eight dimensions, $\delta_\mathrm{tip} = \delta_\mathrm{D}(y + L)/f$ (the Dirac delta function at the tip with the proper length of the hidden direction). Unit: 1/length. Status: DERIVED.
- v_Σ: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$); on the static block it fixes the Robin parameter β_W. Unit: $\mathrm{GeV}^{5/2}$. Value: fixed through $\beta_\mathrm{W} = 0.011912$. Status: INVENTED.
- B_M, B_{MN}: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- Σ: The tip sigma field: a unitary 2×2 matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal W^+, W^- and Z, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.

The terms. The SU(2) equation (matrix form) has, besides the isospin current, the tip current of the sigma field in the unitary gauge $\Sigma = \mathbf{1}$ (N runs over the tip directions): a mass term for the charged components W^1, W^2 and for the combination $W^3 - B$ only. Status: DERIVED. On the static block it is solved by the x_0 reduction (FE-50) with the Robin tip condition (FE-51).

$$\nabla_M\Big(\frac{B^{MN}}{\hat{g}^2_{1,\mathrm{eff}}}\Big) = J^N_1 - \delta_\mathrm{tip}\,\frac{v_\Sigma^2}{4}\left(W^{3N} - B^N\right) \qquad (\Sigma = \mathbf{1}) \tag{FE-24}$$

Symbols in (FE-24).

- ∇: The covariant derivative of a tensor with the Levi-Civita connection of the metric g_{MN} (Christoffel symbols $\Gamma^N{}_{MP}$); ∇_M acts on the index M, $\nabla^M = g^{MN}\nabla_N$. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- B_M, B_{MN}: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\hat{g}_{3,\mathrm{eff}}$, $\hat{g}_{2,\mathrm{eff}}$, $\hat{g}_{1,\mathrm{eff}}$: The field-dependent eight-dimensional couplings produced by I12: $1/\hat{g}^2_{i,\mathrm{eff}} = 1/\hat{g}^2_i + 4c\,\bar{\Psi}\Psi/\Lambda^3_\star$ with $c = c_\mathrm{G}, c_\mathrm{W}, c_\mathrm{B}$. Unit: GeV^{-2}. Status: DERIVED.
- J^N_1: The hypercharge current of all fermions (the source of B). Unit: GeV^7. Status: STANDARD.
- δ_tip: The density that places a tip term in the field equations of eight dimensions, $\delta_\mathrm{tip} = \delta_\mathrm{D}(y + L)/f$ (the Dirac delta function at the tip with the proper length of the hidden direction). Unit: 1/length. Status: DERIVED.
- v_Σ: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$); on the static block it fixes the Robin parameter β_W. Unit: $\mathrm{GeV}^{5/2}$. Value: fixed through $\beta_\mathrm{W} = 0.011912$. Status: INVENTED.

- W^I_M, W_M, W_{MN}, W^I_{MN}: The SU(2) gauge field (geometric normalisation): the components W^I_M ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength W_{MN} and its components W^I_{MN}; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- Σ: The tip sigma field: a unitary 2×2 matrix field on the tip ($\Sigma^\dagger \Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal W^+, W^- and Z, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.

The terms. The hypercharge equation is abelian (no commutator); its source is the hypercharge current, and the tip term (with the opposite sign of FE-23) acts on $W^3 - B$ only. Status: DERIVED.

To linear order in the fields the commutators drop out, and the combinations of IN-10 decouple. For the photon, for example,

$$\nabla_M A^{MN} = \frac{\hat{g}_1^2\, \nabla_M W^{3MN} + \hat{g}_2^2\, \nabla_M B^{MN}}{\hat{g}_1 \hat{g}_2 \hat{g}_Z} = \frac{\hat{g}_1 \hat{g}_2}{\hat{g}_Z} \left(J_2^{3N} + J_1^N \right),$$

Symbols in this equation.

- ∇: The covariant derivative of a tensor with the Levi-Civita connection of the metric g_{MN} (Christoffel symbols $\Gamma^N{}_{MP}$); ∇_M acts on the index M, $\nabla^M = g^{MN} \nabla_N$. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- A_M: The photon field, canonically normalised, $A_M = (\hat{g}_1^2 W^3_M + \hat{g}_2^2 B_M)/(\hat{g}_1 \hat{g}_2 \hat{g}_Z)$. Unit: GeV^3. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\hat{g}_1$: The eight-dimensional $\mathrm{U}(1)_Y$ coupling; $g_1^2 = \hat{g}_1^2 / V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- W^I_M, W_M, W_{MN}, W^I_{MN}: The SU(2) gauge field (geometric normalisation): the components W^I_M ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength W_{MN} and its components W^I_{MN}; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2 / V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- B_M, B_{MN}: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- $\hat{g}_Z$: The eight-dimensional coupling of the Z, $\hat{g}_Z = (\hat{g}_2^2 + \hat{g}_1^2)^{1/2}$. Unit: GeV^{-2}. Status: STANDARD.
- J_2^{IN}: The weak-isospin current of the doublets Q_i, ℓ_i (the source of W^I). Unit: GeV^7. Status: STANDARD.
- J_1^N: The hypercharge current of all fermions (the source of B). Unit: GeV^7. Status: STANDARD.

because the tip terms of the third component of FE-23 and of FE-24, multiplied by $\hat{g}_2^2$ and $\hat{g}_1^2$, cancel. The same steps for $W^\pm$ and Z give

$$\begin{aligned}
&\nabla_M W^{\pm MN} = \hat{g}_2\, J_2^{\pm N} + \delta_{\mathrm{tip}}\, \frac{v_\Sigma^2}{4}\, \hat{g}_2^2\, W^{\pm N}, \qquad \nabla_M A^{MN} = \hat{e}\, J^N_{\mathrm{em}}, \qquad \hat{e} = \frac{\hat{g}_2 \hat{g}_1}{\hat{g}_Z}, \\
&\nabla_M Z^{MN} = \hat{g}_Z \left(J_2^{3N} - \sin^2 \theta_{\mathrm{W}}\, J^N_{\mathrm{em}} \right) + \delta_{\mathrm{tip}}\, \frac{v_\Sigma^2}{4}\, \hat{g}_Z^2\, Z^N, \\
&W^{\pm MN} = \nabla^M W^{\pm N} - \nabla^N W^{\pm M}, \qquad Z^{MN} = \nabla^M Z^N - \nabla^N Z^M, \qquad A^{MN} = \nabla^M A^N - \nabla^N A^M
\end{aligned} \tag{FE-25}$$

Symbols in (FE-25).

- ∇: The covariant derivative of a tensor with the Levi-Civita connection of the metric g_{MN} (Christoffel symbols $\Gamma^N{}_{MP}$); ∇_M acts on the index M, $\nabla^M = g^{MN} \nabla_N$. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

- $W_M^\pm$, W^+, W^-: The charged weak bosons, canonically normalised: $W_M^\pm = (W_M^1 \mp \mathrm{i}W_M^2)/(\sqrt{2}\,\hat{g}_2)$; W^+ carries the electric charge $+e$ and W^- the charge $-e$ (they are each other's antiparticles and have one mass). Unit: GeV^3. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- $J_2^{\pm N}$: The charged combinations of the isospin current, $J_2^\pm = (J_2^1 \mp \mathrm{i}J_2^2)/\sqrt{2}$ (the sources of $W^\pm$). Unit: GeV^7. Status: STANDARD.
- δ_{tip}: The density that places a tip term in the field equations of eight dimensions, $\delta_{\mathrm{tip}} = \delta_{\mathrm{D}}(y + L)/f$ (the Dirac delta function at the tip with the proper length of the hidden direction). Unit: 1/length. Status: DERIVED.
- v_Σ: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$); on the static block it fixes the Robin parameter β_{W}. Unit: $\mathrm{GeV}^{5/2}$. Value: fixed through $\beta_{\mathrm{W}} = 0.011912$. Status: INVENTED.
- A_M: The photon field, canonically normalised, $A_M = (\hat{g}_1^2 W_M^3 + \hat{g}_2^2 B_M)/(\hat{g}_1\hat{g}_2\hat{g}_Z)$. Unit: GeV^3. Status: STANDARD.
- $\hat{e}$: The eight-dimensional electric coupling, $\hat{e} = \hat{g}_2\hat{g}_1/\hat{g}_Z$ ($e^2 = \hat{e}^2/V_\perp$). Unit: GeV^{-2}. Status: STANDARD.
- J_{em}^N: The electromagnetic current, $J_{\mathrm{em}} = J_2^3 + J_1$ (in units of the elementary charge). Unit: GeV^7. Status: STANDARD.
- $\hat{g}_1$: The eight-dimensional $\mathrm{U}(1)_Y$ coupling; $g_1^2 = \hat{g}_1^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- $\hat{g}_Z$: The eight-dimensional coupling of the Z, $\hat{g}_Z = (\hat{g}_2^2 + \hat{g}_1^2)^{1/2}$. Unit: GeV^{-2}. Status: STANDARD.
- Z_M: The Z boson field, canonically normalised, $Z_M = (W_M^3 - B_M)/\hat{g}_Z$. Unit: GeV^3. Status: STANDARD.
- J_2^{IN}: The weak-isospin current of the doublets Q_i, ℓ_i (the source of W^I). Unit: GeV^7. Status: STANDARD.
- θ_{W}: The weak mixing angle of the couplings, $\cos\theta_{\mathrm{W}} = \hat{g}_2/\hat{g}_Z$, $\sin\theta_{\mathrm{W}} = \hat{g}_1/\hat{g}_Z$; fixed by the measured W and Z masses with $\rho_{\mathrm{EW},0}$. Unit: dimensionless. Value: $\cos^2\theta_{\mathrm{W}} = 0.776368$. Status: DERIVED.

The terms. W^+ and W^- are sourced by the charged currents and have the tip mass term with $\hat{g}_2^2$; the Z is sourced by the neutral current $J_2^3 - \sin^2\theta_{\mathrm{W}} J_{\mathrm{em}}$ and has the tip mass term with $\hat{g}_Z^2$; the photon is sourced by the electromagnetic current with the coupling $\hat{e}$ and has no tip term, so it stays massless. Status: DERIVED (exact at linear order). The tip terms become the Robin conditions FE-51; the $W^\pm$ and Z masses and the tree-level Z widths of three generations reproduce the measured values (residual 2.7×10^{-12} in the recomputed widths; the neglected electroweak loop corrections are 4×10^{-3} of the total width).

4.4 The tip sigma field

$$\nabla^\mu j_\mu - \mathrm{i}\big[W^\mu,\ j_\mu\big] = -\mathrm{i}\,\frac{\tau^I}{2}\sum_{i,j=1}^{3}\Big[\bar{\ell}_i\,\frac{\tau^I}{2}\,\Sigma\begin{pmatrix}(\mathcal{Y}_\nu)_{ij}\,\nu_{Rj}\\(\mathcal{Y}_e)_{ij}\,e_{Rj}\end{pmatrix} + \bar{Q}_i\,\frac{\tau^I}{2}\,\Sigma\begin{pmatrix}(\mathcal{Y}_u)_{ij}\,u_{Rj}\\(\mathcal{Y}_d)_{ij}\,d_{Rj}\end{pmatrix} - \mathrm{h.c.}\Big],$$
$$j_\mu = j_\mu^I\,\frac{\tau^I}{2}, \qquad j_\mu^I = \frac{\mathrm{i}v_\Sigma^2}{4}\,\mathrm{tr}\big(\tau^I(D_\mu\Sigma)\Sigma^\dagger\big), \qquad j_\mu^I\big|_{\Sigma=\mathbf{1}} = \frac{v_\Sigma^2}{4}\big(W_\mu^I - \delta^{I3}B_\mu\big) \tag{FE-26}$$

Symbols in (FE-26).

- ∇: The covariant derivative of a tensor with the Levi-Civita connection of the metric g_{MN} (Christoffel symbols $\Gamma^N{}_{MP}$); ∇_M acts on the index M, $\nabla^M = g^{MN}\nabla_N$. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- j_μ, j_μ^I: The current of the sigma field on the tip, $j_\mu^I = \frac{\mathrm{i}v_\Sigma^2}{4}\mathrm{tr}(\tau^I(D_\mu\Sigma)\Sigma^\dagger)$, and the matrix $j_\mu = j_\mu^I\tau^I/2$. Unit: GeV^6. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- W_M^I, W_M, W_{MN}, W_{MN}^I: The SU(2) gauge field (geometric normalisation): the components W_M^I ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W_M^I\tau^I/2$, the field strength W_{MN} and its components W_{MN}^I; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- τ^I: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.

- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- j: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- ℓ_i, $\bar{\ell}_i$: The lepton doublet of the generation i (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- Σ: The tip sigma field: a unitary 2×2 matrix field on the tip ($\Sigma^\dagger \Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal W^+, W^- and Z, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $\mathcal{Y}_\nu$, $\mathcal{Y}_e$, $\mathcal{Y}_u$, $\mathcal{Y}_d$: The complex 3×3 tip Yukawa matrices of the neutrino, charged-lepton, up-quark and down-quark sectors (generation indices i, j); on the static block they become the tip mixing matrices $\mathcal{K}$ (the normalisation of that relation is not fixed: OPEN). Unit: dimensionless. Status: INVENTED.
- ν_R, ν_{Ri}: The right-handed neutrino (a gauge singlet): ν_R the multiplet, ν_{Ri} that of the generation i; its zero modes give Dirac neutrino masses. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- e_R, e_{Ri}: The right-handed charged-lepton singlet: e_R the multiplet, e_{Ri} that of the generation i (e_R, μ_R, τ_R). Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- Q_i, $\bar{Q}_i$: The quark doublet of the generation i (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- u_R, u_{Ri}, $\bar{u}_{Ri}$: The up-type quark singlet: u_R the multiplet, u_{Ri} that of the generation i (u_R, c_R, t_R), $\bar{u}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- d_R, d_{Ri}, $\bar{d}_{Ri}$: The down-type quark singlet: d_R the multiplet, d_{Ri} that of the generation i (d_R, s_R, b_R), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- v_Σ: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu \Sigma)$); on the static block it fixes the Robin parameter β_{W}. Unit: $\mathrm{GeV}^{5/2}$. Value: fixed through $\beta_{\mathrm{W}} = 0.011912$. Status: INVENTED.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{\alpha e}$, δ_{l0}, δ^{I3}); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1 \cdots M_n}_{N_1 \cdots N_n}$, the determinant of the $n \times n$ matrix of Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- B_M, B_{MN}: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.

The terms. The sigma field carries the current j_μ (in the unitary gauge the mass current of W^I and B). Its field equation states that this current is covariantly conserved up to the tip Yukawa terms, which transfer isospin between the doublets and the singlets. It is not an independent equation: the covariant divergence of the left side of FE-23 vanishes identically, so the covariant divergence of its right side, evaluated with the fermion equations FE-14 to FE-19, gives exactly this equation (the Noether identity of the gauge symmetry). In the unitary gauge it therefore holds automatically, and the three modes of Σ are the longitudinal W^+, W^- and Z. Status: STANDARD (for a gauged non-linear sigma model).

4.5 The boundary conditions

The hidden direction is periodic with the period $\pi/(6H)$; each half of a period is the interval $-L \le y \le 0$, joined to the other half at the fold $y = 0$ by the reflection $z \to \pi - z$ and closed at the regular tip $y = -L$. The variational principle with these boundaries gives:

$$\begin{aligned} &z \to \pi - z: \qquad \partial_y F\big|_{y=0} = 0 \ \ \text{(even)} \quad \text{or} \quad F\big|_{y=0} = 0 \ \ \text{(odd)}; \\ &\text{at } y = -L\text{: the natural (Robin) condition of the tip term} \end{aligned} \tag{FE-27}$$

Symbols in (FE-27).

- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$. Unit: dimensionless. Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- y: The label y in ∂_y: the derivative with respect to the hidden coordinate y. Status: STANDARD.
- $F(\xi)$, F_n: The x_0 profile of a boson mode on one half of the period (a function of y or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; F_n is the profile of the n-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.

The terms. At the fold an x_0 profile F is even (zero derivative) or odd (zero value) under the reflection; at the tip the boundary term of the variation gives the natural (Robin) condition of the tip term (zero derivative where there is no tip term). Status: DERIVED. For an x_0 profile they are FE-51.

$$\big(1-\gamma^{x_0}\big)\Psi\big|_{y=0} = 0 \ \ \text{(even)} \quad \text{or} \quad \big(1+\gamma^{x_0}\big)\Psi\big|_{y=0} = 0 \ \ \text{(odd)}, \qquad \big(1-\gamma^{x_0}\big)\Psi\big|_{y=-L} = 0 \tag{FE-28}$$

Symbols in (FE-28).

- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, ... the matrices along x_0, x_4, ...), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.

The terms. At the fold the even modes of dirac16complex satisfy $(1-\gamma^{x_0})\Psi = 0$ and the odd ones $(1+\gamma^{x_0})\Psi = 0$ (both parities are modes of the full period); at the tip $(1-\gamma^{x_0})\Psi = 0$. The conditions are Lagrangian for the boundary form of the Dirac operator, so the operator is self-adjoint. Status: DERIVED. In the block form they are FE-69.

$$\begin{aligned}
&P_{\mathrm{sel}} = \frac{1+\Gamma_4\gamma^{x_0}}{2}\,\frac{1-\mathrm{i}\gamma^{x_5}\gamma^{x_6}}{2};\\
&\text{doublets } (Q,\ell): \quad P_{\mathrm{sel}}\big(1-\gamma^{x_0}\big)\psi\big|_{y=0} = \big(1-P_{\mathrm{sel}}\big)\big(1+\gamma^{x_0}\big)\psi\big|_{y=0} = 0, \qquad \big(1-\gamma^{x_0}\big)\psi\big|_{y=-L} = 0;\\
&\text{singlets } (u_R, d_R, e_R, \nu_R): \quad P_{\mathrm{sel}}\big(1+\gamma^{x_0}\big)\psi\big|_{y=0} = \big(1-P_{\mathrm{sel}}\big)\big(1-\gamma^{x_0}\big)\psi\big|_{y=0} = 0, \qquad \big(1+\gamma^{x_0}\big)\psi\big|_{y=-L} = 0
\end{aligned} \tag{FE-29}$$

Symbols in (FE-29).

- P_{sel}: The projector that selects the two blocks of the half $\mathcal{B} = -1$ in which the Standard-Model zero modes live, $P_{\mathrm{sel}} = \frac{1+\Gamma_4\gamma^{x_0}}{2}\,\frac{1-\mathrm{i}\gamma^{x_5}\gamma^{x_6}}{2}$. Unit: dimensionless. Status: INVENTED.
- Γ_4: The four-dimensional chirality matrix, $\Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$; a doublet zero mode has $\Gamma_4 = +1$ (left-handed), a singlet zero mode $\Gamma_4 = -1$ (right-handed). Unit: dimensionless. Status: DERIVED.

- $\gamma^a,\ \gamma^M,\ \gamma^\mu$: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $Q_i,\ \bar{Q}_i$: The quark doublet of the generation i (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\ell_i,\ \bar{\ell}_i$: The lepton doublet of the generation i (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\psi_i,\ \bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: ψ_i is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation i, ψ_f one fermion component f (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $u_R,\ u_{Ri},\ \bar{u}_{Ri}$: The up-type quark singlet: u_R the multiplet, u_{Ri} that of the generation i (u_R, c_R, t_R), $\bar{u}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $d_R,\ d_{Ri},\ \bar{d}_{Ri}$: The down-type quark singlet: d_R the multiplet, d_{Ri} that of the generation i (d_R, s_R, b_R), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $e_R,\ e_{Ri}$: The right-handed charged-lepton singlet: e_R the multiplet, e_{Ri} that of the generation i (e_R, μ_R, τ_R). Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\nu_R,\ \nu_{Ri}$: The right-handed neutrino (a gauge singlet): ν_R the multiplet, ν_{Ri} that of the generation i; its zero modes give Dirac neutrino masses. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.

The terms. P_{sel} selects the two blocks (of the half $\mathcal{B} = -1$) in which the Standard-Model zero modes live. The conditions give every doublet exactly one Weyl zero mode with $\Gamma_4 = +1$ (left-handed) and every singlet exactly one with $\Gamma_4 = -1$ (right-handed), all localised at the fold: the four-dimensional fermion content is exactly that of three Standard-Model generations, each with its right-handed neutrino. Status: INVENTED (the conditions), self-adjoint DERIVED.

$$\begin{aligned}
&\text{one generation:}\quad b_L(-L) = \mathcal{K}\, b_R(-L), \qquad a_R(-L) = -\mathcal{K}\, a_L(-L), \qquad \chi = (a,\ \mathrm{i}b);\\
&\text{three generations:}\quad b_L(-L) = \mathcal{K}\, b_R(-L), \qquad a_R(-L) = -\mathcal{K}^\dagger a_L(-L), \qquad \mathcal{K} \in \{\mathcal{K}_\nu, \mathcal{K}_e, \mathcal{K}_u, \mathcal{K}_d\};\\
&\text{UM-2:}\quad \mathcal{K}_e = \mathcal{K}_u = \mathbf{1}, \qquad \mathcal{K}_\nu = U, \qquad \mathcal{K}_d = V_{\mathrm{CKM}}
\end{aligned} \tag{FE-30}$$

Symbols in (FE-30).

- $a,\ b$: The upper and lower components of a block spinor, $\chi = (a, \mathrm{i}b)$, real at $k = 0$: a_L, b_L of a doublet and a_R, b_R of a singlet solution (a_{Ri}, b_{Ri} of the generation i, $b_{L\alpha}$, $a_{L\alpha}$ of the doublet in the flavour basis), a_o, b_o of the orbital o. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- L: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1+\Gamma_4)/2$. Status: STANDARD.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $\mathcal{K}$: The tip mixing matrix of a sector in the boundary conditions of the fermion zero modes (a number $\mathcal{K}$ for one generation, a 3×3 matrix for three): $\mathcal{K}_\nu$, $\mathcal{K}_e$, $\mathcal{K}_u$, $\mathcal{K}_d$ of the neutrino, charged-lepton, up-quark and down-quark sectors; UM-2 takes $\mathcal{K}_e = \mathcal{K}_u = \mathbf{1}$, $\mathcal{K}_\nu = U$, $\mathcal{K}_d = V_{\mathrm{CKM}}$. Unit: dimensionless. Status: ASSUMED.
- R: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1-\Gamma_4)/2$. Status: STANDARD.
- $\chi,\ \chi_o,\ \chi_1,\ \chi_2$: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi =$

$(\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \, \mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- U: The PMNS matrix, the unitary 3×3 mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states i); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212^\circ$ (NuFIT 6.0). Status: STANDARD.
- V_{CKM}: The CKM matrix, the unitary 3×3 mixing matrix of the quarks (the same parametrisation as U); in UM-2 $V_{\mathrm{CKM}} = \mathcal{K}_u^\dagger \mathcal{K}_d$. Unit: dimensionless. Value: $s_{12} = 0.22501$, $s_{13} = 0.003732$, $s_{23} = 0.04183$, $\delta = 1.147$ (PDG). Status: STANDARD.

The terms. The tip Yukawa terms are equivalent to these mixing conditions between the lower components b and upper components a of the doublet (L) and singlet (R) solutions at the tip, with the tip mixing number $\mathcal{K}$ for one generation and the 3×3 tip matrices of each sector for three generations. UM-2 takes $\mathcal{K}_e = \mathcal{K}_u = \mathbf{1}$, $\mathcal{K}_\nu = U$ (PMNS) and $\mathcal{K}_d = V_{\mathrm{CKM}}$, so the mixing matrices of the leptons and quarks are those measured. Status: DERIVED (self-adjoint for every complex matrix, sympy, 42 real variables), the choice of the tip matrices ASSUMED; the normalisation that relates $\mathcal{K}$ to the Yukawa matrices $\mathcal{Y}$ of the action is not fixed (OPEN). Solved exactly by the mass conditions FE-59 and FE-60.

5 The background

The background is order 0 of the metric power series (FE-87): the homogeneous metric of the author with the block of the hidden direction and the three extra times, sourced by the block-averaged energy-momentum tensor of every field. Its equations are exact; they are solved in closed form for the static block and numerically for the primordial approach to it.

5.1 The order-0 metric and its curvature

$$\begin{aligned}\bar{g} = \mathrm{diag}\Big(&\mathrm{e}^{-2a_4} b_4 \cot^2 z,\ \mathrm{e}^{2a_4},\ \mathrm{e}^{2a_4},\ \mathrm{e}^{2a_4},\ -1,\\ &-\mathrm{e}^{-2a_4} b_4 (\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4} b_4 (\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4} b_4 (\sin^2 z)^{1/6}\Big), \qquad z = 6Hx_0\end{aligned} \tag{FE-31}$$

Symbols in (FE-31).

- $\bar{g}_{MN}$: The order-0 (background) metric of the power series: the corrected primordial metric in the author's coordinates, determined by the two functions $a_4(x_4)$ and $b_4(x_4)$; $|\bar{g}|$ is the absolute value of its determinant. Unit: dimensionless. Status: AUTHOR.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.

The terms. The diagonal entries are, in the order $x_0, x_1, \ldots, x_7$: the hidden direction, whose metric coefficient $\mathrm{e}^{-2a_4} b_4 \cot^2 z$ depends on x_0 through the angle $z = 6Hx_0$ and on the time through $a_4(x_4)$ and $b_4(x_4)$; ordinary space with the scale factor e^{a_4}; the time x_4; and the three extra times, whose coefficient depends on x_0 through $(\sin^2 z)^{1/6}$. The metric is periodic in x_0 with the period $\pi/(6H)$. It is the metric of the gravitational field only: every other field enters through the right side of FE-1. Status: AUTHOR (the corrected primordial metric of the record, in the author's coordinates).

$$
\begin{aligned}
&y = \frac{\ln|\sin z|}{6H}, \qquad \mathrm{d}y = \cot z \, \mathrm{d}x_0, \qquad \xi = \frac{y}{L} \in [-1, 0], \\
&\bar{g}_{MN} \, \mathrm{d}X^M \mathrm{d}X^N = \mathrm{e}^{2a_4} \big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{d}x_4^2 + f^2 \Big[\mathrm{d}y^2 - \mathrm{e}^{2Hy} \big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big)\Big], \\
&f = \mathrm{e}^{-a_4} b_4^{1/2}, \qquad K = \frac{1}{f^2}, \qquad \mathcal{H} = \dot{a}_4, \qquad \mathcal{H}_f = \frac{\dot{f}}{f} = \frac{\dot{b}_4}{2b_4} - \dot{a}_4, \qquad \dot{K} = -2K\mathcal{H}_f, \\
&\sqrt{|\bar{g}|} = \mathrm{e}^{3a_4} f^4 |\cot z| (\sin^2 z)^{1/4} = \mathrm{e}^{3a_4} f^4 \, \mathrm{e}^{3Hy} \Big| \frac{\mathrm{d}y}{\mathrm{d}x_0} \Big|
\end{aligned}
\tag{FE-32}
$$

Symbols in (FE-32).

- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $\bar{g}_{MN}$: The order-0 (background) metric of the power series: the corrected primordial metric in the author's coordinates, determined by the two functions $a_4(x_4)$ and $b_4(x_4)$; $|\bar{g}|$ is the absolute value of its determinant. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8 X = \mathrm{d}x_0 \, \mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space (also the symmetry axis of the fast-flavour analysis). Unit: length. Status: AUTHOR.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.

- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\rm Pl}$). Unit: GeV. Status: DERIVED.
- $\dot{a}_4$: The time derivative $\mathrm{d}a_4/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot{f}/f = \dot{b}_4/(2b_4) - \dot{a}_4$ (negative when the extra times and x_0 deflate). Unit: GeV. Status: DERIVED.
- $\dot{f}$: The time derivative $\mathrm{d}f/\mathrm{d}x_4$ of the block scale. Unit: GeV. Status: DERIVED.
- $\dot{b}_4$: The time derivative $\mathrm{d}b_4/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- $\dot{K}$: The time derivative $\mathrm{d}K/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.

The terms. y is the proper hidden coordinate in units of the block scale: in it the hidden direction and the extra times form the block $f^2[\mathrm{d}y^2 - \mathrm{e}^{2Hy}(\dots)]$, a four-dimensional space of constant curvature $-H^2/f^2$ scaled by one function f, the block scale. The variable $K = 1/f^2$ measures the curvature of the block, $\mathcal{H}$ is the expansion rate of ordinary space (the Hubble rate) and $\mathcal{H}_f$ the rate of change of the block. One period of x_0 is two halves joined at the fold $y = 0$ (where $\sin^2 z = 1$); each half is closed by the regular tip $y = -L$, with $HL = 3$ (ASSUMED). The last line is the volume element in both charts. Status: DERIVED (exact change of chart).

$$R^{x_1x_2}{}_{x_1x_2} = \mathcal{H}^2, \qquad R^{x_1x_4}{}_{x_1x_4} = \mathcal{H}^2 + \dot{\mathcal{H}}, \qquad R^{x_1x_0}{}_{x_1x_0} = \mathcal{H}\mathcal{H}_f, \qquad R^{x_4x_0}{}_{x_4x_0} = \mathcal{H}_f^2 + \dot{\mathcal{H}}_f,$$
$$R^{x_0x_5}{}_{x_0x_5} = \mathcal{H}_f^2 - H^2K; \qquad \text{(FE-33)}$$

the same for every pair of the same kind; all other $R^{ab}{}_{cd}$ with $\{a,b\} \neq \{c,d\}$ vanish

Symbols in (FE-33).

- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor, with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the Ricci (scalar) curvature $R = g^{MN}R_{MN}$; $R^{ab}{}_{ab}$ (no sum) is the sectional curvature of the coordinate plane (a,b). Unit: 1/length^2. Status: STANDARD.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\rm Pl}$). Unit: GeV. Status: DERIVED.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\dot{\mathcal{H}}$: The time derivative $\mathrm{d}\mathcal{H}/\mathrm{d}x_4$. Unit: GeV^2. Status: DERIVED.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot{f}/f = \dot{b}_4/(2b_4) - \dot{a}_4$ (negative when the extra times and x_0 deflate). Unit: GeV. Status: DERIVED.
- $\dot{\mathcal{H}}_f$: The time derivative $\mathrm{d}\mathcal{H}_f/\mathrm{d}x_4$. Unit: GeV^2. Status: DERIVED.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\rm W}$ and $\rho_{\rm EW,0}$). Status: AUTHOR.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.

The terms. The curvature of the order-0 metric is pair-diagonal: the only non-zero components of the Riemann tensor are the sectional curvatures of the coordinate planes, five kinds in all. $\mathcal{H}^2$ is the curvature of a plane of ordinary space, $\mathcal{H}^2 + \dot{\mathcal{H}}$ that of a space-time plane, $\mathcal{H}\mathcal{H}_f$ that of a plane spanned by ordinary space and a block direction, $\mathcal{H}_f^2 + \dot{\mathcal{H}}_f$ that of the plane of the time and a block direction, and $\mathcal{H}_f^2 - H^2K$ that of a plane of two block directions, which contains the constant curvature of the block. None of them depends on x_0: the metric depends on x_0, its curvature does not. Status: DERIVED (sympy, identical in both charts; 25 checks of the geometry).

$$\mathcal{L}_{(1)} = 2\sum_{\text{pairs}} R^{ab}{}_{ab}, \qquad \mathcal{L}_{(2)} = 8\sum_{\text{2 disjoint pairs}}\prod_{\text{pairs}} R^{ab}{}_{ab}, \qquad \mathcal{L}_{(3)} = 48\sum_{\text{3 disjoint pairs}}\prod_{\text{pairs}} R^{ab}{}_{ab},$$
$$\mathcal{E}_{(1)}{}^{c}{}_{c} = -\sum_{\text{pairs avoiding } c} R^{ab}{}_{ab}, \qquad \mathcal{E}_{(2)}{}^{c}{}_{c} = -4\sum_{\text{2 disjoint pairs avoiding } c}\prod_{\text{pairs}} R^{ab}{}_{ab}, \tag{FE-34}$$
$$\mathcal{E}_{(3)}{}^{c}{}_{c} = -24\sum_{\text{3 disjoint pairs avoiding } c}\prod_{\text{pairs}} R^{ab}{}_{ab}$$

Symbols in (FE-34).

- $\mathcal{L}_{(p)}$: The Lovelock density of order p of the metric (a scalar built from p Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\text{length}^{2p}$. Status: STANDARD.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor, with two indices the Ricci tensor $R_{MN} = R^{P}{}_{MPN}$, without indices the Ricci (scalar) curvature $R = g^{MN}R_{MN}$; $R^{ab}{}_{ab}$ (no sum) is the sectional curvature of the coordinate plane (a, b). Unit: $1/\text{length}^2$. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$, aligned with $x_0, \ldots, x_7$; raised and lowered with η_{ab}; a repeated index is summed (except in a sectional curvature $R^{ab}{}_{ab}$). Status: STANDARD.
- b: A frame index running over $0, \ldots, 7$ (as a). Status: STANDARD.
- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order p, the metric variation of $\sqrt{|g|}\mathcal{L}_{(p)}$: $\mathcal{E}_{(1)}$ is the Einstein tensor $R^{M}{}_{N} - \frac{1}{2}\delta^{M}_{N}R$; every $\mathcal{E}_{(p)}$ is divergence free. Unit: $1/\text{length}^{2p}$. Status: STANDARD.
- c: A frame index running over $0, \ldots, 7$ (as a). Status: STANDARD.

The terms. For a pair-diagonal curvature the Lovelock density of order p is $2^p p!$ times the sum over all sets of p disjoint coordinate pairs of the product of their sectional curvatures, and the diagonal component $\mathcal{E}_{(p)}{}^{c}{}_{c}$ is $-2^{p-1}p!$ times the same sum restricted to the pairs that avoid the direction c. This turns the Lovelock tensors of the order-0 metric into polynomials of the five curvatures of FE-33. Status: DERIVED (checked against the Ricci route for $p = 1$, the explicit Gauss-Bonnet tensor for $p = 2$, the Bianchi identities and the minisuperspace variation with a lapse for $p = 1, 2, 3$).

5.2 The Lovelock tensors of the order-0 metric

$$\begin{aligned}
\mathcal{E}_{(1)}{}^{x_4}{}_{x_4} &= -3\,\mathcal{H}^2 - 12\,\mathcal{H}\mathcal{H}_f - 6\,\mathcal{H}_f^2 + 6\,KH^2 \\
\mathcal{E}_{(1)}{}^{x_1}{}_{x_1} &= -3\,\mathcal{H}^2 - 8\,\mathcal{H}\mathcal{H}_f - 10\,\mathcal{H}_f^2 - 2\,\dot{\mathcal{H}} - 4\,\dot{\mathcal{H}}_f + 6\,KH^2 \\
\mathcal{E}_{(1)}{}^{x_0}{}_{x_0} &= -6\,\mathcal{H}^2 - 9\,\mathcal{H}\mathcal{H}_f - 6\,\mathcal{H}_f^2 - 3\,\dot{\mathcal{H}} - 3\,\dot{\mathcal{H}}_f + 3\,KH^2
\end{aligned} \tag{FE-35}$$

Symbols in (FE-35).

- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order p, the metric variation of $\sqrt{|g|}\mathcal{L}_{(p)}$: $\mathcal{E}_{(1)}$ is the Einstein tensor $R^{M}{}_{N} - \frac{1}{2}\delta^{M}_{N}R$; every $\mathcal{E}_{(p)}$ is divergence free. Unit: $1/\text{length}^{2p}$. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\rm Pl}$). Unit: GeV. Status: DERIVED.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot{f}/f = \dot{b}_4/(2b_4) - \dot{a}_4$ (negative when the extra times and x_0 deflate). Unit: GeV. Status: DERIVED.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\rm W}$ and $\rho_{\rm EW,0}$). Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $\dot{\mathcal{H}}$: The time derivative $\mathrm{d}\mathcal{H}/\mathrm{d}x_4$. Unit: GeV^2. Status: DERIVED.
- $\dot{\mathcal{H}}_f$: The time derivative $\mathrm{d}\mathcal{H}_f/\mathrm{d}x_4$. Unit: GeV^2. Status: DERIVED.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.

The terms. The Einstein tensor of the order-0 metric. The x_4 component contains no time derivative of the rates (it is a constraint); the x_1 and x_0 components contain $\dot{\mathcal{H}}$ and $\dot{\mathcal{H}}_f$. The term $6KH^2$ (and $3KH^2$) is the curvature of

the block. Every off-diagonal component vanishes, and the x_0 component equals the x_5, x_6, x_7 components: all four block directions have the same left side. Status: DERIVED (exact).

$$\begin{aligned}
\mathcal{E}_{(2)}{}^{x_4}{}_{x_4} =& -48\,\mathcal{H}^3\mathcal{H}_f - 216\,\mathcal{H}^2\mathcal{H}_f^2 - 144\,\mathcal{H}\mathcal{H}_f^3 - 12\,\mathcal{H}_f^4 + 72\,K\mathcal{H}^2H^2 \\
&+ 144\,K\mathcal{H}\mathcal{H}_fH^2 + 24\,K\mathcal{H}_f^2H^2 - 12\,K^2H^4 \\
\mathcal{E}_{(2)}{}^{x_1}{}_{x_1} =& -32\,\mathcal{H}^3\mathcal{H}_f - 136\,\mathcal{H}^2\mathcal{H}_f^2 - 192\,\mathcal{H}\mathcal{H}_f^3 - 60\,\mathcal{H}_f^4 \\
&- 32\,\dot{\mathcal{H}}\mathcal{H}\mathcal{H}_f - 48\,\dot{\mathcal{H}}\mathcal{H}_f^2 - 16\,\dot{\mathcal{H}}_f\mathcal{H}^2 - 96\,\dot{\mathcal{H}}_f\mathcal{H}\mathcal{H}_f \\
&- 48\,\dot{\mathcal{H}}_f\mathcal{H}_f^2 + 72\,K\mathcal{H}^2H^2 + 96\,K\mathcal{H}\mathcal{H}_fH^2 + 72\,K\mathcal{H}_f^2H^2 + 48\,\dot{\mathcal{H}}KH^2 \\
&+ 48\,\dot{\mathcal{H}}_fKH^2 - 12\,K^2H^4 \\
\mathcal{E}_{(2)}{}^{x_0}{}_{x_0} =& -12\,\mathcal{H}^4 - 108\,\mathcal{H}^3\mathcal{H}_f - 180\,\mathcal{H}^2\mathcal{H}_f^2 - 108\,\mathcal{H}\mathcal{H}_f^3 - 12\,\mathcal{H}_f^4 \\
&- 12\,\dot{\mathcal{H}}\mathcal{H}^2 - 72\,\dot{\mathcal{H}}\mathcal{H}\mathcal{H}_f - 36\,\dot{\mathcal{H}}\mathcal{H}_f^2 - 36\,\dot{\mathcal{H}}_f\mathcal{H}^2 \\
&- 72\,\dot{\mathcal{H}}_f\mathcal{H}\mathcal{H}_f - 12\,\dot{\mathcal{H}}_f\mathcal{H}_f^2 + 72\,K\mathcal{H}^2H^2 + 36\,K\mathcal{H}\mathcal{H}_fH^2 + 12\,K\mathcal{H}_f^2H^2 \\
&+ 36\,\dot{\mathcal{H}}KH^2 + 12\,\dot{\mathcal{H}}_fKH^2
\end{aligned} \tag{FE-36}$$

Symbols in (FE-36).

- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order p, the metric variation of $\sqrt{|g|}\mathcal{L}_{(p)}$: $\mathcal{E}_{(1)}$ is the Einstein tensor $R^{M}{}_{N} - \frac{1}{2}\delta^{M}_{N}R$; every $\mathcal{E}_{(p)}$ is divergence free. Unit: $1/\text{length}^{2p}$. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot{f}/f = \dot{b}_4/(2b_4) - \dot{a}_4$ (negative when the extra times and x_0 deflate). Unit: GeV. Status: DERIVED.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $\dot{\mathcal{H}}$: The time derivative $\mathrm{d}\mathcal{H}/\mathrm{d}x_4$. Unit: GeV^2. Status: DERIVED.
- $\dot{\mathcal{H}}_f$: The time derivative $\mathrm{d}\mathcal{H}_f/\mathrm{d}x_4$. Unit: GeV^2. Status: DERIVED.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.

The terms. The Gauss-Bonnet tensor: products of two sectional curvatures, so every term is of fourth order in the rates and H (with at most one time derivative of a rate). It is weighted by $\alpha_{(2)} = H^{-2}/32$ in FE-1. Status: DERIVED (sympy, four routes).

$$\begin{aligned}
\mathcal{E}_{(3)}{}^{x_4}{}_{x_4} =& -1440\,\mathcal{H}^3\mathcal{H}_f^3 - 1080\,\mathcal{H}^2\mathcal{H}_f^4 + 864\,K\mathcal{H}^3\mathcal{H}_fH^2 + 1296\,K\mathcal{H}^2\mathcal{H}_f^2H^2 \\
&- 216\,K^2\mathcal{H}^2H^4 \\
\mathcal{E}_{(3)}{}^{x_1}{}_{x_1} =& -576\,\mathcal{H}^3\mathcal{H}_f^3 - 1368\,\mathcal{H}^2\mathcal{H}_f^4 - 576\,\mathcal{H}\mathcal{H}_f^5 - 576\,\dot{\mathcal{H}}\mathcal{H}\mathcal{H}_f^3 \\
&- 144\,\dot{\mathcal{H}}\mathcal{H}_f^4 - 864\,\dot{\mathcal{H}}_f\mathcal{H}^2\mathcal{H}_f^2 - 576\,\dot{\mathcal{H}}_f\mathcal{H}\mathcal{H}_f^3 + 576\,K\mathcal{H}^3\mathcal{H}_fH^2 \\
&+ 1008\,K\mathcal{H}^2\mathcal{H}_f^2H^2 + 576\,K\mathcal{H}\mathcal{H}_f^3H^2 + 576\,\dot{\mathcal{H}}K\mathcal{H}\mathcal{H}_fH^2 \\
&+ 288\,\dot{\mathcal{H}}K\mathcal{H}_f^2H^2 + 288\,\dot{\mathcal{H}}_fK\mathcal{H}^2H^2 + 576\,\dot{\mathcal{H}}_fK\mathcal{H}\mathcal{H}_fH^2 - 216\,K^2\mathcal{H}^2H^4 \\
&- 144\,\dot{\mathcal{H}}K^2H^4 \\
\mathcal{E}_{(3)}{}^{x_0}{}_{x_0} =& -648\,\mathcal{H}^4\mathcal{H}_f^2 - 1224\,\mathcal{H}^3\mathcal{H}_f^3 - 648\,\mathcal{H}^2\mathcal{H}_f^4 - 648\,\dot{\mathcal{H}}\mathcal{H}^2\mathcal{H}_f^2 \\
&- 432\,\dot{\mathcal{H}}\mathcal{H}\mathcal{H}_f^3 - 432\,\dot{\mathcal{H}}_f\mathcal{H}^3\mathcal{H}_f - 648\,\dot{\mathcal{H}}_f\mathcal{H}^2\mathcal{H}_f^2 + 216\,K\mathcal{H}^4H^2 \\
&+ 648\,K\mathcal{H}^3\mathcal{H}_fH^2 + 216\,K\mathcal{H}^2\mathcal{H}_f^2H^2 + 216\,\dot{\mathcal{H}}K\mathcal{H}^2H^2 + 432\,\dot{\mathcal{H}}K\mathcal{H}\mathcal{H}_fH^2 \\
&+ 216\,\dot{\mathcal{H}}_fK\mathcal{H}^2H^2
\end{aligned} \tag{FE-37}$$

Symbols in (FE-37).

- ${\mathcal{E}_{(p)}}^{M}{}_{N}$: The Lovelock tensor of order p, the metric variation of $\sqrt{|g|}\mathcal{L}_{(p)}$: $\mathcal{E}_{(1)}$ is the Einstein tensor ${R^{M}}_{N} - \frac{1}{2}\delta^{M}_{N}R$; every $\mathcal{E}_{(p)}$ is divergence free. Unit: 1/length^{2p}. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot{f}/f = \dot{b}_4/(2b_4) - \dot{a}_4$ (negative when the extra times and x_0 deflate). Unit: GeV. Status: DERIVED.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $\dot{\mathcal{H}}$: The time derivative $\mathrm{d}\mathcal{H}/\mathrm{d}x_4$. Unit: GeV^2. Status: DERIVED.
- $\dot{\mathcal{H}}_f$: The time derivative $\mathrm{d}\mathcal{H}_f/\mathrm{d}x_4$. Unit: GeV^2. Status: DERIVED.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.

The terms. The cubic Lovelock tensor: products of three sectional curvatures (sixth order), weighted by $\alpha_{(3)} = H^{-4}/1152$. In eight dimensions it is the highest Lovelock order that contributes to the field equations. Status: DERIVED (sympy, four routes).

5.3 The background equations

On the order-0 metric every source is diagonal, ${\mathcal{T}^{x_4}}_{x_4} = -\rho$, ${\mathcal{T}^{x_1}}_{x_1} = p_3$, ${\mathcal{T}^{x_0}}_{x_0} = p_0$, ${\mathcal{T}^{x_5}}_{x_5} = p_5$ (FE-2), and the field equations FE-1 reduce to one equation for each kind of direction:

$$\sum_{p=1}^{3}\alpha_{(p)}\,{\mathcal{E}_{(p)}}^{x_4}{}_{x_4} + \Lambda = -\kappa\rho, \qquad \sum_{p=1}^{3}\alpha_{(p)}\,{\mathcal{E}_{(p)}}^{x_1}{}_{x_1} + \Lambda = \kappa p_3,$$
$$\sum_{p=1}^{3}\alpha_{(p)}\,{\mathcal{E}_{(p)}}^{x_0}{}_{x_0} + \Lambda = \kappa p_0, \qquad \sum_{p=1}^{3}\alpha_{(p)}\,{\mathcal{E}_{(p)}}^{x_5}{}_{x_5} + \Lambda = \kappa p_5.$$

Symbols in this equation.

- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)}H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- ${\mathcal{E}_{(p)}}^{M}{}_{N}$: The Lovelock tensor of order p, the metric variation of $\sqrt{|g|}\mathcal{L}_{(p)}$: $\mathcal{E}_{(1)}$ is the Einstein tensor ${R^{M}}_{N} - \frac{1}{2}\delta^{M}_{N}R$; every $\mathcal{E}_{(p)}$ is divergence free. Unit: 1/length^{2p}. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -10H^2$. Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_{\mathrm{N}}\mathcal{G}(K_*) = 2.328 \times 10^{-36}$ GeV^{-2}. Status: AUTHOR.
- ρ: An energy density: in eight dimensions $\rho = -{\mathcal{T}^{x_4}}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- p_3: The pressure of ordinary 3-space, $p_3 = {\mathcal{T}^{x_1}}_{x_1} = {\mathcal{T}^{x_2}}_{x_2} = {\mathcal{T}^{x_3}}_{x_3}$ for an isotropic source. Unit: GeV^8. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- p_0: The pressure along the hidden direction x_0, $p_0 = {\mathcal{T}^{x_0}}_{x_0}$. Unit: GeV^8. Status: STANDARD.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

- p_5: The pressure along the extra times, $p_5 = \mathcal{T}^{x_5}{}_{x_5} = \mathcal{T}^{x_6}{}_{x_6} = \mathcal{T}^{x_7}{}_{x_7}$. Unit: GeV^8. Status: STANDARD.

With the tensors of FE-35 to FE-37, the author's weights and $\Lambda = -10H^2$ they are:

$$
\begin{aligned}
x_4: \quad & -\frac{5\,\mathcal{H}^3\mathcal{H}_f^3}{4H^4} - \frac{15\,\mathcal{H}^2\mathcal{H}_f^4}{16H^4} + \frac{3\,K\mathcal{H}^3\mathcal{H}_f}{4H^2} \\
& + \frac{9\,K\mathcal{H}^2\mathcal{H}_f^2}{8H^2} - \frac{3\,\mathcal{H}^3\mathcal{H}_f}{2H^2} - \frac{27\,\mathcal{H}^2\mathcal{H}_f^2}{4H^2} \\
& - \frac{9\,\mathcal{H}\mathcal{H}_f^3}{2H^2} - \frac{3\,\mathcal{H}_f^4}{8H^2} - \frac{3\,K^2\mathcal{H}^2}{16} + \frac{9\,K\mathcal{H}^2}{4} \\
& + \frac{9\,K\mathcal{H}\mathcal{H}_f}{2} + \frac{3\,K\mathcal{H}_f^2}{4} - 3\,\mathcal{H}^2 - 12\,\mathcal{H}\mathcal{H}_f - 6\,\mathcal{H}_f^2 - \frac{3\,K^2H^2}{8} + 6\,KH^2 - 10\,H^2 \\
& = -\kappa\rho
\end{aligned}
\tag{FE-38}
$$

Symbols in (FE-38).

- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot{f}/f = \dot{b}_4/(2b_4) - \dot{a}_4$ (negative when the extra times and x_0 deflate). Unit: GeV. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_{\mathrm{N}}\mathcal{G}(K_*) = 2.328 \times 10^{-36}$ GeV^{-2}. Status: AUTHOR.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.

The terms. The constraint: the terms divided by H^4 come from the cubic Lovelock tensor, those divided by H^2 from the Gauss-Bonnet tensor, the others from the Einstein tensor; the terms with K and K^2 are the curvature of the block entering through the plane curvatures $\mathcal{H}_f^2 - H^2K$; $-10H^2$ is the cosmological constant. The right side is the total energy density of every field (FE-3). Status: DERIVED. The residual of the stored static block in this equation is 3.9×10^{-13} of its largest term (from the twelve stored digits of K_*; with the closed form 1.2×10^{-16}), and the printed polynomial equals the one in the code symbolically.

$$
\begin{aligned}
x_1:\quad & \Big[-\frac{\mathcal{H}\mathcal{H}_f^3}{2H^4} - \frac{\mathcal{H}_f^4}{8H^4} + \frac{K\mathcal{H}\mathcal{H}_f}{2H^2} + \frac{K\mathcal{H}_f^2}{4H^2} \\
& - \frac{\mathcal{H}\mathcal{H}_f}{H^2} - \frac{3\,\mathcal{H}_f^2}{2H^2} - \frac{K^2}{8} + \frac{3\,K}{2} - 2 \Big] \dot{\mathcal{H}} \\
& + \Big[-\frac{3\,\mathcal{H}^2\mathcal{H}_f^2}{4H^4} - \frac{\mathcal{H}\mathcal{H}_f^3}{2H^4} + \frac{K\mathcal{H}^2}{4H^2} \\
& + \frac{K\mathcal{H}\mathcal{H}_f}{2H^2} - \frac{\mathcal{H}^2}{2H^2} - \frac{3\,\mathcal{H}\mathcal{H}_f}{H^2} - \frac{3\,\mathcal{H}_f^2}{2H^2} + \frac{3\,K}{2} - 4 \Big] \dot{\mathcal{H}}_f \\
& - \frac{\mathcal{H}^3\mathcal{H}_f^3}{2H^4} - \frac{19\,\mathcal{H}^2\mathcal{H}_f^4}{16H^4} - \frac{\mathcal{H}\mathcal{H}_f^5}{2H^4} + \frac{K\mathcal{H}^3\mathcal{H}_f}{2H^2} \\
& + \frac{7\,K\mathcal{H}^2\mathcal{H}_f^2}{8H^2} + \frac{K\mathcal{H}\mathcal{H}_f^3}{2H^2} - \frac{\mathcal{H}^3\mathcal{H}_f}{H^2} - \frac{17\,\mathcal{H}^2\mathcal{H}_f^2}{4H^2} \\
& - \frac{6\,\mathcal{H}\mathcal{H}_f^3}{H^2} - \frac{15\,\mathcal{H}_f^4}{8H^2} - \frac{3\,K^2\mathcal{H}^2}{16} + \frac{9\,K\mathcal{H}^2}{4} \\
& + 3\,K\mathcal{H}\mathcal{H}_f + \frac{9\,K\mathcal{H}_f^2}{4} - 3\,\mathcal{H}^2 - 8\,\mathcal{H}\mathcal{H}_f - 10\,\mathcal{H}_f^2 - \frac{3\,K^2H^2}{8} + 6\,KH^2 - 10\,H^2 \\
& = \kappa p_3
\end{aligned}
\tag{FE-39}
$$

Symbols in (FE-39).

- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\rm Pl}$). Unit: GeV. Status: DERIVED.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot{f}/f = \dot{b}_4/(2b_4) - \dot{a}_4$ (negative when the extra times and x_0 deflate). Unit: GeV. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\rm W}$ and $\rho_{\rm EW,0}$). Status: AUTHOR.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- $\dot{\mathcal{H}}$: The time derivative $\mathrm{d}\mathcal{H}/\mathrm{d}x_4$. Unit: GeV^2. Status: DERIVED.
- $\dot{\mathcal{H}}_f$: The time derivative $\mathrm{d}\mathcal{H}_f/\mathrm{d}x_4$. Unit: GeV^2. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_{\rm N}\mathcal{G}(K_*) = 2.328 \times 10^{-36}$ GeV^{-2}. Status: AUTHOR.
- p_3: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: GeV^8. Status: STANDARD.

The terms. The 3-space equation is linear in $\dot{\mathcal{H}}$ and $\dot{\mathcal{H}}_f$ with coefficients that depend on the rates and on K; it is the equation of motion of the scale factor of ordinary space, with the pressure p_3 of all matter along ordinary space as its source. Status: DERIVED. With FE-40 it is solved for the two rates; the integration from the primordial block to the static block preserves the constraint FE-38 to 6×10^{-10}.

$$
\begin{aligned}
x_0:\ &\Big[-\frac{9\,\mathcal{H}^2\mathcal{H}_f^2}{16H^4}-\frac{3\,\mathcal{H}\mathcal{H}_f^3}{8H^4}+\frac{3\,K\mathcal{H}^2}{16H^2}+\frac{3\,K\mathcal{H}\mathcal{H}_f}{8H^2}\\
&-\frac{3\,\mathcal{H}^2}{8H^2}-\frac{9\,\mathcal{H}\mathcal{H}_f}{4H^2}-\frac{9\,\mathcal{H}_f^2}{8H^2}+\frac{9\,K}{8}-3\Big]\,\dot{\mathcal{H}}\\
&+\Big[-\frac{3\,\mathcal{H}^3\mathcal{H}_f}{8H^4}-\frac{9\,\mathcal{H}^2\mathcal{H}_f^2}{16H^4}+\frac{3\,K\mathcal{H}^2}{16H^2}-\frac{9\,\mathcal{H}^2}{8H^2}\\
&-\frac{9\,\mathcal{H}\mathcal{H}_f}{4H^2}-\frac{3\,\mathcal{H}_f^2}{8H^2}+\frac{3\,K}{8}-3\Big]\,\dot{\mathcal{H}}_f\\
&-\frac{9\,\mathcal{H}^4\mathcal{H}_f^2}{16H^4}-\frac{17\,\mathcal{H}^3\mathcal{H}_f^3}{16H^4}-\frac{9\,\mathcal{H}^2\mathcal{H}_f^4}{16H^4}+\frac{3\,K\mathcal{H}^4}{16H^2}\\
&+\frac{9\,K\mathcal{H}^3\mathcal{H}_f}{16H^2}+\frac{3\,K\mathcal{H}^2\mathcal{H}_f^2}{16H^2}-\frac{3\,\mathcal{H}^4}{8H^2}-\frac{27\,\mathcal{H}^3\mathcal{H}_f}{8H^2}\\
&-\frac{45\,\mathcal{H}^2\mathcal{H}_f^2}{8H^2}-\frac{27\,\mathcal{H}\mathcal{H}_f^3}{8H^2}-\frac{3\,\mathcal{H}_f^4}{8H^2}\\
&+\frac{9\,K\mathcal{H}^2}{4}+\frac{9\,K\mathcal{H}\mathcal{H}_f}{8}+\frac{3\,K\mathcal{H}_f^2}{8}-6\,\mathcal{H}^2-9\,\mathcal{H}\mathcal{H}_f\\
&-6\,\mathcal{H}_f^2+3\,KH^2-10\,H^2\\
&=\kappa p_0;\qquad x_5,x_6,x_7:\ \text{the same left side}=\kappa p_5;\\
&\text{block-isotropic order-0 metric: the same left side}=\kappa\bar{p},\qquad \bar{p}=\tfrac{1}{4}\big(p_0+3p_5\big)
\end{aligned}
\tag{FE-40}
$$

Symbols in (FE-40).

- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\rm Pl}$). Unit: GeV. Status: DERIVED.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot{f}/f = \dot{b}_4/(2b_4) - \dot{a}_4$ (negative when the extra times and x_0 deflate). Unit: GeV. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\rm W}$ and $\rho_{\rm EW,0}$). Status: AUTHOR.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- $\dot{\mathcal{H}}$: The time derivative $\mathrm{d}\mathcal{H}/\mathrm{d}x_4$. Unit: GeV^2. Status: DERIVED.
- $\dot{\mathcal{H}}_f$: The time derivative $\mathrm{d}\mathcal{H}_f/\mathrm{d}x_4$. Unit: GeV^2. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_{\rm N}\mathcal{G}(K_*) = 2.328\times10^{-36}$ GeV^{-2}. Status: AUTHOR.
- p_0: The pressure along the hidden direction x_0, $p_0 = \mathcal{T}^{x_0}{}_{x_0}$. Unit: GeV^8. Status: STANDARD.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- p_5: The pressure along the extra times, $p_5 = \mathcal{T}^{x_5}{}_{x_5} = \mathcal{T}^{x_6}{}_{x_6} = \mathcal{T}^{x_7}{}_{x_7}$. Unit: GeV^8. Status: STANDARD.
- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial \ln f$. Unit: GeV^8 (or GeV^4 for a four-dimensional density). Status: DERIVED.

The terms. The block equation is linear in $\dot{\mathcal{H}}$ and $\dot{\mathcal{H}}_f$ as well; it is the equation of motion of the block scale. Its source is the pressure along the block: p_0 for the x_0 component and p_5 for the x_5, x_6, x_7 components. The left sides of the four block components are identical, so the order-0 metric (one scale f for all four) solves them all only for a block-isotropic source, $p_0 = p_5$; in general f obeys the average with $\bar{p}$ (subsection "The block pressures" below). Status: DERIVED. The primordial solutions of FE-38 to FE-40 (with the stabiliser and radiation) have a one-step defect of 1.3×10^{-11} and a constraint residual of 3.6×10^{-12} when they are re-integrated from the stored samples.

$$
\dot{\rho} = -3\mathcal{H}(\rho + p_3) - \mathcal{H}_f(\rho + p_0) - 3\mathcal{H}_f(\rho + p_5) = -3\mathcal{H}(\rho + p_3) - 4\mathcal{H}_f(\rho + \bar{p})
\tag{FE-41}
$$

Symbols in (FE-41).

- $\dot\rho$: The time derivative $\mathrm{d}\rho/\mathrm{d}x_4$. Unit: GeV^9. Status: STANDARD.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H}=\dot a_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H}=(8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- ρ: An energy density: in eight dimensions $\rho=-\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- p_3: The pressure of ordinary 3-space, $p_3=\mathcal{T}^{x_1}{}_{x_1}=\mathcal{T}^{x_2}{}_{x_2}=\mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: GeV^8. Status: STANDARD.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f=\dot f/f=\dot b_4/(2b_4)-\dot a_4$ (negative when the extra times and x_0 deflate). Unit: GeV. Status: DERIVED.
- p_0: The pressure along the hidden direction x_0, $p_0=\mathcal{T}^{x_0}{}_{x_0}$. Unit: GeV^8. Status: STANDARD.
- p_5: The pressure along the extra times, $p_5=\mathcal{T}^{x_5}{}_{x_5}=\mathcal{T}^{x_6}{}_{x_6}=\mathcal{T}^{x_7}{}_{x_7}$. Unit: GeV^8. Status: STANDARD.
- $\bar p$: The hidden pressure, the pressure averaged over the block, $\bar p=(p_0+3p_5)/4$; for a species at fixed temperature $\bar p=\frac{1}{4}\,\partial p/\partial\ln f$. Unit: GeV^8 (or GeV^4 for a four-dimensional density). Status: DERIVED.

The terms. The conservation law FE-11 on the order-0 metric: the energy density decreases by the work done by the pressure along each direction, p_3 along the three directions of ordinary space (rate $\mathcal{H}$), p_0 along the hidden direction and p_5 along the three extra times (rate $\mathcal{H}_f$); the four block terms combine to $4\mathcal{H}_f(\rho+\bar p)$. For the stabiliser ($p_0=p_5=\rho_{\mathrm{stab}}$, $p_3=-\rho_{\mathrm{stab}}$) it gives $\dot\rho_{\mathrm{stab}}=-8\mathcal{H}_f\rho_{\mathrm{stab}}$, so $\rho_{\mathrm{stab}}\propto f^{-8}$ as assumed in IN-4. Status: DERIVED.

5.4 The block pressures

The block scale f multiplies the four block directions together, so the action depends on it through the sum of the four block components. For a change $\Delta\ln f$ of the block scale ($\Delta g^{MM}=-2g^{MM}\Delta\ln f$ for the four block directions), the change of the action is the sum of the four block equations, and because their left sides are equal,

$$\begin{aligned}\frac{\Delta S}{\Delta\ln f}&=-\frac{1}{\kappa}\sqrt{|g|}\sum_{M\in\{x_0,\,x_5,\,x_6,\,x_7\}}\Big(\sum_{p=1}^{3}\alpha_{(p)}\,\mathcal{E}_{(p)}{}^{M}{}_{M}+\Lambda-\kappa\,\mathcal{T}^{M}{}_{M}\Big)\\&=-\frac{4}{\kappa}\sqrt{|g|}\,\Big(\sum_{p=1}^{3}\alpha_{(p)}\,\mathcal{E}_{(p)}{}^{x_0}{}_{x_0}+\Lambda-\kappa\bar p\Big),\qquad \bar p=\tfrac{1}{4}\big(p_0+3p_5\big).\end{aligned}$$

Symbols in this equation.

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21}=m_2^2-m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- S: The action of UM-2 (the integral of the Lagrangian density over spacetime); its variation with respect to every field gives the field equations. Unit: dimensionless ($\hbar=1$). Status: ASSUMED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f=\mathrm{e}^{-a_4}b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa=8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp=8\pi G_{\mathrm{N}}\mathcal{G}(K_*)=2.328\times10^{-36}\ \mathrm{GeV}^{-2}$. Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\dots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N=\delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1,M_2,\dots$): it runs over $x_0,x_1,\dots,x_7$; a repeated index is summed. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z=6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)} H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- ${\mathcal{E}_{(p)}}^M{}_N$: The Lovelock tensor of order p, the metric variation of $\sqrt{|g|}\mathcal{L}_{(p)}$: $\mathcal{E}_{(1)}$ is the Einstein tensor ${R^M}_N - \frac{1}{2}\delta^M_N R$; every $\mathcal{E}_{(p)}$ is divergence free. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -10H^2$. Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.
- ${\mathcal{T}^M}_N$: The energy-momentum tensor of all matter (every field except gravity, the stabiliser included), the source of gravity; defined by the change of the matter action under a change of the metric. Unit: GeV^8. Status: STANDARD.
- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial \ln f$. Unit: GeV^8 (or GeV^4 for a four-dimensional density). Status: DERIVED.
- p_0: The pressure along the hidden direction x_0, $p_0 = {\mathcal{T}^{x_0}}_{x_0}$. Unit: GeV^8. Status: STANDARD.
- p_5: The pressure along the extra times, $p_5 = {\mathcal{T}^{x_5}}_{x_5} = {\mathcal{T}^{x_6}}_{x_6} = {\mathcal{T}^{x_7}}_{x_7}$. Unit: GeV^8. Status: STANDARD.

So the equation of motion of f is the block equation FE-40 with the hidden pressure $\bar{p}$, the average of the block pressures. The difference of the block pressures,

$$p_0 = \bar{p} + \tfrac{3}{4}\big(p_0 - p_5\big), \qquad p_5 = \bar{p} - \tfrac{1}{4}\big(p_0 - p_5\big),$$

Symbols in this equation.

- p_0: The pressure along the hidden direction x_0, $p_0 = {\mathcal{T}^{x_0}}_{x_0}$. Unit: GeV^8. Status: STANDARD.
- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial \ln f$. Unit: GeV^8 (or GeV^4 for a four-dimensional density). Status: DERIVED.
- p_5: The pressure along the extra times, $p_5 = {\mathcal{T}^{x_5}}_{x_5} = {\mathcal{T}^{x_6}}_{x_6} = {\mathcal{T}^{x_7}}_{x_7}$. Unit: GeV^8. Status: STANDARD.

cannot be absorbed by f: it would change the shape of the block (the proper length of the hidden direction against that of the extra times), a mode outside the order-0 metric. This block-shape mode and its response to $p_0 - p_5$ are not computed in UM-2 (OPEN). The physically correct block equations are therefore: the x_0 component with p_0, the extra-time components with p_5, and, for the order-0 metric, their average with $\bar{p}$.

The sources of UM-2 are:

- the stabiliser: block-isotropic, $p_0 = p_5 = \rho_{\mathrm{stab}}$, so $\bar{p} = p_0 = p_5$ and its static block (FE-42, FE-43) is an exact solution of all eight components;
- the Standard-Model plasma and matter, and the dark matter: the masses of the zero modes and of the dark matter depend on the proper length of the hidden direction only (they are x_0 eigenvalues), so their mass terms contribute to p_0 only, $p_0 = -(\rho - 3p_3)\,\mathrm{d}\ln m/\mathrm{d}\ln f$ for an ideal gas, and nothing to p_5 for quanta without momentum along the extra times; the parts of the plasma pressure that depend on the gauge couplings (the four-dimensional couplings scale with the volume $V_\perp$ of all four block directions, FE-82, and with them the QCD scale, FE-84) contribute to p_0 and p_5 alike; for the dark matter at rest $p_0/\rho = -\mathrm{d}\ln\varepsilon_{\mathrm{DM}}/\mathrm{d}\ln f = 0.772034$ and $p_5 = 0$, so $\bar{p} = p_0/4$ and $|p_0 - p_5|/\bar{p} = 4$;
- the Kohn-Sham states of dirac16complex: p_0 and p_5 from FE-67; for the self-interacting test states $|p_0 - p_5|/\bar{p} = 0.90$.

The block-shape source is therefore not small compared with the hidden pressure that is kept; its absolute size is suppressed like that of $\bar{p}$, at most 3×10^{-32} of the stabiliser (at the dark-matter freeze-out), if the shape mode is stable (OPEN).

What the code does. The background code `um2_background.py` writes the block equation once, for "$x_0 = x_5 = x_6 = x_7$", with one symbol for the block pressure; it is used only for block-isotropic sources: the stabiliser ($p_0 = p_5 = \rho_{\mathrm{stab}}$; and the stability family $p_0 = p_5 = \nu\rho_{\mathrm{stab}}$, $\nu = \frac{1}{4}$ to 3), for which that symbol equals p_0, p_5 and $\bar{p}$ alike. Its first-order response to matter (FE-45 to FE-47) is written for $\Delta\bar{p} = \frac{1}{4}(\Delta p_0 + 3\Delta p_5)$. The Kohn-Sham job `um2_ks_jobs.py` (feedback) forms $\bar{p} = \frac{1}{4}(p_0 + 3p_5)$ from the cell averages of FE-67 and stores $p_0 - p_5$ separately; the coupled solution `um2_coupled.py` uses $\bar{p} = \frac{1}{4}\,\mathrm{d}p/\mathrm{d}\ln f$ at fixed temperature (FE-81), which is $\frac{1}{4}(p_0 + 3p_5)$ for every source, because a change of $\ln f$ scales the four block directions together (for the mass terms $p_5 = 0$ and it is

$p_0/4$); and the residual analysis `um2_residuals.py` recomputes the response with $\bar{p} = \frac{1}{4}(p_0 + 3p_5)$ and reports the block-shape source. No stored number rests on an inconsistent choice of the block pressure. Version 3 of the UM-2 documents wrote $\kappa\bar{p}$ in the block equation of the static block and the stabiliser, and in one place κp_0: both are correct there, because the stabiliser is block-isotropic, and $K_\pm$, $\mathcal{G}(K_*)$, $\kappa\rho_{\rm stab}(f_*)$ and $\omega_{\rm r}$ do not depend on the choice.

5.5 The static block and the stabiliser

A static block has $\mathcal{H}_f = 0$ and $\dot{\mathcal{H}}_f = 0$. With $\dot{\mathcal{H}} = 0$ the background equations become:

$$
\begin{aligned}
&x_4: \ -3\mathcal{G}(K)\,\mathcal{H}^2 + 6H^2K\big(1 - 2w_{(2)}K\big) + \Lambda + \kappa\rho = 0, \qquad x_1: \text{ the same with } -\kappa p_3 \text{ in place of } \kappa\rho,\\
&x_0: \ 6\big(12w_{(2)}K - 1\big)\mathcal{H}^2 - \frac{12\big(w_{(2)} - 18w_{(3)}K\big)}{H^2}\,\mathcal{H}^4 + 3H^2K + \Lambda - \kappa p_0 = 0,\\
&x_5, x_6, x_7: \text{ the same with } \kappa p_5 \text{ in place of } \kappa p_0; \qquad \text{block-isotropic order-0 metric: the same with } \kappa\bar{p},\\
&\mathcal{G}(K) = 1 - 24w_{(2)}K + 72w_{(3)}K^2 = \frac{K^2 - 12K + 16}{16}
\end{aligned}
\tag{FE-42}
$$

Symbols in (FE-42).

- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K) = 1 - 24w_{(2)}K + 72w_{(3)}K^2 = (K^2 - 12K + 16)/16$; four-dimensional gravity is attractive for $\mathcal{G} > 0$. Unit: dimensionless. Value: $\mathcal{G}(K_*) = 20/3 + \sqrt{51} = 13.8081$. Status: DERIVED.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\rm Pl}$). Unit: GeV. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\rm W}$ and $\rho_{\rm EW,0}$). Status: AUTHOR.
- $w_{(p)}$: The author's Lovelock weights, $w_{(1)} = 1$, $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$. Unit: dimensionless. Value: $(1, 1/32, 1/1152)$. Status: AUTHOR.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -10H^2$. Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_{\rm N}\mathcal{G}(K_*) = 2.328 \times 10^{-36}$ GeV^{-2}. Status: AUTHOR.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- p_3: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: GeV^8. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- p_0: The pressure along the hidden direction x_0, $p_0 = \mathcal{T}^{x_0}{}_{x_0}$. Unit: GeV^8. Status: STANDARD.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- p_5: The pressure along the extra times, $p_5 = \mathcal{T}^{x_5}{}_{x_5} = \mathcal{T}^{x_6}{}_{x_6} = \mathcal{T}^{x_7}{}_{x_7}$. Unit: GeV^8. Status: STANDARD.
- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial \ln f$. Unit: GeV^8 (or GeV^4 for a four-dimensional density). Status: DERIVED.

The terms. $\mathcal{G}(K)$ multiplies $\mathcal{H}^2$ in the constraint; it is the coefficient of the four-dimensional Einstein-Hilbert term (FE-48), so four-dimensional gravity is attractive where $\mathcal{G} > 0$, i.e. for $K < 6 - 2\sqrt{5}$ or $K > 6 + 2\sqrt{5} = 10.47$. The term $6H^2K(1 - 2w_{(2)}K)$ is the curvature energy of the block (Einstein and Gauss-Bonnet); in the block

equation $3H^2K$. The sources are $\kappa\rho$ (constraint), κp_3 (3-space), κp_0 (hidden direction) and κp_5 (extra times). Without sources and with $\mathcal{H}=0$ the only static block is $K=8$, which would need $\Lambda=-24H^2$ (not the author's $-10H^2$) and has $\mathcal{G}=-1$, repulsive gravity: with the author's weights the block must be held by a source. Status: DERIVED (exact).

At $\mathcal{H}=0$ the constraint and the block equation read

$$x_4:\ 6H^2K-\tfrac{3}{8}H^2K^2-10H^2=-\kappa\rho,\qquad x_0:\ 3H^2K-10H^2=\kappa p_0,$$

Symbols in this equation.

- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda=-10H^2$ and the mass parameter $m_\Psi=H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- K: The block variable, $K=1/f^2$. Unit: dimensionless. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa=8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp=8\pi G_\mathrm{N}\mathcal{G}(K_*)=2.328\times10^{-36}\ \mathrm{GeV}^{-2}$. Status: AUTHOR.
- ρ: An energy density: in eight dimensions $\rho=-\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z=6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- p_0: The pressure along the hidden direction x_0, $p_0=\mathcal{T}^{x_0}{}_{x_0}$. Unit: GeV^8. Status: STANDARD.

and with the stabiliser as the only source ($\rho=p_0=p_5=\rho_\mathrm{stab}$, $p_3=-\rho_\mathrm{stab}$, FE-4) they require

$$10H^2-6H^2K+\tfrac{3}{8}H^2K^2=3H^2K-10H^2\quad\Leftrightarrow\quad 3K^2-72K+160=0\quad\Leftrightarrow\quad K=12\pm\tfrac{4}{3}\sqrt{51}:$$

Symbols in this equation.

- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda=-10H^2$ and the mass parameter $m_\Psi=H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- K: The block variable, $K=1/f^2$. Unit: dimensionless. Status: DERIVED.

$$\begin{aligned}&\kappa\rho_\mathrm{stab}=10H^2-6H^2K+\tfrac{3}{8}H^2K^2\ \ (x_4),\qquad \kappa p_0=\kappa p_5=\kappa\rho_\mathrm{stab}=3H^2K-10H^2\ \ (x_0;\ x_5,x_6,x_7),\\ &\Rightarrow\ K_\pm=12\pm\tfrac{4}{3}\sqrt{51},\qquad K_*=12+\tfrac{4}{3}\sqrt{51}=21.5219,\qquad f_*=K_*^{-1/2}=0.215556,\\ &\mathcal{G}(K_*)=\tfrac{20}{3}+\sqrt{51}=13.8081,\qquad \kappa\rho_\mathrm{stab}(f_*)=\big(26+4\sqrt{51}\big)H^2\end{aligned}\tag{FE-43}$$

Symbols in (FE-43).

- κ: The eight-dimensional gravitational coupling, $\kappa=8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp=8\pi G_\mathrm{N}\mathcal{G}(K_*)=2.328\times10^{-36}\ \mathrm{GeV}^{-2}$. Status: AUTHOR.
- ρ_stab: The energy density of the block stabiliser, a function of the block scale, $\rho_\mathrm{stab}\propto f^{-8}$; its value at the static block is $\rho_\mathrm{stab}(f_*)$. Unit: GeV^8. Value: $\kappa\rho_\mathrm{stab}(f_*)=(26+4\sqrt{51})H^2=54.566\,H^2$. Status: ASSUMED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda=-10H^2$ and the mass parameter $m_\Psi=H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- K: The block variable, $K=1/f^2$. Unit: dimensionless. Status: DERIVED.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- p_0: The pressure along the hidden direction x_0, $p_0=\mathcal{T}^{x_0}{}_{x_0}$. Unit: GeV^8. Status: STANDARD.
- p_5: The pressure along the extra times, $p_5=\mathcal{T}^{x_5}{}_{x_5}=\mathcal{T}^{x_6}{}_{x_6}=\mathcal{T}^{x_7}{}_{x_7}$. Unit: GeV^8. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z=6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.

- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $K_\pm$: The two static blocks with the stabiliser, $K_\pm = 12 \pm \frac{4}{3}\sqrt{51}$. Unit: dimensionless. Value: 21.5219 and 2.4781. Status: DERIVED.
- K_*: The block variable of the stabilised block, $K_* = 12 + \frac{4}{3}\sqrt{51}$ (the upper root K_+). Unit: dimensionless. Value: 21.5219. Status: DERIVED.
- f_*: The block scale of the stabilised (static) block, $f_* = K_*^{-1/2}$. Unit: dimensionless. Value: 0.215556. Status: DERIVED.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K) = 1 - 24w_{(2)}K + 72w_{(3)}K^2 = (K^2 - 12K + 16)/16$; four-dimensional gravity is attractive for $\mathcal{G} > 0$. Unit: dimensionless. Value: $\mathcal{G}(K_*) = 20/3 + \sqrt{51} = 13.8081$. Status: DERIVED.

The terms. The first line gives the stabiliser density required by the constraint and by the block equations; the stabiliser is block-isotropic, so the x_0 and the extra-time equations coincide and the static block is an exact solution of all components (the 3-space equation is the constraint with $-\kappa p_3 = \kappa\rho_{\rm stab}$). Of the two roots, K_* (upper) is the vacuum of UM-2: four-dimensional gravity is attractive there ($\mathcal{G}(K_*) = 13.8081$) and the stabiliser energy is positive, $\kappa\rho_{\rm stab} = 54.566\,H^2$; the lower root $K_- = 2.4781$ has $\mathcal{G} = -0.4748$. The block scale $f_* = 0.215556$ fixes the proper size of the hidden direction, $f_* L = 3f_*/H = 8.926 \times 10^{-21}$ m on each half. Status: DERIVED (closed form; the stabiliser itself ASSUMED).

5.6 The radion and the response of the block to matter

$$\Delta\dot{\mathcal{H}} = \Xi_{\mathcal{H}K}\,\Delta K, \qquad \Delta\dot{\mathcal{H}}_f = \Xi_{fK}\,\Delta K, \qquad \Delta\dot{K} = -2K_*\,\Delta\mathcal{H}_f,$$
$$\omega_{\rm r}^2 = 2K_*\,\Xi_{fK} = \Big(\frac{31552}{2249} - \frac{2416\sqrt{51}}{6747}\Big)H^2 = (3.38705\,H)^2 \tag{FE-44}$$

Symbols in (FE-44).

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, $\Delta N_{\rm eff}$ an addition to $N_{\rm eff}$, $\Delta S_{\rm m}$ the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- $\dot{\mathcal{H}}$: The time derivative $\mathrm{d}\mathcal{H}/\mathrm{d}x_4$. Unit: GeV^2. Status: DERIVED.
- $\Xi_{\mathcal{H}K}$: The coefficient of ΔK in the first-order equation for $\Delta\dot{\mathcal{H}}$ (linearised block about K_*). Unit: GeV^2. Value: $6(238 - 19\sqrt{51})H^2/2249 = 0.272956\,H^2$. Status: DERIVED.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- $\dot{\mathcal{H}}_f$: The time derivative $\mathrm{d}\mathcal{H}_f/\mathrm{d}x_4$. Unit: GeV^2. Status: DERIVED.
- Ξ_{fK}: The coefficient of ΔK in the first-order equation for $\Delta\dot{\mathcal{H}}_f$. Unit: GeV^2. Value: $(4063 - 485\sqrt{51})H^2/2249 = 0.266522\,H^2$. Status: DERIVED.
- $\dot{K}$: The time derivative $\mathrm{d}K/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- K_*: The block variable of the stabilised block, $K_* = 12 + \frac{4}{3}\sqrt{51}$ (the upper root K_+). Unit: dimensionless. Value: 21.5219. Status: DERIVED.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot{f}/f = \dot{b}_4/(2b_4) - \dot{a}_4$ (negative when the extra times and x_0 deflate). Unit: GeV. Status: DERIVED.
- $\omega_{\rm r}$: The angular frequency of the oscillation of the block about the static point (the radion), equal to the radion mass. Unit: GeV. Value: $3.38705\,H = 48.42$ TeV. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\rm W}$ and $\rho_{\rm EW,0}$). Status: AUTHOR.

The terms. Linearising FE-38 to FE-40 about the stabilised block (with $\rho_{\rm stab} \propto K^4$, from FE-41) gives a closed linear system for the changes of the two rates and of K: $\Xi_{\mathcal{H}K}$ and Ξ_{fK} are the restoring coefficients, and $\Delta\dot{K} = -2K_*\Delta\mathcal{H}_f$ is the definition of $\mathcal{H}_f$. The block oscillates with the frequency $\omega_{\rm r}$: the radion, the quantum of the block scale, with the mass $m_{\rm r} = \omega_{\rm r} = 3.38705\,H = 48.42$ TeV. The eigenvalues are 0 and $\pm\mathrm{i}\omega_{\rm r}$: the block is stable, and it is stable for every block-isotropic stabiliser law $p_0 = p_5 = \nu\rho_{\rm stab}$ with $\nu = \frac{1}{4}$ to 3 ($\omega_{\rm r} = 3.33$ to $4.10\,H$). The full nonlinear evolution gives $3.3868\,H$ (linear and nonlinear frequencies agree to 7.9×10^{-5}). Status: DERIVED.

$$\begin{aligned}
\Delta\dot{\mathcal{H}} &= \Xi_{\mathcal{H}K}\,\Delta K + \Xi_{\mathcal{H}3}\,\kappa\Delta p_3 + \Xi_{\mathcal{H}p}\,\kappa\Delta\bar{p} + \Xi_{\mathcal{H}\rho}\,\kappa\Delta\rho, \qquad 3\mathcal{G}(K_*)\,\mathcal{H}^2 = \kappa\Delta\rho,\\
\Delta\dot{\mathcal{H}}_f &= \Xi_{fK}\,\Delta K + \Xi_{f3}\,\kappa\Delta p_3 + \Xi_{fp}\,\kappa\Delta\bar{p} + \Xi_{f\rho}\,\kappa\Delta\rho,\\
\Delta\ddot{K} + \omega_{\mathrm{r}}^2\,\Delta K &= -2K_*\big(\Xi_{f3}\,\kappa\Delta p_3 + \Xi_{fp}\,\kappa\Delta\bar{p} + \Xi_{f\rho}\,\kappa\Delta\rho\big)
\end{aligned} \tag{FE-45}$$

Symbols in (FE-45).

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- $\dot{\mathcal{H}}$: The time derivative $\mathrm{d}\mathcal{H}/\mathrm{d}x_4$. Unit: GeV^2. Status: DERIVED.
- $\Xi_{\mathcal{H}K}$: The coefficient of ΔK in the first-order equation for $\Delta\dot{\mathcal{H}}$ (linearised block about K_*). Unit: GeV^2. Value: $6(238-19\sqrt{51})H^2/2249 = 0.272956\,H^2$. Status: DERIVED.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- $\Xi_{\mathcal{H}3}$: The coefficient of $\kappa\Delta p_3$ in the equation for $\Delta\dot{\mathcal{H}}$. Unit: dimensionless. Value: $2283/4498 - 162\sqrt{51}/2249 = -0.006853$. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_{\mathrm{N}}\mathcal{G}(K_*) = 2.328\times10^{-36}$ GeV^{-2}. Status: AUTHOR.
- p_3: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: GeV^8. Status: STANDARD.
- $\Xi_{\mathcal{H}p}$: The coefficient of $\kappa\Delta\bar{p}$ in the equation for $\Delta\dot{\mathcal{H}}$. Unit: dimensionless. Value: $(28\sqrt{51}-114)/2249 = 0.038221$. Status: DERIVED.
- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial\ln f$. Unit: GeV^8 (or GeV^4 for a four-dimensional density). Status: DERIVED.
- $\Xi_{\mathcal{H}\rho}$: The coefficient of $\kappa\Delta\rho$ in the equation for $\Delta\dot{\mathcal{H}}$. Unit: dimensionless. Value: $135021/265382 - 10308\sqrt{51}/132691 = -0.045997$. Status: DERIVED.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K) = 1 - 24w_{(2)}K + 72w_{(3)}K^2 = (K^2 - 12K + 16)/16$; four-dimensional gravity is attractive for $\mathcal{G} > 0$. Unit: dimensionless. Value: $\mathcal{G}(K_*) = 20/3 + \sqrt{51} = 13.8081$. Status: DERIVED.
- K_*: The block variable of the stabilised block, $K_* = 12 + \frac{4}{3}\sqrt{51}$ (the upper root K_+). Unit: dimensionless. Value: 21.5219. Status: DERIVED.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- $\dot{\mathcal{H}}_f$: The time derivative $\mathrm{d}\mathcal{H}_f/\mathrm{d}x_4$. Unit: GeV^2. Status: DERIVED.
- Ξ_{fK}: The coefficient of ΔK in the first-order equation for $\Delta\dot{\mathcal{H}}_f$. Unit: GeV^2. Value: $(4063 - 485\sqrt{51})H^2/2249 = 0.266522\,H^2$. Status: DERIVED.
- Ξ_{f3}: The coefficient of $\kappa\Delta p_3$ in the equation for $\Delta\dot{\mathcal{H}}_f$. Unit: dimensionless. Value: $21\sqrt{51}/2249 - 171/4498 = 0.028666$. Status: DERIVED.
- Ξ_{fp}: The coefficient of $\kappa\Delta\bar{p}$ in the equation for $\Delta\dot{\mathcal{H}}_f$. Unit: dimensionless. Value: $239\sqrt{51}/6747 - 485/2249 = 0.037320$. Status: DERIVED.
- $\Xi_{f\rho}$: The coefficient of $\kappa\Delta\rho$ in the equation for $\Delta\dot{\mathcal{H}}_f$. Unit: dimensionless. Value: $57/4498 - 7\sqrt{51}/2249 = -0.009555$. Status: DERIVED.
- $\ddot{K}$: The second time derivative $\mathrm{d}^2K/\mathrm{d}x_4^2$. Unit: GeV^2. Status: DERIVED.
- ω_{r}: The angular frequency of the oscillation of the block about the static point (the radion), equal to the radion mass. Unit: GeV. Value: $3.38705\,H = 48.42$ TeV. Status: DERIVED.

The terms. Matter on top of the stabiliser changes the sources by $\kappa\Delta\rho$, $\kappa\Delta p_3$ and $\kappa\Delta\bar{p}$ (the block-averaged changes; the shape source $\Delta p_0 - \Delta p_5$ is outside the order-0 metric). Each enters both rates with the coefficients of FE-46; $\mathcal{H}^2$ is of first order and is fixed by the constraint, $3\mathcal{G}(K_*)\mathcal{H}^2 = \kappa\Delta\rho$. The last line is the radion equation with matter as its source. Status: DERIVED (exact to first order). The residual of the adiabatic solution in the full nonlinear equations is 1.4×10^{-7} for sources of $10^{-5}\,H^2$ and falls by the factor 10 with the sources (it is of second order).

$$\Xi_{\mathcal{H}K}=\frac{6\big(238-19\sqrt{51}\big)}{2249}H^2,\qquad \Xi_{fK}=\frac{4063-485\sqrt{51}}{2249}H^2,\qquad \Xi_{\mathcal{H}3}=\frac{2283}{4498}-\frac{162\sqrt{51}}{2249},$$
$$\Xi_{\mathcal{H}p}=\frac{28\sqrt{51}-114}{2249},\qquad \Xi_{\mathcal{H}\rho}=\frac{135021}{265382}-\frac{10308\sqrt{51}}{132691},\qquad \Xi_{f3}=\frac{21\sqrt{51}}{2249}-\frac{171}{4498},\tag{FE-46}$$
$$\Xi_{fp}=\frac{239\sqrt{51}}{6747}-\frac{485}{2249},\qquad \Xi_{f\rho}=\frac{57}{4498}-\frac{7\sqrt{51}}{2249}$$

Symbols in (FE-46).

- $\Xi_{\mathcal{H}K}$: The coefficient of ΔK in the first-order equation for $\Delta\dot{\mathcal{H}}$ (linearised block about K_*). Unit: GeV^2. Value: $6(238-19\sqrt{51})H^2/2249 = 0.272956\,H^2$. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda=-10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\rm W}$ and $\rho_{\rm EW,0}$). Status: AUTHOR.
- Ξ_{fK}: The coefficient of ΔK in the first-order equation for $\Delta\dot{\mathcal{H}}_f$. Unit: GeV^2. Value: $(4063-485\sqrt{51})H^2/2249 = 0.266522\,H^2$. Status: DERIVED.
- $\Xi_{\mathcal{H}3}$: The coefficient of $\kappa\Delta p_3$ in the equation for $\Delta\dot{\mathcal{H}}$. Unit: dimensionless. Value: $2283/4498-162\sqrt{51}/2249 = -0.006853$. Status: DERIVED.
- $\Xi_{\mathcal{H}p}$: The coefficient of $\kappa\Delta\bar{p}$ in the equation for $\Delta\dot{\mathcal{H}}$. Unit: dimensionless. Value: $(28\sqrt{51}-114)/2249 = 0.038221$. Status: DERIVED.
- $\Xi_{\mathcal{H}\rho}$: The coefficient of $\kappa\Delta\rho$ in the equation for $\Delta\dot{\mathcal{H}}$. Unit: dimensionless. Value: $135021/265382 - 10308\sqrt{51}/132691 = -0.045997$. Status: DERIVED.
- Ξ_{f3}: The coefficient of $\kappa\Delta p_3$ in the equation for $\Delta\dot{\mathcal{H}}_f$. Unit: dimensionless. Value: $21\sqrt{51}/2249-171/4498 = 0.028666$. Status: DERIVED.
- Ξ_{fp}: The coefficient of $\kappa\Delta\bar{p}$ in the equation for $\Delta\dot{\mathcal{H}}_f$. Unit: dimensionless. Value: $239\sqrt{51}/6747-485/2249 = 0.037320$. Status: DERIVED.
- $\Xi_{f\rho}$: The coefficient of $\kappa\Delta\rho$ in the equation for $\Delta\dot{\mathcal{H}}_f$. Unit: dimensionless. Value: $57/4498-7\sqrt{51}/2249 = -0.009555$. Status: DERIVED.

The terms. The exact coefficients (rational numbers and $\sqrt{51}$, from $K_* = 12+\frac{4}{3}\sqrt{51}$): numerically $\Xi_{\mathcal{H}K} = 0.272956\,H^2$, $\Xi_{fK} = 0.266522\,H^2$, $\Xi_{\mathcal{H}3} = -0.006853$, $\Xi_{\mathcal{H}p} = 0.038221$, $\Xi_{\mathcal{H}\rho} = -0.045997$, $\Xi_{f3} = 0.028666$, $\Xi_{fp} = 0.037320$, $\Xi_{f\rho} = -0.009555$. Status: DERIVED (sympy).

For a source that changes slowly compared with $1/\omega_{\rm r}$ the radion follows it adiabatically, $\Delta\ddot{K}=0$:

$$\Delta K=-\frac{2K_*}{\omega_{\rm r}^2}\big(\Xi_{f3}\,\kappa\Delta p_3+\Xi_{fp}\,\kappa\Delta\bar{p}+\Xi_{f\rho}\,\kappa\Delta\rho\big)\,:$$
$$c_3=-\frac{2K_*\Xi_{f3}H^2}{\omega_{\rm r}^2},\qquad c_p=-\frac{2K_*\Xi_{fp}H^2}{\omega_{\rm r}^2},\qquad c_\rho=-\frac{2K_*\Xi_{f\rho}H^2}{\omega_{\rm r}^2}.$$

Symbols in this equation.

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21} = m_2^2 - m_1^2$ a difference of squared masses, $\Delta N_{\rm eff}$ an addition to $N_{\rm eff}$, $\Delta S_{\rm m}$ the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- K_*: The block variable of the stabilised block, $K_* = 12+\frac{4}{3}\sqrt{51}$ (the upper root K_+). Unit: dimensionless. Value: 21.5219. Status: DERIVED.
- $\omega_{\rm r}$: The angular frequency of the oscillation of the block about the static point (the radion), equal to the radion mass. Unit: GeV. Value: $3.38705\,H = 48.42$ TeV. Status: DERIVED.
- Ξ_{f3}: The coefficient of $\kappa\Delta p_3$ in the equation for $\Delta\dot{\mathcal{H}}_f$. Unit: dimensionless. Value: $21\sqrt{51}/2249-171/4498 = 0.028666$. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_{\rm N}\mathcal{G}(K_*) = 2.328\times10^{-36}$ GeV^{-2}. Status: AUTHOR.
- p_3: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: GeV^8. Status: STANDARD.

- Ξ_{fp}: The coefficient of $\kappa\Delta\bar{p}$ in the equation for $\Delta\dot{\mathcal{H}}_f$. Unit: dimensionless. Value: $239\sqrt{51}/6747 - 485/2249 = 0.037320$. Status: DERIVED.
- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial \ln f$. Unit: GeV^8 (or GeV^4 for a four-dimensional density). Status: DERIVED.
- $\Xi_{f\rho}$: The coefficient of $\kappa\Delta\rho$ in the equation for $\Delta\dot{\mathcal{H}}_f$. Unit: dimensionless. Value: $57/4498 - 7\sqrt{51}/2249 = -0.009555$. Status: DERIVED.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- c_3: The coefficient of $\kappa\Delta p_3$ in the adiabatic block response, $c_3 = -3c_\rho$ (so radiation drops out). Unit: dimensionless. Value: -0.107556. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- c_p: The coefficient of $\kappa\Delta\bar{p}$ in the adiabatic block response. Unit: dimensionless. Value: -0.140028. Status: DERIVED.
- c_ρ: The coefficient of $\kappa\Delta\rho$ in the adiabatic block response ΔK. Unit: dimensionless. Value: 0.0358521. Status: DERIVED.

The hidden-pressure coefficient can be read off the block equation directly: with $\Delta\rho = \Delta p_3 = 0$ (so $\mathcal{H} = 0$) the block equation at $K_* + \Delta K$ with the stabiliser $\rho_\mathrm{stab} \propto K^4$ and the extra hidden pressure gives

$$3H^2\Delta K - \kappa\,\frac{\mathrm{d}\rho_\mathrm{stab}}{\mathrm{d}K}\,\Delta K = \kappa\Delta\bar{p}, \qquad \frac{\mathrm{d}\rho_\mathrm{stab}}{\mathrm{d}K} = \frac{4\rho_\mathrm{stab}}{K} \quad \Rightarrow \quad c_p = \frac{H^2}{3H^2 - 4\kappa\rho_\mathrm{stab}/K_*} = -0.140028,$$

Symbols in this equation.

- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_eff an addition to N_eff, ΔS_m the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_\mathrm{N}\mathcal{G}(K_*) = 2.328 \times 10^{-36}\ \mathrm{GeV}^{-2}$. Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- ρ_stab: The energy density of the block stabiliser, a function of the block scale, $\rho_\mathrm{stab} \propto f^{-8}$; its value at the static block is $\rho_\mathrm{stab}(f_*)$. Unit: GeV^8. Value: $\kappa\rho_\mathrm{stab}(f_*) = (26 + 4\sqrt{51})H^2 = 54.566\,H^2$. Status: ASSUMED.
- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial \ln f$. Unit: GeV^8 (or GeV^4 for a four-dimensional density). Status: DERIVED.
- c_p: The coefficient of $\kappa\Delta\bar{p}$ in the adiabatic block response. Unit: dimensionless. Value: -0.140028. Status: DERIVED.
- K_*: The block variable of the stabilised block, $K_* = 12 + \frac{4}{3}\sqrt{51}$ (the upper root K_+). Unit: dimensionless. Value: 21.5219. Status: DERIVED.

with the stiffness $3H^2 - 4\kappa\rho_\mathrm{stab}/K_* = -7.1414\,H^2$.

$$\Delta K = \frac{c_\rho\,\kappa\Delta\rho + c_3\,\kappa\Delta p_3 + c_p\,\kappa\Delta\bar{p}}{H^2}, \qquad \Delta\ln f = -\frac{\Delta K}{2K_*}, \qquad c_3 = -3c_\rho, \qquad \Delta\dot{\mathcal{H}} = -\frac{\kappa(\Delta\rho + \Delta p_3)}{2\mathcal{G}(K_*)} \qquad \text{(FE-47)}$$

Symbols in (FE-47).

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- c_ρ: The coefficient of $\kappa\Delta\rho$ in the adiabatic block response ΔK. Unit: dimensionless. Value: 0.0358521. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_{\text{N}}\mathcal{G}(K_*) = 2.328\times10^{-36}\ \text{GeV}^{-2}$. Status: AUTHOR.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- c_3: The coefficient of $\kappa\Delta p_3$ in the adiabatic block response, $c_3 = -3c_\rho$ (so radiation drops out). Unit: dimensionless. Value: -0.107556. Status: DERIVED.
- p_3: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: GeV^8. Status: STANDARD.
- c_p: The coefficient of $\kappa\Delta\bar{p}$ in the adiabatic block response. Unit: dimensionless. Value: -0.140028. Status: DERIVED.
- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial \ln f$. Unit: GeV^8 (or GeV^4 for a four-dimensional density). Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\text{EW},0}$). Status: AUTHOR.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- K_*: The block variable of the stabilised block, $K_* = 12 + \frac{4}{3}\sqrt{51}$ (the upper root K_+). Unit: dimensionless. Value: 21.5219. Status: DERIVED.
- $\dot{\mathcal{H}}$: The time derivative $\mathrm{d}\mathcal{H}/\mathrm{d}x_4$. Unit: GeV^2. Status: DERIVED.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K) = 1 - 24w_{(2)}K + 72w_{(3)}K^2 = (K^2 - 12K + 16)/16$; four-dimensional gravity is attractive for $\mathcal{G} > 0$. Unit: dimensionless. Value: $\mathcal{G}(K_*) = 20/3 + \sqrt{51} = 13.8081$. Status: DERIVED.

The terms. The adiabatic response of the block: $c_\rho = 0.0358521$, $c_3 = -0.107556$, $c_p = -0.140028$. Because $c_3 = -3c_\rho$ exactly, the block responds only to the trace $\rho - 3p_3$ and to the hidden pressure $\bar{p}$: radiation (traceless, $\bar{p} = 0$) drops out. A positive hidden pressure lowers K, i.e. enlarges the block. The last relation is the four-dimensional Friedmann acceleration equation: with $\kappa\Delta\rho = 8\pi G_{\text{N}}\mathcal{G}(K_*)\rho_4$ (FE-80) it is $\dot{\mathcal{H}} = -4\pi G_{\text{N}}(\rho_4 + p_4)$ of general relativity, and the constraint is $\mathcal{H}^2 = \frac{8\pi G_{\text{N}}}{3}\rho_4$. Status: DERIVED (exact for a slowly varying source; the neglected inertia is $(\mathcal{H}/m_{\text{r}})^2 \approx 3\times10^{-33}$ at the freeze-out). The Kohn-Sham sources enter through it with residual 1.4×10^{-12} (the response recomputed from the stored cell averages); the block-shape source $|p_0 - p_5|$ is up to 4 times $\bar{p}$ (OPEN, see "The block pressures").

5.7 Four-dimensional gravity

On the static block the Lovelock densities split into a four-dimensional part and a part of the block. For a block of constant curvature the internal densities are numbers:

$$
\begin{aligned}
&\mathcal{L}_{(1)} = \mathcal{L}^{(4)}_{(1)} + \mathcal{L}^{(\text{int})}_{(1)}, \qquad \mathcal{L}_{(2)} = \mathcal{L}^{(4)}_{(2)} + 2\mathcal{L}^{(4)}_{(1)}\mathcal{L}^{(\text{int})}_{(1)} + \mathcal{L}^{(\text{int})}_{(2)}, \qquad \mathcal{L}_{(3)} = 3\mathcal{L}^{(4)}_{(2)}\mathcal{L}^{(\text{int})}_{(1)} + 3\mathcal{L}^{(4)}_{(1)}\mathcal{L}^{(\text{int})}_{(2)},\\
&\mathcal{L}^{(4)}_{(1)} = R^{(4)}, \qquad \mathcal{L}^{(4)}_{(3)} = 0, \qquad \mathcal{L}^{(\text{int})}_{(1)} = -12H^2K, \qquad \mathcal{L}^{(\text{int})}_{(2)} = 24H^4K^2, \qquad \mathcal{L}^{(\text{int})}_{(3)} = 0,\\
&\sum_{p=1}^{3}\alpha_{(p)}\,\mathcal{L}_{(p)} - 2\Lambda - 2\kappa\rho_{\text{stab}} = \mathcal{G}(K)\,R^{(4)} + (\text{topological}) - 2\mathcal{G}(K)\,\Lambda_4,\\
&\Lambda_4 = \frac{6H^2K\big(1 - 2w_{(2)}K\big) + \Lambda + \kappa\rho_{\text{stab}}}{\mathcal{G}(K)}
\end{aligned}
\qquad \text{(FE-48)}
$$

Symbols in (FE-48).

- $\mathcal{L}_{(p)}$: The Lovelock density of order p of the metric (a scalar built from p Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\text{length}^{2p}$. Status: STANDARD.
- $\mathcal{L}^{(4)}_{(r)}$: The Lovelock density of order r of the four-dimensional metric $g^{(4)}_{\mu\nu}$ ($\mathcal{L}^{(4)}_{(1)} = R^{(4)}$; the order 2 is topological and the order 3 vanishes in four dimensions). Unit: $1/\text{length}^{2r}$. Status: STANDARD.
- $\mathcal{L}^{(\text{int})}_{(r)}$: The Lovelock density of order r of the static block (a four-dimensional space of constant curvature $-H^2K$): 1, $-12H^2K$, $24H^4K^2$, 0 for $r = 0, 1, 2, 3$. Unit: $1/\text{length}^{2r}$. Status: DERIVED.
- $R^{(4)}$: The Ricci scalar of the four-dimensional metric $g^{(4)}_{\mu\nu}$. Unit: $1/\text{length}^2$. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\text{EW},0}$). Status: AUTHOR.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)}H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -10H^2$. Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_\text{N}\mathcal{G}(K_*) = 2.328 \times 10^{-36}$ GeV^{-2}. Status: AUTHOR.
- ρ_stab: The energy density of the block stabiliser, a function of the block scale, $\rho_\text{stab} \propto f^{-8}$; its value at the static block is $\rho_\text{stab}(f_*)$. Unit: GeV^8. Value: $\kappa\rho_\text{stab}(f_*) = (26 + 4\sqrt{51})H^2 = 54.566\,H^2$. Status: ASSUMED.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K) = 1 - 24w_{(2)}K + 72w_{(3)}K^2 = (K^2 - 12K + 16)/16$; four-dimensional gravity is attractive for $\mathcal{G} > 0$. Unit: dimensionless. Value: $\mathcal{G}(K_*) = 20/3 + \sqrt{51} = 13.8081$. Status: DERIVED.
- Λ_4: The four-dimensional cosmological constant of the effective theory on the block, $\Lambda_4 = [6H^2K(1 - 2w_{(2)}K) + \Lambda + \kappa\rho_\text{stab}]/\mathcal{G}(K)$; zero at the static point, the observed dark energy a tuned residual. Unit: GeV^2. Value: 0 at the static point; observed $\Lambda_4/H^2 \sim 10^{-92}$. Status: DERIVED.
- $w_{(p)}$: The author's Lovelock weights, $w_{(1)} = 1$, $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$. Unit: dimensionless. Value: $(1, 1/32, 1/1152)$. Status: AUTHOR.

The terms. $\mathcal{L}^{(4)}_{(p)}$ are the Lovelock densities of the four observable dimensions ($R^{(4)}$ the scalar curvature; the four-dimensional Gauss-Bonnet density is topological, the cubic one vanishes), $\mathcal{L}^{(\text{int})}_{(p)}$ those of the block. The coefficient of $R^{(4)}$ is $1 + \alpha_{(2)}(-24H^2K) + \alpha_{(3)}72H^4K^2 = \mathcal{G}(K)$, and the constant terms combine into the four-dimensional cosmological constant Λ_4, whose numerator is the constraint at $\mathcal{H} = 0$: $\Lambda_4 = 0$ exactly on the stabilised block. The observed dark energy needs $\Lambda_4/H^2 \sim 10^{-92}$, a tuned residual (OPEN). Status: DERIVED. Residual 6.9×10^{-13} (the stored block in Λ_4 and in the stationarity of the constraint).

$$
\begin{aligned}
&G^{(4)}_{\mu\nu} + \Lambda_4\, g^{(4)}_{\mu\nu} = 8\pi G_\text{N}\,\mathcal{T}^{(4)}_{\mu\nu}, \qquad 8\pi G_\text{N} = \frac{\kappa}{\mathcal{G}(K_*)\,V_\perp}, \qquad V_\perp = \frac{2f^4 v_\text{t}\big(1 - \text{e}^{-3HL}\big)}{3H}, \\
&\mathcal{T}^{(4)}_{\mu\nu} = \int \text{d}x_0\,\text{d}x_5\,\text{d}x_6\,\text{d}x_7\,\sqrt{|g_\perp|}\,\mathcal{T}_{\mu\nu}, \qquad g_3^2 = \frac{\hat{g}_3^2}{V_\perp}, \qquad g_2^2 = \frac{\hat{g}_2^2}{V_\perp}, \qquad g_1^2 = \frac{\hat{g}_1^2}{V_\perp}
\end{aligned}
\tag{FE-49}
$$

Symbols in (FE-49).

- $G^{(4)}_{\mu\nu}$: The Einstein tensor of the four-dimensional metric, $R^{(4)}_{\mu\nu} - \frac{1}{2}g^{(4)}_{\mu\nu}R^{(4)}$. Unit: $1/\text{length}^2$. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index running over $x_1, \ldots, x_4$ (as μ). Status: STANDARD.
- Λ_4: The four-dimensional cosmological constant of the effective theory on the block, $\Lambda_4 = [6H^2K(1 - 2w_{(2)}K) + \Lambda + \kappa\rho_\text{stab}]/\mathcal{G}(K)$; zero at the static point, the observed dark energy a tuned residual. Unit: GeV^2. Value: 0 at the static point; observed $\Lambda_4/H^2 \sim 10^{-92}$. Status: DERIVED.
- $g^{(4)}_{\mu\nu}$: The four-dimensional metric of the observable coordinates $x_1, \ldots, x_4$ (the metric of the effective four-dimensional general relativity on the stabilised block). Unit: dimensionless. Status: DERIVED.

- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_{N}: Newton's constant of four-dimensional gravity, $8\pi G_{\mathrm{N}} = \kappa/(\mathcal{G}(K_*)V_\perp)$. Unit: GeV^{-2}. Value: 6.709×10^{-39} GeV^{-2} ($M_{\mathrm{Pl}} = G_{\mathrm{N}}^{-1/2} = 1.22091 \times 10^{19}$ GeV). Status: STANDARD.
- $\mathcal{T}^{(4)}_{\mu\nu}$: The four-dimensional energy-momentum tensor: $\mathcal{T}_{\mu\nu}$ integrated over the block (both halves of a period of x_0 and the extra times). Unit: GeV^4. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_{\mathrm{N}}\mathcal{G}(K_*) = 2.328 \times 10^{-36}$ GeV^{-2}. Status: AUTHOR.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K) = 1 - 24w_{(2)}K + 72w_{(3)}K^2 = (K^2 - 12K + 16)/16$; four-dimensional gravity is attractive for $\mathcal{G} > 0$. Unit: dimensionless. Value: $\mathcal{G}(K_*) = 20/3 + \sqrt{51} = 13.8081$. Status: DERIVED.
- K_*: The block variable of the stabilised block, $K_* = 12 + \frac{4}{3}\sqrt{51}$ (the upper root K_+). Unit: dimensionless. Value: 21.5219. Status: DERIVED.
- $V_\perp$: The proper volume of the block per unit four-volume (both halves of a period of x_0 and the extra times), $V_\perp = 2f^4 v_{\mathrm{t}}(1 - \mathrm{e}^{-3HL})/(3H)$. Unit: GeV^{-4}. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- v_{t}: The coordinate volume of the compact extra times x_5, x_6, x_7. Unit: GeV^{-3}. Value: not fixed; $v_{\mathrm{t}} = H^{-3}$ in the quoted conversions. Status: ASSUMED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $g_\perp$: The metric of the four transverse directions x_0, x_5, x_6, x_7 (the block); $|g_\perp|$ is the absolute value of its determinant. Unit: dimensionless. Status: DERIVED.
- $\mathcal{T}^M{}_N$: The energy-momentum tensor of all matter (every field except gravity, the stabiliser included), the source of gravity; defined by the change of the matter action under a change of the metric. Unit: GeV^8. Status: STANDARD.
- g_3: The four-dimensional strong coupling ($g_3^2 = 4\pi\alpha_3$). Unit: dimensionless. Value: $g_3^2 = 0.7786$ at $2M_{\mathrm{DM}}$. Status: DERIVED.
- $\hat{g}_3$: The eight-dimensional SU(3) coupling (it multiplies the gluon kinetic term as $1/\hat{g}_3^2$); $g_3^2 = \hat{g}_3^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- g_2: The four-dimensional SU(2) coupling ($g_2^2 = 4\pi\alpha_2$). Unit: dimensionless. Value: $g_2^2 = 0.3781$ at $2M_{\mathrm{DM}}$. Status: DERIVED.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- g_1: The four-dimensional hypercharge coupling ($g_1^2 = 4\pi\alpha_1$, Standard-Model normalisation without the GUT factor). Unit: dimensionless. Value: $g_1^2 = 0.1380$ at $2M_{\mathrm{DM}}$. Status: DERIVED.
- $\hat{g}_1$: The eight-dimensional $\mathrm{U}(1)_Y$ coupling; $g_1^2 = \hat{g}_1^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.

The terms. Four-dimensional gravity on the stabilised block is general relativity with Newton's constant $8\pi G_{\mathrm{N}} = \kappa/(\mathcal{G}(K_*)V_\perp)$, where $V_\perp$ is the proper volume of the block (both halves of the period, the four block scale factors f^4, the coordinate volume v_{t} of the extra times); $\kappa/V_\perp = 2.328 \times 10^{-36}$ GeV^{-2} reproduces $M_{\mathrm{Pl}} = 1.22091 \times 10^{19}$ GeV. The four-dimensional energy-momentum tensor is the eight-dimensional one integrated over the block, and the four-dimensional gauge couplings are the eight-dimensional ones divided by the same volume (they run below m_{KK} as in the Standard Model without a Higgs particle, FE-77, FE-83; at the scale $2M_{\mathrm{DM}} = 88.7$ TeV of the

portal I12, $g_3^2 = 0.7786$, $g_2^2 = 0.3781$, $g_1^2 = 0.1380$). Status: DERIVED. Its linear perturbations are FE-90.

6 The masses from the hidden coordinate

On the static block every field is a sum of modes, each a four-dimensional field times a profile in the hidden coordinate. The four-dimensional mass of a mode is an eigenvalue of the x_0 problem of its field: the profile depends on x_0 periodically, with the period $\pi/(6H)$, through $y(z)$. With $\xi = y/L$ on each half of a period, the eigenvalue of a boson is $\tilde{m} = m\,fL$ and that of a fermion $\tilde{\varepsilon} = \varepsilon fL$. There is no Higgs field: the gauge bosons get their masses from the tip, the fermions from their localisation in x_0 and from the tip mixing.

6.1 The gauge bosons and the graviton

A four-dimensional component of a gauge field (of the gluons, of W or of B) is a four-dimensional field times a profile $F(y)$, in the gauge where the y component vanishes. Inserted in the gauge action IN-8 on the static block, the field strengths along the four observable directions carry F^2 and those along y carry $(\partial_y F)^2/f^2$; integrated over the block they give, for each half of the period, the reduced action and its eigenvalue equation:

$$\begin{aligned} &\int_{-L}^{0} \mathrm{d}y\, f^4 \mathrm{e}^{3Hy}\Big[\frac{(\partial_y F)^2}{f^2} - m^2F^2\Big] \text{ stationary :} \qquad -\frac{1}{f^2}\,\mathrm{e}^{-3Hy}\,\partial_y\big(\mathrm{e}^{3Hy}\,\partial_y F\big) = m^2 F, \\ &\Leftrightarrow \qquad -\mathrm{e}^{-3HL\xi}\,\frac{\mathrm{d}}{\mathrm{d}\xi}\Big(\mathrm{e}^{3HL\xi}\,\frac{\mathrm{d}F}{\mathrm{d}\xi}\Big) = \tilde{m}^2 F, \qquad \tilde{m} = m\,fL \end{aligned} \tag{FE-50}$$

Symbols in (FE-50).

- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- y: The label y in ∂_y: the derivative with respect to the hidden coordinate y. Status: STANDARD.
- $F(\xi)$, F_n: The x_0 profile of a boson mode on one half of the period (a function of y or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; F_n is the profile of the n-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- m, m_s, m_i: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $\tilde{m}$, $\tilde{m}_n$: The dimensionless x_0 eigenvalue of a boson mode, $\tilde{m} = m\,fL$; $\tilde{m}_n$ the n-th even Kaluza-Klein level of the tower with $\beta = 0$ ($\tilde{m}_1$ the first). Unit: dimensionless. Value: $\tilde{m}_1^2 = 30.1196$, $\tilde{m}_2^2 = 59.7284$, $\tilde{m}_3^2 = 109.0764$ ($HL = 3$), i.e. $\tilde{m}_1 = 5.48813$, $\tilde{m}_2 = 7.72842$, $\tilde{m}_3 = 10.44397$. Status: DERIVED.

The terms. $f^4\mathrm{e}^{3Hy}$ is the proper volume of the block per unit y, $(\partial_y F)^2/f^2$ the energy of the variation of the profile along the proper hidden length, and m^2F^2 the four-dimensional mass term. Stationarity gives a Sturm-Liouville problem whose eigenvalue is the squared four-dimensional mass; in the variable ξ the block scale

and the length L combine into the dimensionless eigenvalue $\tilde{m}$. The same operator holds for every four-dimensional component of the gluons, the W and the B field and, in Einstein gravity, for the transverse-traceless graviton; with the Gauss-Bonnet and cubic terms the four-dimensional and the hidden parts of the graviton operator acquire curvature-dependent coefficients, which are not computed: the graviton tower (FE-53) is that of Einstein gravity (OPEN). Status: DERIVED. Residual of the stored eigenvalues 5.9×10^{-12}.

$$\begin{aligned} &\frac{\mathrm{d}F}{\mathrm{d}\xi}\Big|_{\xi=0} = 0 \;\; \text{(even)} \quad \text{or} \quad F(0) = 0 \;\; \text{(odd)}, \qquad \frac{\mathrm{d}F}{\mathrm{d}\xi}\Big|_{\xi=-1} = \beta\, F(-1), \\ &\beta = 0 \;\; \text{(photon, gluons, graviton)}, \qquad \beta = \beta_{\mathrm{W}} \;\; (W^{\pm}), \qquad \beta = \beta_{\mathrm{Z}} = \frac{\beta_{\mathrm{W}}}{\cos^2\theta_{\mathrm{W}}} \;\; (Z) \end{aligned} \tag{FE-51}$$

Symbols in (FE-51).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $F(\xi)$, F_n: The x_0 profile of a boson mode on one half of the period (a function of y or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; F_n is the profile of the n-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- β: The Robin parameter of the tip condition of a boson profile, $\mathrm{d}F/\mathrm{d}\xi = \beta F$ at $\xi = -1$ (zero for the massless gauge fields and the graviton). Unit: dimensionless. Status: DERIVED.
- β_{W}: The Robin parameter of W^+ and W^- (set by the tip mass term of the sigma field). Unit: dimensionless. Value: 0.011912 (fitted). Status: DERIVED.
- $W_M^{\pm}$, W^+, W^-: The charged weak bosons, canonically normalised: $W_M^{\pm} = (W_M^1 \mp \mathrm{i}W_M^2)/(\sqrt{2}\,\hat{g}_2)$; W^+ carries the electric charge $+e$ and W^- the charge $-e$ (they are each other's antiparticles and have one mass). Unit: GeV^3. Status: STANDARD.
- β_{Z}: The Robin parameter of the Z, $\beta_{\mathrm{Z}} = \beta_{\mathrm{W}}/\cos^2\theta_{\mathrm{W}}$. Unit: dimensionless. Value: 0.015343. Status: DERIVED.
- θ_{W}: The weak mixing angle of the couplings, $\cos\theta_{\mathrm{W}} = \hat{g}_2/\hat{g}_Z$, $\sin\theta_{\mathrm{W}} = \hat{g}_1/\hat{g}_Z$; fixed by the measured W and Z masses with $\rho_{\mathrm{EW},0}$. Unit: dimensionless. Value: $\cos^2\theta_{\mathrm{W}} = 0.776368$. Status: DERIVED.
- Z_M: The Z boson field, canonically normalised, $Z_M = (W_M^3 - B_M)/\hat{g}_Z$. Unit: GeV^3. Status: STANDARD.

The terms. At the fold the profile is even or odd under the reflection $z \to \pi - z$ (FE-27). At the tip the boundary term of the variation, $2f^2\mathrm{e}^{3Hy}\partial_y F\,\Delta F$, is balanced by the variation of the tip mass term IN-15, which gives the Robin condition with the parameter β: zero for the photon, the gluons and the graviton (no tip mass term), β_{W} for W^+ and W^- (one value for both, which are particle and antiparticle), and β_{Z} for the Z. The ratio $\beta_{\mathrm{Z}}/\beta_{\mathrm{W}} = \hat{g}_Z^2/\hat{g}_2^2 = 1/\cos^2\theta_{\mathrm{W}}$ is the ratio of the coefficients of the tip mass terms in IN-15; $\beta_{\mathrm{W}} = 0.011912$ (proportional to v_Σ^2) is fitted to the measured masses (FE-55). Status: DERIVED.

With $F = \mathrm{e}^{-3HL\xi/2}\tilde{F}$ the operator loses its first-derivative term and the conditions become

$$\begin{aligned} &-\frac{\mathrm{d}^2\tilde{F}}{\mathrm{d}\xi^2} + \frac{9H^2L^2}{4}\,\tilde{F} = \tilde{m}^2\tilde{F}, \qquad \frac{\mathrm{d}\tilde{F}}{\mathrm{d}\xi} - \frac{3HL}{2}\,\tilde{F} = 0 \;\; (\xi = 0,\ \text{even}), \qquad \tilde{F} = 0 \;\; (\xi = 0,\ \text{odd}), \\ &\frac{\mathrm{d}\tilde{F}}{\mathrm{d}\xi} - \Big(\beta + \frac{3HL}{2}\Big)\tilde{F} = 0 \;\; (\xi = -1), \end{aligned}$$

Symbols in this equation.

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $\tilde{F}$: The rescaled boson profile, $F = \mathrm{e}^{-3HL\xi/2}\tilde{F}$. Unit: dimensionless. Status: DERIVED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.

- L: The depth of the regular tip: each half of a period of x_0 ends at $y=-L$; $HL=3$. Unit: length. Value: $L=3/H$ ($fL=8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $\tilde m$, $\tilde m_n$: The dimensionless x_0 eigenvalue of a boson mode, $\tilde m = m\,fL$; $\tilde m_n$ the n-th even Kaluza-Klein level of the tower with $\beta=0$ ($\tilde m_1$ the first). Unit: dimensionless. Value: $\tilde m_1^2=30.1196$, $\tilde m_2^2=59.7284$, $\tilde m_3^2=109.0764$ ($HL=3$), i.e. $\tilde m_1=5.48813$, $\tilde m_2=7.72842$, $\tilde m_3=10.44397$. Status: DERIVED.
- β: The Robin parameter of the tip condition of a boson profile, $\mathrm{d}F/\mathrm{d}\xi=\beta F$ at $\xi=-1$ (zero for the massless gauge fields and the graviton). Unit: dimensionless. Status: DERIVED.

whose even solution is $\tilde F=\cos\varpi\xi+\frac{3HL}{2\varpi}\sin\varpi\xi$ and whose odd solution is $\tilde F=\sin\varpi\xi$; the tip condition then gives:

$$F=\mathrm{e}^{-3HL\xi/2}\,\tilde F,\qquad \varpi^2=\tilde m^2-\frac{9H^2L^2}{4}:$$
$$\text{even: }\Big[\varpi^2+\frac{3HL}{2}\Big(\beta+\frac{3HL}{2}\Big)\Big]\sin\varpi-\beta\,\varpi\cos\varpi=0,\qquad \text{odd: }\varpi\cos\varpi+\Big(\beta+\frac{3HL}{2}\Big)\sin\varpi=0; \tag{FE-52}$$
$$\beta=0:\quad \tilde m=0\ \text{(flat profile)},\qquad \tilde m_n^2=\frac{9H^2L^2}{4}+\varpi_n^2\ \text{(even)},\qquad \varpi\cot\varpi=-\frac{3HL}{2}\ \text{(odd)}$$

Symbols in (FE-52).

- $F(\xi)$, F_n: The x_0 profile of a boson mode on one half of the period (a function of y or of $\xi=y/L$): of a gauge field (gauge $A_y=0$) or of the transverse-traceless graviton; F_n is the profile of the n-th Kaluza-Klein level of the graviton tower ($F_0=1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda=-10H^2$ and the mass parameter $m_\Psi=H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y=-L$; $HL=3$. Unit: length. Value: $L=3/H$ ($fL=8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- ξ: The scaled hidden coordinate, $\xi=y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $\tilde F$: The rescaled boson profile, $F=\mathrm{e}^{-3HL\xi/2}\tilde F$. Unit: dimensionless. Status: DERIVED.
- ϖ, ϖ_n: The wave number of an oscillating x_0 profile on $-1\le\xi\le0$ (of a boson after the rescaling, or of a bulk level of dirac16complex); ϖ_n that of the n-th even level ($\varpi_n=n\pi$ for $\beta=0$). Unit: dimensionless. Value: 1.89901 for the lowest odd level of dirac16complex. Status: DERIVED.
- $\tilde m$, $\tilde m_n$: The dimensionless x_0 eigenvalue of a boson mode, $\tilde m = m\,fL$; $\tilde m_n$ the n-th even Kaluza-Klein level of the tower with $\beta=0$ ($\tilde m_1$ the first). Unit: dimensionless. Value: $\tilde m_1^2=30.1196$, $\tilde m_2^2=59.7284$, $\tilde m_3^2=109.0764$ ($HL=3$), i.e. $\tilde m_1=5.48813$, $\tilde m_2=7.72842$, $\tilde m_3=10.44397$. Status: DERIVED.
- β: The Robin parameter of the tip condition of a boson profile, $\mathrm{d}F/\mathrm{d}\xi=\beta F$ at $\xi=-1$ (zero for the massless gauge fields and the graviton). Unit: dimensionless. Status: DERIVED.
- n: The level number of the Kaluza-Klein tower ($n=1,2,3,\ldots$; $n=0$ the flat zero mode). Status: DERIVED.

The terms. ϖ is the wave number of the rescaled profile; $9H^2L^2/4=20.25$ is a gap: every excited level lies above it. For $\beta=0$ the even condition has the root $\tilde m=0$ with the flat profile $F=1$: the graviton, the photon and the gluons are exactly massless, and their profile is the same at every point of the hidden direction. The excited even levels are $\tilde m_n^2=20.25+n^2\pi^2$ ($\tilde m_1^2=30.1196$) and the first odd level is $\tilde m^2=27.0890$. For the small β of $W^\pm$ and Z the lowest even root is far below the gap: $\tilde m_\mathrm{W}^2=1.32149\times10^{-5}$ and $\tilde m_\mathrm{Z}^2=1.70150\times10^{-5}$, approximately $\tilde m^2\simeq\beta\,\mathrm{e}^{-3HL}/N_0$ with $N_0=(1-\mathrm{e}^{-3HL})/(3HL)$ (integrate the equation over ξ with $F\simeq1$). Status: DERIVED (mpmath; finite elements to 3×10^{-14}).

$$F_n(\xi)=\mathrm{e}^{-3HL\xi/2}\Big(\cos\varpi_n\xi+\frac{3HL}{2\varpi_n}\sin\varpi_n\xi\Big),\qquad \varpi_n=\pi,\ 2\pi,\ 3\pi,\ \ldots,$$
$$\tilde m_n^2=\frac{9H^2L^2}{4}+\varpi_n^2,\qquad m_n^{\mathrm{KK}}=\frac{\tilde m_n}{fL},\qquad N_n=\int_{-1}^{0}\mathrm{e}^{3HL\xi}F_n^2\,\mathrm{d}\xi=\frac12\Big(1+\frac{9H^2L^2}{4\varpi_n^2}\Big),\qquad N_0=\frac{1-\mathrm{e}^{-3HL}}{3HL} \tag{FE-53}$$

Symbols in (FE-53).

- $F(\xi)$, F_n: The x_0 profile of a boson mode on one half of the period (a function of y or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; F_n is the profile of the n-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- n: The level number of the Kaluza-Klein tower ($n = 1, 2, 3, \ldots$; $n = 0$ the flat zero mode). Status: DERIVED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- ϖ, ϖ_n: The wave number of an oscillating x_0 profile on $-1 \le \xi \le 0$ (of a boson after the rescaling, or of a bulk level of dirac16complex); ϖ_n that of the n-th even level ($\varpi_n = n\pi$ for $\beta = 0$). Unit: dimensionless. Value: 1.89901 for the lowest odd level of dirac16complex. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\tilde{m}$, $\tilde{m}_n$: The dimensionless x_0 eigenvalue of a boson mode, $\tilde{m} = m\,fL$; $\tilde{m}_n$ the n-th even Kaluza-Klein level of the tower with $\beta = 0$ ($\tilde{m}_1$ the first). Unit: dimensionless. Value: $\tilde{m}_1^2 = 30.1196$, $\tilde{m}_2^2 = 59.7284$, $\tilde{m}_3^2 = 109.0764$ ($HL = 3$), i.e. $\tilde{m}_1 = 5.48813$, $\tilde{m}_2 = 7.72842$, $\tilde{m}_3 = 10.44397$. Status: DERIVED.
- m_n^KK, m_KK: The mass of the n-th Kaluza-Klein graviton (and gauge) level, $m_n^\mathrm{KK} = \tilde{m}_n/(fL)$; $m_\mathrm{KK} = m_1^\mathrm{KK}$ is the Kaluza-Klein scale. Unit: GeV. Value: $m_\mathrm{KK} = 121.32$ TeV. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- N_n, N_0: The normalisation integral of the profile F_n of the n-th Kaluza-Klein level, $N_n = \int_{-1}^{0} \mathrm{e}^{3HL\xi} F_n^2 \,\mathrm{d}\xi$; N_0 that of the flat zero mode. Unit: dimensionless. Value: $N_0 = (1 - \mathrm{e}^{-3HL})/(3HL) = 0.1111$. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.

The terms. The tower of Kaluza-Klein gravitons, the terms $n \ge 1$ of the expansion of the metric perturbation in x_0 profiles (FE-87): F_n is the even profile with $\varpi_n = n\pi$, m_n^KK the four-dimensional mass and N_n the norm of the profile in the measure $\mathrm{e}^{3HL\xi}\,\mathrm{d}\xi$ (N_0 that of the flat zero mode, the massless four-dimensional graviton). The first level sets the Kaluza-Klein scale $m_\mathrm{KK} = m_1^\mathrm{KK} = 121.32$ TeV; $\tilde{m}_n^2 = 30.1196$, 59.7284 and 109.0764 for $n = 1, 2, 3$. The same levels are the Kaluza-Klein modes of the gauge fields. Status: DERIVED (finite elements agree to 3.9×10^{-9}; Einstein gravity, see FE-50). Residual of the stored tower 1.2×10^{-12}.

$$V_\mathrm{N}(r) = -\frac{G_\mathrm{N} M_1 M_2}{r}\Big[1 + \sum_{n\ge1} \mathcal{A}_n\, \mathrm{e}^{-m_n^\mathrm{KK} r}\Big], \qquad \mathcal{A}_n = \frac{N_0}{N_n}\Big(\int_{-1}^{0} \varrho_1 F_n \,\mathrm{d}\xi\Big)\Big(\int_{-1}^{0} \varrho_2 F_n \,\mathrm{d}\xi\Big) \tag{FE-54}$$

Symbols in (FE-54).

- $V_\mathrm{N}(r)$: The potential energy of two static masses at the distance r (Newton potential with the Kaluza-Klein corrections). Unit: GeV. Status: DERIVED.
- r: A distance: between the two masses in $V_\mathrm{N}(r)$, from the centre of the star in the adiabaticity condition. Unit: length. Status: STANDARD.
- G_N: Newton's constant of four-dimensional gravity, $8\pi G_\mathrm{N} = \kappa/(\mathcal{G}(K_*)V_\perp)$. Unit: GeV^{-2}. Value: 6.709×10^{-39} GeV^{-2} ($M_\mathrm{Pl} = G_\mathrm{N}^{-1/2} = 1.22091 \times 10^{19}$ GeV). Status: STANDARD.
- M_1, M_2: The masses of the two static sources in the potential V_N. Unit: GeV. Status: STANDARD.
- n: The level number of the Kaluza-Klein tower ($n = 1, 2, 3, \ldots$; $n = 0$ the flat zero mode). Status: DERIVED.
- $\mathcal{A}_n$: The relative strength of the Yukawa force of the n-th Kaluza-Klein graviton between two sources with the profiles ϱ_1, ϱ_2. Unit: dimensionless. Value: 0.0904, 0.0136, 0.0057 ($n = 1, 2, 3$, two quark-doublet profiles). Status: DERIVED.

- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- m_n^{KK}, m_{KK}: The mass of the n-th Kaluza-Klein graviton (and gauge) level, $m_n^{\mathrm{KK}} = \tilde{m}_n/(fL)$; $m_{\mathrm{KK}} = m_1^{\mathrm{KK}}$ is the Kaluza-Klein scale. Unit: GeV. Value: $m_{\mathrm{KK}} = 121.32$ TeV. Status: DERIVED.
- N_n, N_0: The normalisation integral of the profile F_n of the n-th Kaluza-Klein level, $N_n = \int_{-1}^{0} \mathrm{e}^{3HL\xi} F_n^2 \,\mathrm{d}\xi$; N_0 that of the flat zero mode. Unit: dimensionless. Value: $N_0 = (1 - \mathrm{e}^{-3HL})/(3HL) = 0.1111$. Status: DERIVED.
- ϱ_1, ϱ_2: The x_0 profiles of the two sources of V_{N} as probability densities in ξ ($\int_{-1}^{0} \varrho_1 \,\mathrm{d}\xi = 1$): L times the profile ϱ_f of the zero mode of each source. Unit: dimensionless. Status: DERIVED.
- $F(\xi)$, F_n: The x_0 profile of a boson mode on one half of the period (a function of y or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; F_n is the profile of the n-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.

The terms. Two static masses M_1, M_2 whose matter is distributed in x_0 with the normalised densities ϱ_1, ϱ_2 exchange the massless graviton (Newton's potential) and the tower: each level adds a Yukawa potential with the range $1/m_n^{\mathrm{KK}}$ and the strength $\mathcal{A}_n$, the product of the overlaps of the two densities with the profile of the level. For the zero-mode profiles of quark doublets $\mathcal{A}_n = 0.0904$, 0.0136, 0.0057; for the flat profile of the gauge zero modes all $\mathcal{A}_n$ vanish (the profiles are orthogonal). The range $1/m_{\mathrm{KK}} = 1.6 \times 10^{-21}$ m is far below every measured distance, so Newton's law is exact wherever it is tested. Status: DERIVED.

$$\rho_{\mathrm{EW}} = \frac{\tilde{m}_{\mathrm{W}}^2}{\tilde{m}_{\mathrm{Z}}^2 \cos^2\theta_{\mathrm{W}}} = \rho_{\mathrm{EW},0}, \qquad \cos^2\theta_{\mathrm{W}} = \frac{m_{\mathrm{W}}^2/m_{\mathrm{Z}}^2}{\rho_{\mathrm{EW},0}}, \qquad m_{\mathrm{KK}} = m_{\mathrm{W}}\,\frac{\tilde{m}_1}{\tilde{m}_{\mathrm{W}}}, \qquad m_{\mathrm{Z}} = m_{\mathrm{W}}\,\frac{\tilde{m}_{\mathrm{Z}}}{\tilde{m}_{\mathrm{W}}} \tag{FE-55}$$

Symbols in (FE-55).

- ρ_{EW}: The electroweak parameter, $\rho_{\mathrm{EW}} = \tilde{m}_{\mathrm{W}}^2/(\tilde{m}_{\mathrm{Z}}^2 \cos^2\theta_{\mathrm{W}})$. Unit: dimensionless. Status: DERIVED.
- $\tilde{m}_{\mathrm{W}}$: The dimensionless x_0 eigenvalue of the lowest $W^{\pm}$ mode ($\beta = \beta_{\mathrm{W}}$). Unit: dimensionless. Value: $\tilde{m}_{\mathrm{W}}^2 = 1.32149 \times 10^{-5}$. Status: DERIVED.
- $\tilde{m}_{\mathrm{Z}}$: The dimensionless x_0 eigenvalue of the lowest Z mode ($\beta = \beta_{\mathrm{Z}}$). Unit: dimensionless. Value: $\tilde{m}_{\mathrm{Z}}^2 = 1.70150 \times 10^{-5}$. Status: DERIVED.
- θ_{W}: The weak mixing angle of the couplings, $\cos\theta_{\mathrm{W}} = \hat{g}_2/\hat{g}_Z$, $\sin\theta_{\mathrm{W}} = \hat{g}_1/\hat{g}_Z$; fixed by the measured W and Z masses with $\rho_{\mathrm{EW},0}$. Unit: dimensionless. Value: $\cos^2\theta_{\mathrm{W}} = 0.776368$. Status: DERIVED.
- $\rho_{\mathrm{EW},0}$: The measured value of the electroweak parameter (PDG fit). Unit: dimensionless. Value: 1.00038. Status: STANDARD.
- m_{W}: The mass of W^+ and W^-. Unit: GeV. Value: 80.3625 GeV (measured, input). Status: STANDARD.
- m_{Z}: The mass of the Z. Unit: GeV. Value: 91.1879 GeV (reproduced). Status: STANDARD.
- m_n^{KK}, m_{KK}: The mass of the n-th Kaluza-Klein graviton (and gauge) level, $m_n^{\mathrm{KK}} = \tilde{m}_n/(fL)$; $m_{\mathrm{KK}} = m_1^{\mathrm{KK}}$ is the Kaluza-Klein scale. Unit: GeV. Value: $m_{\mathrm{KK}} = 121.32$ TeV. Status: DERIVED.
- $\tilde{m}$, $\tilde{m}_n$: The dimensionless x_0 eigenvalue of a boson mode, $\tilde{m} = m\,fL$; $\tilde{m}_n$ the n-th even Kaluza-Klein level of the tower with $\beta = 0$ ($\tilde{m}_1$ the first). Unit: dimensionless. Value: $\tilde{m}_1^2 = 30.1196$, $\tilde{m}_2^2 = 59.7284$, $\tilde{m}_3^2 = 109.0764$ ($HL = 3$), i.e. $\tilde{m}_1 = 5.48813$, $\tilde{m}_2 = 7.72842$, $\tilde{m}_3 = 10.44397$. Status: DERIVED.

The terms. ρ_{EW} is the electroweak parameter of the tree-level masses; at leading order in β it is exactly 1 (because $\beta_{\mathrm{Z}}/\beta_{\mathrm{W}} = 1/\cos^2\theta_{\mathrm{W}}$, the custodial relation), and the next order makes it larger than 1. β_{W} is fitted so that ρ_{EW} equals the measured value, $\rho_{\mathrm{EW},0} - 1 = 0.00038$; with the measured m_{W} and m_{Z} this fixes the weak angle of the couplings, $\cos^2\theta_{\mathrm{W}} = 0.776368$. Then m_{Z} follows from the ratio of the eigenvalues: the measured $m_{\mathrm{W}} = 80.3625$ GeV gives the measured $m_{\mathrm{Z}} = 91.1879$ GeV (to 10^{-8}, a consequence of the fit), and the first even level gives the Kaluza-Klein scale $m_{\mathrm{KK}} = 121.32$ TeV and with it the author's constant $H = 14.296$ TeV. The tree-level Z widths of three generations agree with the measured total width to 0.4 % (the size of the neglected electroweak loop corrections). Status: DERIVED (fit).

6.2 The fermions

Every fermion of UM-2 obeys an eight-dimensional Dirac equation of the form $\gamma^M D_M \Psi = (M(y) - \mathrm{i}v(y)\gamma^{x_4})\Psi$, with a mass function $M(y)$ and a vector potential $v(y)$ along the time: $M = m_\Psi$ for free dirac16complex, $M = \mu_{\psi i}$, $v = 0$ for a Standard-Model multiplet, and the Kohn-Sham mean fields for the interacting dirac16complex (FE-63). It reduces exactly:

$$
\begin{aligned}
&\Psi = \mathrm{e}^{\mathrm{i}kx_1}\,\mathrm{e}^{-\frac{3}{2}Hy}\,\mathrm{e}^{a_4/2}\,b_4^{-1}\,\mathcal{V}\chi(y,x_4)\,:\\
&\gamma^M D_M\Psi - \big(M(y) - \mathrm{i}v(y)\gamma^{x_4}\big)\Psi = \mathrm{e}^{\mathrm{i}kx_1}\,\mathrm{e}^{-\frac{3}{2}Hy}\,\mathrm{e}^{a_4/2}\,b_4^{-1}\,\gamma^{x_4}\mathcal{V}\big[\partial_{x_4}\chi + \mathrm{i}h\chi\big],\\
&h = \varsigma\Big[-\frac{\mathrm{i}}{f}\sigma_1\partial_y + M(y)\,\sigma_2 + \mathrm{e}^{-a_4}k\,\sigma_3\Big] + v(y), \qquad \mathrm{i}\,\partial_{x_4}\chi = h\chi;\\
&\text{in each block: } \gamma^{x_0} = \sigma_3, \qquad \mathcal{B} = \varsigma\varsigma_2, \qquad C = \varsigma_2\sigma_2, \qquad \Gamma_4 = -\varsigma\varsigma_2\sigma_3
\end{aligned}
\tag{FE-56}
$$

Symbols in (FE-56).

- Ψ, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{V}$: The constant 16×2 matrix of the reduction ansatz that places the two-component spinor χ of one block into the 16 components of the eight-dimensional spinor. Unit: dimensionless. Status: DERIVED.
- χ, χ_o, χ_1, χ_2: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1,\chi_2)^T = (a,\mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^0 \chi_o^\dagger\chi_o\,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- $M(y)$: The mass function of the block Hamiltonian (the scalar potential): $M = \mu_{\psi i}$ for a Standard-Model fermion, $M = M_\mathrm{eff}$ for dirac16complex. Unit: GeV. Status: DERIVED.
- $v(y)$: The vector potential of the block Hamiltonian (the mean field along γ^{x_4}): for dirac16complex the contact term and the mean field of I11; zero for a Standard-Model fermion. Unit: GeV. Status: DERIVED.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status:

STANDARD.

- h: The block Hamiltonian: the 2×2 differential operator of one block that generates the time evolution of χ, $\mathrm{i}\,\partial_{x_4}\chi = h\chi$; its eigenvalues are the energies ε. Unit: GeV. Status: DERIVED.
- ς: The first of the three signs $(\varsigma, \varsigma_2, \varsigma_3)$, each ± 1, that label the eight blocks of the reduction. Unit: dimensionless. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\sigma_1,\ \sigma_2,\ \sigma_3$: The Pauli matrices acting on the two components of a block spinor χ ($\sigma_3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- y: The label y in ∂_y: the derivative with respect to the hidden coordinate y. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $\mathcal{B}$: The author's operator $\mathcal{B} = -\mathrm{i}C\gamma^{x_4}$; its eigenvalues ∓ 1 define the halves Ψ_A and Ψ_B, and it commutes with every interaction of the good sector. Unit: dimensionless. Status: AUTHOR.
- ς_2: The second sign that labels the blocks (± 1). Unit: dimensionless. Status: DERIVED.
- C: The matrix of the Dirac adjoint, $C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}$ ($\bar{\Psi} = \Psi^\dagger C$); in each block of the reduction $C = \varsigma_2\sigma_2$. Unit: dimensionless. Status: AUTHOR.
- Γ_4: The four-dimensional chirality matrix, $\Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$; a doublet zero mode has $\Gamma_4 = +1$ (left-handed), a singlet zero mode $\Gamma_4 = -1$ (right-handed). Unit: dimensionless. Status: DERIVED.

The terms. The ansatz has a plane wave $\mathrm{e}^{\mathrm{i}kx_1}$ along ordinary space, the warp factor $\mathrm{e}^{-\frac{3}{2}Hy}$ (which removes the spin-connection term $\frac{3}{2}(H/f)\gamma^{x_0}$ of FE-13), the time-dependent factor $\mathrm{e}^{a_4/2} b_4^{-1}$ (which removes the term along γ^{x_4}), the constant 16×2 matrix $\mathcal{V}$ (two columns of the record's unitary block basis, which splits the sixteen components into eight blocks of two) and the two-component block spinor χ. The eight-dimensional equation becomes the Schrödinger form $\mathrm{i}\,\partial_{x_4}\chi = h\chi$ with the block Hamiltonian h: $-\frac{\mathrm{i}}{f}\sigma_1\partial_y$ is the momentum along the proper hidden length, $M(y)\sigma_2$ the mass, $\mathrm{e}^{-a_4}k\sigma_3$ the physical momentum along ordinary space, and $v(y)$ the potential; $\varsigma = \pm 1$ is the first of the three signs that label the block. In each block γ^{x_0}, $\mathcal{B}$, C and Γ_4 are the matrices or numbers in the last line. Status: DERIVED (exact in all eight blocks, for general a_4, b_4, M, v; checked with the author's matrices).

$$\chi = \big(\mathrm{e}^{c\xi},\ 0\big)^T \ \text{(doublet, } \Gamma_4 = +1\text{)}, \qquad \chi = \big(0,\ \mathrm{e}^{-c\xi}\big)^T \ \text{(singlet, } \Gamma_4 = -1\text{)}, \qquad c = f\mu L,$$
$$h\big(\mathrm{e}^{c\xi},\ 0\big)^T = \varsigma\,\mathrm{e}^{-a_4} k \big(\mathrm{e}^{c\xi},\ 0\big)^T \tag{FE-57}$$

Symbols in (FE-57).

- $\chi,\ \chi_o,\ \chi_1,\ \chi_2$: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $c,\ c_L,\ c_R$: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); c_L of a doublet (localised at the fold for $c_L > 0$), c_R of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets c_R: up -15.4300, down -14.6258, strange -11.5158, charm -8.7683, bottom -7.5005, top -3.3823, electron -14.5539, muon -8.9809, tau -5.9529, neutrinos -35.0452, -32.8481, -31.0702 (fixed by the measured masses). Status: DERIVED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- ^{T}: The transpose: a row written with ^{T} is a column. Status: STANDARD.
- Γ_4: The four-dimensional chirality matrix, $\Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$; a doublet zero mode has $\Gamma_4 = +1$ (left-handed), a singlet zero mode $\Gamma_4 = -1$ (right-handed). Unit: dimensionless. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\mu_{\psi i}$: The bulk mass of a Standard-Model fermion: $\mu_{\psi i}$ of the multiplet ψ of the generation i; with the hidden direction it fixes the localisation $c = f\mu L$ of the zero mode. Unit: GeV. Value: $\mu_{\psi i} = c/(fL)$ with the

localisation c of the zero mode (doublets: $\mu_Q = 5.2304\,H$, $\mu_\ell = 9.2055\,H$). Status: ASSUMED.

- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- h: The block Hamiltonian: the 2×2 differential operator of one block that generates the time evolution of χ, $\mathrm{i}\,\partial_{x_4}\chi = h\chi$; its eigenvalues are the energies ε. Unit: GeV. Status: DERIVED.
- ς: The first of the three signs $(\varsigma, \varsigma_2, \varsigma_3)$, each ± 1, that label the eight blocks of the reduction. Unit: dimensionless. Status: DERIVED.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.

The terms. For a constant bulk mass μ the block Hamiltonian has, at every momentum, the exact solutions $\chi = (\mathrm{e}^{c\xi}, 0)$ and $(0, \mathrm{e}^{-c\xi})$ with the localisation parameter $c = f\mu L$; the y derivative and the mass term cancel, and only the momentum term remains: the energy is $\mathrm{e}^{-a_4}|k|$, a massless four-dimensional particle. The fold conditions FE-29 keep the first for the doublets ($\Gamma_4 = +1$, left-handed) and the second for the singlets ($\Gamma_4 = -1$, right-handed); they are localised at the fold for $c_L > 0$ and $c_R < 0$. The doublets have $c_L = 3.38230$ (quarks) and 5.95288 (leptons). Status: DERIVED (exact).

At $k = 0$ in a block with $\varsigma = 1$ the eigenvalue equation $h\chi = \varepsilon\chi$ is a pair of first-order equations:

$$\varsigma = 1,\quad k = 0,\quad \chi = (a,\ \mathrm{i}b): \qquad h\chi = \varepsilon\chi \quad\Leftrightarrow\quad \frac{1}{f}\,\partial_y b + M\,b = \varepsilon\,a, \qquad -\frac{1}{f}\,\partial_y a + M\,a = \varepsilon\,b,$$

Symbols in this equation.

- ς: The first of the three signs $(\varsigma, \varsigma_2, \varsigma_3)$, each ± 1, that label the eight blocks of the reduction. Unit: dimensionless. Status: DERIVED.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- χ, χ_o, χ_1, χ_2: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- a, b: The upper and lower components of a block spinor, $\chi = (a, \mathrm{i}b)$, real at $k = 0$: a_L, b_L of a doublet and a_R, b_R of a singlet solution (a_{Ri}, b_{Ri} of the generation i, $b_{L\alpha}$, $a_{L\alpha}$ of the doublet in the flavour basis), a_o, b_o of the orbital o. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- h: The block Hamiltonian: the 2×2 differential operator of one block that generates the time evolution of χ, $\mathrm{i}\,\partial_{x_4}\chi = h\chi$; its eigenvalues are the energies ε. Unit: GeV. Status: DERIVED.
- ε, ε_o: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); ε_o the Kohn-Sham level of the orbital o. Unit: GeV. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- y: The label y in ∂_y: the derivative with respect to the hidden coordinate y. Status: STANDARD.
- $M(y)$: The mass function of the block Hamiltonian (the scalar potential): $M = \mu_{\psi i}$ for a Standard-Model fermion, $M = M_{\mathrm{eff}}$ for dirac16complex. Unit: GeV. Status: DERIVED.

which in ξ, with $c = fML$ and $\tilde{\varepsilon} = \varepsilon f L$, are the first line of FE-58. Their solutions that satisfy the fold conditions of the doublet and of the singlet reach the tip with:

$$\frac{\mathrm{d}a}{\mathrm{d}\xi} = c\,a - \tilde{\varepsilon}\,b, \qquad \frac{\mathrm{d}b}{\mathrm{d}\xi} = \tilde{\varepsilon}\,a - c\,b, \qquad \chi = (a,\ \mathrm{i}b), \qquad \tilde{\varepsilon} = \varepsilon f L, \qquad \upsilon_L = \sqrt{c_L^2 - \tilde{\varepsilon}^2}, \qquad \upsilon_R = \sqrt{c_R^2 - \tilde{\varepsilon}^2},$$
$$(a_L,\ b_L)\big|_{\xi=-1} = \Big(\cosh\upsilon_L - c_L\frac{\sinh\upsilon_L}{\upsilon_L}, -\tilde{\varepsilon}\,\frac{\sinh\upsilon_L}{\upsilon_L}\Big), \qquad (a_R,\ b_R)\big|_{\xi=-1} = \Big(\tilde{\varepsilon}\,\frac{\sinh\upsilon_R}{\upsilon_R}, \cosh\upsilon_R + c_R\frac{\sinh\upsilon_R}{\upsilon_R}\Big) \tag{FE-58}$$

Symbols in (FE-58).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $a,\ b$: The upper and lower components of a block spinor, $\chi = (a, \mathrm{i}b)$, real at $k = 0$: a_L, b_L of a doublet and a_R, b_R of a singlet solution (a_{Ri}, b_{Ri} of the generation i, $b_{L\alpha}$, $a_{L\alpha}$ of the doublet in the flavour basis), a_o, b_o of the orbital o. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $c,\ c_L,\ c_R$: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); c_L of a doublet (localised at the fold for $c_L > 0$), c_R of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets c_R: up -15.4300, down -14.6258, strange -11.5158, charm -8.7683, bottom -7.5005, top -3.3823, electron -14.5539, muon -8.9809, tau -5.9529, neutrinos -35.0452, -32.8481, -31.0702 (fixed by the measured masses). Status: DERIVED.
- $\tilde{\varepsilon}$: The dimensionless energy (mass) of a fermion mode, $\tilde{\varepsilon} = \varepsilon f L$. Unit: dimensionless. Status: DERIVED.
- $\chi,\ \chi_o,\ \chi_1,\ \chi_2$: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\varepsilon,\ \varepsilon_o$: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); ε_o the Kohn-Sham level of the orbital o. Unit: GeV. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $\upsilon_L,\ \upsilon_R$: The auxiliary roots $\upsilon = (c^2 - \tilde{\varepsilon}^2)^{1/2}$ of the doublet (υ_L) and singlet (υ_R) solutions at $k = 0$. Unit: dimensionless. Status: DERIVED.
- L: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1 + \Gamma_4)/2$. Status: STANDARD.
- R: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1 - \Gamma_4)/2$. Status: STANDARD.

The terms. a and b are the upper and lower components of the block spinor. The doublet solution starts at the fold with $b = 0$, the singlet solution with $a = 0$; υ_L and υ_R are the decay rates in ξ (real for $\tilde{\varepsilon} < |c|$, the case of every Standard-Model fermion). The tip values of the four components enter the tip conditions FE-30. Status: DERIVED (exact).

For one generation the two tip conditions of FE-30, applied to the doublet solution and the singlet solution each multiplied by an amplitude, are two linear equations for the two amplitudes; a non-zero solution needs

$$\det\begin{pmatrix} b_L & -\mathcal{K}\,b_R \\ \mathcal{K}\,a_L & a_R \end{pmatrix} = b_L a_R + \mathcal{K}^2 a_L b_R = -\mathcal{D}(\tilde{\varepsilon}) = 0 :$$

Symbols in this equation.

- $a,\ b$: The upper and lower components of a block spinor, $\chi = (a, \mathrm{i}b)$, real at $k = 0$: a_L, b_L of a doublet and a_R, b_R of a singlet solution (a_{Ri}, b_{Ri} of the generation i, $b_{L\alpha}$, $a_{L\alpha}$ of the doublet in the flavour basis), a_o, b_o of the orbital o. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.

- L: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1+\Gamma_4)/2$. Status: STANDARD.
- $\mathcal{K}$: The tip mixing matrix of a sector in the boundary conditions of the fermion zero modes (a number $\mathcal{K}$ for one generation, a 3×3 matrix for three): $\mathcal{K}_\nu$, $\mathcal{K}_e$, $\mathcal{K}_u$, $\mathcal{K}_d$ of the neutrino, charged-lepton, up-quark and down-quark sectors; UM-2 takes $\mathcal{K}_e = \mathcal{K}_u = \mathbf{1}$, $\mathcal{K}_\nu = U$, $\mathcal{K}_d = V_{\mathrm{CKM}}$. Unit: dimensionless. Status: ASSUMED.
- R: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1-\Gamma_4)/2$. Status: STANDARD.
- $\mathcal{D}(\tilde\varepsilon)$: The mass function of one generation: its zeros are the dimensionless masses $\tilde\varepsilon$. Unit: dimensionless. Status: DERIVED.
- $\tilde\varepsilon$: The dimensionless energy (mass) of a fermion mode, $\tilde\varepsilon = \varepsilon f L$. Unit: dimensionless. Status: DERIVED.

$$\mathcal{D}(\tilde\varepsilon) = \tilde\varepsilon^2\,\frac{\sinh\upsilon_L}{\upsilon_L}\frac{\sinh\upsilon_R}{\upsilon_R} - \mathcal{K}^2\Big(\cosh\upsilon_L - c_L\frac{\sinh\upsilon_L}{\upsilon_L}\Big)\Big(\cosh\upsilon_R + c_R\frac{\sinh\upsilon_R}{\upsilon_R}\Big) = 0,$$
$$\tilde\varepsilon \simeq \mathcal{K}\,\mathcal{Z}(c_L)\,\mathcal{Z}(-c_R), \qquad \mathcal{Z}(c)^2 = \frac{c\,\mathrm{e}^{-c}}{\sinh c} \tag{FE-59}$$

Symbols in (FE-59).

- $\mathcal{D}(\tilde\varepsilon)$: The mass function of one generation: its zeros are the dimensionless masses $\tilde\varepsilon$. Unit: dimensionless. Status: DERIVED.
- $\tilde\varepsilon$: The dimensionless energy (mass) of a fermion mode, $\tilde\varepsilon = \varepsilon f L$. Unit: dimensionless. Status: DERIVED.
- $\upsilon_L,\ \upsilon_R$: The auxiliary roots $\upsilon = (c^2 - \tilde\varepsilon^2)^{1/2}$ of the doublet (υ_L) and singlet (υ_R) solutions at $k = 0$. Unit: dimensionless. Status: DERIVED.
- L: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1+\Gamma_4)/2$. Status: STANDARD.
- R: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1-\Gamma_4)/2$. Status: STANDARD.
- $\mathcal{K}$: The tip mixing matrix of a sector in the boundary conditions of the fermion zero modes (a number $\mathcal{K}$ for one generation, a 3×3 matrix for three): $\mathcal{K}_\nu$, $\mathcal{K}_e$, $\mathcal{K}_u$, $\mathcal{K}_d$ of the neutrino, charged-lepton, up-quark and down-quark sectors; UM-2 takes $\mathcal{K}_e = \mathcal{K}_u = \mathbf{1}$, $\mathcal{K}_\nu = U$, $\mathcal{K}_d = V_{\mathrm{CKM}}$. Unit: dimensionless. Status: ASSUMED.
- $c,\ c_L,\ c_R$: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); c_L of a doublet (localised at the fold for $c_L > 0$), c_R of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets c_R: up -15.4300, down -14.6258, strange -11.5158, charm -8.7683, bottom -7.5005, top -3.3823, electron -14.5539, muon -8.9809, tau -5.9529, neutrinos -35.0452, -32.8481, -31.0702 (fixed by the measured masses). Status: DERIVED.
- $\mathcal{Z}(c)$: The tip amplitude of a normalised zero mode with the localisation c, $\mathcal{Z}(c)^2 = c\,\mathrm{e}^{-c}/\sinh c$. Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.

The terms. The exact mass condition of one generation: $\mathcal{D}$ vanishes at the dimensionless mass $\tilde\varepsilon$. For strongly localised modes the root is the product of the tip mixing number and the two tip amplitudes $\mathcal{Z}(c_L)$ and $\mathcal{Z}(-c_R)$, the values of the normalised zero-mode profiles at the tip: a fermion is light when its right-handed mode is sharply localised at the fold. The localisations reproduce the measured masses of the twelve fermions (for example the quarks 2.16 MeV, 4.70 MeV, 93.5 MeV, 1.273 GeV, 4.183 GeV, 172.56 GeV with $c_R = -15.4300$, -14.6258, -11.5158, -8.7683, -7.5005, -3.3823; the neutrinos with $c_R = -35.0452$, -32.8481, -31.0702). Status: DERIVED (exact roots; shooting agrees to 6×10^{-10}; the first-order formula agrees to 1.1×10^{-3} for the top quark and better than 3×10^{-4} otherwise). Residual of the stored roots 5.4×10^{-11}; the masses are far below the Kaluza-Klein scale, $(m/m_{\mathrm{KK}})^2 \le 2.0\times10^{-6}$.

For three generations the amplitudes are generation vectors, $\boldsymbol{\alpha}$ of the doublets and $\boldsymbol{\beta}$ of the singlets, and the tip conditions are

$$\mathrm{diag}\big(b_{L\alpha}\big)\,\boldsymbol{\alpha} = \mathcal{K}\,\mathrm{diag}\big(b_{Ri}\big)\,\boldsymbol{\beta}, \qquad \mathrm{diag}\big(a_{Ri}\big)\,\boldsymbol{\beta} = -\mathcal{K}^\dagger\,\mathrm{diag}\big(a_{L\alpha}\big)\,\boldsymbol{\alpha};$$

Symbols in this equation.

- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- a, b: The upper and lower components of a block spinor, $\chi = (a, ib)$, real at $k = 0$: a_L, b_L of a doublet and a_R, b_R of a singlet solution (a_{Ri}, b_{Ri} of the generation i, $b_{L\alpha}$, $a_{L\alpha}$ of the doublet in the flavour basis), a_o, b_o of the orbital o. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- L: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1 + \Gamma_4)/2$. Status: STANDARD.
- α: A flavour index running over e, μ, τ (the flavour states of the leptons and neutrinos). Status: STANDARD.
- $\boldsymbol{\alpha}$, $\boldsymbol{\alpha}^{(i)}$: The vector of the three doublet amplitudes at the tip in the mass condition of three generations; $\boldsymbol{\alpha}^{(i)}$ its null vector for the mass state i (the left-handed flavour content). Unit: dimensionless. Status: DERIVED.
- $\mathcal{K}$: The tip mixing matrix of a sector in the boundary conditions of the fermion zero modes (a number $\mathcal{K}$ for one generation, a 3×3 matrix for three): $\mathcal{K}_\nu$, $\mathcal{K}_e$, $\mathcal{K}_u$, $\mathcal{K}_d$ of the neutrino, charged-lepton, up-quark and down-quark sectors; UM-2 takes $\mathcal{K}_e = \mathcal{K}_u = \mathbf{1}$, $\mathcal{K}_\nu = U$, $\mathcal{K}_d = V_{\mathrm{CKM}}$. Unit: dimensionless. Status: ASSUMED.
- R: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1 - \Gamma_4)/2$. Status: STANDARD.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $\boldsymbol{\beta}^{(i)}$: The vector of the three singlet amplitudes of the mass state i. Unit: dimensionless. Status: DERIVED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.

eliminating $\boldsymbol{\beta}$ gives:

$$\Big[\mathrm{diag}\big(b_{L\alpha}\big) + \mathcal{K}\,\mathrm{diag}\Big(\frac{b_{Ri}}{a_{Ri}}\Big)\mathcal{K}^\dagger\,\mathrm{diag}\big(a_{L\alpha}\big)\Big]\boldsymbol{\alpha} = 0; \qquad \det\big[\,\cdots\,\big] = \prod_{i=1}^{3}\frac{b_L a_{Ri} + a_L b_{Ri}}{a_{Ri}},$$
$$b_L a_R + a_L b_R = -\mathcal{D}(\tilde{\varepsilon})\big|_{\mathcal{K}=1}, \qquad \boldsymbol{\alpha}^{(i)} = \mathcal{K}\,\mathbf{u}_i, \qquad \boldsymbol{\beta}^{(i)} \propto \mathbf{u}_i, \qquad U = \mathcal{K}_e^\dagger \mathcal{K}_\nu, \qquad V_{\mathrm{CKM}} = \mathcal{K}_u^\dagger \mathcal{K}_d \tag{FE-60}$$

Symbols in (FE-60).

- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- a, b: The upper and lower components of a block spinor, $\chi = (a, ib)$, real at $k = 0$: a_L, b_L of a doublet and a_R, b_R of a singlet solution (a_{Ri}, b_{Ri} of the generation i, $b_{L\alpha}$, $a_{L\alpha}$ of the doublet in the flavour basis), a_o, b_o of the orbital o. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- L: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1 + \Gamma_4)/2$. Status: STANDARD.
- α: A flavour index running over e, μ, τ (the flavour states of the leptons and neutrinos). Status: STANDARD.
- $\mathcal{K}$: The tip mixing matrix of a sector in the boundary conditions of the fermion zero modes (a number $\mathcal{K}$ for one generation, a 3×3 matrix for three): $\mathcal{K}_\nu$, $\mathcal{K}_e$, $\mathcal{K}_u$, $\mathcal{K}_d$ of the neutrino, charged-lepton, up-quark and down-quark sectors; UM-2 takes $\mathcal{K}_e = \mathcal{K}_u = \mathbf{1}$, $\mathcal{K}_\nu = U$, $\mathcal{K}_d = V_{\mathrm{CKM}}$. Unit: dimensionless. Status: ASSUMED.
- R: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1 - \Gamma_4)/2$. Status: STANDARD.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\boldsymbol{\alpha}$, $\boldsymbol{\alpha}^{(i)}$: The vector of the three doublet amplitudes at the tip in the mass condition of three generations; $\boldsymbol{\alpha}^{(i)}$ its null vector for the mass state i (the left-handed flavour content). Unit: dimensionless. Status: DERIVED.
- $\mathcal{D}(\tilde{\varepsilon})$: The mass function of one generation: its zeros are the dimensionless masses $\tilde{\varepsilon}$. Unit: dimensionless. Status: DERIVED.
- $\tilde{\varepsilon}$: The dimensionless energy (mass) of a fermion mode, $\tilde{\varepsilon} = \varepsilon f L$. Unit: dimensionless. Status: DERIVED.

- $\mathbf{u}_i$: The unit vector of the generation i (the i-th column of the 3×3 identity). Unit: dimensionless. Status: STANDARD.
- $\boldsymbol{\beta}^{(i)}$: The vector of the three singlet amplitudes of the mass state i. Unit: dimensionless. Status: DERIVED.
- U: The PMNS matrix, the unitary 3×3 mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states i); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212^\circ$ (NuFIT 6.0). Status: STANDARD.
- V_{CKM}: The CKM matrix, the unitary 3×3 mixing matrix of the quarks (the same parametrisation as U); in UM-2 $V_{\mathrm{CKM}} = \mathcal{K}_u^\dagger \mathcal{K}_d$. Unit: dimensionless. Value: $s_{12} = 0.22501$, $s_{13} = 0.003732$, $s_{23} = 0.04183$, $\delta = 1.147$ (PDG). Status: STANDARD.

The terms. The bracket is the 3×3 mass matrix condition of one sector (neutrinos, charged leptons, up-type or down-type quarks): the masses are the roots of its determinant, $\boldsymbol{\alpha}$ is the left-handed flavour content of each mass state. For one doublet bulk mass per sector and a unitary tip matrix the determinant factorises into the one-generation conditions with $\mathcal{K} = 1$: the masses are the one-generation roots, the left-handed rotation of the sector is its tip matrix ($\boldsymbol{\alpha}^{(i)} = \mathcal{K}\mathbf{u}_i$, $\mathbf{u}_i$ the unit vectors) and the right-handed rotations are trivial. The lepton mixing matrix is therefore $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$ and the quark mixing matrix $V_{\mathrm{CKM}} = \mathcal{K}_u^\dagger \mathcal{K}_d$. Status: DERIVED (mpmath at 50 digits: the roots equal the one-generation roots to 3.0×10^{-42}, the null vectors the columns of the tip matrix to 1.1×10^{-16}); residual of the stored three-generation roots 1.2×10^{-11}.

$$U = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}\mathrm{e}^{-\mathrm{i}\delta_{\mathrm{CP}}} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}\mathrm{e}^{\mathrm{i}\delta_{\mathrm{CP}}} & c_{12}c_{23} - s_{12}s_{23}s_{13}\mathrm{e}^{\mathrm{i}\delta_{\mathrm{CP}}} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}\mathrm{e}^{\mathrm{i}\delta_{\mathrm{CP}}} & -c_{12}s_{23} - s_{12}c_{23}s_{13}\mathrm{e}^{\mathrm{i}\delta_{\mathrm{CP}}} & c_{23}c_{13} \end{pmatrix}, \qquad J_{\mathrm{CP}} = c_{12}s_{12}c_{23}s_{23}c_{13}^2 s_{13}\sin\delta_{\mathrm{CP}} \tag{FE-61}$$

Symbols in (FE-61).

- U: The PMNS matrix, the unitary 3×3 mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states i); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212^\circ$ (NuFIT 6.0). Status: STANDARD.
- $c_{12},\ c_{13},\ c_{23}$: The cosines of the mixing angles, $c_{ij} = \cos\theta_{ij}$. Unit: dimensionless. Status: STANDARD.
- $s_{12},\ s_{13},\ s_{23}$: The sines of the mixing angles, $s_{ij} = \sin\theta_{ij}$. Unit: dimensionless. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- δ_{CP}: The CP-violating phase of a mixing matrix. Unit: dimensionless. Value: 212° (PMNS), 1.147 rad (CKM). Status: STANDARD.
- J_{CP}: The Jarlskog invariant of a mixing matrix (the measure of CP violation). Unit: dimensionless. Value: -0.01777 (PMNS), 3.117×10^{-5} (CKM). Status: STANDARD.

The terms. The standard parametrisation of a unitary 3×3 mixing matrix by three angles ($c_{ij} = \cos\theta_{ij}$, $s_{ij} = \sin\theta_{ij}$) and the CP-violating phase δ_{CP}; J_{CP} is the Jarlskog invariant, the measure of CP violation in flavour oscillations (FE-105). UM-2 uses it for both tip matrices: $\mathcal{K}_\nu = U$ with the NuFIT 6.0 values (normal ordering) and $\mathcal{K}_d = V_{\mathrm{CKM}}$ with the PDG values. Status: STANDARD (inputs; unitary to 1.1×10^{-16}).

6.3 dirac16complex on the static block

Without its self-interaction and the portals dirac16complex has the constant mass function $M = m_\Psi$ in every block, with $c_\Psi = f m_\Psi L$. Its levels follow from the same pair of equations with the fold condition of each parity (FE-69) and the tip condition $b = 0$; for $\tilde{\varepsilon}^2 = c_\Psi^2 + \varpi^2$ the solutions that satisfy the fold conditions are

$$\text{odd: } a = \sin\varpi\xi, \quad b = \frac{c_\Psi \sin\varpi\xi - \varpi\cos\varpi\xi}{\tilde{\varepsilon}}; \qquad \text{even: } a = \cos\varpi\xi + \frac{c_\Psi}{\varpi}\sin\varpi\xi, \quad b = \frac{\tilde{\varepsilon}}{\varpi}\sin\varpi\xi,$$

Symbols in this equation.

- $a,\ b$: The upper and lower components of a block spinor, $\chi = (a, \mathrm{i}b)$, real at $k = 0$: a_L, b_L of a doublet and a_R, b_R of a singlet solution (a_{Ri}, b_{Ri} of the generation i, $b_{L\alpha}$, $a_{L\alpha}$ of the doublet in the flavour basis), a_o, b_o of the orbital o. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.

- ϖ, ϖ_n: The wave number of an oscillating x_0 profile on $-1 \le \xi \le 0$ (of a boson after the rescaling, or of a bulk level of dirac16complex); ϖ_n that of the n-th even level ($\varpi_n = n\pi$ for $\beta = 0$). Unit: dimensionless. Value: 1.89901 for the lowest odd level of dirac16complex. Status: DERIVED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- c_Ψ: The localisation parameter of dirac16complex, $c_\Psi = f\, m_\Psi L$. Unit: dimensionless. Value: 0.646667 at the stabilised block. Status: DERIVED.
- $\tilde{\varepsilon}$: The dimensionless energy (mass) of a fermion mode, $\tilde{\varepsilon} = \varepsilon f L$. Unit: dimensionless. Status: DERIVED.

and $b = 0$ at $\xi = -1$ gives $\tan\varpi = -\varpi/c_\Psi$ (odd) and $\sin\varpi = 0$ (even):

$$\begin{aligned}&\text{even: } \chi = \left(\mathrm{e}^{c_\Psi \xi},\ 0\right)^T,\ \varepsilon = 0 \text{ (all eight blocks)}, \qquad \varepsilon = \pm\sqrt{m_\Psi^2 + \left(\frac{\varpi_n}{fL}\right)^2},\ \varpi_n = \pi,\ 2\pi,\ \ldots;\\ &\text{odd: } \varepsilon = \pm\sqrt{m_\Psi^2 + \left(\frac{\varpi}{fL}\right)^2}, \quad \tan\varpi = -\frac{\varpi}{c_\Psi}; \qquad c_\Psi = f\, m_\Psi L; \qquad \text{fold band: } \varepsilon = \mathrm{e}^{-a_4}|k|\end{aligned} \tag{FE-62}$$

Symbols in (FE-62).

- χ, χ_o, χ_1, χ_2: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- c_Ψ: The localisation parameter of dirac16complex, $c_\Psi = f\, m_\Psi L$. Unit: dimensionless. Value: 0.646667 at the stabilised block. Status: DERIVED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- ^{T}: The transpose: a row written with ^{T} is a column. Status: STANDARD.
- ε, ε_o: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); ε_o the Kohn-Sham level of the orbital o. Unit: GeV. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- ϖ, ϖ_n: The wave number of an oscillating x_0 profile on $-1 \le \xi \le 0$ (of a boson after the rescaling, or of a bulk level of dirac16complex); ϖ_n that of the n-th even level ($\varpi_n = n\pi$ for $\beta = 0$). Unit: dimensionless. Value: 1.89901 for the lowest odd level of dirac16complex. Status: DERIVED.
- n: The level number of the Kaluza-Klein tower ($n = 1, 2, 3, \ldots$; $n = 0$ the flat zero mode). Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.

The terms. The even zero mode $(\mathrm{e}^{c_\Psi \xi}, 0)$ exists in all eight blocks and at every momentum has the energy $\mathrm{e}^{-a_4}|k|$: the fold band, massless four-dimensional quanta (eight states per momentum, two massless Dirac fermions). The odd levels and the even bulk levels are massive: the lowest odd level, $\varepsilon = 3.102204\, m_\Psi = 44.35$ TeV ($\varpi = 1.89901$), is the dark matter, the lowest odd Dirac fermion of the two halves Ψ_A and Ψ_B; the lowest even bulk level is $4.959981\, m_\Psi$. The negative levels are the antiparticles. At the stabilised block $c_\Psi = 3f_* = 0.646667$. Status: DERIVED (exact; the Kohn-Sham solver reproduces the levels to 10^{-12}). Residual of the stored levels 8.5×10^{-13}.

7 dirac16complex coupled to all fields

This section closes the coupled system. dirac16complex is treated by Kohn-Sham theory: its many-body state on the block is described by orbitals of the block Hamiltonian of FE-56 with mean fields, which contain its own self-interaction and the densities and condensates of every Standard-Model field. The Kohn-Sham energy-momentum tensor is the source of gravity; the dark matter acts back on the Standard Model through the field-dependent couplings; and the block responds to every species.

7.1 The Kohn-Sham equations

$$h\,\chi_o = \varepsilon_o\,\chi_o, \qquad h = \varsigma\Big[-\frac{\mathrm{i}}{f}\sigma_1\partial_y + M_{\mathrm{eff}}(y)\,\sigma_2 + \mathrm{e}^{-a_4}k\,\sigma_3\Big] + v(y) + \frac{1-\sigma_3}{2}\,v_{\mathrm{b}}(y) \tag{FE-63}$$

Symbols in (FE-63).

- h: The block Hamiltonian: the 2×2 differential operator of one block that generates the time evolution of χ, $\mathrm{i}\,\partial_{x_4}\chi = h\chi$; its eigenvalues are the energies ε. Unit: GeV. Status: DERIVED.
- χ, χ_o, χ_1, χ_2: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1,\chi_2)^T = (a,\mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0}\chi_o^\dagger\chi_o\,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- o: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over o runs over the orbitals. Status: RECORD.
- ε, ε_o: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k=0$ the four-dimensional mass); ε_o the Kohn-Sham level of the orbital o. Unit: GeV. Status: DERIVED.
- ς: The first of the three signs $(\varsigma,\varsigma_2,\varsigma_3)$, each ±1, that label the eight blocks of the reduction. Unit: dimensionless. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- σ_1, σ_2, σ_3: The Pauli matrices acting on the two components of a block spinor χ ($\sigma_3 = \mathrm{diag}(1,-1)$). Unit: dimensionless. Status: STANDARD.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- y: The label y in ∂_y: the derivative with respect to the hidden coordinate y. Status: STANDARD.
- $M_{\mathrm{eff}}(y)$: The effective (scalar) mass field of the Kohn-Sham Hamiltonian: m_Ψ plus the mean fields of the contact term and of I12. Unit: GeV. Status: DERIVED.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y=0$ at the fold ($z=\pi/2$) and $y=-L$ at the regular tip; each half of a period of x_0 is the interval $-L\le y\le 0$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- $v(y)$: The vector potential of the block Hamiltonian (the mean field along γ^{x_4}): for dirac16complex the contact term and the mean field of I11; zero for a Standard-Model fermion. Unit: GeV. Status: DERIVED.
- $v_{\mathrm{b}}(y)$: The mean field of I13 (baryon densities), acting on the lower component of χ only. Unit: GeV. Status: DERIVED.

The terms. χ_o is the Kohn-Sham orbital o (a two-component block spinor, FE-56) with the level ε_o. The block Hamiltonian has the effective mass function $M_{\mathrm{eff}}(y)$ in place of m_Ψ, the vector potential $v(y)$ along the time, and, on the lower component only ($\frac{1}{2}(1-\sigma_3) = P_b$ in the block form, since $\gamma^{x_0} = \sigma_3$), the I13 potential $v_{\mathrm{b}}(y)$. These mean fields couple dirac16complex to itself, to every fermion of the three generations and to every gauge field

(FE-64); gravity enters through f, e^{-a_4} and the warp factor. Status: DERIVED (from FE-12 with the mean fields of the record's Kohn-Sham treatment, RECORD). Solved to self-consistency (60 to 240 exact free eigenfunctions per sector, tolerance 10^{-13}, Anderson mixing, Richardson extrapolation in the basis size); an independent solver reproduces the record's reference states (energies to 10^{-13}, integrated hidden pressures to 4×10^{-10}). Residual of the stored self-consistent states $7.4\times10^{-14}\,H$ (one further Kohn-Sham map of the stored densities moves them by 1.8×10^{-14}).

$$M_{\mathrm{eff}}(y)=m_\Psi+\frac{15}{16}\,\lambda\,n^{\mathrm{sc}}(y)+\frac{\langle\mathcal{F}\rangle}{\Lambda_\star^3},\qquad v(y)=-\frac{\lambda}{16}\,n(y)+\frac{c_{\mathrm{V}}}{\Lambda_\star^6}\sum_{i=1}^{3}\sum_{\psi}q_\psi\,n_{\psi i}(y),$$
$$v_{\mathrm{b}}(y)=\frac{c_{\mathrm{A}}}{\Lambda_\star^6}\sum_{i=1}^{3}\sum_{\psi}q_\psi^B\,n_{\psi i}(y)\qquad\text{(FE-64)}$$

Symbols in (FE-64).

- $M_{\mathrm{eff}}(y)$: The effective (scalar) mass field of the Kohn-Sham Hamiltonian: m_Ψ plus the mean fields of the contact term and of I12. Unit: GeV. Status: DERIVED.
- y: The hidden coordinate of the y chart, $y=\ln|\sin z|/(6H)$, with $y=0$ at the fold ($z=\pi/2$) and $y=-L$ at the regular tip; each half of a period of x_0 is the interval $-L\le y\le0$. Unit: length. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex, $m_\Psi=3H\mathcal{M}=H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- λ: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ of the record). Unit: GeV^{-6}. Value: not fixed; Kohn-Sham states computed for $\lambda=0$ and $\pm7\times10^{-5}H^{-6}$; the fold-band radiation requires $|\lambda|\lesssim1.4\times10^{-10}H^{-6}$ (for $\Delta N_{\mathrm{eff}}\le0.001$). Status: RECORD.
- n^{sc}, n_o^{sc}: The scalar density of dirac16complex (the Kohn-Sham expectation of $\bar\Psi\Psi$, per proper 7-volume) and n_o^{sc} the contribution of the orbital o. Unit: GeV^7. Status: RECORD.
- $\mathcal{F}$: The gauge-kinetic operator of the portal I12, $\mathcal{F}=c_{\mathrm{G}}G^A_{MN}G^{AMN}+c_{\mathrm{W}}W^I_{MN}W^{IMN}+c_{\mathrm{B}}B_{MN}B^{MN}$; $\langle\mathcal{F}\rangle$ its mean value in an environment. Unit: GeV^4. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $v(y)$: The vector potential of the block Hamiltonian (the mean field along γ^{x_4}): for dirac16complex the contact term and the mean field of I11; zero for a Standard-Model fermion. Unit: GeV. Status: DERIVED.
- n, n_o, $n_{\psi i}$, n_f, n_i: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and n_o the contribution of the orbital o, $n_{\psi i}(y)$ the density of the Standard-Model fermion ψ_i (per proper 7-volume of eight dimensions); n_f the density of the fermion component f and n_i that of the neutrino mass state i (per 3-volume of four dimensions). Unit: GeV^7 (eight dimensions), GeV^3 (four dimensions). Status: STANDARD.
- c_{V}: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_{\mathrm{V}}=4.38\times10^{-17}$ GeV^{-2}. Status: INVENTED.
- i: The generation index ($i=1,2,3$), which also labels the neutrino mass states ν_1,ν_2,ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- ψ: The label of a Standard-Model multiplet: $\psi\in\{Q,u_R,d_R,\ell,e_R,\nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over ψ runs over the six. Status: STANDARD.
- q_ψ, q_f: The $B-L$ charge: q_ψ of the multiplet ψ ($q_Q=q_{u_R}=q_{d_R}=\frac13$, $q_\ell=q_{e_R}=q_{\nu_R}=-1$), q_f of the fermion component f. Unit: dimensionless. Status: STANDARD.
- $v_{\mathrm{b}}(y)$: The mean field of I13 (baryon densities), acting on the lower component of χ only. Unit: GeV. Status: DERIVED.
- c_{A}: The coefficient of the x_0-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_{\mathrm{A}}=1.278\times10^{-9}$ GeV^{-2} (dark-matter relic). Status: INVENTED.
- q_ψ^B, q_f^B: The baryon number of the multiplet ψ or of the fermion component f ($\frac13$ for quarks, 0 for leptons). Unit: dimensionless. Status: STANDARD.

The terms. M_{eff}: the bare mass m_Ψ, the local exchange mean field of the contact self-interaction proportional to the scalar density n^{sc} (coefficient $\frac{15}{16}\lambda$, from the record), and the I12 term: the condensate $\langle\mathcal{F}\rangle$ of the gauge-field invariants of the environment acts as a uniform mass shift. v: the exchange mean field of the number density n

(coefficient $-\frac{1}{16}\lambda$) and the I11 term, the $B-L$ weighted number densities $n_{\psi i}$ of the Standard-Model fermions of every generation. v_{b}: the I13 term, the baryon-number weighted densities ($q^B_\psi = \frac{1}{3}$ for quarks, 0 for leptons). Status: RECORD (self-interaction), DERIVED (portals); the portal fields are applied with the densities and condensates of each environment (FE-72).

$$
\begin{aligned}
&n = \sum_o \tfrac{1}{2}\, d_o\, \phi_o\, n_o, \qquad n^{\mathrm{sc}} = \sum_o \tfrac{1}{2}\, d_o\, \phi_o\, n^{\mathrm{sc}}_o, \qquad n_o = \frac{\mathrm{e}^{-3Hy}}{\mathrm{Vol}_7}\, \chi_o^\dagger \chi_o, \\
&n^{\mathrm{sc}}_o = \frac{\mathrm{e}^{-3Hy}}{\mathrm{Vol}_7}\, \chi_o^\dagger \varsigma\sigma_2 \chi_o, \qquad n^{\mathrm{t}}_o = \frac{\mathrm{e}^{-3Hy}}{\mathrm{Vol}_7}\, \chi_o^\dagger \varsigma\sigma_3 \chi_o, \qquad d_o = 4\, r_3\big(|\vec{n}_o|^2\big), \qquad \vec{k}_o = k_{\mathrm{lat}}\, \vec{n}_o, \\
&\mathrm{Vol}_7 = \Big(\frac{2\pi}{k_{\mathrm{lat}}}\Big)^3 v_{\mathrm{t}}\, f^4, \qquad \phi_o = \frac{1}{\mathrm{e}^{(\varepsilon_o - \mu_{\mathrm{chem}})/T} + 1} \quad \text{(Mermin; or 0 and 1 for the aufbau)}
\end{aligned}
\tag{FE-65}
$$

Symbols in (FE-65).

- $n,\ n_o,\ n_{\psi i},\ n_f,\ n_i$: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and n_o the contribution of the orbital o, $n_{\psi i}(y)$ the density of the Standard-Model fermion ψ_i (per proper 7-volume of eight dimensions); n_f the density of the fermion component f and n_i that of the neutrino mass state i (per 3-volume of four dimensions). Unit: GeV^7 (eight dimensions), GeV^3 (four dimensions). Status: STANDARD.
- o: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over o runs over the orbitals. Status: RECORD.
- $d_o,\ d_s$: A degeneracy: $d_o = 4r_3(|\vec{n}_o|^2)$ the number of states of the momentum shell of the orbital o; d_s the number of internal states (spin, colour, particle and antiparticle) of the species s. Unit: dimensionless. Status: STANDARD.
- ϕ_o: The occupation of the Kohn-Sham orbital o (Fermi-Dirac for the Mermin functional, 0 or 1 for the aufbau). Unit: dimensionless. Status: RECORD.
- $n^{\mathrm{sc}},\ n^{\mathrm{sc}}_o$: The scalar density of dirac16complex (the Kohn-Sham expectation of $\bar{\Psi}\Psi$, per proper 7-volume) and n^{sc}_o the contribution of the orbital o. Unit: GeV^7. Status: RECORD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- Vol_7: The normalisation volume of a Kohn-Sham cell, $\mathrm{Vol}_7 = (2\pi/k_{\mathrm{lat}})^3 v_{\mathrm{t}} f^4$ (the proper seven-volume of a cell is $2\,\mathrm{Vol}_7(1-\mathrm{e}^{-3HL})/(3H)$). Unit: GeV^{-6}. Value: 34.27 in units $H = 1$, $v_{\mathrm{t}} = 1$, $f = f_*$. Status: RECORD.
- $\chi,\ \chi_o,\ \chi_1,\ \chi_2$: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- ς: The first of the three signs $(\varsigma, \varsigma_2, \varsigma_3)$, each ± 1, that label the eight blocks of the reduction. Unit: dimensionless. Status: DERIVED.
- $\sigma_1,\ \sigma_2,\ \sigma_3$: The Pauli matrices acting on the two components of a block spinor χ ($\sigma_3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- n^{t}_o: The density of the orbital o along the momentum term ($\chi_o^\dagger \varsigma\sigma_3 \chi_o$ times the weight $\mathrm{e}^{-3Hy}/\mathrm{Vol}_7$); it carries the 3-space pressure. Unit: GeV^7. Status: RECORD.
- r_3: The number of representations of an integer as an ordered sum of three squares of integers (signs counted). Unit: dimensionless. Status: STANDARD.
- $\vec{n}_o$: The integer lattice vector of the momentum shell of the orbital o ($\vec{n} \in \mathbb{Z}^3$). Unit: dimensionless. Status: RECORD.
- $\vec{k},\ \vec{k}_o$: The wave vector (momentum) in ordinary 3-space; $\vec{k}_o$ that of the orbital o, a point of the lattice $k_{\mathrm{lat}}\vec{n}_o$. Unit: GeV. Status: STANDARD.

- k_{lat}: The spacing of the momentum lattice of the Kohn-Sham cell. Unit: GeV. Value: $0.25\,H$. Status: RECORD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- v_{t}: The coordinate volume of the compact extra times x_5, x_6, x_7. Unit: GeV^{-3}. Value: not fixed; $v_{\mathrm{t}} = H^{-3}$ in the quoted conversions. Status: ASSUMED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- ε, ε_o: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); ε_o the Kohn-Sham level of the orbital o. Unit: GeV. Status: DERIVED.
- μ_{chem}: The chemical potential of the Mermin occupations. Unit: GeV. Status: RECORD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.

The terms. n, n^{sc} are the number and scalar densities, sums over the occupied orbitals with the weight $\frac{1}{2}$ of each fold parity, the shell degeneracy d_o and the occupation ϕ_o; n_o, n_o^{sc}, n_o^{t} are the orbital densities (number, scalar, and the one weighted with $\varsigma\sigma_3$ that gives the momentum flux), with the factor $\mathrm{e}^{-3Hy}/\mathrm{Vol}_7$ that turns $\chi^\dagger\chi$ into a proper density. The momenta form a cubic lattice $\vec{k}_o = k_{\mathrm{lat}}\vec{n}_o$ ($k_{\mathrm{lat}} = 0.25\,H$); r_3 counts the representations of $|\vec{n}|^2$ as a sum of three squares, and the factor 4 the remaining degeneracy. Vol_7 is the normalisation volume of a cell (the coordinate volume v_{t} of the extra times times f^4 times the cube of the lattice period). The occupations are Fermi-Dirac at the temperature T and the chemical potential μ_{chem} (Mermin) or 0 and 1 (aufbau). Status: RECORD (exact definitions).

$$E_{\mathrm{KS}} = \sum_o d_o\,\phi_o\,\varepsilon_o - 2\,\mathrm{Vol}_7 \int_{-L}^{0} \mathrm{e}^{3Hy}\, e_{\mathrm{int}}\,\mathrm{d}y, \qquad e_{\mathrm{int}} = \frac{15}{32}\,\lambda\left(n^{\mathrm{sc}}\right)^2 - \frac{1}{32}\,\lambda\,n^2 \tag{FE-66}$$

Symbols in (FE-66).

- E_{KS}: The Kohn-Sham energy of a cell. Unit: GeV. Status: RECORD.
- o: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over o runs over the orbitals. Status: RECORD.
- d_o, d_s: A degeneracy: $d_o = 4r_3(|\vec{n}_o|^2)$ the number of states of the momentum shell of the orbital o; d_s the number of internal states (spin, colour, particle and antiparticle) of the species s. Unit: dimensionless. Status: STANDARD.
- ϕ_o: The occupation of the Kohn-Sham orbital o (Fermi-Dirac for the Mermin functional, 0 or 1 for the aufbau). Unit: dimensionless. Status: RECORD.
- ε, ε_o: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); ε_o the Kohn-Sham level of the orbital o. Unit: GeV. Status: DERIVED.
- Vol_7: The normalisation volume of a Kohn-Sham cell, $\mathrm{Vol}_7 = (2\pi/k_{\mathrm{lat}})^3 v_{\mathrm{t}} f^4$ (the proper seven-volume of a cell is $2\,\mathrm{Vol}_7(1 - \mathrm{e}^{-3HL})/(3H)$). Unit: GeV^{-6}. Value: 34.27 in units $H = 1$, $v_{\mathrm{t}} = 1$, $f = f_*$. Status: RECORD.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- e_{int}: The interaction energy density of the contact term in the Kohn-Sham energy and energy-momentum tensor. Unit: GeV^8. Status: RECORD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- λ: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar{\Psi}\Psi)^2$ of the record). Unit: GeV^{-6}. Value:

not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm 7 \times 10^{-5} H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4 \times 10^{-10} H^{-6}$ (for $\Delta N_{\text{eff}} \leq 0.001$). Status: RECORD.

- n^{sc}, n_o^{sc}: The scalar density of dirac16complex (the Kohn-Sham expectation of $\bar{\Psi}\Psi$, per proper 7-volume) and n_o^{sc} the contribution of the orbital o. Unit: GeV^7. Status: RECORD.
- n, n_o, $n_{\psi i}$, n_f, n_i: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and n_o the contribution of the orbital o, $n_{\psi i}(y)$ the density of the Standard-Model fermion ψ_i (per proper 7-volume of eight dimensions); n_f the density of the fermion component f and n_i that of the neutrino mass state i (per 3-volume of four dimensions). Unit: GeV^7 (eight dimensions), GeV^3 (four dimensions). Status: STANDARD.

The terms. The Kohn-Sham energy is the sum of the occupied levels minus the double-counted interaction energy e_{int}, integrated over the proper hidden length of both halves. Status: RECORD (computed for every state; Richardson errors below 2.2×10^{-9} relative).

$$\begin{aligned} \rho &= \sum_o \tfrac{1}{2}\, d_o \phi_o\, \varepsilon_o\, n_o - e_{\text{int}}, \qquad p_3 = \sum_o \tfrac{1}{2}\, d_o \phi_o\, \frac{\mathrm{e}^{-a_4 |\vec{k}_o|}}{3}\, n_o^{\text{t}} + e_{\text{int}}, \qquad p_5 = e_{\text{int}}, \\ p_0 &= \sum_o \tfrac{1}{2}\, d_o \phi_o \Big[(\varepsilon_o - v)\, n_o - M_{\text{eff}}\, n_o^{\text{sc}} - \mathrm{e}^{-a_4 |\vec{k}_o|}\, n_o^{\text{t}} \Big] + e_{\text{int}} \end{aligned} \tag{FE-67}$$

Symbols in (FE-67).

- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- o: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over o runs over the orbitals. Status: RECORD.
- d_o, d_s: A degeneracy: $d_o = 4 r_3(|\vec{n}_o|^2)$ the number of states of the momentum shell of the orbital o; d_s the number of internal states (spin, colour, particle and antiparticle) of the species s. Unit: dimensionless. Status: STANDARD.
- ϕ_o: The occupation of the Kohn-Sham orbital o (Fermi-Dirac for the Mermin functional, 0 or 1 for the aufbau). Unit: dimensionless. Status: RECORD.
- ε, ε_o: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); ε_o the Kohn-Sham level of the orbital o. Unit: GeV. Status: DERIVED.
- n, n_o, $n_{\psi i}$, n_f, n_i: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and n_o the contribution of the orbital o, $n_{\psi i}(y)$ the density of the Standard-Model fermion ψ_i (per proper 7-volume of eight dimensions); n_f the density of the fermion component f and n_i that of the neutrino mass state i (per 3-volume of four dimensions). Unit: GeV^7 (eight dimensions), GeV^3 (four dimensions). Status: STANDARD.
- e_{int}: The interaction energy density of the contact term in the Kohn-Sham energy and energy-momentum tensor. Unit: GeV^8. Status: RECORD.
- p_3: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: GeV^8. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- $\vec{k}$, $\vec{k}_o$: The wave vector (momentum) in ordinary 3-space; $\vec{k}_o$ that of the orbital o, a point of the lattice $k_{\text{lat}}\vec{n}_o$. Unit: GeV. Status: STANDARD.
- n_o^{t}: The density of the orbital o along the momentum term ($\chi_o^\dagger \varsigma \sigma_3 \chi_o$ times the weight $\mathrm{e}^{-3Hy}/\text{Vol}_7$); it carries the 3-space pressure. Unit: GeV^7. Status: RECORD.
- p_5: The pressure along the extra times, $p_5 = \mathcal{T}^{x_5}{}_{x_5} = \mathcal{T}^{x_6}{}_{x_6} = \mathcal{T}^{x_7}{}_{x_7}$. Unit: GeV^8. Status: STANDARD.
- p_0: The pressure along the hidden direction x_0, $p_0 = \mathcal{T}^{x_0}{}_{x_0}$. Unit: GeV^8. Status: STANDARD.
- $v(y)$: The vector potential of the block Hamiltonian (the mean field along γ^{x_4}): for dirac16complex the contact term and the mean field of I11; zero for a Standard-Model fermion. Unit: GeV. Status: DERIVED.

- $M_{\mathrm{eff}}(y)$: The effective (scalar) mass field of the Kohn-Sham Hamiltonian: m_Ψ plus the mean fields of the contact term and of I12. Unit: GeV. Status: DERIVED.
- n^{sc}, n_o^{sc}: The scalar density of dirac16complex (the Kohn-Sham expectation of $\bar\Psi\Psi$, per proper 7-volume) and n_o^{sc} the contribution of the orbital o. Unit: GeV^7. Status: RECORD.

The terms. The energy-momentum tensor of a Kohn-Sham state, the source of the background equations: ρ the energy density, p_3 the pressure along ordinary space (the momentum flux of the occupied momenta), p_5 the pressure along the extra times (only the interaction energy: the orbitals have no momentum along them), and p_0 the pressure along the hidden direction, which contains the kinetic energy of the hidden motion, $(\varepsilon_o - v)n_o - M_{\mathrm{eff}} n_o^{\mathrm{sc}} - \mathrm{e}^{-a_4}|\vec{k}_o| n_o^{\mathrm{t}}$. So $p_0 \neq p_5$: the Kohn-Sham source is not block-isotropic. Its block averages enter the block response FE-47 with the hidden pressure $\bar p = \frac{1}{4}(p_0 + 3p_5)$; the difference $p_0 - p_5$ (up to $0.90\,\bar p$ for the self-interacting test states and $4\,\bar p$ for dark matter at rest, whose $p_3 = 0$ and $p_0/\rho = 0.772034 = -\mathrm{d}\ln\varepsilon_{\mathrm{DM}}/\mathrm{d}\ln f$) drives the block-shape mode (OPEN; section "The background", subsection "The block pressures"). Status: DERIVED. The response of the block recomputed from the stored cell averages agrees with the stored one to 1.4×10^{-12}, and for the free fold-band gas (radiation) the response cancels to 9×10^{-13}.

$$E_{\mathrm{KS}} = 2\,\mathrm{Vol}_7 \int_{-L}^{0} \mathrm{e}^{3Hy}\rho\,\mathrm{d}y, \qquad p_0\big|_{y=0} - \mathrm{e}^{-3HL}\,p_0\big|_{y=-L} = 3H\int_{-L}^{0}\mathrm{e}^{3Hy}p_5\,\mathrm{d}y \tag{FE-68}$$

Symbols in (FE-68).

- E_{KS}: The Kohn-Sham energy of a cell. Unit: GeV. Status: RECORD.
- Vol_7: The normalisation volume of a Kohn-Sham cell, $\mathrm{Vol}_7 = (2\pi/k_{\mathrm{lat}})^3 v_{\mathrm{t}} f^4$ (the proper seven-volume of a cell is $2\,\mathrm{Vol}_7(1-\mathrm{e}^{-3HL})/(3H)$). Unit: GeV^{-6}. Value: 34.27 in units $H=1$, $v_{\mathrm{t}}=1$, $f=f_*$. Status: RECORD.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y=-L$; $HL=3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y=0$ at the fold ($z=\pi/2$) and $y=-L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- p_0: The pressure along the hidden direction x_0, $p_0 = \mathcal{T}^{x_0}{}_{x_0}$. Unit: GeV^8. Status: STANDARD.
- p_5: The pressure along the extra times, $p_5 = \mathcal{T}^{x_5}{}_{x_5} = \mathcal{T}^{x_6}{}_{x_6} = \mathcal{T}^{x_7}{}_{x_7}$. Unit: GeV^8. Status: STANDARD.

The terms. Two identities of every self-consistent state: the Kohn-Sham energy equals the integrated energy density, and the y component of the conservation law FE-11, integrated over one half of the period, relates the hidden pressure at the fold and at the tip to the extra-time pressure in between. Status: DERIVED (verified for every stored state to 4×10^{-15} and 9×10^{-8}).

$$\chi = (\chi_1,\ \chi_2)^T: \qquad \chi_2(0) = 0 \ \ \text{(even)} \quad \text{or} \quad \chi_1(0) = 0 \ \ \text{(odd)}, \qquad \chi_2(-L) = 0 \tag{FE-69}$$

Symbols in (FE-69).

- χ, χ_o, χ_1, χ_2: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1,\chi_2)^T = (a, \mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0}\chi_o^\dagger\chi_o\,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- ^{T}: The transpose: a row written with ^{T} is a column. Status: STANDARD.

- L: The depth of the regular tip: each half of a period of x_0 ends at $y=-L$; $HL=3$. Unit: length. Value: $L=3/H$ ($fL=8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.

The terms. The block form of the boundary conditions FE-28 ($\gamma^{x_0}=\sigma_3$ in every block): at the fold the lower component vanishes for the even orbitals and the upper one for the odd orbitals; at the tip the lower component vanishes. With them the block Hamiltonian is self-adjoint. Status: DERIVED.

7.2 The portals on the zero modes

A four-dimensional particle of number density n_f (per unit 3-volume) in a zero mode with the normalised profile ϱ_f has, on each half of the period, the proper eight-dimensional density of the last line of FE-70 (the proper block volume element is $f^4\mathrm{e}^{3Hy}\,\mathrm{d}y$ times the extra-time volume, and there are two halves). Inserted into the mean fields v and v_b of FE-64, and integrated with the density of an orbital of dirac16complex, the portals I11 and I13 become four-dimensional contact interactions:

$$\mathcal{L}_4=\big(G_\mathrm{V}\,q_f\,\tilde{I}_{of}+G_\mathrm{A}\,q_f^B\,\tilde{I}^{(b)}_{of}\big)\big(\mathrm{i}\bar{\Psi}_o\gamma^\mu\Psi_o\big)\big(\mathrm{i}\bar{\psi}_f\gamma_\mu\psi_f\big),$$
$$\tilde{I}_{of}=\frac{1}{H}\int_{-L}^{0}\big(a_o^2+b_o^2\big)\varrho_f\,\mathrm{e}^{-3Hy}\,\mathrm{d}y,\qquad \tilde{I}^{(b)}_{of}=\frac{1}{H}\int_{-L}^{0}b_o^2\,\varrho_f\,\mathrm{e}^{-3Hy}\,\mathrm{d}y,\qquad G_\mathrm{V}=\frac{c_\mathrm{V}H}{2f^4v_\mathrm{t}\Lambda_\star^6},\qquad G_\mathrm{A}=\frac{c_\mathrm{A}H}{2f^4v_\mathrm{t}\Lambda_\star^6},$$
$$n_{\psi i}(y)=\frac{n_f\,\varrho_f(y)\,\mathrm{e}^{-3Hy}}{2f^4v_\mathrm{t}},\qquad \varrho(y;c)=\frac{2c}{L}\,\frac{\mathrm{e}^{2cy/L}}{1-\mathrm{e}^{-2c}}$$

(FE-70)

Symbols in (FE-70).

- $\mathcal{L}_4$: The four-dimensional contact Lagrangian of I11 and I13 between a state o of dirac16complex and a Standard-Model fermion component f. Unit: GeV^4. Status: DERIVED.
- G_V: The four-dimensional strength of the vector portal I11 on the zero modes, $G_\mathrm{V}=c_\mathrm{V}H/(2f^4v_\mathrm{t}\Lambda_\star^6)$. Unit: GeV^{-2}. Value: benchmark 4.38×10^{-17} GeV^{-2} ($\Delta N_\mathrm{eff}=0.001$). Status: DERIVED.
- q_ψ, q_f: The $B-L$ charge: q_ψ of the multiplet ψ ($q_Q=q_{u_R}=q_{d_R}=\frac{1}{3}$, $q_\ell=q_{e_R}=q_{\nu_R}=-1$), q_f of the fermion component f. Unit: dimensionless. Status: STANDARD.
- f: The label of a fermion species: a Standard-Model multiplet of one generation (its N_c Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over f runs over these species. Status: STANDARD.
- $\tilde{I}_{of}$: The overlap of the state o of dirac16complex with the zero mode of the fermion component f, $\tilde{I}_{of}=H^{-1}\int_{-L}^{0}(a_o^2+b_o^2)\varrho_f\mathrm{e}^{-3Hy}\mathrm{d}y$ (the integral in units of H); $\tilde{I}_Q$, $\tilde{I}_{u_R}$, $\tilde{I}_{d_R}$, $\tilde{I}_\ell$, $\tilde{I}_{e_R}$ those of the dark matter with Q, u_R, d_R, ℓ, e_R, $\tilde{I}_{\nu_R,i}$ with the right-handed neutrino of the generation i. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q=8.2722$, $\tilde{I}_\ell=1.2524$, $\tilde{I}_{u_R}=0.3854$, $\tilde{I}_{d_R}=0.3961$, $\tilde{I}_{e_R}=0.3972$. Status: DERIVED.
- o: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over o runs over the orbitals. Status: RECORD.
- G_A: The four-dimensional strength of the portal I13 on the zero modes, $G_\mathrm{A}=c_\mathrm{A}H/(2f^4v_\mathrm{t}\Lambda_\star^6)$. Unit: GeV^{-2}. Value: 1.278×10^{-9} GeV^{-2} (dark-matter relic density). Status: DERIVED.
- q_ψ^B, q_f^B: The baryon number of the multiplet ψ or of the fermion component f ($\frac{1}{3}$ for quarks, 0 for leptons). Unit: dimensionless. Status: STANDARD.
- $\tilde{I}^{(b)}_{of}$: The overlap of the lower component of the state o with the zero mode of f, $\tilde{I}^{(b)}_{of}=H^{-1}\int_{-L}^{0}b_o^2\varrho_f\mathrm{e}^{-3Hy}\mathrm{d}y$ (the overlap of I13); $\tilde{I}^{(b)}_Q$, $\tilde{I}^{(b)}_{u_R}$, $\tilde{I}^{(b)}_{d_R}$, $\tilde{I}^{(b)}_{\nu_R}$ those of the dark matter. Unit: dimensionless. Value: dark matter: $\tilde{I}^{(b)}_Q=2.6135$. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi}=\Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M=\mathbf{e}_a{}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M=g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.

- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- ψ_i, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: ψ_i is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation i, ψ_f one fermion component f (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- a, b: The upper and lower components of a block spinor, $\chi = (a, ib)$, real at $k = 0$: a_L, b_L of a doublet and a_R, b_R of a singlet solution (a_{Ri}, b_{Ri} of the generation i, $b_{L\alpha}$, $a_{L\alpha}$ of the doublet in the flavour basis), a_o, b_o of the orbital o. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- $\varrho_f(y)$, $\varrho(y;c)$: The x_0 profile of a zero mode as a probability density on one half, $\int_{-L}^{0} \varrho \,\mathrm{d}y = 1$: ϱ_f that of the fermion component f, $\varrho(y;c)$ its form for the localisation c. Unit: 1/length (GeV). Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- c_V: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_\mathrm{V} = 4.38 \times 10^{-17}$ GeV^{-2}. Status: INVENTED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- v_t: The coordinate volume of the compact extra times x_5, x_6, x_7. Unit: GeV^{-3}. Value: not fixed; $v_\mathrm{t} = H^{-3}$ in the quoted conversions. Status: ASSUMED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- c_A: The coefficient of the x_0-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_\mathrm{A} = 1.278 \times 10^{-9}$ GeV^{-2} (dark-matter relic). Status: INVENTED.
- n, n_o, $n_{\psi i}$, n_f, n_i: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and n_o the contribution of the orbital o, $n_{\psi i}(y)$ the density of the Standard-Model fermion ψ_i (per proper 7-volume of eight dimensions); n_f the density of the fermion component f and n_i that of the neutrino mass state i (per 3-volume of four dimensions). Unit: GeV^7 (eight dimensions), GeV^3 (four dimensions). Status: STANDARD.
- ψ: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over ψ runs over the six. Status: STANDARD.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- c, c_L, c_R: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); c_L of a doublet (localised at the fold for $c_L > 0$), c_R of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets c_R: up -15.4300, down -14.6258, strange -11.5158, charm -8.7683, bottom -7.5005, top -3.3823, electron -14.5539, muon -8.9809, tau -5.9529, neutrinos -35.0452, -32.8481, -31.0702 (fixed by the measured masses). Status: DERIVED.

The terms. Ψ_o is the four-dimensional field of the orbital o (the dark matter, or a fold band) and ψ_f that of the chirality component f of a Standard-Model fermion. G_V and G_A are the four-dimensional Fermi-type couplings of I11 and I13 (the eight-dimensional coefficients divided by the block volume); q_f is the $B - L$ charge and q_f^B the baryon number. The dimensionless overlaps $\tilde{I}_{of}$ (of the whole orbital density $a_o^2 + b_o^2$) and $\tilde{I}_{of}^{(b)}$ (of its lower component only) with the zero-mode density ϱ_f measure how much the two particles share the hidden direction; $\varrho(y;c)$ is the normalised zero-mode density of the localisation c. For the dark matter $\tilde{I}_Q = 8.2722$, $\tilde{I}_\ell = 1.2524$, $\tilde{I}_{u_R} = 0.3854$, $\tilde{I}_{d_R} = 0.3961$, $\tilde{I}_{e_R} = 0.3972$, $\tilde{I}_Q^{(b)} = 2.6135$; I13 vanishes on the fold bands (to 7×10^{-14}), whose lower component is zero. Both interactions are vector-vector contact terms: they couple the dark matter coherently to the

baryon and lepton numbers of matter. Status: DERIVED. The overlaps equal the derivatives of the self-consistent Kohn-Sham levels with respect to the mean-field amplitudes (Hellmann-Feynman) to 3.8×10^{-12}.

$$\mathcal{L}_4^{\mathrm{I12}} = -\mathcal{O}_{\mathrm{DM}}\,\frac{\bar{\Psi}_{\mathrm{DM}}\Psi_{\mathrm{DM}}}{\Lambda_\star^3}\big(c_{\mathrm{G}}\,G^A_{\mu\nu}G^{A\mu\nu} + c_{\mathrm{W}}\,W^I_{\mu\nu}W^{I\mu\nu} + c_{\mathrm{B}}\,B_{\mu\nu}B^{\mu\nu}\big), \qquad \mathcal{O}_{\mathrm{DM}} = \frac{\partial\varepsilon_{\mathrm{DM}}}{\partial m_\Psi},$$
$$G_{\mathrm{g}} = \mathcal{O}_{\mathrm{DM}}\,\frac{c_{\mathrm{G}}\,g_3^2}{\Lambda_\star^3}, \qquad G_{\mathrm{W}} = \mathcal{O}_{\mathrm{DM}}\,\frac{c_{\mathrm{W}}\,g_2^2}{\Lambda_\star^3}, \qquad G_{\mathrm{B}} = \mathcal{O}_{\mathrm{DM}}\,\frac{c_{\mathrm{B}}\,g_1^2}{\Lambda_\star^3} \tag{FE-71}$$

Symbols in (FE-71).

- $\mathcal{L}_4^{\mathrm{I12}}$: The four-dimensional Lagrangian of I12 on the dark matter. Unit: GeV^4. Status: DERIVED.
- $\mathcal{O}_o$, $\mathcal{O}_{\mathrm{DM}}$: The scalar overlap of a state of dirac16complex, $\mathcal{O}_o = \partial\varepsilon_o/\partial m_\Psi$ (Hellmann-Feynman: the integrated scalar density of the orbital); $\mathcal{O}_{\mathrm{DM}}$ that of the dark matter. Unit: dimensionless. Value: $\mathcal{O}_{\mathrm{DM}} = 0.7072$. Status: DERIVED.
- Ψ_{DM}, $\bar{\Psi}_{\mathrm{DM}}$: The four-dimensional field of a dark-matter quantum (the lowest odd level of dirac16complex on the static block, in either half) and its Dirac adjoint. Unit: GeV$^{3/2}$. Status: DERIVED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- c_{G}: The coefficient of the gluon term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{G}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- G^A_M, G_M, G_{MN}, G^A_{MN}: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components G^A_M ($A = 1, \ldots, 8$), the matrix $G_M = G^A_M t^A$, the field strength G_{MN} (matrix) and its components G^A_{MN}; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in x_0). Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index running over $x_1, \ldots, x_4$ (as μ). Status: STANDARD.
- c_{W}: The coefficient of the SU(2) term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{W}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- W^I_M, W_M, W_{MN}, W^I_{MN}: The SU(2) gauge field (geometric normalisation): the components W^I_M ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength W_{MN} and its components W^I_{MN}; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c_{B}: The coefficient of the hypercharge term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{B}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- B_M, B_{MN}: The hypercharge gauge field U(1)$_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- $\varepsilon_{\mathrm{DM}}$: The energy of the dark-matter level at rest (its mass M_{DM} on the static block). Unit: GeV. Value: $3.102204\, m_\Psi = 44.35$ TeV. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- G_{g}: The coupling of a dark-matter pair to canonically normalised gluon pairs through I12. Unit: GeV^{-3}. Value: 6.09×10^{-15} GeV^{-3} (benchmark). Status: DERIVED.
- g_3: The four-dimensional strong coupling ($g_3^2 = 4\pi\alpha_3$). Unit: dimensionless. Value: $g_3^2 = 0.7786$ at $2M_{\mathrm{DM}}$. Status: DERIVED.
- G_{W}: The coupling of a dark-matter pair to canonically normalised SU(2) gauge pairs through I12. Unit: GeV^{-3}. Value: 2.96×10^{-15} GeV^{-3}. Status: DERIVED.
- g_2: The four-dimensional SU(2) coupling ($g_2^2 = 4\pi\alpha_2$). Unit: dimensionless. Value: $g_2^2 = 0.3781$ at $2M_{\mathrm{DM}}$. Status: DERIVED.
- G_{B}: The coupling of a dark-matter pair to canonically normalised hypercharge pairs through I12. Unit: GeV^{-3}. Value: 1.08×10^{-15} GeV^{-3}. Status: DERIVED.

- g_1: The four-dimensional hypercharge coupling ($g_1^2 = 4\pi\alpha_1$, Standard-Model normalisation without the GUT factor). Unit: dimensionless. Value: $g_1^2 = 0.1380$ at $2M_{\mathrm{DM}}$. Status: DERIVED.

The terms. On the dark matter I12 becomes a scalar coupling of $\bar{\Psi}_{\mathrm{DM}}\Psi_{\mathrm{DM}}$ to the squared field strengths of the gluons, of W and of B; the zero modes of the gauge fields are flat, so the overlap is the response $\mathcal{O}_{\mathrm{DM}} = 0.7072$ of the dark-matter level to a uniform change of the mass. G_{g}, G_{W}, G_{B} are the couplings to the canonically normalised fields (field divided by its four-dimensional coupling), used in the annihilation cross-section FE-127; at the unitarity-safe benchmark $c/\Lambda_\star^3 = 1.106 \times 10^{-14}$ GeV^{-3} they are 6.09, 2.96 and 1.08×10^{-15} GeV^{-3}. Status: DERIVED (the response checked to 10^{-8}).

For an orbital normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \,\mathrm{d}y = 1$, first-order perturbation theory (the Hellmann-Feynman theorem) gives the shift of its level by small mean fields as their expectation value; with $\chi_o = (a_o, \mathrm{i}b_o)$, $\chi_o^\dagger\chi_o = a_o^2 + b_o^2$, $\chi_o^\dagger \frac{1}{2}(1-\sigma_3)\chi_o = b_o^2$ and $\chi_o^\dagger \varsigma\sigma_2\chi_o = 2\varsigma a_o b_o$:

$$\Delta\varepsilon_o = \int_{-L}^{0} \chi_o^\dagger \Big[v + \frac{1-\sigma_3}{2}\, v_{\mathrm{b}} + \varsigma\sigma_2 \,\frac{\langle\mathcal{F}\rangle}{\Lambda_\star^3}\Big]\chi_o \,\mathrm{d}y, \qquad \mathcal{O}_o = \frac{\partial\varepsilon_o}{\partial m_\Psi} = \int_{-L}^{0} 2\varsigma\, a_o b_o \,\mathrm{d}y.$$

Symbols in this equation.

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- ε, ε_o: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); ε_o the Kohn-Sham level of the orbital o. Unit: GeV. Status: DERIVED.
- o: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over o runs over the orbitals. Status: RECORD.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- χ, χ_o, χ_1, χ_2: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0} \chi_o^\dagger\chi_o \,\mathrm{d}y = 1$. Unit: GeV$^{1/2}$ (normalised orbital). Status: DERIVED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $v(y)$: The vector potential of the block Hamiltonian (the mean field along γ^{x_4}): for dirac16complex the contact term and the mean field of I11; zero for a Standard-Model fermion. Unit: GeV. Status: DERIVED.
- σ_1, σ_2, σ_3: The Pauli matrices acting on the two components of a block spinor χ ($\sigma_3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- $v_{\mathrm{b}}(y)$: The mean field of I13 (baryon densities), acting on the lower component of χ only. Unit: GeV. Status: DERIVED.
- ς: The first of the three signs $(\varsigma, \varsigma_2, \varsigma_3)$, each ± 1, that label the eight blocks of the reduction. Unit: dimensionless. Status: DERIVED.
- $\mathcal{F}$: The gauge-kinetic operator of the portal I12, $\mathcal{F} = c_{\mathrm{G}} G^A_{MN} G^{AMN} + c_{\mathrm{W}} W^I_{MN} W^{IMN} + c_{\mathrm{B}} B_{MN} B^{MN}$; $\langle\mathcal{F}\rangle$ its mean value in an environment. Unit: GeV^4. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- $\mathcal{O}_o$, $\mathcal{O}_{\mathrm{DM}}$: The scalar overlap of a state of dirac16complex, $\mathcal{O}_o = \partial\varepsilon_o/\partial m_\Psi$ (Hellmann-Feynman: the integrated scalar density of the orbital); $\mathcal{O}_{\mathrm{DM}}$ that of the dark matter. Unit: dimensionless. Value: $\mathcal{O}_{\mathrm{DM}} = 0.7072$. Status: DERIVED.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.

- a, b: The upper and lower components of a block spinor, $\chi = (a, ib)$, real at $k = 0$: a_L, b_L of a doublet and a_R, b_R of a singlet solution (a_{Ri}, b_{Ri} of the generation i, $b_{L\alpha}$, $a_{L\alpha}$ of the doublet in the flavour basis), a_o, b_o of the orbital o. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.

With the portal parts of v and v_b (FE-64) and the densities of FE-70 this is:

$$\Delta\varepsilon_o = \sum_f \left(G_\mathrm{V}\, q_f\, \tilde{I}_{of} + G_\mathrm{A}\, q_f^B\, \tilde{I}_{of}^{(b)}\right) n_f + \mathcal{O}_o\, \frac{c_{\mathrm{I}12}}{\Lambda_\star^3}\left[\left\langle G^A_{\mu\nu} G^{A\mu\nu}\right\rangle + \left\langle W^I_{\mu\nu} W^{I\mu\nu}\right\rangle + \left\langle B_{\mu\nu} B^{\mu\nu}\right\rangle\right], \qquad \mathcal{O}_o = \frac{\partial\varepsilon_o}{\partial m_\Psi} \text{ (FE-72)}$$

Symbols in (FE-72).

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_eff an addition to N_eff, ΔS_m the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- ε, ε_o: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); ε_o the Kohn-Sham level of the orbital o. Unit: GeV. Status: DERIVED.
- o: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over o runs over the orbitals. Status: RECORD.
- f: The label of a fermion species: a Standard-Model multiplet of one generation (its N_c Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over f runs over these species. Status: STANDARD.
- G_V: The four-dimensional strength of the vector portal I11 on the zero modes, $G_\mathrm{V} = c_\mathrm{V} H/(2 f^4 v_\mathrm{t} \Lambda_\star^6)$. Unit: GeV^{-2}. Value: benchmark 4.38×10^{-17} GeV^{-2} ($\Delta N_\mathrm{eff} = 0.001$). Status: DERIVED.
- q_ψ, q_f: The $B - L$ charge: q_ψ of the multiplet ψ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), q_f of the fermion component f. Unit: dimensionless. Status: STANDARD.
- $\tilde{I}_{of}$: The overlap of the state o of dirac16complex with the zero mode of the fermion component f, $\tilde{I}_{of} = H^{-1}\int_{-L}^0 (a_o^2 + b_o^2)\varrho_f \mathrm{e}^{-3Hy}\mathrm{d}y$ (the integral in units of H); $\tilde{I}_Q$, $\tilde{I}_{u_R}$, $\tilde{I}_{d_R}$, $\tilde{I}_\ell$, $\tilde{I}_{e_R}$ those of the dark matter with Q, u_R, d_R, ℓ, e_R, $\tilde{I}_{\nu_R,i}$ with the right-handed neutrino of the generation i. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q = 8.2722$, $\tilde{I}_\ell = 1.2524$, $\tilde{I}_{u_R} = 0.3854$, $\tilde{I}_{d_R} = 0.3961$, $\tilde{I}_{e_R} = 0.3972$. Status: DERIVED.
- G_A: The four-dimensional strength of the portal I13 on the zero modes, $G_\mathrm{A} = c_\mathrm{A} H/(2 f^4 v_\mathrm{t} \Lambda_\star^6)$. Unit: GeV^{-2}. Value: 1.278×10^{-9} GeV^{-2} (dark-matter relic density). Status: DERIVED.
- q_ψ^B, q_f^B: The baryon number of the multiplet ψ or of the fermion component f ($\frac{1}{3}$ for quarks, 0 for leptons). Unit: dimensionless. Status: STANDARD.
- $\tilde{I}_{of}^{(b)}$: The overlap of the lower component of the state o with the zero mode of f, $\tilde{I}_{of}^{(b)} = H^{-1}\int_{-L}^0 b_o^2 \varrho_f \mathrm{e}^{-3Hy}\mathrm{d}y$ (the overlap of I13); $\tilde{I}_Q^{(b)}$, $\tilde{I}_{u_R}^{(b)}$, $\tilde{I}_{d_R}^{(b)}$, $\tilde{I}_{\nu_R}^{(b)}$ those of the dark matter. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q^{(b)} = 2.6135$. Status: DERIVED.
- n, n_o, $n_{\psi i}$, n_f, n_i: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and n_o the contribution of the orbital o, $n_{\psi i}(y)$ the density of the Standard-Model fermion ψ_i (per proper 7-volume of eight dimensions); n_f the density of the fermion component f and n_i that of the neutrino mass state i (per 3-volume of four dimensions). Unit: GeV^7 (eight dimensions), GeV^3 (four dimensions). Status: STANDARD.
- $\mathcal{O}_o$, $\mathcal{O}_\mathrm{DM}$: The scalar overlap of a state of dirac16complex, $\mathcal{O}_o = \partial\varepsilon_o/\partial m_\Psi$ (Hellmann-Feynman: the integrated scalar density of the orbital); $\mathcal{O}_\mathrm{DM}$ that of the dark matter. Unit: dimensionless. Value: $\mathcal{O}_\mathrm{DM} = 0.7072$. Status: DERIVED.
- $c_{\mathrm{I}12}$: The common coefficient of the three terms of I12 when they are universal ($c_\mathrm{G} = c_\mathrm{W} = c_\mathrm{B} = c_{\mathrm{I}12}$). Unit: dimensionless. Value: $c_{\mathrm{I}12}/\Lambda_\star^3 = 1.106 \times 10^{-14}$ GeV^{-3} (unitarity-safe benchmark). Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- G^A_M, G_M, G_{MN}, G^A_{MN}: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components G^A_M ($A = 1, \ldots, 8$), the matrix $G_M = G^A_M t^A$, the field strength G_{MN} (matrix) and its components G^A_{MN}; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in x_0). Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+, +, +, -$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.

- ν: A four-dimensional coordinate index running over $x_1, \ldots, x_4$ (as μ). Status: STANDARD.
- W^I_M, W_M, W_{MN}, W^I_{MN}: The SU(2) gauge field (geometric normalisation): the components W^I_M ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength W_{MN} and its components W^I_{MN}; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- B_M, B_{MN}: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.

The terms. The first sum is the shift by the I11 and I13 mean fields of the fermion densities n_f of the environment (each with its charge and overlap); the second term is the I12 shift by the condensates of the three gauge groups, with the response $\mathcal{O}_o$ and the universal I12 coefficient c_{I12}. The level of dirac16complex thus depends on every Standard-Model field present. Status: DERIVED (first order; in the interacting Kohn-Sham state the response is screened by only 3.5×10^{-4}). The largest shift over the ten environments of the coupled solution is 1.5×10^{-6} of the dark-matter mass, at the dark-matter freeze-out, from the condensates through I12.

$$\begin{aligned} \Delta\varepsilon_{\mathrm{I11}} &= G_{\mathrm{V}}\Big[\frac{1}{3}\,n_{\mathrm{u}}\,\frac{\tilde{I}_Q + \tilde{I}_{u_R}}{2} + \frac{1}{3}\,n_{\mathrm{d}}\,\frac{\tilde{I}_Q + \tilde{I}_{d_R}}{2} - n_{\mathrm{e}}\,\frac{\tilde{I}_\ell + \tilde{I}_{e_R}}{2}\Big], \\ \Delta\varepsilon_{\mathrm{I13}} &= G_{\mathrm{A}}\Big[\frac{1}{3}\,n_{\mathrm{u}}\,\frac{\tilde{I}^{(b)}_Q + \tilde{I}^{(b)}_{u_R}}{2} + \frac{1}{3}\,n_{\mathrm{d}}\,\frac{\tilde{I}^{(b)}_Q + \tilde{I}^{(b)}_{d_R}}{2}\Big], \qquad n_{\mathrm{u}} = 2n_{\mathrm{p}} + n_{\mathrm{n}}, \qquad n_{\mathrm{d}} = n_{\mathrm{p}} + 2n_{\mathrm{n}} \end{aligned} \tag{FE-73}$$

Symbols in (FE-73).

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21} = m^2_2 - m^2_1$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- $\Delta\varepsilon_{\mathrm{I11}}$, $\Delta\varepsilon_{\mathrm{I13}}$: The shifts of the dark-matter level by the mean fields of I11 and of I13 in matter. Unit: GeV. Value: at most 8.2×10^{-10} GeV (I13, at the freeze-out). Status: DERIVED.
- G_{V}: The four-dimensional strength of the vector portal I11 on the zero modes, $G_{\mathrm{V}} = c_{\mathrm{V}} H/(2f^4 v_{\mathrm{t}} \Lambda^6_\star)$. Unit: GeV^{-2}. Value: benchmark 4.38×10^{-17} GeV^{-2} ($\Delta N_{\mathrm{eff}} = 0.001$). Status: DERIVED.
- n_{u}, n_{d}: The number densities of up and down quarks bound in nucleons, $n_{\mathrm{u}} = 2n_{\mathrm{p}} + n_{\mathrm{n}}$, $n_{\mathrm{d}} = n_{\mathrm{p}} + 2n_{\mathrm{n}}$. Unit: GeV^3. Status: STANDARD.
- $\tilde{I}_{of}$: The overlap of the state o of dirac16complex with the zero mode of the fermion component f, $\tilde{I}_{of} = H^{-1}\int_{-L}^{0}(a_o^2 + b_o^2)\varrho_f \mathrm{e}^{-3Hy}\mathrm{d}y$ (the integral in units of H); $\tilde{I}_Q$, $\tilde{I}_{u_R}$, $\tilde{I}_{d_R}$, $\tilde{I}_\ell$, $\tilde{I}_{e_R}$ those of the dark matter with Q, u_R, d_R, ℓ, e_R, $\tilde{I}_{\nu_R,i}$ with the right-handed neutrino of the generation i. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q = 8.2722$, $\tilde{I}_\ell = 1.2524$, $\tilde{I}_{u_R} = 0.3854$, $\tilde{I}_{d_R} = 0.3961$, $\tilde{I}_{e_R} = 0.3972$. Status: DERIVED.
- n_{e}: The number density of electrons (net of positrons); in the Thomson terms of the photon and baryon equations only free (ionised) electrons count. Unit: GeV^3. Status: STANDARD.
- G_{A}: The four-dimensional strength of the portal I13 on the zero modes, $G_{\mathrm{A}} = c_{\mathrm{A}} H/(2f^4 v_{\mathrm{t}} \Lambda^6_\star)$. Unit: GeV^{-2}. Value: 1.278×10^{-9} GeV^{-2} (dark-matter relic density). Status: DERIVED.
- $\tilde{I}^{(b)}_{of}$: The overlap of the lower component of the state o with the zero mode of f, $\tilde{I}^{(b)}_{of} = H^{-1}\int_{-L}^{0} b_o^2 \varrho_f \mathrm{e}^{-3Hy}\mathrm{d}y$ (the overlap of I13); $\tilde{I}^{(b)}_Q$, $\tilde{I}^{(b)}_{u_R}$, $\tilde{I}^{(b)}_{d_R}$, $\tilde{I}^{(b)}_{\nu_R}$ those of the dark matter. Unit: dimensionless. Value: dark matter: $\tilde{I}^{(b)}_Q = 2.6135$. Status: DERIVED.
- n_{p}: The number density of protons (free and bound). Unit: GeV^3. Status: STANDARD.
- n_{n}: The number density of neutrons (free and bound). Unit: GeV^3. Status: STANDARD.

The terms. In neutral matter of protons, neutrons and electrons the quark densities are $n_{\mathrm{u}} = 2n_{\mathrm{p}} + n_{\mathrm{n}}$ and $n_{\mathrm{d}} = n_{\mathrm{p}} + 2n_{\mathrm{n}}$; each chirality carries half of the density of an unpolarised fermion, hence the averages of the doublet and singlet overlaps. I11 acts with $B - L$ ($\frac{1}{3}$ per quark, -1 per electron), I13 with the baryon number on the lower component. Status: DERIVED (evaluated in ten environments; at most 8.2×10^{-10} GeV, from I13 at the freeze-out).

7.3 The condensates of the Standard-Model plasma

The I12 shift needs the expectation values of the squared field strengths in each environment. They follow from the trace of the energy-momentum tensor and from the dependence of the pressure on the couplings.

$$\big\langle G^A_{\mu\nu}G^{A\mu\nu}\big\rangle(T) = \big\langle G^A_{\mu\nu}G^{A\mu\nu}\big\rangle(0) - \frac{32\pi^2}{b_3}\,\Theta_{\rm QCD}, \qquad \big\langle G^A_{\mu\nu}G^{A\mu\nu}\big\rangle(0) = 4\pi^2\times 0.012\ \text{GeV}^4, \qquad b_3 = 11 - \frac{2}{3}\,N_{\rm f} \tag{FE-74}$$

Symbols in (FE-74).

- G^A_M, G_M, G_{MN}, G^A_{MN}: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components G^A_M ($A = 1,\dots,8$), the matrix $G_M = G^A_M t^A$, the field strength G_{MN} (matrix) and its components G^A_{MN}; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in x_0). Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- A: A colour (adjoint SU(3)) index running over $1,\dots,8$; a repeated index is summed. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index running over $x_1,\dots,x_4$ (as μ). Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- b_3, b_2, b_1, $b_{\rm G}$: The one-loop coefficients of the running couplings, $\mathrm{d}\alpha/\mathrm{d}\ln\mu_{\rm ren} = -b\,\alpha^2/(2\pi)$: $b_3 = 11 - \frac{2}{3}N_{\rm f}$ (SU(3)), $b_2 = \frac{10}{3}$ (SU(2) without a Higgs particle), $b_1 = -\frac{20}{3}$ ($\mathrm{U}(1)_Y$), $b_{\rm G}$ that of the group G (not related to the metric function b_4). Unit: dimensionless. Status: STANDARD.
- $\Theta_{\rm QCD}$, $\Theta_{\rm G}$, Θ_V: The trace $\rho - 3p$ of the energy-momentum tensor (interaction measure) of a plasma component: $\Theta_{\rm QCD}$ of the quark-gluon plasma or hadron gas, $\Theta_{\rm G}$ of the gauge plasma of the group G, Θ_V of the thermal gas of the massive vector V ($\Theta_{\rm W}$ of $W^\pm$, $\Theta_{\rm Z}$ of Z). Unit: GeV^4. Status: STANDARD.
- $N_{\rm f}$: The number of active quark flavours (quarks lighter than the scale). Unit: dimensionless. Status: STANDARD.

The terms. The gluon condensate at the temperature T is the vacuum value minus the thermal part, which the trace anomaly of QCD relates to the interaction measure $\Theta_{\rm QCD} = \rho - 3p$ of the quark-gluon plasma; b_3 is the first coefficient of the QCD beta function with $N_{\rm f}$ active flavours. The field strength is in the geometric normalisation of the action (the canonical field times g_3), so the vacuum value is $4\pi^2$ times the standard condensate $0.012\ \text{GeV}^4$. Status: STANDARD (the relation does not depend on the signature convention).

$$\begin{aligned}
&\frac{\Theta_{\rm QCD}}{T^4} = 4\exp\Big[-\frac{\ln^2(T/0.2\,\text{GeV})}{0.5}\Big]\ \ (T<1\,\text{GeV}), \qquad \frac{\Theta_{\rm QCD}}{T^4} = \frac{b_3}{3}\Big(1+\frac{5N_{\rm f}}{12}\Big)\alpha_3^2\ \ (T>1\,\text{GeV}),\\
&p_{\rm QCD} = \frac{8\pi^2}{45}T^4\Big[1+\frac{21N_{\rm f}}{32}-\frac{15}{4}\Big(1+\frac{5N_{\rm f}}{12}\Big)\frac{\alpha_3}{\pi}\Big], \qquad \frac{\mathrm{d}\alpha_3}{\mathrm{d}\ln T} = -\frac{b_3}{2\pi}\,\alpha_3^2, \qquad \Theta_{\rm QCD} = T^5\,\frac{\mathrm{d}}{\mathrm{d}T}\Big(\frac{p_{\rm QCD}}{T^4}\Big)
\end{aligned} \tag{FE-75}$$

Symbols in (FE-75).

- $\Theta_{\rm QCD}$, $\Theta_{\rm G}$, Θ_V: The trace $\rho - 3p$ of the energy-momentum tensor (interaction measure) of a plasma component: $\Theta_{\rm QCD}$ of the quark-gluon plasma or hadron gas, $\Theta_{\rm G}$ of the gauge plasma of the group G, Θ_V of the thermal gas of the massive vector V ($\Theta_{\rm W}$ of $W^\pm$, $\Theta_{\rm Z}$ of Z). Unit: GeV^4. Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- b_3, b_2, b_1, $b_{\rm G}$: The one-loop coefficients of the running couplings, $\mathrm{d}\alpha/\mathrm{d}\ln\mu_{\rm ren} = -b\,\alpha^2/(2\pi)$: $b_3 = 11 - \frac{2}{3}N_{\rm f}$ (SU(3)), $b_2 = \frac{10}{3}$ (SU(2) without a Higgs particle), $b_1 = -\frac{20}{3}$ ($\mathrm{U}(1)_Y$), $b_{\rm G}$ that of the group G (not related to the metric function b_4). Unit: dimensionless. Status: STANDARD.
- $N_{\rm f}$: The number of active quark flavours (quarks lighter than the scale). Unit: dimensionless. Status: STANDARD.
- α_3: The strong fine-structure constant, $\alpha_3 = g_3^2/(4\pi)$, at the scale given in brackets. Unit: dimensionless. Value: $\alpha_3(m_{\rm Z}) = 0.1180$; $\alpha_3(m_{\rm KK}) = 0.0606$. Status: STANDARD.

- p_{QCD}: The pressure of the quark-gluon plasma (leading orders). Unit: GeV^4. Status: ESTIMATE.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.

The terms. The interaction measure of QCD: below 1 GeV a smooth lattice-like shape peaked near 200 MeV (ESTIMATE), above 1 GeV the leading-order perturbative result; p_{QCD} is the pressure of the quark-gluon plasma to first order in α_3, the running of α_3 is one loop (at the scale $2\pi T$), and $\Theta_{\mathrm{QCD}} = T^5\,\mathrm{d}(p_{\mathrm{QCD}}/T^4)/\mathrm{d}T$ is the definition of the interaction measure through the pressure. Status: ESTIMATE (below 1 GeV), STANDARD (above).

$$\begin{aligned}
&\left\langle F^V_{\mu\nu} F^{V\,\mu\nu} \right\rangle(T) = -2\,\Theta_V \quad (\text{for } V = W^\pm, Z), \qquad \left\langle F^2_{\mathrm{geo}} \right\rangle = -4\,\frac{\partial p}{\partial\left(1/g^2_{\mathrm{G}}\right)} = 4 g^2_{\mathrm{G}}\, p_{\alpha,\mathrm{G}}, \\
&p_{\alpha,\mathrm{G}} = -\frac{g^2_{\mathrm{G}} T^4}{144}\, d_{\mathrm{adj}} \Big[C_{\mathrm{adj}} + \frac{5}{2}\, T_{\mathrm{F}} N_{\mathrm{f}} \Big]
\end{aligned} \tag{FE-76}$$

Symbols in (FE-76).

- $F^V_{\mu\nu}$: The field strength of the canonically normalised massive vector $V = W^\pm$ or Z, $F^V_{\mu\nu} = \partial_\mu V_\nu - \partial_\nu V_\mu$. Unit: GeV^2. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index running over $x_1, \ldots, x_4$ (as μ). Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- Θ_{QCD}, Θ_{G}, Θ_V: The trace $\rho - 3p$ of the energy-momentum tensor (interaction measure) of a plasma component: Θ_{QCD} of the quark-gluon plasma or hadron gas, Θ_{G} of the gauge plasma of the group G, Θ_V of the thermal gas of the massive vector V (Θ_{W} of $W^\pm$, Θ_{Z} of Z). Unit: GeV^4. Status: STANDARD.
- F_{geo}: The field strength of the gauge group G in geometric normalisation; $\langle F^2_{\mathrm{geo}} \rangle$ is its condensate ($G^A_{\mu\nu} G^{A\mu\nu}$, $W^I_{\mu\nu} W^{I\mu\nu}$ or $B_{\mu\nu} B^{\mu\nu}$). Unit: GeV^2. Status: STANDARD.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- p: A pressure: $p_M = \mathcal{T}^M{}_M$ (no sum) for the direction $M \neq x_4$, p_s the pressure of the species s; Δp the pressure perturbation in the cosmological perturbation equations; in the Feynman-Hellmann relation the pressure of a gauge plasma. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- g_{G}: The four-dimensional coupling of the gauge group G (one of g_3, g_2, g_1). Unit: dimensionless. Status: STANDARD.
- $p_{\alpha,\mathrm{G}}$, $p_{\alpha,2}$, $p_{\alpha,1}$: The order-α (interaction) part of the pressure of the gauge plasma of the group G; $p_{\alpha,2}$ of SU(2), $p_{\alpha,1}$ of $\mathrm{U}(1)_Y$. Unit: GeV^4. Status: STANDARD.
- d_{adj}: The dimension of the adjoint representation of the gauge group (8, 3, 1). Unit: dimensionless. Status: STANDARD.
- C_{adj}: The quadratic Casimir of the adjoint representation (3, 2, 0). Unit: dimensionless. Status: STANDARD.
- T_{F}: The Dynkin index of the fermion representation ($\frac{1}{2}$ per Dirac fundamental; for $\mathrm{U}(1)_Y$ the sum of the squared hypercharges). Unit: dimensionless. Status: STANDARD.
- N_{f}: The number of active quark flavours (quarks lighter than the scale). Unit: dimensionless. Status: STANDARD.

The terms. For a massive vector boson V ($W^\pm$, Z) the thermal condensate of its canonical field strength is -2 times the trace $\Theta_V = \rho_V - 3p_V$ of its ideal gas: on shell the Proca Lagrangian is a total derivative, so the kinetic and mass terms have opposite expectation values. For a gauge plasma the Feynman-Hellmann theorem gives the condensate of the geometric field strength as the derivative of the pressure with respect to $1/g^2_{\mathrm{G}}$, i.e. $4g^2_{\mathrm{G}}$ times the order-α_{G} pressure $p_{\alpha,\mathrm{G}}$ of a plasma with the adjoint dimension d_{adj}, the Casimir C_{adj} and N_{f} fermions of index T_{F}. Status: STANDARD (for QCD it reproduces the trace-anomaly condensate FE-74 to 10^{-16}); no scalar term, since there is no Higgs particle (ESTIMATE for the electroweak plasma).

$$\frac{1}{\alpha_2(\mu_{\rm ren})}=\frac{\sin^2\theta_{\overline{\rm MS}}}{\alpha_{\rm em}}+\frac{b_2}{2\pi}\ln\frac{\mu_{\rm ren}}{m_{\rm Z}},\qquad \frac{1}{\alpha_1(\mu_{\rm ren})}=\frac{1-\sin^2\theta_{\overline{\rm MS}}}{\alpha_{\rm em}}+\frac{b_1}{2\pi}\ln\frac{\mu_{\rm ren}}{m_{\rm Z}},\qquad b_2=\frac{10}{3},\quad b_1=-\frac{20}{3}\ \text{(FE-77)}$$

Symbols in (FE-77).

- α_2: The SU(2) fine-structure constant, $\alpha_2=g_2^2/(4\pi)$. Unit: dimensionless. Status: STANDARD.
- $\mu_{\rm ren}$: The renormalisation scale of a running coupling (taken as $2\pi T$ for the condensates of a plasma). Unit: GeV. Status: STANDARD.
- $\theta_{\overline{\rm MS}}$: The weak mixing angle in the $\overline{\rm MS}$ scheme at $m_{\rm Z}$. Unit: dimensionless. Value: $\sin^2\theta_{\overline{\rm MS}}=0.23122$. Status: STANDARD.
- $\alpha_{\rm em}$: The electromagnetic fine-structure constant ($\overline{\rm MS}$, at $m_{\rm Z}$). Unit: dimensionless. Value: 1/127.95. Status: STANDARD.
- $b_3,\ b_2,\ b_1,\ b_{\rm G}$: The one-loop coefficients of the running couplings, $\mathrm{d}\alpha/\mathrm{d}\ln\mu_{\rm ren}=-b\,\alpha^2/(2\pi)$: $b_3=11-\frac{2}{3}N_{\rm f}$ (SU(3)), $b_2=\frac{10}{3}$ (SU(2) without a Higgs particle), $b_1=-\frac{20}{3}$ ($\mathrm{U}(1)_Y$), $b_{\rm G}$ that of the group G (not related to the metric function b_4). Unit: dimensionless. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $m_{\rm Z}$: The mass of the Z. Unit: GeV. Value: 91.1879 GeV (reproduced). Status: STANDARD.
- α_1: The hypercharge fine-structure constant, $\alpha_1=g_1^2/(4\pi)$. Unit: dimensionless. Status: STANDARD.

The terms. The one-loop running of the SU(2) and hypercharge couplings from their $\overline{\rm MS}$ values at $m_{\rm Z}$, with the coefficients of three generations of fermions and the gauge bosons and without the contribution of a Higgs doublet ($b_2=\frac{10}{3}$, $b_1=-\frac{20}{3}$). Status: STANDARD.

$$\langle W^I_{\mu\nu}W^{I\mu\nu}\rangle=g_2^2\Big[2\big(-2\Theta_{\rm W}\big)+\cos^2\theta_{\rm W}\big(-2\Theta_{\rm Z}\big)\Big]+4g_2^2\,p_{\alpha,2},\qquad \langle B_{\mu\nu}B^{\mu\nu}\rangle=g_1^2\sin^2\theta_{\rm W}\big(-2\Theta_{\rm Z}\big)+4g_1^2\,p_{\alpha,1}\ \text{(FE-78)}$$

Symbols in (FE-78).

- $W^I_M,\ W_M,\ W_{MN},\ W^I_{MN}$: The SU(2) gauge field (geometric normalisation): the components W^I_M ($I=1,2,3$; W^3 the third component), the matrix $W_M=W^I_M\tau^I/2$, the field strength W_{MN} and its components W^I_{MN}; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- I: A weak-isospin (adjoint SU(2)) index running over $1,2,3$; a repeated index is summed. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1,x_2,x_3,x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index running over $x_1,\ldots,x_4$ (as μ). Status: STANDARD.
- g_2: The four-dimensional SU(2) coupling ($g_2^2=4\pi\alpha_2$). Unit: dimensionless. Value: $g_2^2=0.3781$ at $2M_{\rm DM}$. Status: DERIVED.
- $\Theta_{\rm QCD},\ \Theta_{\rm G},\ \Theta_V$: The trace $\rho-3p$ of the energy-momentum tensor (interaction measure) of a plasma component: $\Theta_{\rm QCD}$ of the quark-gluon plasma or hadron gas, $\Theta_{\rm G}$ of the gauge plasma of the group G, Θ_V of the thermal gas of the massive vector V ($\Theta_{\rm W}$ of $W^\pm$, $\Theta_{\rm Z}$ of Z). Unit: GeV^4. Status: STANDARD.
- $\theta_{\rm W}$: The weak mixing angle of the couplings, $\cos\theta_{\rm W}=\hat g_2/\hat g_Z$, $\sin\theta_{\rm W}=\hat g_1/\hat g_Z$; fixed by the measured W and Z masses with $\rho_{\rm EW,0}$. Unit: dimensionless. Value: $\cos^2\theta_{\rm W}=0.776368$. Status: DERIVED.
- $p_{\alpha,\rm G},\ p_{\alpha,2},\ p_{\alpha,1}$: The order-$\alpha$ (interaction) part of the pressure of the gauge plasma of the group G; $p_{\alpha,2}$ of SU(2), $p_{\alpha,1}$ of $\mathrm{U}(1)_Y$. Unit: GeV^4. Status: STANDARD.
- $B_M,\ B_{MN}$: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- g_1: The four-dimensional hypercharge coupling ($g_1^2=4\pi\alpha_1$, Standard-Model normalisation without the GUT factor). Unit: dimensionless. Value: $g_1^2=0.1380$ at $2M_{\rm DM}$. Status: DERIVED.

The terms. The electroweak condensates of a hot plasma: the free parts of $W^\pm$ (two charge states) and of the Z (with the projections $\cos^2\theta_{\rm W}$ onto W^3 and $\sin^2\theta_{\rm W}$ onto B; the photon is massless and contributes no free part) and the interaction parts $4g^2p_\alpha$ of FE-76. Status: ESTIMATE (below $m_{\rm W}$ the interaction part is multiplied by the thermal suppression of a massive vector); at the freeze-out the W condensate is $-5.0\times10^{11}\ \mathrm{GeV}^4$, almost all from the interaction part.

7.4 dirac16complex acting on the Standard Model

In four dimensions the I12 term of FE-21 shifts the inverse squared couplings by the I12 coefficient times the scalar density of the dark matter, whose block integral is $\mathcal{O}_{\mathrm{DM}}$ times its number density (both species and antiparticles):

$$\frac{1}{g_3^2} \;\to\; \frac{1}{g_3^2} + \frac{4c_{\mathrm{G}}\,\mathcal{O}_{\mathrm{DM}}\,n_{\mathrm{DM,tot}}}{\Lambda_\star^3} \qquad \Rightarrow \qquad \frac{\Delta\alpha_3}{\alpha_3} = -g_3^2\,\Delta\Big(\frac{1}{g_3^2}\Big) = -4G_{\mathrm{g}}\,n_{\mathrm{DM,tot}},$$

Symbols in this equation.

- g_3: The four-dimensional strong coupling ($g_3^2 = 4\pi\alpha_3$). Unit: dimensionless. Value: $g_3^2 = 0.7786$ at $2M_{\mathrm{DM}}$. Status: DERIVED.
- c_{G}: The coefficient of the gluon term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{G}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- $\mathcal{O}_o$, $\mathcal{O}_{\mathrm{DM}}$: The scalar overlap of a state of dirac16complex, $\mathcal{O}_o = \partial\varepsilon_o/\partial m_\Psi$ (Hellmann-Feynman: the integrated scalar density of the orbital); $\mathcal{O}_{\mathrm{DM}}$ that of the dark matter. Unit: dimensionless. Value: $\mathcal{O}_{\mathrm{DM}} = 0.7072$. Status: DERIVED.
- $n_{\mathrm{DM,tot}}$: The total number density of dark-matter quanta: both species (the lowest odd levels of Ψ_A and Ψ_B), particles and antiparticles, $n_{\mathrm{DM,tot}} = 2(n_{\mathrm{DM}} + \bar{n}_{\mathrm{DM}})$. Unit: GeV^3. Status: STANDARD.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- α_3: The strong fine-structure constant, $\alpha_3 = g_3^2/(4\pi)$, at the scale given in brackets. Unit: dimensionless. Value: $\alpha_3(m_{\mathrm{Z}}) = 0.1180$; $\alpha_3(m_{\mathrm{KK}}) = 0.0606$. Status: STANDARD.
- G_{g}: The coupling of a dark-matter pair to canonically normalised gluon pairs through I12. Unit: GeV^{-3}. Value: 6.09×10^{-15} GeV^{-3} (benchmark). Status: DERIVED.

and likewise for α_2 and α_1. Within the block the same term multiplies the kinetic term of the gauge fields by a function of y, which changes their x_0 problem:

$$\begin{aligned}
&\frac{\Delta\alpha_3}{\alpha_3} = -4G_{\mathrm{g}}\,n_{\mathrm{DM,tot}}, \qquad \frac{\Delta\alpha_2}{\alpha_2} = -4G_{\mathrm{W}}\,n_{\mathrm{DM,tot}}, \qquad \frac{\Delta\alpha_1}{\alpha_1} = -4G_{\mathrm{B}}\,n_{\mathrm{DM,tot}},\\
&-\frac{1}{\Upsilon\,\mathrm{e}^{3HL\xi}}\,\frac{\mathrm{d}}{\mathrm{d}\xi}\Big(\Upsilon\,\mathrm{e}^{3HL\xi}\,\frac{\mathrm{d}F}{\mathrm{d}\xi}\Big) = \tilde{m}^2 F, \qquad \Upsilon(\xi) = 1 + A_{\mathrm{mf}}\,n_{\mathrm{DM}}^{\mathrm{sc}}, \qquad n_{\mathrm{DM}}^{\mathrm{sc}} = 2a_{\mathrm{DM}}b_{\mathrm{DM}},\\
&\frac{\mathrm{d}F}{\mathrm{d}\xi}(0) = 0, \qquad \Upsilon(-1)\,\frac{\mathrm{d}F}{\mathrm{d}\xi}(-1) = \beta_{\mathrm{W}}F(-1)
\end{aligned} \tag{FE-79}$$

Symbols in (FE-79).

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- α_3: The strong fine-structure constant, $\alpha_3 = g_3^2/(4\pi)$, at the scale given in brackets. Unit: dimensionless. Value: $\alpha_3(m_{\mathrm{Z}}) = 0.1180$; $\alpha_3(m_{\mathrm{KK}}) = 0.0606$. Status: STANDARD.
- G_{g}: The coupling of a dark-matter pair to canonically normalised gluon pairs through I12. Unit: GeV^{-3}. Value: 6.09×10^{-15} GeV^{-3} (benchmark). Status: DERIVED.
- $n_{\mathrm{DM,tot}}$: The total number density of dark-matter quanta: both species (the lowest odd levels of Ψ_A and Ψ_B), particles and antiparticles, $n_{\mathrm{DM,tot}} = 2(n_{\mathrm{DM}} + \bar{n}_{\mathrm{DM}})$. Unit: GeV^3. Status: STANDARD.
- α_2: The SU(2) fine-structure constant, $\alpha_2 = g_2^2/(4\pi)$. Unit: dimensionless. Status: STANDARD.
- G_{W}: The coupling of a dark-matter pair to canonically normalised SU(2) gauge pairs through I12. Unit: GeV^{-3}. Value: 2.96×10^{-15} GeV^{-3}. Status: DERIVED.
- α_1: The hypercharge fine-structure constant, $\alpha_1 = g_1^2/(4\pi)$. Unit: dimensionless. Status: STANDARD.
- G_{B}: The coupling of a dark-matter pair to canonically normalised hypercharge pairs through I12. Unit: GeV^{-3}. Value: 1.08×10^{-15} GeV^{-3}. Status: DERIVED.
- $\Upsilon(\xi)$: The factor by which the scalar density of the dark matter changes the gauge kinetic term along x_0 (I12), $\Upsilon = 1 + A_{\mathrm{mf}} n_{\mathrm{DM}}^{\mathrm{sc}}$. Unit: dimensionless. Status: DERIVED.

- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $F(\xi)$, F_n: The x_0 profile of a boson mode on one half of the period (a function of y or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; F_n is the profile of the n-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- $\tilde{m}$, $\tilde{m}_n$: The dimensionless x_0 eigenvalue of a boson mode, $\tilde{m} = m\,fL$; $\tilde{m}_n$ the n-th even Kaluza-Klein level of the tower with $\beta = 0$ ($\tilde{m}_1$ the first). Unit: dimensionless. Value: $\tilde{m}_1^2 = 30.1196$, $\tilde{m}_2^2 = 59.7284$, $\tilde{m}_3^2 = 109.0764$ ($HL = 3$), i.e. $\tilde{m}_1 = 5.48813$, $\tilde{m}_2 = 7.72842$, $\tilde{m}_3 = 10.44397$. Status: DERIVED.
- A_mf: The amplitude of the I12 mean field in the x_0 problem of the gauge fields (a bookkeeping parameter: the result is linear in it). Unit: GeV^{-1} (with the orbital normalisation). Status: DERIVED.
- $n_\mathrm{DM}^\mathrm{sc}$: The scalar density of the dark-matter orbital, $n_\mathrm{DM}^\mathrm{sc} = 2a_\mathrm{DM}b_\mathrm{DM}$. Unit: GeV (normalised orbital). Status: DERIVED.
- a_DM, b_DM: The upper and lower components of the dark-matter orbital (block spinor $(a, \mathrm{i}b)$, normalised on one half). Unit: $\mathrm{GeV}^{1/2}$. Status: DERIVED.
- β_W: The Robin parameter of W^+ and W^- (set by the tip mass term of the sigma field). Unit: dimensionless. Value: 0.011912 (fitted). Status: DERIVED.

The terms. The first line: the dark matter weakens every gauge coupling in proportion to its density. The second line: the x_0 problem of FE-50 with the kinetic function $\Upsilon = 1 + A_\mathrm{mf} n_\mathrm{DM}^\mathrm{sc}$, where $n_\mathrm{DM}^\mathrm{sc} = 2a_\mathrm{DM}b_\mathrm{DM}$ is the scalar density of the dark-matter orbital and A_mf the amplitude of the mean field; the tip condition carries $\Upsilon(-1)$. The flat profile still solves the problem for every Υ (the photon and the gluons stay exactly massless), while the W eigenvalue changes by $\mathrm{d}\ln\tilde{m}_\mathrm{W}^2/\mathrm{d}A_\mathrm{mf} = -0.111253$: the W mass follows the four-dimensional coupling. Status: DERIVED (first-order perturbation theory with the exact W eigenfunction; shooting agrees). The physical amplitude is $|\Delta\alpha/\alpha| \le 1.7 \times 10^{-16}$ (at the freeze-out). Residual 3.3×10^{-12}.

7.5 Gravity closes the loop

$$\Delta K = \frac{c_\rho\,\kappa\Delta\rho + c_3\,\kappa\Delta p_3 + c_p\,\kappa\Delta\bar{p}}{H^2}, \qquad \Delta\ln f = -\frac{\Delta K}{2K_*},$$
$$\Delta\ln m_s = \frac{\mathrm{d}\ln m_s}{\mathrm{d}\ln f}\,\Delta\ln f, \qquad \kappa\Delta\rho = 8\pi G_\mathrm{N}\,\mathcal{G}(K_*)\,\rho_4 \tag{FE-80}$$

Symbols in (FE-80).

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_eff an addition to N_eff, ΔS_m the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- c_ρ: The coefficient of $\kappa\Delta\rho$ in the adiabatic block response ΔK. Unit: dimensionless. Value: 0.0358521. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_\mathrm{N}\mathcal{G}(K_*) = 2.328 \times 10^{-36}$ GeV^{-2}. Status: AUTHOR.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- c_3: The coefficient of $\kappa\Delta p_3$ in the adiabatic block response, $c_3 = -3c_\rho$ (so radiation drops out). Unit: dimensionless. Value: -0.107556. Status: DERIVED.

- p_3: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: GeV^8. Status: STANDARD.
- c_p: The coefficient of $\kappa\Delta\bar{p}$ in the adiabatic block response. Unit: dimensionless. Value: -0.140028. Status: DERIVED.
- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial \ln f$. Unit: GeV^8 (or GeV^4 for a four-dimensional density). Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- K_*: The block variable of the stabilised block, $K_* = 12 + \frac{4}{3}\sqrt{51}$ (the upper root K_+). Unit: dimensionless. Value: 21.5219. Status: DERIVED.
- m, m_s, m_i: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.
- s: The label of a particle species of the plasma or of matter (photon, gluons, $W^\pm$, Z, quarks, leptons, nucleons, dark matter); a sum over s runs over the species present. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_N: Newton's constant of four-dimensional gravity, $8\pi G_\mathrm{N} = \kappa/(\mathcal{G}(K_*)V_\perp)$. Unit: GeV^{-2}. Value: 6.709×10^{-39} GeV^{-2} ($M_\mathrm{Pl} = G_\mathrm{N}^{-1/2} = 1.22091 \times 10^{19}$ GeV). Status: STANDARD.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K) = 1 - 24w_{(2)}K + 72w_{(3)}K^2 = (K^2 - 12K + 16)/16$; four-dimensional gravity is attractive for $\mathcal{G} > 0$. Unit: dimensionless. Value: $\mathcal{G}(K_*) = 20/3 + \sqrt{51} = 13.8081$. Status: DERIVED.
- ρ_4: A four-dimensional energy density (per 3-volume); its block average is $\kappa\Delta\rho = 8\pi G_\mathrm{N}\mathcal{G}(K_*)\rho_4$. Unit: GeV^4. Status: DERIVED.

The terms. The adiabatic block response FE-47, applied to the block-averaged sources of every species of an environment: the four-dimensional densities ρ_4 (and likewise the pressures) enter as $\kappa\Delta\rho = 8\pi G_\mathrm{N}\mathcal{G}(K_*)\rho_4$; the change of the block scale changes every mass m_s in proportion to its block sensitivity $\mathrm{d}\ln m_s/\mathrm{d}\ln f$ (FE-83 to FE-85). The masses change the sources, so the loop mass, source, block, mass is iterated to self-consistency. Status: DERIVED. The loop gain is at most 1.05×10^{-33} (at the freeze-out), so one step suffices; the block moves by at most $\Delta\ln f = 1.1 \times 10^{-34}$ (at the freeze-out) and by 7.3×10^{-94} for the cosmic mean today. Residual of the stored environments 2.0×10^{-11}.

The hidden pressure is the response of the pressure to the block scale at fixed temperature (a change of $\ln f$ changes the proper lengths of the four block directions together): for an ideal gas of species of mass m_s, for the QCD plasma whose pressure depends on Λ_QCD, and for the order-α pressure of the electroweak plasma,

$$\bar{p} = \frac{1}{4}\,\frac{\mathrm{d}p}{\mathrm{d}\ln f}, \qquad \frac{\partial p_s}{\partial \ln m_s} = -\big(\rho_s - 3p_s\big), \qquad \frac{\partial p_\mathrm{QCD}}{\partial \ln \Lambda_\mathrm{QCD}} = -\Theta_\mathrm{QCD}, \qquad \frac{\partial p_{\alpha,\mathrm{G}}}{\partial \ln \alpha_\mathrm{G}} = p_{\alpha,\mathrm{G}}\,:$$

Symbols in this equation.

- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial \ln f$. Unit: GeV^8 (or GeV^4 for a four-dimensional density). Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- p: A pressure: $p_M = \mathcal{T}^M{}_M$ (no sum) for the direction $M \neq x_4$, p_s the pressure of the species s; Δp the pressure perturbation in the cosmological perturbation equations; in the Feynman-Hellmann relation the pressure of a gauge plasma. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.

- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- s: The label of a particle species of the plasma or of matter (photon, gluons, $W^\pm$, Z, quarks, leptons, nucleons, dark matter); a sum over s runs over the species present. Status: STANDARD.
- m, m_s, m_i: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- p_{QCD}: The pressure of the quark-gluon plasma (leading orders). Unit: GeV^4. Status: ESTIMATE.
- $\Lambda^{(N_\mathrm{f})}_{\mathrm{QCD}}$: The QCD scale with N_f active flavours ($\Lambda^{(6)}_{\mathrm{QCD}}$, $^{(5)}$, $^{(4)}$, $^{(3)}$), defined by the one-loop running. Unit: GeV. Status: STANDARD.
- Θ_{QCD}, Θ_G, Θ_V: The trace $\rho - 3p$ of the energy-momentum tensor (interaction measure) of a plasma component: Θ_{QCD} of the quark-gluon plasma or hadron gas, Θ_G of the gauge plasma of the group G, Θ_V of the thermal gas of the massive vector V (Θ_W of $W^\pm$, Θ_Z of Z). Unit: GeV^4. Status: STANDARD.
- $p_{\alpha,\mathrm{G}}$, $p_{\alpha,2}$, $p_{\alpha,1}$: The order-α (interaction) part of the pressure of the gauge plasma of the group G; $p_{\alpha,2}$ of SU(2), $p_{\alpha,1}$ of $\mathrm{U}(1)_Y$. Unit: GeV^4. Status: STANDARD.
- α_G: The fine-structure constant of the gauge group G (one of α_3, α_2, α_1). Unit: dimensionless. Status: STANDARD.

$$\bar{p} = -\frac{1}{4}\sum_s \left(\rho_s - 3p_s\right)\frac{\mathrm{d}\ln m_s}{\mathrm{d}\ln f} - \frac{1}{4}\,\Theta_{\mathrm{QCD}}\,\frac{\mathrm{d}\ln\Lambda^{(N_\mathrm{f})}_{\mathrm{QCD}}}{\mathrm{d}\ln f} + \frac{1}{4}\,p_{\alpha,2}\,\frac{\mathrm{d}\ln\alpha_2}{\mathrm{d}\ln f} + \frac{1}{4}\,p_{\alpha,1}\,\frac{\mathrm{d}\ln\alpha_1}{\mathrm{d}\ln f} \qquad \text{(FE-81)}$$

Symbols in (FE-81).

- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial \ln f$. Unit: GeV^8 (or GeV^4 for a four-dimensional density). Status: DERIVED.
- s: The label of a particle species of the plasma or of matter (photon, gluons, $W^\pm$, Z, quarks, leptons, nucleons, dark matter); a sum over s runs over the species present. Status: STANDARD.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- p: A pressure: $p_M = \mathcal{T}^M{}_M$ (no sum) for the direction $M \neq x_4$, p_s the pressure of the species s; Δp the pressure perturbation in the cosmological perturbation equations; in the Feynman-Hellmann relation the pressure of a gauge plasma. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- m, m_s, m_i: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-\alpha_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- Θ_{QCD}, Θ_G, Θ_V: The trace $\rho - 3p$ of the energy-momentum tensor (interaction measure) of a plasma component: Θ_{QCD} of the quark-gluon plasma or hadron gas, Θ_G of the gauge plasma of the group G, Θ_V of the thermal gas of the massive vector V (Θ_W of $W^\pm$, Θ_Z of Z). Unit: GeV^4. Status: STANDARD.
- $\Lambda^{(N_\mathrm{f})}_{\mathrm{QCD}}$: The QCD scale with N_f active flavours ($\Lambda^{(6)}_{\mathrm{QCD}}$, $^{(5)}$, $^{(4)}$, $^{(3)}$), defined by the one-loop running. Unit: GeV. Status: STANDARD.
- $p_{\alpha,\mathrm{G}}$, $p_{\alpha,2}$, $p_{\alpha,1}$: The order-α (interaction) part of the pressure of the gauge plasma of the group G; $p_{\alpha,2}$ of SU(2), $p_{\alpha,1}$ of $\mathrm{U}(1)_Y$. Unit: GeV^4. Status: STANDARD.
- α_2: The SU(2) fine-structure constant, $\alpha_2 = g_2^2/(4\pi)$. Unit: dimensionless. Status: STANDARD.
- α_1: The hypercharge fine-structure constant, $\alpha_1 = g_1^2/(4\pi)$. Unit: dimensionless. Status: STANDARD.

The terms. The first sum runs over the species s with block-dependent masses (the plasma species as ideal

gases with their x_0 masses; nucleons and electrons as degenerate Fermi gases in the cold environments; the dark matter): non-relativistic matter gives $-\frac{1}{4}\rho\,\mathrm{d}\ln m/\mathrm{d}\ln f$, radiation nothing. The second term is the QCD plasma through the block dependence of its scale, the last two the electroweak plasma through the block dependence of its couplings. A change of $\ln f$ scales the four block directions together, so FE-81 is the average $\bar{p} = \frac{1}{4}(p_0 + 3p_5)$: the mass terms depend on the hidden length only (they contribute to p_0, not to p_5), the coupling terms on the volume of all four block directions (equal parts in p_0 and p_5); the difference $p_0 - p_5$ is the block-shape source (OPEN). Status: DERIVED.

$$\frac{\mathrm{d}\ln\alpha_\mathrm{G}(\mu_\mathrm{ren})}{\mathrm{d}\ln f} = -\alpha_\mathrm{G}(\mu_\mathrm{ren})\Big[\frac{4}{\alpha_\mathrm{G}(m_\mathrm{KK})} + \frac{b_\mathrm{G}}{2\pi}\Big], \qquad \Theta_\mathrm{G} = -\frac{b_\mathrm{G}\,\alpha_\mathrm{G}}{2\pi}\,p_{\alpha,\mathrm{G}} \tag{FE-82}$$

Symbols in (FE-82).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- α_G: The fine-structure constant of the gauge group G (one of α_3, α_2, α_1). Unit: dimensionless. Status: STANDARD.
- μ_ren: The renormalisation scale of a running coupling (taken as $2\pi T$ for the condensates of a plasma). Unit: GeV. Status: STANDARD.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-\alpha_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- m_n^KK, m_KK: The mass of the n-th Kaluza-Klein graviton (and gauge) level, $m_n^\mathrm{KK} = \tilde{m}_n/(fL)$; $m_\mathrm{KK} = m_1^\mathrm{KK}$ is the Kaluza-Klein scale. Unit: GeV. Value: $m_\mathrm{KK} = 121.32$ TeV. Status: DERIVED.
- b_3, b_2, b_1, b_G: The one-loop coefficients of the running couplings, $\mathrm{d}\alpha/\mathrm{d}\ln\mu_\mathrm{ren} = -b\,\alpha^2/(2\pi)$: $b_3 = 11 - \frac{2}{3}N_\mathrm{f}$ (SU(3)), $b_2 = \frac{10}{3}$ (SU(2) without a Higgs particle), $b_1 = -\frac{20}{3}$ ($\mathrm{U}(1)_Y$), b_G that of the group G (not related to the metric function b_4). Unit: dimensionless. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- Θ_QCD, Θ_G, Θ_V: The trace $\rho - 3p$ of the energy-momentum tensor (interaction measure) of a plasma component: Θ_QCD of the quark-gluon plasma or hadron gas, Θ_G of the gauge plasma of the group G, Θ_V of the thermal gas of the massive vector V (Θ_W of $W^\pm$, Θ_Z of Z). Unit: GeV^4. Status: STANDARD.
- $p_{\alpha,\mathrm{G}}$, $p_{\alpha,2}$, $p_{\alpha,1}$: The order-α (interaction) part of the pressure of the gauge plasma of the group G; $p_{\alpha,2}$ of SU(2), $p_{\alpha,1}$ of $\mathrm{U}(1)_Y$. Unit: GeV^4. Status: STANDARD.

The terms. The eight-dimensional couplings are constants, so the four-dimensional couplings at the Kaluza-Klein scale scale as f^{-4} ($g_\mathrm{G}^2 = \hat{g}^2/V_\perp$), and $m_\mathrm{KK} \propto 1/f$; running down to μ_ren with the coefficient b_G gives the first relation. The second is the leading-order trace anomaly of a gauge plasma, the electroweak analogue of Θ_QCD. Status: DERIVED.

$$\frac{1}{\alpha_3(m_\mathrm{KK})} = \frac{1}{\alpha_3(m_\mathrm{Z})} + \frac{7}{2\pi}\ln\frac{m_\mathrm{KK}}{m_\mathrm{Z}}, \qquad \Lambda_\mathrm{QCD}^{(6)} = m_\mathrm{KK}\,\mathrm{e}^{-2\pi/(7\alpha_3(m_\mathrm{KK}))}, \qquad \Lambda_\mathrm{QCD}^{(3)} = \big(\Lambda_\mathrm{QCD}^{(6)}\big)^{21/27}\big(m_\mathrm{c}m_\mathrm{b}m_\mathrm{t}\big)^{2/27},$$
$$\frac{\mathrm{d}\ln m_\mathrm{p}}{\mathrm{d}\ln f} = \Big(1 - \frac{\sigma_{\pi N} + \sigma_\mathrm{s}}{m_\mathrm{p}}\Big)\frac{\mathrm{d}\ln\Lambda_\mathrm{QCD}^{(3)}}{\mathrm{d}\ln f} + \frac{\sigma_{\pi N}}{2m_\mathrm{p}}\Big(\frac{\mathrm{d}\ln m_\mathrm{u}}{\mathrm{d}\ln f} + \frac{\mathrm{d}\ln m_\mathrm{d}}{\mathrm{d}\ln f}\Big) + \frac{\sigma_\mathrm{s}}{m_\mathrm{p}}\,\frac{\mathrm{d}\ln m_\mathrm{s}}{\mathrm{d}\ln f} \tag{FE-83}$$

Symbols in (FE-83).

- α_3: The strong fine-structure constant, $\alpha_3 = g_3^2/(4\pi)$, at the scale given in brackets. Unit: dimensionless. Value: $\alpha_3(m_\mathrm{Z}) = 0.1180$; $\alpha_3(m_\mathrm{KK}) = 0.0606$. Status: STANDARD.
- m_n^KK, m_KK: The mass of the n-th Kaluza-Klein graviton (and gauge) level, $m_n^\mathrm{KK} = \tilde{m}_n/(fL)$; $m_\mathrm{KK} = m_1^\mathrm{KK}$ is the Kaluza-Klein scale. Unit: GeV. Value: $m_\mathrm{KK} = 121.32$ TeV. Status: DERIVED.
- m_Z: The mass of the Z. Unit: GeV. Value: 91.1879 GeV (reproduced). Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\Lambda_\mathrm{QCD}^{(N_\mathrm{f})}$: The QCD scale with N_f active flavours ($\Lambda_\mathrm{QCD}^{(6)}$, $^{(5)}$, $^{(4)}$, $^{(3)}$), defined by the one-loop running. Unit: GeV. Status: STANDARD.

- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- m_u, m_d, m_s, m_c, m_b, m_t: The masses of the up, down, strange, charm, bottom and top quarks (the upright subscript names the quark). Unit: GeV. Value: 2.16 MeV, 4.70 MeV, 93.5 MeV, 1.273 GeV, 4.183 GeV, 172.56 GeV (the x_0 masses, fitted to the PDG values). Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- m_p, m_n: The masses of the proton and the neutron. Unit: GeV. Value: 0.938272 GeV, 0.939565 GeV. Status: STANDARD.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\sigma_{\pi N}$: The pion-nucleon sigma term (the contribution of the up and down quark masses to the nucleon mass). Unit: GeV. Value: 45 MeV. Status: STANDARD.
- σ_s: The strange sigma term of the nucleon. Unit: GeV. Value: 40 MeV. Status: STANDARD.

The terms. The strong coupling at the Kaluza-Klein scale (one loop, six flavours, $\alpha_3(m_\mathrm{KK}) = 0.0606$), the QCD scale of six flavours and its one-loop matching to three flavours at the heavy-quark masses, and the block dependence of the proton mass: the part of m_p that is not the pion-nucleon and strange sigma terms ($\sigma_{\pi N} = 45$ MeV, $\sigma_\mathrm{s} = 40$ MeV) scales with $\Lambda^{(3)}_\mathrm{QCD}$, the sigma terms with the light-quark masses. Result: $\mathrm{d}\ln m_\mathrm{p}/\mathrm{d}\ln f = -46.1$, the most block-sensitive mass of UM-2. Status: ESTIMATE (one loop).

$$
\begin{aligned}
&\frac{\mathrm{d}\ln\Lambda^{(6)}_\mathrm{QCD}}{\mathrm{d}\ln f} = -1 - \frac{8\pi}{7\alpha_3(m_\mathrm{KK})}, \qquad \frac{\mathrm{d}\ln\Lambda^{(5)}_\mathrm{QCD}}{\mathrm{d}\ln f} = \frac{21}{23}\frac{\mathrm{d}\ln\Lambda^{(6)}_\mathrm{QCD}}{\mathrm{d}\ln f} + \frac{2}{23}\frac{\mathrm{d}\ln m_\mathrm{t}}{\mathrm{d}\ln f},\\
&\frac{\mathrm{d}\ln\Lambda^{(4)}_\mathrm{QCD}}{\mathrm{d}\ln f} = \frac{21}{25}\frac{\mathrm{d}\ln\Lambda^{(6)}_\mathrm{QCD}}{\mathrm{d}\ln f} + \frac{2}{25}\Big(\frac{\mathrm{d}\ln m_\mathrm{t}}{\mathrm{d}\ln f} + \frac{\mathrm{d}\ln m_\mathrm{b}}{\mathrm{d}\ln f}\Big),\\
&\frac{\mathrm{d}\ln\Lambda^{(3)}_\mathrm{QCD}}{\mathrm{d}\ln f} = \frac{21}{27}\frac{\mathrm{d}\ln\Lambda^{(6)}_\mathrm{QCD}}{\mathrm{d}\ln f} + \frac{2}{27}\Big(\frac{\mathrm{d}\ln m_\mathrm{c}}{\mathrm{d}\ln f} + \frac{\mathrm{d}\ln m_\mathrm{b}}{\mathrm{d}\ln f} + \frac{\mathrm{d}\ln m_\mathrm{t}}{\mathrm{d}\ln f}\Big)
\end{aligned}
\tag{FE-84}
$$

Symbols in (FE-84).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $\Lambda^{(N_\mathrm{f})}_\mathrm{QCD}$: The QCD scale with N_f active flavours ($\Lambda^{(6)}_\mathrm{QCD}$, $^{(5)}$, $^{(4)}$, $^{(3)}$), defined by the one-loop running. Unit: GeV. Status: STANDARD.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- α_3: The strong fine-structure constant, $\alpha_3 = g_3^2/(4\pi)$, at the scale given in brackets. Unit: dimensionless. Value: $\alpha_3(m_\mathrm{Z}) = 0.1180$; $\alpha_3(m_\mathrm{KK}) = 0.0606$. Status: STANDARD.
- m_n^KK, m_KK: The mass of the n-th Kaluza-Klein graviton (and gauge) level, $m_n^\mathrm{KK} = \tilde{m}_n/(fL)$; $m_\mathrm{KK} = m_1^\mathrm{KK}$ is the Kaluza-Klein scale. Unit: GeV. Value: $m_\mathrm{KK} = 121.32$ TeV. Status: DERIVED.
- m_u, m_d, m_s, m_c, m_b, m_t: The masses of the up, down, strange, charm, bottom and top quarks (the upright subscript names the quark). Unit: GeV. Value: 2.16 MeV, 4.70 MeV, 93.5 MeV, 1.273 GeV, 4.183 GeV, 172.56 GeV (the x_0 masses, fitted to the PDG values). Status: DERIVED.

The terms. The block dependence of the QCD scale with six flavours, from $\Lambda^{(6)}_\mathrm{QCD} = m_\mathrm{KK}\mathrm{e}^{-2\pi/(7\alpha_3)}$ with $\alpha_3 \propto f^{-4}$ and $m_\mathrm{KK} \propto 1/f$, and the matching through the top, bottom and charm thresholds: -60.2 (six flavours), -55.6, -52.0 and -49.0 (three flavours). Status: ESTIMATE (one loop).

$$
\frac{\mathrm{d}\ln m}{\mathrm{d}\ln f} = -1 + \frac{1}{2}\Big[\frac{\mathrm{d}\ln\mathcal{Z}(c_L)^2}{\mathrm{d}\ln c_L} + \frac{\mathrm{d}\ln\mathcal{Z}(-c_R)^2}{\mathrm{d}\ln(-c_R)}\Big]
\tag{FE-85}
$$

Symbols in (FE-85).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- m, m_s, m_i: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\mathcal{Z}(c)$: The tip amplitude of a normalised zero mode with the localisation c, $\mathcal{Z}(c)^2 = c\,\mathrm{e}^{-c}/\sinh c$. Unit: dimensionless. Status: DERIVED.
- c, c_L, c_R: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); c_L of a doublet (localised at the fold for $c_L > 0$), c_R of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets c_R: up -15.4300, down -14.6258, strange -11.5158, charm -8.7683, bottom -7.5005, top -3.3823, electron -14.5539, muon -8.9809, tau -5.9529, neutrinos -35.0452, -32.8481, -31.0702 (fixed by the measured masses). Status: DERIVED.
- L: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1+\Gamma_4)/2$. Status: STANDARD.
- R: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1-\Gamma_4)/2$. Status: STANDARD.

The terms. A fermion mass is $m = \tilde{\varepsilon}/(fL)$ with the first-order root of FE-59, and every localisation $c = f\mu L$ is proportional to f: the -1 is the scaling of the hidden length, the bracket the change of the tip amplitudes. For the quarks -18.82, -18.01, -14.90, -12.15, -10.89, -6.77 (up, down, strange, charm, bottom, top): the lighter the quark, the sharper its localisation and the stronger its block dependence. Status: DERIVED.

$$
\begin{aligned}
&n_\mathrm{p} + n_\mathrm{n} = \eta_\mathrm{b}\, n_\gamma, \qquad n_\gamma = \frac{2\zeta(3)}{\pi^2}\, T^3, \qquad \frac{n_\mathrm{n}}{n_\mathrm{p}} = \mathrm{e}^{-(m_\mathrm{n}-m_\mathrm{p})/T}, \qquad n_\mathrm{e} = n_\mathrm{p}, \\
&n_{\mathrm{DM,tot}} = 2\big(n_\mathrm{DM} + \bar{n}_\mathrm{DM}\big) = 2\cdot 4\left(\frac{M_\mathrm{DM} T}{2\pi}\right)^{3/2} \mathrm{e}^{-M_\mathrm{DM}/T} \quad \left(\frac{M_\mathrm{DM}}{T} < 40\right)
\end{aligned}
\tag{FE-86}
$$

Symbols in (FE-86).

- n_p: The number density of protons (free and bound). Unit: GeV^3. Status: STANDARD.
- n_n: The number density of neutrons (free and bound). Unit: GeV^3. Status: STANDARD.
- η_b: The ratio of the baryon density to the photon density. Unit: dimensionless. Value: 6.1×10^{-10}. Status: STANDARD.
- n_γ: The number density of photons. Unit: GeV^3. Status: STANDARD.
- ζ: The Riemann zeta function; only its value $\zeta(3)$ occurs, the sum of $1/n^3$ over the positive integers n. Unit: dimensionless. Value: $\zeta(3) = 1.2020569$. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- m_p, m_n: The masses of the proton and the neutron. Unit: GeV. Value: 0.938272 GeV, 0.939565 GeV. Status: STANDARD.
- n_e: The number density of electrons (net of positrons); in the Thomson terms of the photon and baryon equations only free (ionised) electrons count. Unit: GeV^3. Status: STANDARD.
- $n_{\mathrm{DM,tot}}$: The total number density of dark-matter quanta: both species (the lowest odd levels of Ψ_A and Ψ_B), particles and antiparticles, $n_{\mathrm{DM,tot}} = 2(n_\mathrm{DM} + \bar{n}_\mathrm{DM})$. Unit: GeV^3. Status: STANDARD.
- n_DM, $\bar{n}_\mathrm{DM}$: The number densities of dark-matter particles and antiparticles of one species. Unit: GeV^3. Status: STANDARD.
- M_DM: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.

The terms. The densities of the hot environments: the net baryon density from the baryon-to-photon ratio $\eta_{\rm b} = 6.1 \times 10^{-10}$ shared between protons and neutrons in equilibrium, as many electrons as protons, and the dark matter in equilibrium (two species and their antiparticles, four states each, hence $2 \cdot 4$) while $M_{\rm DM}/T < 40$, its relic density afterwards. The cold environments (the cosmic mean today, the Galaxy at the Sun, the Earth, the centre of the Sun, a white dwarf, a neutron-star core) use measured densities. Status: ASSUMED.

8 The linear perturbations

8.1 The metric as a power series

$$g_{MN} = \bar{g}_{MN} + \epsilon\, h^{(1)}_{MN} + \epsilon^2\, h^{(2)}_{MN} + \ldots, \qquad h^{(1)}_{MN}(X^\mu, x_0) = \sum_{n\geq 0} h^{(1,n)}_{MN}(X^\mu)\, F_n(x_0), \qquad \epsilon \sim 10^{-5} \qquad \text{(FE-87)}$$

Symbols in (FE-87).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\, \mathrm{d}X^M \mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\, \mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\bar{g}_{MN}$: The order-0 (background) metric of the power series: the corrected primordial metric in the author's coordinates, determined by the two functions $a_4(x_4)$ and $b_4(x_4)$; $|\bar{g}|$ is the absolute value of its determinant. Unit: dimensionless. Status: AUTHOR.
- ϵ: The bookkeeping parameter of the power series of the metric: the term of order q is multiplied by ϵ^q (the size of the first-order potentials). Unit: dimensionless. Value: $\epsilon \sim 10^{-5}$. Status: DERIVED.
- $h^{(1)}_{MN}$, $h^{(2)}_{MN}$: The first- and second-order terms of the power series of the metric. Unit: dimensionless. Status: DERIVED.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\, \mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- n: The level number of the Kaluza-Klein tower ($n = 1, 2, 3, \ldots$; $n = 0$ the flat zero mode). Status: DERIVED.
- $h^{(1,n)}_{MN}$: The four-dimensional coefficient of the n-th x_0 profile F_n in the first-order term ($n = 0$: the flat zero mode, the perturbations of four-dimensional gravity; $n \geq 1$: the Kaluza-Klein gravitons). Unit: dimensionless. Status: DERIVED.
- $F(\xi)$, F_n: The x_0 profile of a boson mode on one half of the period (a function of y or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; F_n is the profile of the n-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.

The terms. The metric of UM-2 is the order-0 metric (FE-31) plus terms of decreasing magnitude, counted by the bookkeeping parameter $\epsilon \sim 10^{-5}$ (the size of the cosmological potentials). Each term is expanded in the x_0 profiles F_n of FE-50: $n = 0$ is the flat profile (four-dimensional gravity, the cosmological perturbations below), $n \geq 1$ are the Kaluza-Klein gravitons of FE-53. Status: DERIVED. The terms decrease strictly: order 0 is 1, order 1 is at most 2.4×10^{-5} on cosmological scales (the rms potential, FE-98), the Kaluza-Klein terms have ranges below 1.6×10^{-21} m, the radion term is at most 1.1×10^{-34}, and order 2 is estimated as the square of order 1, 5.8×10^{-10} (ESTIMATE). Residual (the first omitted order relative to the kept one) 2.4×10^{-5}; at a neutron star order 2 is 0.03 of order 0, and there full nonlinear general relativity applies.

$$g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N = \mathrm{e}^{2a_4}\Big[\big(1-2\Psi_\mathrm{N}\big)\big(\mathrm{d}x_1^2+\mathrm{d}x_2^2+\mathrm{d}x_3^2\big)+\mathrm{d}\vec{x}\cdot h^\mathrm{TT}\cdot\mathrm{d}\vec{x}\Big]-\big(1+2\Phi\big)\,\mathrm{d}x_4^2 + f^2\big(1+2\sigma_\mathrm{r}\big)\Big[\mathrm{d}y^2-\mathrm{e}^{2Hy}\big(\mathrm{d}x_5^2+\mathrm{d}x_6^2+\mathrm{d}x_7^2\big)\Big] \tag{FE-88}$$

Symbols in (FE-88).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N=\delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0,\ldots,x_7$; a repeated index is summed. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X=\{x_0,\ldots,x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X=\{x_0,x_1,\ldots,x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X=\mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1,\ldots,x_4$. Unit: length (each coordinate). Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- Ψ_N: The curvature potential of the Newtonian gauge (the perturbation of the 3-space metric, $1-2\Psi_\mathrm{N}$); not related to the field Ψ. Unit: dimensionless. Value: at most 2.4×10^{-5}. Status: STANDARD.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space (also the symmetry axis of the fast-flavour analysis). Unit: length. Status: AUTHOR.
- $\vec{x}$: The position in ordinary 3-space, $\vec{x}=(x_1,x_2,x_3)$; $\mathrm{d}\vec{x}^{\,2}=\mathrm{d}x_1^2+\mathrm{d}x_2^2+\mathrm{d}x_3^2$. Unit: length. Status: STANDARD.
- h^TT: The transverse-traceless (tensor) part of the first-order metric of 3-space (gravitational waves). Unit: dimensionless. Value: an input (tensor-to-scalar ratio below 0.036). Status: STANDARD.
- Φ: The lapse potential of the Newtonian gauge (the perturbation of the time component, $1+2\Phi$). Unit: dimensionless. Value: at most 2.4×10^{-5}. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f=\mathrm{e}^{-a_4}b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- σ_r: The radion perturbation: the first-order change of the block scale, $f^2\to f^2(1+2\sigma_\mathrm{r})$. Unit: dimensionless. Value: at most 1.1×10^{-34} (static response). Status: DERIVED.
- y: The hidden coordinate of the y chart, $y=\ln|\sin z|/(6H)$, with $y=0$ at the fold ($z=\pi/2$) and $y=-L$ at the regular tip; each half of a period of x_0 is the interval $-L\le y\le0$. Unit: length. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda=-10H^2$ and the mass parameter $m_\Psi=H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

The terms. The order-1, $n=0$ term in eight dimensions, in the Newtonian gauge: Φ is the Newtonian potential (in the time component), Ψ_N the curvature potential of ordinary space, h^TT the transverse-traceless tensor of gravitational waves, and σ_r the radion, a common change of the block scale. The radion is massive (48.42 TeV, FE-44): its static response to matter is at most 1.1×10^{-34}, so it is absent on every observable scale; the tensor amplitude is an input, bounded by the tensor-to-scalar ratio below 0.036. Status: DERIVED.

$$g^{(4)}_{\mu\nu}\,\mathrm{d}X^\mu \mathrm{d}X^\nu = \mathrm{e}^{2a_4}\Big[-\big(1+2\Phi\big)\,\mathrm{d}\tau_\mathrm{c}^2 + \big(1-2\Psi_\mathrm{N}\big)\,\mathrm{d}\vec{x}^{\,2}\Big], \qquad \mathrm{d}x_4 = \mathrm{e}^{a_4}\,\mathrm{d}\tau_\mathrm{c} \tag{FE-89}$$

Symbols in (FE-89).

- $g^{(4)}_{\mu\nu}$: The four-dimensional metric of the observable coordinates $x_1, \ldots, x_4$ (the metric of the effective four-dimensional general relativity on the stabilised block). Unit: dimensionless. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index running over $x_1, \ldots, x_4$ (as μ). Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- Φ: The lapse potential of the Newtonian gauge (the perturbation of the time component, $1+2\Phi$). Unit: dimensionless. Value: at most 2.4×10^{-5}. Status: STANDARD.
- τ_c: The conformal time, $\mathrm{d}x_4 = \mathrm{e}^{a_4}\mathrm{d}\tau_\mathrm{c}$; primes denote $\mathrm{d}/\mathrm{d}\tau_\mathrm{c}$. Unit: length. Status: STANDARD.
- Ψ_N: The curvature potential of the Newtonian gauge (the perturbation of the 3-space metric, $1-2\Psi_\mathrm{N}$); not related to the field Ψ. Unit: dimensionless. Value: at most 2.4×10^{-5}. Status: STANDARD.
- $\vec{x}$: The position in ordinary 3-space, $\vec{x} = (x_1, x_2, x_3)$; $\mathrm{d}\vec{x}^{\,2} = \mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2$. Unit: length. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.

The terms. The four-dimensional part of FE-88 in the conformal time τ_c of the expanding Universe ($\mathrm{d}x_4 = \mathrm{e}^{a_4}\mathrm{d}\tau_\mathrm{c}$), the form used by the Einstein-Boltzmann codes. A prime denotes the derivative with respect to τ_c; a_4' is the conformal expansion rate. Status: STANDARD (Newtonian gauge, Ma and Bertschinger).

8.2 The Einstein-Boltzmann equations

The order-1, $n = 0$ part of FE-1, reduced on the stabilised block (FE-49), is general relativity with the four-dimensional energy-momentum tensor of all species; for one Fourier mode k:

$$\begin{aligned}
&k^2\Psi_\mathrm{N} + 3a_4'\big(\Psi_\mathrm{N}' + a_4'\Phi\big) = -4\pi G_\mathrm{N}\,\mathrm{e}^{2a_4}\,\Delta\rho, \qquad k^2\big(\Psi_\mathrm{N}' + a_4'\Phi\big) = 4\pi G_\mathrm{N}\,\mathrm{e}^{2a_4}\big(\rho_\mathrm{tot} + p_\mathrm{tot}\big)\theta_\mathrm{tot},\\
&\Psi_\mathrm{N}'' + a_4'\big(\Phi' + 2\Psi_\mathrm{N}'\big) + \big(2a_4'' + a_4'^2\big)\Phi + \frac{k^2}{3}\big(\Psi_\mathrm{N} - \Phi\big) = 4\pi G_\mathrm{N}\,\mathrm{e}^{2a_4}\,\Delta p,\\
&k^2\big(\Psi_\mathrm{N} - \Phi\big) = 12\pi G_\mathrm{N}\,\mathrm{e}^{2a_4}\big(\rho_\mathrm{tot} + p_\mathrm{tot}\big)\sigma_\mathrm{an}
\end{aligned} \tag{FE-90}$$

Symbols in (FE-90).

- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- Ψ_N: The curvature potential of the Newtonian gauge (the perturbation of the 3-space metric, $1-2\Psi_\mathrm{N}$); not related to the field Ψ. Unit: dimensionless. Value: at most 2.4×10^{-5}. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- $'$: The prime on a function of the conformal time τ_c is its derivative $\mathrm{d}/\mathrm{d}\tau_\mathrm{c}$ (it is used only in the cosmological perturbation equations; two primes are the second derivative). Status: STANDARD.
- Φ: The lapse potential of the Newtonian gauge (the perturbation of the time component, $1+2\Phi$). Unit: dimensionless. Value: at most 2.4×10^{-5}. Status: STANDARD.

- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_N: Newton's constant of four-dimensional gravity, $8\pi G_\mathrm{N} = \kappa/(\mathcal{G}(K_*)V_\perp)$. Unit: GeV^{-2}. Value: 6.709×10^{-39} GeV^{-2} ($M_\mathrm{Pl} = G_\mathrm{N}^{-1/2} = 1.22091 \times 10^{19}$ GeV). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_eff an addition to N_eff, ΔS_m the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- ρ_tot, p_tot: The total background energy density and pressure of four-dimensional cosmology (all species). Unit: GeV^4. Status: STANDARD.
- θ_tot: The velocity divergence of the total matter, weighted with $\rho + p$ of each species. Unit: 1/length. Status: STANDARD.
- p: A pressure: $p_M = \mathcal{T}^M{}_M$ (no sum) for the direction $M \neq x_4$, p_s the pressure of the species s; Δp the pressure perturbation in the cosmological perturbation equations; in the Feynman-Hellmann relation the pressure of a gauge plasma. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- σ_an: The anisotropic stress (shear) of the total matter (the quantity σ of Ma and Bertschinger), weighted with $\rho + p$. Unit: dimensionless. Status: STANDARD.

The terms. The four linearised Einstein equations: the energy constraint (the change $\Delta\rho$ of the total energy density sources Ψ_N, the Poisson equation inside the horizon), the momentum constraint (the velocity divergence θ_tot of the total fluid), the pressure equation (the change Δp of the total pressure) and the anisotropic-stress equation (σ_an makes Φ and Ψ_N differ; it comes from the free-streaming neutrinos and photons). All species of UM-2 contribute to the sources: the dark matter, the baryons, the photons, the three neutrino mass states and the fold bands produced by freeze-in. Status: STANDARD (solved with CAMB 2.0.5 at the Planck 2018 parameters; the TT and EE spectra agree with Planck ΛCDM to 2.5×10^{-4} and 2.7×10^{-4}).

$$\delta'_\mathrm{DM} = -\theta_\mathrm{DM} + 3\Psi'_\mathrm{N}, \qquad \theta'_\mathrm{DM} = -a'_4\,\theta_\mathrm{DM} + k^2\Phi \tag{FE-91}$$

Symbols in (FE-91).

- δ_DM, θ_DM: The density contrast and the velocity divergence of the dark matter. Unit: dimensionless; 1/length. Status: STANDARD.
- $'$: The prime on a function of the conformal time τ_c is its derivative $\mathrm{d}/\mathrm{d}\tau_\mathrm{c}$ (it is used only in the cosmological perturbation equations; two primes are the second derivative). Status: STANDARD.
- Ψ_N: The curvature potential of the Newtonian gauge (the perturbation of the 3-space metric, $1 - 2\Psi_\mathrm{N}$); not related to the field Ψ. Unit: dimensionless. Value: at most 2.4×10^{-5}. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- Φ: The lapse potential of the Newtonian gauge (the perturbation of the time component, $1 + 2\Phi$). Unit: dimensionless. Value: at most 2.4×10^{-5}. Status: STANDARD.

The terms. The dark matter (the lowest odd quanta of Ψ_A and Ψ_B) is cold: its density contrast δ_DM changes with its velocity divergence θ_DM and the change of the curvature potential, and its velocity responds only to the expansion and to the gradient of Φ; it has no pressure (Kohn-Sham: $p_3 = 0$ for quanta at rest) and no interaction at these energies (its portal couplings are of order 10^{-9} GeV^{-2} and below). Status: STANDARD.

$$\delta'_\mathrm{b} = -\theta_\mathrm{b} + 3\Psi'_\mathrm{N}, \qquad \theta'_\mathrm{b} = -a'_4\,\theta_\mathrm{b} + c_\mathrm{s}^2 k^2 \delta_\mathrm{b} + R_\mathrm{b}\,\mathrm{e}^{a_4} n_\mathrm{e} \sigma_\mathrm{T} \big(\theta_\gamma - \theta_\mathrm{b}\big) + k^2\Phi, \qquad R_\mathrm{b} = \frac{4\rho_\gamma}{3\rho_\mathrm{b}} \tag{FE-92}$$

Symbols in (FE-92).

- δ_b, θ_b: The density contrast and the velocity divergence of the baryons (nucleons and electrons). Unit: dimensionless; 1/length. Status: STANDARD.
- $'$: The prime on a function of the conformal time τ_c is its derivative $\mathrm{d}/\mathrm{d}\tau_\mathrm{c}$ (it is used only in the cosmological perturbation equations; two primes are the second derivative). Status: STANDARD.
- Ψ_N: The curvature potential of the Newtonian gauge (the perturbation of the 3-space metric, $1 - 2\Psi_\mathrm{N}$); not related to the field Ψ. Unit: dimensionless. Value: at most 2.4×10^{-5}. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- c_s: The sound speed of the baryon fluid. Unit: dimensionless (c = 1). Status: STANDARD.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- R_b: The photon-to-baryon momentum ratio of the Thomson drag, $R_\mathrm{b} = 4\rho_\gamma/(3\rho_\mathrm{b})$. Unit: dimensionless. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- n_e: The number density of electrons (net of positrons); in the Thomson terms of the photon and baryon equations only free (ionised) electrons count. Unit: GeV^3. Status: STANDARD.
- σ_T: The Thomson cross-section. Unit: GeV^{-2}. Value: $6.6524587 \times 10^{-29}$ m^2. Status: STANDARD.
- δ_γ, θ_γ: The density contrast and the velocity divergence of the photons. Unit: dimensionless; 1/length. Status: STANDARD.
- Φ: The lapse potential of the Newtonian gauge (the perturbation of the time component, $1 + 2\Phi$). Unit: dimensionless. Value: at most 2.4×10^{-5}. Status: STANDARD.
- ρ_γ: The energy density of the photons, $\rho_\gamma = \pi^2 T_\gamma^4/15$. Unit: GeV^4. Status: STANDARD.
- ρ_b: The energy density of the baryons. Unit: GeV^4. Status: STANDARD.

The terms. The baryons (the up and down quarks confined in protons and neutrons, and the electrons): the same continuity equation, with the sound speed c_s of the baryon fluid and the Thomson drag of the photons (n_e the free-electron density, σ_T the Thomson cross-section, R_b the ratio of the photon and baryon inertia). Status: STANDARD.

$$\delta_\gamma' = -\frac{4}{3}\,\theta_\gamma + 4\Psi_\mathrm{N}', \qquad \theta_\gamma' = k^2\Big(\frac{1}{4}\,\delta_\gamma - \sigma_{\mathrm{an},\gamma}\Big) + k^2\Phi + \mathrm{e}^{a_4} n_\mathrm{e}\sigma_\mathrm{T}\big(\theta_\mathrm{b} - \theta_\gamma\big) \tag{FE-93}$$

Symbols in (FE-93).

- δ_γ, θ_γ: The density contrast and the velocity divergence of the photons. Unit: dimensionless; 1/length. Status: STANDARD.
- $'$: The prime on a function of the conformal time τ_c is its derivative $\mathrm{d}/\mathrm{d}\tau_\mathrm{c}$ (it is used only in the cosmological perturbation equations; two primes are the second derivative). Status: STANDARD.
- Ψ_N: The curvature potential of the Newtonian gauge (the perturbation of the 3-space metric, $1 - 2\Psi_\mathrm{N}$); not related to the field Ψ. Unit: dimensionless. Value: at most 2.4×10^{-5}. Status: STANDARD.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- $\sigma_{\mathrm{an},\gamma}$: The anisotropic stress (shear) of the photons, $F_2^\gamma/2$. Unit: dimensionless. Status: STANDARD.
- Φ: The lapse potential of the Newtonian gauge (the perturbation of the time component, $1 + 2\Phi$). Unit: dimensionless. Value: at most 2.4×10^{-5}. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- n_e: The number density of electrons (net of positrons); in the Thomson terms of the photon and baryon equations only free (ionised) electrons count. Unit: GeV^3. Status: STANDARD.

- σ_T: The Thomson cross-section. Unit: GeV^{-2}. Value: $6.6524587 \times 10^{-29}$ m^2. Status: STANDARD.
- δ_b, θ_b: The density contrast and the velocity divergence of the baryons (nucleons and electrons). Unit: dimensionless; 1/length. Status: STANDARD.

The terms. The photon density contrast and velocity divergence: radiation ($\frac{4}{3}$), the pressure gradient $\frac{1}{4}\delta_\gamma$, the anisotropic stress $\sigma_{\mathrm{an},\gamma}$, gravity, and the Thomson scattering on the free electrons. Status: STANDARD.

$$\begin{aligned} F_l^{\gamma\prime} &= \frac{k}{2l+1}\left[l\,F_{l-1}^\gamma - (l+1)F_{l+1}^\gamma\right] - \mathrm{e}^{a_4} n_\mathrm{e}\sigma_\mathrm{T}\Big[F_l^\gamma - \frac{\delta_{l2}}{10}\big(F_2^\gamma + G_0^\gamma + G_2^\gamma\big)\Big] \quad (l \geq 2), \\ F_0^\gamma &= \delta_\gamma, \qquad F_1^\gamma = \frac{4\theta_\gamma}{3k}, \qquad F_2^\gamma = 2\sigma_{\mathrm{an},\gamma} \end{aligned} \tag{FE-94}$$

Symbols in (FE-94).

- F_l^γ: The multipole l of the photon temperature perturbation (the brightness function of Ma and Bertschinger). Unit: dimensionless. Status: STANDARD.
- l: The multipole (angular moment) number $l = 0, 1, 2, \ldots$ of a Boltzmann hierarchy or of an angular moment $\int u^l \cdots$; as a coefficient it is the same number. Status: STANDARD.
- $'$: The prime on a function of the conformal time τ_c is its derivative $\mathrm{d}/\mathrm{d}\tau_\mathrm{c}$ (it is used only in the cosmological perturbation equations; two primes are the second derivative). Status: STANDARD.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- n_e: The number density of electrons (net of positrons); in the Thomson terms of the photon and baryon equations only free (ionised) electrons count. Unit: GeV^3. Status: STANDARD.
- σ_T: The Thomson cross-section. Unit: GeV^{-2}. Value: $6.6524587 \times 10^{-29}$ m^2. Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta_N^M = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{\alpha e}$, δ_{l0}, δ^{I3}); with upper and lower index strings the generalised Kronecker delta $\delta_{N_1\cdots N_n}^{M_1\cdots M_n}$, the determinant of the $n \times n$ matrix of Kronecker deltas $\delta_{N_b}^{M_a}$ (+1 or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- G_l^γ: The multipole l of the photon polarisation perturbation. Unit: dimensionless. Status: STANDARD.
- δ_γ, θ_γ: The density contrast and the velocity divergence of the photons. Unit: dimensionless; 1/length. Status: STANDARD.
- $\sigma_{\mathrm{an},\gamma}$: The anisotropic stress (shear) of the photons, $F_2^\gamma/2$. Unit: dimensionless. Status: STANDARD.

The terms. The Legendre multipoles F_l^γ of the photon temperature perturbation: free streaming couples each multipole to its neighbours, Thomson scattering damps them, and the quadrupole is fed by the polarisation (G_0^γ, G_2^γ). Status: STANDARD.

$$G_l^{\gamma\prime} = \frac{k}{2l+1}\left[l\,G_{l-1}^\gamma - (l+1)G_{l+1}^\gamma\right] + \mathrm{e}^{a_4} n_\mathrm{e}\sigma_\mathrm{T}\Big[-G_l^\gamma + \frac{1}{2}\big(F_2^\gamma + G_0^\gamma + G_2^\gamma\big)\Big(\delta_{l0} + \frac{\delta_{l2}}{5}\Big)\Big] \tag{FE-95}$$

Symbols in (FE-95).

- G_l^γ: The multipole l of the photon polarisation perturbation. Unit: dimensionless. Status: STANDARD.
- l: The multipole (angular moment) number $l = 0, 1, 2, \ldots$ of a Boltzmann hierarchy or of an angular moment $\int u^l \cdots$; as a coefficient it is the same number. Status: STANDARD.
- $'$: The prime on a function of the conformal time τ_c is its derivative $\mathrm{d}/\mathrm{d}\tau_\mathrm{c}$ (it is used only in the cosmological perturbation equations; two primes are the second derivative). Status: STANDARD.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.

- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- n_e: The number density of electrons (net of positrons); in the Thomson terms of the photon and baryon equations only free (ionised) electrons count. Unit: GeV^3. Status: STANDARD.
- σ_T: The Thomson cross-section. Unit: GeV^{-2}. Value: $6.6524587 \times 10^{-29}$ m^2. Status: STANDARD.
- F_l^γ: The multipole l of the photon temperature perturbation (the brightness function of Ma and Bertschinger). Unit: dimensionless. Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta_N^M = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{\alpha e}$, δ_{l0}, δ^{I3}); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1 \cdots M_n}_{N_1 \cdots N_n}$, the determinant of the $n \times n$ matrix of Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.

The terms. The multipoles G_l^γ of the photon polarisation: free streaming and Thomson scattering, which generates polarisation from the temperature quadrupole. Status: STANDARD.

$$\begin{aligned} \mathcal{N}_0^{(i)\prime} &= -\frac{k_\nu k}{E_i^\mathrm{c}} \mathcal{N}_1^{(i)} - \Psi_\mathrm{N}' \frac{\mathrm{d}\ln f_\mathrm{FD}}{\mathrm{d}\ln k_\nu}, \qquad \mathcal{N}_1^{(i)\prime} = \frac{k_\nu k}{3E_i^\mathrm{c}} \left(\mathcal{N}_0^{(i)} - 2\mathcal{N}_2^{(i)}\right) - \frac{E_i^\mathrm{c} k}{3k_\nu} \Phi \frac{\mathrm{d}\ln f_\mathrm{FD}}{\mathrm{d}\ln k_\nu}, \\ \mathcal{N}_l^{(i)\prime} &= \frac{k_\nu k}{(2l+1)E_i^\mathrm{c}} \left[l\,\mathcal{N}_{l-1}^{(i)} - (l+1)\mathcal{N}_{l+1}^{(i)}\right] \quad (l \geq 2), \qquad E_i^\mathrm{c} = \sqrt{k_\nu^2 + \mathrm{e}^{2a_4} m_i^2} \end{aligned} \tag{FE-96}$$

Symbols in (FE-96).

- $\mathcal{N}_l^{(i)}$: The multipole l of the perturbation of the distribution of the neutrino mass state i (a function of k, k_ν and τ_c; the quantity Ψ_l of Ma and Bertschinger). Unit: dimensionless. Status: STANDARD.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $'$: The prime on a function of the conformal time τ_c is its derivative $\mathrm{d}/\mathrm{d}\tau_\mathrm{c}$ (it is used only in the cosmological perturbation equations; two primes are the second derivative). Status: STANDARD.
- k_ν: The comoving momentum of a neutrino (the physical momentum times e^{a_4}). Unit: GeV. Status: STANDARD.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4} k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- E_i^c: The comoving energy of a neutrino of the mass state i, $E_i^\mathrm{c} = (k_\nu^2 + \mathrm{e}^{2a_4} m_i^2)^{1/2}$. Unit: GeV. Status: STANDARD.
- Ψ_N: The curvature potential of the Newtonian gauge (the perturbation of the 3-space metric, $1 - 2\Psi_\mathrm{N}$); not related to the field Ψ. Unit: dimensionless. Value: at most 2.4×10^{-5}. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- f_FD: The Fermi-Dirac distribution of the relic neutrinos, $f_\mathrm{FD} = 1/(\mathrm{e}^{k/T_\nu} + 1)$. Unit: dimensionless. Status: STANDARD.
- Φ: The lapse potential of the Newtonian gauge (the perturbation of the time component, $1 + 2\Phi$). Unit: dimensionless. Value: at most 2.4×10^{-5}. Status: STANDARD.
- l: The multipole (angular moment) number $l = 0, 1, 2, \ldots$ of a Boltzmann hierarchy or of an angular moment $\int u^l \cdots$; as a coefficient it is the same number. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- m, m_s, m_i: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.

The terms. Each neutrino mass state ν_i (after decoupling the flavour coherence is lost and each mass state propagates on its own) has the multipoles $\mathcal{N}_l^{(i)}$ of its perturbed distribution at the comoving momentum k_ν; E_i^c is

its comoving energy with the mass m_i (FE-100), f_{FD} the Fermi-Dirac distribution. The masses make the neutrinos slow at late times (they then cluster like matter). The fold bands of dirac16complex obey the massless limit. Status: STANDARD (each mass state with the degeneracy $N_{\mathrm{eff}}/3 = 1.0144$; the masses in CAMB agree with the x_0 masses to 7.8×10^{-12}).

$$a_4'^2 = \frac{8\pi G_{\mathrm{N}}}{3}\,\mathrm{e}^{2a_4}\rho_{\mathrm{tot}} + \frac{\Lambda_4}{3}\,\mathrm{e}^{2a_4}, \qquad T_\nu = \Big(\frac{4}{11}\Big)^{1/3} T_\gamma, \qquad P_{\mathcal{R}}(k) = A_{\mathrm{s}}\Big(\frac{k}{k_0}\Big)^{n_{\mathrm{s}}-1} \tag{FE-97}$$

Symbols in (FE-97).

- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- $'$: The prime on a function of the conformal time τ_{c} is its derivative $\mathrm{d}/\mathrm{d}\tau_{\mathrm{c}}$ (it is used only in the cosmological perturbation equations; two primes are the second derivative). Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_{N}: Newton's constant of four-dimensional gravity, $8\pi G_{\mathrm{N}} = \kappa/(\mathcal{G}(K_*)V_\perp)$. Unit: GeV^{-2}. Value: 6.709×10^{-39} GeV^{-2} ($M_{\mathrm{Pl}} = G_{\mathrm{N}}^{-1/2} = 1.22091 \times 10^{19}$ GeV). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $\rho_{\mathrm{tot}},\ p_{\mathrm{tot}}$: The total background energy density and pressure of four-dimensional cosmology (all species). Unit: GeV^4. Status: STANDARD.
- Λ_4: The four-dimensional cosmological constant of the effective theory on the block, $\Lambda_4 = [6H^2K(1 - 2w_{(2)}K) + \Lambda + \kappa\rho_{\mathrm{stab}}]/\mathcal{G}(K)$; zero at the static point, the observed dark energy a tuned residual. Unit: GeV^2. Value: 0 at the static point; observed $\Lambda_4/H^2 \sim 10^{-92}$. Status: DERIVED.
- T_ν: The temperature of the relic neutrinos (one common temperature). Unit: GeV. Value: 1.9454 K today. Status: STANDARD.
- T_γ: The temperature of the photons (equal to the plasma temperature T; the subscript is written where the neutrinos have their own temperature). Unit: GeV. Value: 2.7255 K today. Status: STANDARD.
- $P_{\mathcal{R}}(k)$: The dimensionless power spectrum of the primordial curvature perturbation $\mathcal{R}$. Unit: dimensionless. Status: STANDARD.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- A_{s}: The amplitude of the primordial spectrum at k_0. Unit: dimensionless. Value: 2.099×10^{-9} ($\ln(10^{10}A_{\mathrm{s}}) = 3.044$, Planck 2018). Status: STANDARD.
- k_0: The pivot wavenumber of the primordial spectrum. Unit: 1/length. Value: 0.05 Mpc^{-1}. Status: STANDARD.
- n_{s}: The spectral index of the primordial spectrum. Unit: dimensionless. Value: 0.9649 (Planck 2018). Status: STANDARD.

The terms. The four-dimensional Friedmann equation in conformal time (from FE-38 at first order with FE-80; Λ_4 the tuned dark energy, FE-48), the temperature of the neutrinos after the electron-positron annihilation, and the primordial curvature spectrum with the amplitude $A_{\mathrm{s}} = 2.099 \times 10^{-9}$ and the index $n_{\mathrm{s}} = 0.9649$ at the pivot k_0. Status: STANDARD (inputs: the Planck 2018 primordial parameters; $N_{\mathrm{eff}} = 3.0432$ in the neutrinos and 0.0010 in the fold bands; results: $\mathcal{H}_0 = 67.38$ km/s/Mpc, the sound horizon at the drag epoch 147.10 Mpc, BAO $\chi^2 = 31.5$ for 13 DESI DR2 points).

$$\begin{aligned}
&\Phi_{\mathrm{rms}}(k) = \Big[\frac{k^3 P_{\Phi_{\mathrm{W}}}(k)}{2\pi^2}\Big]^{1/2}, \qquad \Phi_{\mathrm{W}} = \frac{\Phi + \Psi_{\mathrm{N}}}{2}, \qquad \Phi_{\mathrm{W}} = \frac{3}{2}\,\Omega_{\mathrm{m}}\mathcal{H}_0^2\,\frac{\delta_{\mathrm{m}}}{\mathrm{e}^{a_4}k^2} \quad \text{(inside the horizon)},\\
&\Phi_{\mathrm{W}} = \frac{3}{5}\,\mathcal{R}\,\mathrm{e}^{-a_4}D_+ \quad \text{(largest scales)}, \qquad D_+ = \frac{5}{2}\,\Omega_{\mathrm{m}}\,\frac{\mathcal{H}}{\mathcal{H}_0}\int_{-\infty}^{a_4}\Big(\frac{\mathcal{H}_0}{\mathcal{H}}\Big)^3 \mathrm{e}^{-2\tilde{a}_4}\,\mathrm{d}\tilde{a}_4,\\
&\frac{\mathcal{H}}{\mathcal{H}_0} = \big(\Omega_{\mathrm{m}}\mathrm{e}^{-3a_4} + 1 - \Omega_{\mathrm{m}}\big)^{1/2}
\end{aligned} \tag{FE-98}$$

Symbols in (FE-98).

- $\Phi_{\rm rms}(k)$: The root-mean-square amplitude of the Weyl potential per logarithmic interval of k. Unit: dimensionless. Value: 2.42×10^{-5} at $k = 10^{-4}$ Mpc^{-1}. Status: STANDARD.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- $P_{\Phi_{\rm W}}(k)$: The power spectrum of the Weyl potential. Unit: length^3. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\Phi_{\rm W}$: The Weyl potential, $\Phi_{\rm W} = (\Phi + \Psi_{\rm N})/2$ (it deflects light). Unit: dimensionless. Status: STANDARD.
- Φ: The lapse potential of the Newtonian gauge (the perturbation of the time component, $1 + 2\Phi$). Unit: dimensionless. Value: at most 2.4×10^{-5}. Status: STANDARD.
- $\Psi_{\rm N}$: The curvature potential of the Newtonian gauge (the perturbation of the 3-space metric, $1 - 2\Psi_{\rm N}$); not related to the field Ψ. Unit: dimensionless. Value: at most 2.4×10^{-5}. Status: STANDARD.
- $\Omega_{\rm m}$: The matter density today in units of the critical density. Unit: dimensionless. Value: 0.3150. Status: STANDARD.
- $\mathcal{H}_0$: The Hubble rate today (the Hubble constant). Unit: GeV (km/s/Mpc). Value: 67.38 km/s/Mpc. Status: DERIVED.
- $\delta_{\rm m}$: The density contrast of the total matter (dark matter and baryons). Unit: dimensionless. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{R}$: The primordial comoving curvature perturbation. Unit: dimensionless. Status: STANDARD.
- D_+: The linear growth factor of the matter perturbations, normalised as $D_+ = \mathrm{e}^{a_4}$ in the matter era. Unit: dimensionless. Value: $D_+\mathrm{e}^{-a_4} = 0.7878$ today. Status: STANDARD.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\rm Pl}$). Unit: GeV. Status: DERIVED.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no upper bound. Status: STANDARD.
- $\tilde{a}_4$: An earlier value of a_4 (the integration variable of the growth integral). Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.

The terms. The size of the first-order potentials: $\Phi_{\rm rms}$ is the rms Weyl potential $\Phi_{\rm W}$ per logarithmic wavenumber, from its power spectrum $P_{\Phi_{\rm W}}$. Inside the horizon it follows from the matter density contrast $\delta_{\rm m}$ (the Poisson equation, from FE-90), on the largest scales from the primordial curvature perturbation $\mathcal{R}$ with the growth factor D_+, which the dark energy suppresses at late times. The relations are written for the magnitudes: in the sign conventions of FE-89 and FE-90 the potential of an overdensity is negative; only the rms value is used. Status: STANDARD. $\Phi_{\rm rms} = 2.42 \times 10^{-5}$ at $k = 10^{-4}$ Mpc^{-1} down to 5.03×10^{-8} at 1 Mpc^{-1}; the Poisson form agrees with the CAMB solution to 1.5×10^{-6} for $k \geq 0.01$ Mpc^{-1} (the residual of the stored first-order solution in this equation) and the large-scale form to 4.4×10^{-5} ($D_+\mathrm{e}^{-a_4} = 0.7878$ today, with $\Omega_{\rm m} = 0.3150$).

$$\sigma_8^2 = \int_0^\infty \frac{\mathrm{d}k}{k}\,\frac{k^3 P_{\rm m}(k)}{2\pi^2}\,W_{\rm th}(kR_8)^2, \qquad W_{\rm th}(kR_8) = \frac{3\big[\sin(kR_8) - kR_8\cos(kR_8)\big]}{(kR_8)^3}, \qquad R_8 = 8\,h_{100}^{-1}\,\mathrm{Mpc} \quad \text{(FE-99)}$$

Symbols in (FE-99).

- σ_8: The root-mean-square linear matter fluctuation in spheres of radius R_8 today. Unit: dimensionless. Value: 0.8118. Status: DERIVED.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no upper bound. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.

- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4} k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- $P_{\mathrm{m}}(k)$: The linear matter power spectrum today. Unit: length^3. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- W_{th}: The Fourier transform of a spherical top-hat window. Unit: dimensionless. Status: STANDARD.
- R_8: The radius of the spheres of σ_8, $R_8 = 8\,h_{100}^{-1}$ Mpc. Unit: length. Status: STANDARD.
- h_{100}: The reduced Hubble constant, $\mathcal{H}_0/(100\ \mathrm{km\ s^{-1}\ Mpc^{-1}})$. Unit: dimensionless. Value: 0.6738. Status: STANDARD.

The terms. The rms density fluctuation in spheres of radius $8\,h_{100}^{-1}$ Mpc, from the linear matter power spectrum P_{m} and the top-hat window W_{th}. Status: STANDARD. $\sigma_8 = 0.8118$ for UM-2 (0.8119 without the fold bands) against 0.8109 for ΛCDM.

9 The three neutrino flavours

The three flavours of left-handed neutrinos are the upper members of the lepton doublets ℓ_i. Their masses come from the tip mixing with the right-handed neutrinos (FE-59, FE-60), their mixing from the tip matrix $\mathcal{K}_\nu = U$, and their interactions from the exchange of W^+, W^- and Z. This section collects the equations of their masses, of their interactions at low energy and of their flavour evolution in vacuum, in matter, in the early Universe and in dense neutrino gases, and the cosmic neutrino background.

9.1 The masses and the interactions at low energy

$$m_1 = 1.000\ \mathrm{meV}, \qquad m_2 = \sqrt{m_1^2 + \Delta m_{21}^2} = 8.712\ \mathrm{meV}, \qquad m_3 = \sqrt{m_1^2 + \Delta m_{31}^2} = 50.140\ \mathrm{meV} \tag{FE-100}$$

Symbols in (FE-100).

- $m_1,\ m_2,\ m_3$: The masses of the neutrino mass states ν_1, ν_2, ν_3 (normal ordering). Unit: GeV (quoted in meV). Value: 1.000, 8.712, 50.140 meV (m_1 ASSUMED). Status: DERIVED.
- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- $\Delta m_{21}^2,\ \Delta m_{31}^2,\ \Delta m_{32}^2$: The differences of the squared neutrino masses, $\Delta m_{ij}^2 = m_i^2 - m_j^2$. Unit: GeV^2 (quoted in eV^2). Value: $\Delta m_{21}^2 = 7.49 \times 10^{-5}$ eV^2, $\Delta m_{31}^2 = 2.513 \times 10^{-3}$ eV^2 (NuFIT 6.0). Status: STANDARD.

The terms. The masses of the three mass states ν_1, ν_2, ν_3: the lightest mass m_1 is chosen and the ordering is normal (ASSUMED), the two splittings $\Delta m_{21}^2 = 7.49 \times 10^{-5}$ eV^2 and $\Delta m_{31}^2 = 2.513 \times 10^{-3}$ eV^2 are the measured ones (NuFIT 6.0). In UM-2 they are Dirac masses and x_0 eigenvalues of FE-59: the doublet localisation $c_L = 5.95288$ of the leptons and the right-handed localisations $c_R = -35.0452$, -32.8481, -31.0702, with tip mixing of order one; the smallness of the masses comes from the sharp localisation of the right-handed neutrinos at the fold. Status: ASSUMED (m_1, ordering), STANDARD (splittings), DERIVED (as x_0 eigenvalues). The same masses enter the flavour evolution, the cosmic neutrino background and the CMB computation (they agree across the three to 5.7×10^{-12}).

$$\mathcal{L}_{\mathrm{CC}} = -2\sqrt{2}\,G_{\mathrm{F}}\big(\bar{\nu}_\alpha \gamma^\mu P_L\, \ell_\alpha^-\big)\big(\bar{\ell}_\beta^- \gamma_\mu P_L\, \nu_\beta\big), \qquad \mathcal{L}_{\mathrm{NC}} = -2\sqrt{2}\,G_{\mathrm{F}}\big(\bar{\nu}_\alpha \gamma^\mu P_L\, \nu_\alpha\big) \sum_f \bar{\psi}_f \gamma_\mu \big(\tilde{g}_L^f P_L + \tilde{g}_R^f P_R\big)\psi_f,$$

$$\nu_\alpha = \sum_{i=1}^{3} U_{\alpha i}\, \nu_i, \qquad P_L = \frac{1+\Gamma_4}{2}, \qquad P_R = \frac{1-\Gamma_4}{2} \tag{FE-101}$$

Symbols in (FE-101).

- $\mathcal{L}_{\rm CC}$, $\mathcal{L}_{\rm NC}$: The charged-current and neutral-current four-fermion Lagrangians (exchange of $W^\pm$ and of Z far below their masses). Unit: GeV^4. Status: STANDARD.
- $G_{\rm F}$: The Fermi constant. Unit: GeV^{-2}. Value: 1.1663788×10^{-5} GeV^{-2} (measured). Status: STANDARD.
- $\bar\nu_\alpha$: The antineutrino of the flavour α; in a Lagrangian the Dirac adjoint of the neutrino field (it annihilates antineutrinos). Unit: GeV$^{3/2}$. Status: STANDARD.
- α: A flavour index running over e, μ, τ (the flavour states of the leptons and neutrinos). Status: STANDARD.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- P_L, P_R: The projectors on the four-dimensional chiralities, $P_L = (1+\Gamma_4)/2$ (left-handed) and $P_R = (1-\Gamma_4)/2$ (right-handed). Unit: dimensionless. Status: STANDARD.
- L: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1+\Gamma_4)/2$. Status: STANDARD.
- ℓ^-_α, $\bar\ell^-_\alpha$: The charged lepton of the flavour α (e^-, μ^-, τ^-; the lower member of the lepton doublet) at low energy, and its Dirac adjoint. Unit: GeV$^{3/2}$. Status: STANDARD.
- β: A second flavour index running over e, μ, τ. Status: STANDARD.
- ν_α, ν_i: A left-handed neutrino: ν_α of the flavour $\alpha = e, \mu, \tau$ (ν_e, ν_μ, ν_τ), ν_i of the mass state i (mass m_i). Unit: GeV$^{3/2}$. Status: STANDARD.
- f: The label of a fermion species: a Standard-Model multiplet of one generation (its $N_{\rm c}$ Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over f runs over these species. Status: STANDARD.
- ψ_i, $\bar\psi_i$: A Standard-Model fermion field of eight dimensions: ψ_i is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation i, ψ_f one fermion component f (in four-dimensional formulas its zero mode); $\bar\psi = \psi^\dagger C$ the Dirac adjoint. Unit: GeV$^{7/2}$ (eight dimensions), GeV$^{3/2}$ (four dimensions). Status: ASSUMED.
- $\tilde g^f_L$, $\tilde g^f_R$, $\tilde g^f_V$: The neutral-current couplings of the fermion f to the Z (left-handed, right-handed and vector), $\tilde g^f_L = T^f_3 - Q^f_{\rm em}\sin^2\theta_{\rm W}$. Unit: dimensionless. Status: STANDARD.
- R: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1-\Gamma_4)/2$. Status: STANDARD.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- U: The PMNS matrix, the unitary 3×3 mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states i); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\rm CP} = 212^\circ$ (NuFIT 6.0). Status: STANDARD.
- Γ_4: The four-dimensional chirality matrix, $\Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$; a doublet zero mode has $\Gamma_4 = +1$ (left-handed), a singlet zero mode $\Gamma_4 = -1$ (right-handed). Unit: dimensionless. Status: DERIVED.

The terms. At momentum transfers far below $m_{\rm W}$ and $m_{\rm Z}$ the exchange of $W^\pm$ (FE-25) between the zero modes of the lepton doublets gives the charged-current contact interaction $\mathcal{L}_{\rm CC}$ between the neutrino and the charged lepton of the same flavour, and the exchange of the Z the neutral-current interaction $\mathcal{L}_{\rm NC}$ of the neutrinos with every fermion f of the medium, with the chiral couplings $\tilde g^f_L$ and $\tilde g^f_R$. P_L and P_R project on the left-handed and right-handed zero modes. The flavour states ν_α ($\alpha = e, \mu, \tau$) are the combinations of the mass states with the mixing matrix U, the tip matrix of the neutrino sector. Status: STANDARD.

$$\frac{G_{\rm F}}{\sqrt{2}} = \frac{g_2^2}{8m_{\rm W}^2}, \qquad \tilde g^f_L = T^f_3 - Q^f_{\rm em}\sin^2\theta_{\rm W}, \qquad \tilde g^f_R = -Q^f_{\rm em}\sin^2\theta_{\rm W}, \qquad \tilde g^f_V = \tilde g^f_L + \tilde g^f_R = T^f_3 - 2Q^f_{\rm em}\sin^2\theta_{\rm W} \tag{FE-102}$$

Symbols in (FE-102).

- $G_{\rm F}$: The Fermi constant. Unit: GeV^{-2}. Value: 1.1663788×10^{-5} GeV^{-2} (measured). Status: STANDARD.
- g_2: The four-dimensional SU(2) coupling ($g_2^2 = 4\pi\alpha_2$). Unit: dimensionless. Value: $g_2^2 = 0.3781$ at $2M_{\rm DM}$. Status: DERIVED.
- $m_{\rm W}$: The mass of W^+ and W^-. Unit: GeV. Value: 80.3625 GeV (measured, input). Status: STANDARD.

- $\tilde{g}_L^f$, $\tilde{g}_R^f$, $\tilde{g}_V^f$: The neutral-current couplings of the fermion f to the Z (left-handed, right-handed and vector), $\tilde{g}_L^f = T_3^f - Q_{\mathrm{em}}^f \sin^2\theta_{\mathrm{W}}$. Unit: dimensionless. Status: STANDARD.
- f: The label of a fermion species: a Standard-Model multiplet of one generation (its N_{c} Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over f runs over these species. Status: STANDARD.
- L: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1+\Gamma_4)/2$. Status: STANDARD.
- T_3: The third component of the weak isospin ($+\frac{1}{2}$ for the upper and $-\frac{1}{2}$ for the lower member of a doublet, 0 for a singlet); T_3^f that of the fermion f. Unit: dimensionless. Status: STANDARD.
- Q_{em}: The electric charge in units of the elementary charge, $Q_{\mathrm{em}} = T_3 + Y$; Q_{em}^f that of the fermion f. Unit: dimensionless. Status: STANDARD.
- θ_{W}: The weak mixing angle of the couplings, $\cos\theta_{\mathrm{W}} = \hat{g}_2/\hat{g}_Z$, $\sin\theta_{\mathrm{W}} = \hat{g}_1/\hat{g}_Z$; fixed by the measured W and Z masses with $\rho_{\mathrm{EW},0}$. Unit: dimensionless. Value: $\cos^2\theta_{\mathrm{W}} = 0.776368$. Status: DERIVED.
- R: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1-\Gamma_4)/2$. Status: STANDARD.
- V: The label V of the vector coupling of a fermion to the Z, $\tilde{g}_V^f = \tilde{g}_L^f + \tilde{g}_R^f$. Status: STANDARD.

The terms. The Fermi constant is fixed by the four-dimensional SU(2) coupling and the W mass; the neutral-current couplings of a fermion with the weak isospin T_3^f and the charge Q_{em}^f contain the weak angle; $\tilde{g}_V^f$ is the vector coupling that acts in an unpolarised medium. Status: STANDARD; UM-2 uses the measured $G_{\mathrm{F}} = 1.1663788 \times 10^{-5}\ \mathrm{GeV}^{-2}$. The weak angle has two values in UM-2: the tree-level angle of the x_0 masses ($\sin^2\theta_{\mathrm{W}} = 1 - 0.776368 = 0.22363$, FE-55) and the $\overline{\mathrm{MS}}$ angle $\sin^2\theta_{\overline{\mathrm{MS}}} = 0.23122$ used in the couplings of the Z widths; the difference changes the total Z width by 8.3×10^{-3}, the size of the electroweak loop corrections that UM-2 does not compute.

9.2 Vacuum and matter

$$V_\alpha = \sqrt{2}\,G_{\mathrm{F}}\Big[\delta_{\alpha e}\big(n_{e^-} - n_{e^+}\big) + \sum_f \tilde{g}_V^f\big(n_f - n_{\bar{f}}\big)\Big],$$
$$V = \mathrm{diag}\big(V_{\mathrm{CC}},\ 0,\ 0\big) + V_{\mathrm{NC}}\,\mathbf{1}, \qquad V_{\mathrm{CC}} = \sqrt{2}\,G_{\mathrm{F}}\,n_{\mathrm{e}}, \qquad V_{\mathrm{NC}} = -\frac{G_{\mathrm{F}}\,n_{\mathrm{n}}}{\sqrt{2}} \tag{FE-103}$$

Symbols in (FE-103).

- V_α, V: The matter potential of forward scattering: V_α of the flavour α (V_e, V_μ), V the 3×3 matrix of the potentials in the flavour basis; antineutrinos have $-V$. Unit: GeV. Status: STANDARD.
- α: A flavour index running over e, μ, τ (the flavour states of the leptons and neutrinos). Status: STANDARD.
- G_{F}: The Fermi constant. Unit: GeV^{-2}. Value: $1.1663788 \times 10^{-5}\ \mathrm{GeV}^{-2}$ (measured). Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta_N^M = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{\alpha e}$, δ_{l0}, δ^{I3}); with upper and lower index strings the generalised Kronecker delta $\delta_{N_1\cdots N_n}^{M_1\cdots M_n}$, the determinant of the $n \times n$ matrix of Kronecker deltas $\delta_{N_b}^{M_a}$ (+1 or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- e: The electron flavour as a value of a flavour index ($\delta_{\alpha e}$, U_{e2}, V_e, $\mathcal{P}_{ee}$). Status: STANDARD.
- n_{e^-}, n_{e^+}: The number densities of electrons and of positrons. Unit: GeV^3. Status: STANDARD.
- f: The label of a fermion species: a Standard-Model multiplet of one generation (its N_{c} Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over f runs over these species. Status: STANDARD.
- $\tilde{g}_L^f$, $\tilde{g}_R^f$, $\tilde{g}_V^f$: The neutral-current couplings of the fermion f to the Z (left-handed, right-handed and vector), $\tilde{g}_L^f = T_3^f - Q_{\mathrm{em}}^f \sin^2\theta_{\mathrm{W}}$. Unit: dimensionless. Status: STANDARD.
- V: The label V of the vector coupling of a fermion to the Z, $\tilde{g}_V^f = \tilde{g}_L^f + \tilde{g}_R^f$. Status: STANDARD.
- n, n_o, $n_{\psi i}$, n_f, n_i: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and n_o the contribution of the orbital o, $n_{\psi i}(y)$ the density of the Standard-Model fermion ψ_i (per proper 7-volume of eight dimensions); n_f the density of the fermion component f and n_i that of the neutrino mass state i (per 3-volume of four dimensions). Unit: GeV^7 (eight dimensions), GeV^3 (four dimensions). Status: STANDARD.
- $n_{\bar{f}}$: The number density of the antiparticles of the fermion f. Unit: GeV^3. Status: STANDARD.

- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- V_{CC}, V_{NC}: The charged-current potential of electron neutrinos and the neutral-current potential common to all flavours. Unit: GeV. Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- n_{e}: The number density of electrons (net of positrons); in the Thomson terms of the photon and baryon equations only free (ionised) electrons count. Unit: GeV^3. Status: STANDARD.
- n_{n}: The number density of neutrons (free and bound). Unit: GeV^3. Status: STANDARD.

The terms. Coherent forward scattering on the medium gives each flavour the potential V_α: the charged current acts on ν_e only, with the net electron density; the neutral current acts on every flavour with the net densities $n_f - n_{\bar{f}}$ of the fermions of the medium weighted by $\tilde{g}_V^f$. In neutral matter of electrons, protons and neutrons the neutral-current terms of the electrons and the protons cancel ($\tilde{g}_V$ of the proton, two up quarks and one down quark, is minus that of the electron) and the neutrons leave $V_{\mathrm{NC}} = -G_{\mathrm{F}} n_{\mathrm{n}}/\sqrt{2}$, common to all flavours. Status: STANDARD (an unpolarised medium at rest; antineutrinos have $-V$).

$$\mathrm{i}\,\frac{\mathrm{d}\boldsymbol{\psi}}{\mathrm{d}x_4} = H_{\mathrm{fl}}\,\boldsymbol{\psi}, \qquad H_{\mathrm{fl}} = \frac{1}{2E}\,U\,\mathrm{diag}\big(m_1^2,\ m_2^2,\ m_3^2\big)\,U^\dagger + V, \qquad \mathcal{U}(L_{\mathrm{b}}) = \mathrm{e}^{-\mathrm{i}H_{\mathrm{fl}}L_{\mathrm{b}}}, \qquad \mathcal{P}(\nu_\alpha \to \nu_\beta) = |\mathcal{U}_{\beta\alpha}|^2 \tag{FE-104}$$

Symbols in (FE-104).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $\boldsymbol{\psi}$: The vector of the three flavour amplitudes of a neutrino. Unit: dimensionless. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- H_{fl}: The flavour Hamiltonian of a neutrino of energy E (vacuum mass term plus matter potential). Unit: GeV. Status: STANDARD.
- E: The energy of a particle: of the neutrino in the flavour equations, $E = (k^2 + m^2)^{1/2}$ of a particle of momentum k in the thermal integrals. Unit: GeV. Status: STANDARD.
- U: The PMNS matrix, the unitary 3×3 mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states i); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212^\circ$ (NuFIT 6.0). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- m_1, m_2, m_3: The masses of the neutrino mass states ν_1, ν_2, ν_3 (normal ordering). Unit: GeV (quoted in meV). Value: 1.000, 8.712, 50.140 meV (m_1 ASSUMED). Status: DERIVED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- V_α, V: The matter potential of forward scattering: V_α of the flavour α (V_e, V_μ), V the 3×3 matrix of the potentials in the flavour basis; antineutrinos have $-V$. Unit: GeV. Status: STANDARD.
- $\mathcal{U}$: The flavour evolution operator over the baseline L_{b}; $\mathcal{U}_{\beta\alpha}$ the amplitude of $\nu_\alpha \to \nu_\beta$. Unit: dimensionless. Status: STANDARD.
- L_{b}: The baseline: the distance travelled by the neutrino. Unit: length. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $\mathcal{P}$: The probability of a flavour transition, $\mathcal{P}(\nu_\alpha \to \nu_\beta)$; $\mathcal{P}_{ee}$ the survival probability of electron neutrinos. Unit: dimensionless. Status: STANDARD.
- ν_α, ν_i: A left-handed neutrino: ν_α of the flavour $\alpha = e, \mu, \tau$ (ν_e, ν_μ, ν_τ), ν_i of the mass state i (mass m_i). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- α: A flavour index running over e, μ, τ (the flavour states of the leptons and neutrinos). Status: STANDARD.
- β: A second flavour index running over e, μ, τ. Status: STANDARD.

The terms. $\boldsymbol{\psi}$ is the vector of the three flavour amplitudes of a neutrino of energy E; the flavour Hamiltonian H_{fl} contains the vacuum part (the squared masses rotated with U) and the matter potential V. For a constant density

the evolution over the baseline L_b is the matrix exponential $\mathcal{U}$, and $\mathcal{P}$ is the probability of the flavour change. The common part $V_\mathrm{NC}\mathbf{1}$ is a phase and drops out. Status: STANDARD (solved exactly: matrix exponential for constant density, ordered products of layers otherwise; the vacuum formula FE-105 is reproduced to 3.9×10^{-15}, unitarity to 1.8×10^{-15}). Residual of the stored probabilities (unitarity, vacuum formula, CPT, the T2K, NOvA and DUNE probabilities, the solar survival probability) 7.1×10^{-12}.

$$\mathcal{P}(\nu_\alpha\to\nu_\beta)=\Big|\sum_{i=1}^{3}U_{\beta i}U_{\alpha i}^{*}\,\mathrm{e}^{-\mathrm{i}m_i^2L_\mathrm{b}/(2E)}\Big|^2,\qquad \mathcal{P}(\nu_\alpha\to\nu_\beta)=\mathcal{P}(\bar\nu_\beta\to\bar\nu_\alpha),$$

$$\mathcal{P}(\nu_\mu\to\nu_e)-\mathcal{P}(\nu_e\to\nu_\mu)=-16J_\mathrm{CP}\sin\Delta_{21}\sin\Delta_{31}\sin\Delta_{32},\qquad \Delta_{ij}=\frac{\Delta m_{ij}^2L_\mathrm{b}}{4E},\qquad J_\mathrm{CP}=\mathrm{Im}\big(U_{e2}U_{\mu3}U_{e3}^{*}U_{\mu2}^{*}\big) \tag{FE-105}$$

Symbols in (FE-105).

- $\mathcal{P}$: The probability of a flavour transition, $\mathcal{P}(\nu_\alpha\to\nu_\beta)$; $\mathcal{P}_{ee}$ the survival probability of electron neutrinos. Unit: dimensionless. Status: STANDARD.
- ν_α, ν_i: A left-handed neutrino: ν_α of the flavour $\alpha=e,\mu,\tau$ (ν_e, ν_μ, ν_τ), ν_i of the mass state i (mass m_i). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- α: A flavour index running over e,μ,τ (the flavour states of the leptons and neutrinos). Status: STANDARD.
- β: A second flavour index running over e,μ,τ. Status: STANDARD.
- i: The generation index ($i=1,2,3$), which also labels the neutrino mass states ν_1,ν_2,ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- U: The PMNS matrix, the unitary 3×3 mixing matrix of the leptons, $\nu_\alpha=\sum_i U_{\alpha i}\nu_i$ (rows $\alpha=e,\mu,\tau$, columns the mass states i); in UM-2 $U=\mathcal{K}_e^\dagger\mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12}=0.308$, $\sin^2\theta_{13}=0.02215$, $\sin^2\theta_{23}=0.470$, $\delta_\mathrm{CP}=212^\circ$ (NuFIT 6.0). Status: STANDARD.
- *: Complex conjugation of a number or of each entry of a matrix. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- m, m_s, m_i: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.
- L_b: The baseline: the distance travelled by the neutrino. Unit: length. Status: STANDARD.
- E: The energy of a particle: of the neutrino in the flavour equations, $E=(k^2+m^2)^{1/2}$ of a particle of momentum k in the thermal integrals. Unit: GeV. Status: STANDARD.
- $\bar\nu_\alpha$: The antineutrino of the flavour α; in a Lagrangian the Dirac adjoint of the neutrino field (it annihilates antineutrinos). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- e: The electron flavour as a value of a flavour index ($\delta_{\alpha e}$, U_{e2}, V_e, $\mathcal{P}_{ee}$). Status: STANDARD.
- J_CP: The Jarlskog invariant of a mixing matrix (the measure of CP violation). Unit: dimensionless. Value: -0.01777 (PMNS), 3.117×10^{-5} (CKM). Status: STANDARD.
- Δ_{ij}: The oscillation phase of the pair of mass states (i,j), $\Delta_{ij}=\Delta m_{ij}^2L_\mathrm{b}/(4E)$. Unit: dimensionless. Status: STANDARD.
- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2=m_2^2-m_1^2$ a difference of squared masses, ΔN_eff an addition to N_eff, ΔS_m the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- j: A second generation index ($j=1,2,3$). Status: STANDARD.
- Im: The imaginary part of a complex number. Status: STANDARD.

The terms. In vacuum the probability is the squared sum of three amplitudes with the phases $m_i^2L_\mathrm{b}/(2E)$; CPT symmetry relates neutrinos and antineutrinos; the difference of the probabilities of a process and its time reverse is proportional to the Jarlskog invariant J_CP (CP violation of the tip matrix) and to the three oscillation phases Δ_{ij}. Status: STANDARD (CPT checked to 2.8×10^{-16}, the T-odd difference to 5.9×10^{-14}).

In the two-flavour approximation of one sector (for example the 1-2 sector with $\theta_{13}=0$) the flavour Hamiltonian of FE-104 is, up to a multiple of $\mathbf{1}$,

$$H_{\mathrm{fl}} = \frac{\Delta m^2_{21}}{4E}\begin{pmatrix} -\cos 2\theta_{12} & \sin 2\theta_{12} \\ \sin 2\theta_{12} & \cos 2\theta_{12} \end{pmatrix} + \begin{pmatrix} V_{\mathrm{CC}} & 0 \\ 0 & 0 \end{pmatrix},$$

Symbols in this equation.

- H_{fl}: The flavour Hamiltonian of a neutrino of energy E (vacuum mass term plus matter potential). Unit: GeV. Status: STANDARD.
- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21} = m^2_2 - m^2_1$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- Δm^2_{21}, Δm^2_{31}, Δm^2_{32}: The differences of the squared neutrino masses, $\Delta m^2_{ij} = m^2_i - m^2_j$. Unit: GeV^2 (quoted in eV^2). Value: $\Delta m^2_{21} = 7.49 \times 10^{-5}$ eV^2, $\Delta m^2_{31} = 2.513 \times 10^{-3}$ eV^2 (NuFIT 6.0). Status: STANDARD.
- E: The energy of a particle: of the neutrino in the flavour equations, $E = (k^2 + m^2)^{1/2}$ of a particle of momentum k in the thermal integrals. Unit: GeV. Status: STANDARD.
- θ_{ij}: The mixing angles of the standard parametrisation of U (or of V_{CKM}): θ_{12}, θ_{13}, θ_{23}; in two-flavour formulas θ_{ij} is the angle of the sector (i, j). Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$ (PMNS). Status: STANDARD.
- V_{CC}, V_{NC}: The charged-current potential of electron neutrinos and the neutral-current potential common to all flavours. Unit: GeV. Status: STANDARD.

whose two diagonal entries are equal, and the mixing in matter maximal, at $2EV_{\mathrm{CC}} = \Delta m^2_{21}\cos 2\theta_{12}$; with the third flavour the conditions become:

$$2E\,V_{\mathrm{CC}} = \Delta m^2_{ee}\cos 2\theta_{13}, \qquad \Delta m^2_{ee} = c^2_{12}\,\Delta m^2_{31} + s^2_{12}\,\Delta m^2_{32}, \qquad 2E\,V_{\mathrm{CC}}\,c^2_{13} = \Delta m^2_{21}\cos 2\theta_{12} \tag{FE-106}$$

Symbols in (FE-106).

- E: The energy of a particle: of the neutrino in the flavour equations, $E = (k^2 + m^2)^{1/2}$ of a particle of momentum k in the thermal integrals. Unit: GeV. Status: STANDARD.
- V_{CC}, V_{NC}: The charged-current potential of electron neutrinos and the neutral-current potential common to all flavours. Unit: GeV. Status: STANDARD.
- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21} = m^2_2 - m^2_1$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- Δm^2_{ee}: The effective squared-mass difference of electron-neutrino disappearance, $\Delta m^2_{ee} = c^2_{12}\Delta m^2_{31} + s^2_{12}\Delta m^2_{32}$. Unit: GeV^2. Status: STANDARD.
- θ_{ij}: The mixing angles of the standard parametrisation of U (or of V_{CKM}): θ_{12}, θ_{13}, θ_{23}; in two-flavour formulas θ_{ij} is the angle of the sector (i, j). Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$ (PMNS). Status: STANDARD.
- c_{12}, c_{13}, c_{23}: The cosines of the mixing angles, $c_{ij} = \cos\theta_{ij}$. Unit: dimensionless. Status: STANDARD.
- Δm^2_{21}, Δm^2_{31}, Δm^2_{32}: The differences of the squared neutrino masses, $\Delta m^2_{ij} = m^2_i - m^2_j$. Unit: GeV^2 (quoted in eV^2). Value: $\Delta m^2_{21} = 7.49 \times 10^{-5}$ eV^2, $\Delta m^2_{31} = 2.513 \times 10^{-3}$ eV^2 (NuFIT 6.0). Status: STANDARD.
- s_{12}, s_{13}, s_{23}: The sines of the mixing angles, $s_{ij} = \sin\theta_{ij}$. Unit: dimensionless. Status: STANDARD.

The terms. The resonance conditions of the 1-3 sector (with the effective splitting Δm^2_{ee}, the ν_e-weighted average of Δm^2_{31} and Δm^2_{32}) and of the 1-2 sector (with the factor c^2_{13}, the ν_e content of the 1-2 subspace). Status: STANDARD. The 1-3 resonance lies at 11.135 GeV in rock of 2.8 g/cm^3 (the exact three-flavour evolution gives 11.133 GeV); the 1-2 resonance at the centre of the Sun at 1.93 MeV.

$$\begin{aligned} \mathcal{P}_{ee} &= c^4_{13}\Big(\frac{1}{2} + \frac{1}{2}\cos 2\theta_{12}\cos 2\theta^{\mathrm{m}}_{12}\Big) + s^4_{13}, \qquad A_{\mathrm{MSW}} = \frac{2E\,V_{\mathrm{CC}}\,c^2_{13}}{\Delta m^2_{21}}, \\ \cos 2\theta^{\mathrm{m}}_{12} &= \frac{\cos 2\theta_{12} - A_{\mathrm{MSW}}}{\sqrt{\big(\cos 2\theta_{12} - A_{\mathrm{MSW}}\big)^2 + \sin^2 2\theta_{12}}} \end{aligned} \tag{FE-107}$$

Symbols in (FE-107).

- $\mathcal{P}$: The probability of a flavour transition, $\mathcal{P}(\nu_\alpha \to \nu_\beta)$; $\mathcal{P}_{ee}$ the survival probability of electron neutrinos. Unit: dimensionless. Status: STANDARD.
- e: The electron flavour as a value of a flavour index ($\delta_{\alpha e}$, U_{e2}, V_e, $\mathcal{P}_{ee}$). Status: STANDARD.
- $c_{12},\ c_{13},\ c_{23}$: The cosines of the mixing angles, $c_{ij} = \cos\theta_{ij}$. Unit: dimensionless. Status: STANDARD.
- θ_{ij}: The mixing angles of the standard parametrisation of U (or of V_{CKM}): θ_{12}, θ_{13}, θ_{23}; in two-flavour formulas θ_{ij} is the angle of the sector (i,j). Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$ (PMNS). Status: STANDARD.
- θ_{12}^{m}: The effective 1-2 mixing angle in matter at the production point of a solar neutrino. Unit: dimensionless. Status: STANDARD.
- $s_{12},\ s_{13},\ s_{23}$: The sines of the mixing angles, $s_{ij} = \sin\theta_{ij}$. Unit: dimensionless. Status: STANDARD.
- A_{MSW}: The ratio of the matter term to the vacuum term in the 1-2 sector. Unit: dimensionless. Status: STANDARD.
- E: The energy of a particle: of the neutrino in the flavour equations, $E = (k^2 + m^2)^{1/2}$ of a particle of momentum k in the thermal integrals. Unit: GeV. Status: STANDARD.
- $V_{\mathrm{CC}},\ V_{\mathrm{NC}}$: The charged-current potential of electron neutrinos and the neutral-current potential common to all flavours. Unit: GeV. Status: STANDARD.
- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21} = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- $\Delta m^2_{21},\ \Delta m^2_{31},\ \Delta m^2_{32}$: The differences of the squared neutrino masses, $\Delta m^2_{ij} = m_i^2 - m_j^2$. Unit: GeV^2 (quoted in eV^2). Value: $\Delta m^2_{21} = 7.49 \times 10^{-5}\ \mathrm{eV}^2$, $\Delta m^2_{31} = 2.513 \times 10^{-3}\ \mathrm{eV}^2$ (NuFIT 6.0). Status: STANDARD.

The terms. The survival probability of solar electron neutrinos produced at a density where A_{MSW} is the ratio of the matter term to the vacuum term: the mixing angle in matter θ_{12}^{m} at the production point, adiabatic evolution to the surface, and decoherence on the way to the Earth; the 1-3 sector enters through c_{13}^4 and s_{13}^4. Status: STANDARD (the numerical three-flavour evolution through the solar density agrees to 1.3×10^{-3}, the size of the non-adiabatic and production-region effects that the formula drops).

$$n_{\mathrm{e}}^{\mathrm{res}} = \frac{\Delta m^2_{ij}\cos 2\theta_{ij}}{2\sqrt{2}\,G_{\mathrm{F}}E}, \qquad \gamma_{\mathrm{ad}} = \frac{\Delta m^2_{ij}\sin^2 2\theta_{ij}}{2E\cos 2\theta_{ij}\,|\mathrm{d}\ln n_{\mathrm{e}}/\mathrm{d}r|} \gg 1, \qquad \mathcal{P}_{\mathrm{LZ}} = \mathrm{e}^{-\pi\gamma_{\mathrm{ad}}/2} \tag{FE-108}$$

Symbols in (FE-108).

- $n_{\mathrm{e}}^{\mathrm{res}}$: The electron density of an MSW resonance. Unit: GeV^3. Status: STANDARD.
- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21} = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- $m,\ m_s,\ m_i$: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- j: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- θ_{ij}: The mixing angles of the standard parametrisation of U (or of V_{CKM}): θ_{12}, θ_{13}, θ_{23}; in two-flavour formulas θ_{ij} is the angle of the sector (i,j). Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$ (PMNS). Status: STANDARD.
- G_{F}: The Fermi constant. Unit: GeV^{-2}. Value: $1.1663788 \times 10^{-5}\ \mathrm{GeV}^{-2}$ (measured). Status: STANDARD.
- E: The energy of a particle: of the neutrino in the flavour equations, $E = (k^2 + m^2)^{1/2}$ of a particle of momentum k in the thermal integrals. Unit: GeV. Status: STANDARD.
- γ_{ad}: The adiabaticity parameter of a resonance (adiabatic for $\gamma_{\mathrm{ad}} \gg 1$). Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- n_{e}: The number density of electrons (net of positrons); in the Thomson terms of the photon and baryon equations only free (ionised) electrons count. Unit: GeV^3. Status: STANDARD.
- r: A distance: between the two masses in $V_{\mathrm{N}}(r)$, from the centre of the star in the adiabaticity condition. Unit: length. Status: STANDARD.

- $\mathcal{P}_{\rm LZ}$: The Landau-Zener probability of a jump between the matter eigenstates at a resonance. Unit: dimensionless. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

The terms. The electron density at the resonance, the adiabaticity parameter $\gamma_{\rm ad}$ (the ratio of the oscillation length at the resonance to the width of the resonance region; the evolution is adiabatic for $\gamma_{\rm ad} \gg 1$), and the Landau-Zener probability $\mathcal{P}_{\rm LZ}$ of a jump between the matter eigenstates. Status: STANDARD (two-flavour conditions; in a supernova envelope both resonances are adiabatic, $\mathcal{P}_{\rm LZ} \leq 2.6 \times 10^{-77}$).

9.3 The early Universe

$$V_e - V_\mu = -\frac{8\sqrt{2}\,G_{\rm F}|\vec{k}|}{3m_{\rm W}^2}\,\rho_{\rm e}, \qquad \rho_{\rm e} \simeq \frac{7\pi^2}{60}\,T^4 \quad (T \gg m_{\rm e}) \tag{FE-109}$$

Symbols in (FE-109).

- V_α, V: The matter potential of forward scattering: V_α of the flavour α (V_e, V_μ), V the 3×3 matrix of the potentials in the flavour basis; antineutrinos have $-V$. Unit: GeV. Status: STANDARD.
- e: The electron flavour as a value of a flavour index ($\delta_{\alpha e}$, U_{e2}, V_e, $\mathcal{P}_{ee}$). Status: STANDARD.
- $G_{\rm F}$: The Fermi constant. Unit: GeV^{-2}. Value: 1.1663788×10^{-5} GeV^{-2} (measured). Status: STANDARD.
- $\vec{k}$, $\vec{k}_o$: The wave vector (momentum) in ordinary 3-space; $\vec{k}_o$ that of the orbital o, a point of the lattice $k_{\rm lat}\vec{n}_o$. Unit: GeV. Status: STANDARD.
- $m_{\rm W}$: The mass of W^+ and W^-. Unit: GeV. Value: 80.3625 GeV (measured, input). Status: STANDARD.
- $\rho_{\rm e}$, $p_{\rm e}$: The energy density and the pressure of the electrons and positrons together (ideal gas). Unit: GeV^4. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- $m_{\rm e}$: The electron mass. Unit: GeV. Value: 0.51099895 MeV. Status: STANDARD.

The terms. In the hot plasma the net electron density is tiny (of the order of $\eta_{\rm b} = 6.1 \times 10^{-10}$ times the photon density), and the leading potential is the thermal one: the W propagator expanded to the next order in $|\vec{k}|T/m_{\rm W}^2$, proportional to the energy density $\rho_{\rm e}$ of the electrons and positrons. It suppresses the flavour mixing at high temperature. Status: STANDARD (leading order in $T/m_{\rm W}$; it falls below the vacuum term at $T = 4.99$ MeV for the atmospheric splitting and 2.39 MeV for the solar one, at $|\vec{k}| = 3.15\,T$). Residual 8.7×10^{-12}; the neglected CP-asymmetric potential of the net electron density is 1.1×10^{-2} of the thermal one at the onset (ESTIMATE).

$$\begin{aligned}
&\mathrm{i}\Big(\partial_{x_4} + \frac{\vec{k}}{|\vec{k}|}\cdot\vec{\nabla}\Big)\boldsymbol{\rho}_{\vec{k}} = \big[H^{\rm vac}_{\vec{k}} + V + V_{\nu\nu},\ \boldsymbol{\rho}_{\vec{k}}\big] + \mathrm{i}\,C_{\rm coll}[\boldsymbol{\rho}], \qquad H^{\rm vac}_{\vec{k}} = \frac{U\,\mathrm{diag}\big(m_1^2,\ m_2^2,\ m_3^2\big)\,U^\dagger}{2|\vec{k}|},\\
&V_{\nu\nu} = \sqrt{2}\,G_{\rm F}\int\frac{\mathrm{d}^3\vec{k}_2}{(2\pi)^3}\Big(1 - \frac{\vec{k}\cdot\vec{k}_2}{|\vec{k}||\vec{k}_2|}\Big)\big(\boldsymbol{\rho}_{\vec{k}_2} - \bar{\boldsymbol{\rho}}_{\vec{k}_2}\big), \qquad \partial_{x_4} \to \partial_{x_4} - \mathcal{H}|\vec{k}|\,\partial_{|\vec{k}|} \ \ \text{(expanding Universe)}
\end{aligned} \tag{FE-110}$$

Symbols in (FE-110).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\vec{k}$, $\vec{k}_o$: The wave vector (momentum) in ordinary 3-space; $\vec{k}_o$ that of the orbital o, a point of the lattice $k_{\rm lat}\vec{n}_o$. Unit: GeV. Status: STANDARD.

- $\vec{\nabla}$: The gradient in ordinary 3-space, $(\partial/\partial x_1, \partial/\partial x_2, \partial/\partial x_3)$. Unit: 1/length. Status: STANDARD.
- $\boldsymbol{\rho}_{\vec{k}}$, $\bar{\boldsymbol{\rho}}_{\vec{k}}$: The 3×3 flavour density matrices of the neutrinos and of the antineutrinos of momentum $\vec{k}$. Unit: dimensionless. Status: STANDARD.
- $H^{\mathrm{vac}}_{\vec{k}}$: The vacuum flavour Hamiltonian of a neutrino of momentum $\vec{k}$. Unit: GeV. Status: STANDARD.
- V_α, V: The matter potential of forward scattering: V_α of the flavour α (V_e, V_μ), V the 3×3 matrix of the potentials in the flavour basis; antineutrinos have $-V$. Unit: GeV. Status: STANDARD.
- $V_{\nu\nu}$: The potential of neutrino-neutrino forward scattering (a matrix in flavour space). Unit: GeV. Status: STANDARD.
- C_{coll}: The collision term of the quantum kinetic equations. Unit: GeV. Status: STANDARD.
- U: The PMNS matrix, the unitary 3×3 mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states i); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212^\circ$ (NuFIT 6.0). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- m_1, m_2, m_3: The masses of the neutrino mass states ν_1, ν_2, ν_3 (normal ordering). Unit: GeV (quoted in meV). Value: 1.000, 8.712, 50.140 meV (m_1 ASSUMED). Status: DERIVED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- G_{F}: The Fermi constant. Unit: GeV^{-2}. Value: 1.1663788×10^{-5} GeV^{-2} (measured). Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- k_2, $\vec{k}_2$: A second momentum (variable of integration). Unit: GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.

The terms. The density matrix $\boldsymbol{\rho}_{\vec{k}}$ of the neutrinos of momentum $\vec{k}$ (its diagonal the occupation numbers of the flavours, its off-diagonal entries the flavour coherence; $\bar{\boldsymbol{\rho}}$ that of the antineutrinos) evolves by the commutator with the vacuum Hamiltonian H^{vac}, the matter potential and the neutrino-neutrino potential $V_{\nu\nu}$ (forward scattering on the other neutrinos, with the angular factor of the current-current interaction), and by the collision term C_{coll}; in the expanding Universe the momenta redshift. Status: STANDARD. It is used linearised for the fast flavour instability (FE-117) and momentum-averaged for the decoupling (FE-111).

$$\frac{\mathrm{d}T_\gamma}{\mathrm{d}x_4} = -\frac{4\mathcal{H}\rho_\gamma + 3\mathcal{H}\big(\rho_{\mathrm{e}} + p_{\mathrm{e}}\big) + 3\mathcal{H}T_\gamma \frac{\mathrm{d}p_{\mathrm{QED}}}{\mathrm{d}T_\gamma} + C_{\nu_e} + 2C_{\nu_\mu}}{\frac{\mathrm{d}\rho_\gamma}{\mathrm{d}T_\gamma} + \frac{\mathrm{d}\rho_{\mathrm{e}}}{\mathrm{d}T_\gamma} + T_\gamma \frac{\mathrm{d}^2 p_{\mathrm{QED}}}{\mathrm{d}T_\gamma^2}}, \qquad \frac{\mathrm{d}T_{\nu_\alpha}}{\mathrm{d}x_4} = -\mathcal{H}T_{\nu_\alpha} + \frac{C_{\nu_\alpha}}{\mathrm{d}\rho_{\nu_\alpha}/\mathrm{d}T_{\nu_\alpha}},$$
$$\mathcal{H} = \frac{\sqrt{8\pi\rho_{\mathrm{tot}}/3}}{M_{\mathrm{Pl}}}, \qquad \rho_\gamma = \frac{\pi^2}{15}T_\gamma^4, \qquad \rho_{\nu_\alpha} = \frac{7\pi^2}{120}T_{\nu_\alpha}^4 \qquad \text{(FE-111)}$$

Symbols in (FE-111).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- T_γ: The temperature of the photons (equal to the plasma temperature T; the subscript is written where the neutrinos have their own temperature). Unit: GeV. Value: 2.7255 K today. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- ρ_γ: The energy density of the photons, $\rho_\gamma = \pi^2 T_\gamma^4/15$. Unit: GeV^4. Status: STANDARD.
- ρ_{e}, p_{e}: The energy density and the pressure of the electrons and positrons together (ideal gas). Unit: GeV^4. Status: STANDARD.
- p_{QED}, $p^{(2)}_{\mathrm{QED}}$, $p^{(3)}_{\mathrm{QED}}$: The finite-temperature QED correction to the pressure of the electron-photon plasma, and its parts of order e^2 and e^3. Unit: GeV^4. Status: STANDARD.
- C_{ν_α}, C_{ν_e}, C_{ν_μ}: The energy transfer rate per volume into the neutrinos of the flavour α by collisions. Unit: GeV^5. Status: STANDARD.

- T_{ν_α}, T_{ν_e}, T_{ν_μ}: The temperature of the neutrinos of the flavour α (of ν_e, of ν_μ and ν_τ) during decoupling. Unit: GeV. Status: STANDARD.
- ρ_ν, ρ_{ν_α}: The energy density of one neutrino flavour (neutrinos and antineutrinos) at the common temperature T_ν, $\frac{7}{8}\frac{\pi^2}{15}T_\nu^4$ when massless; ρ_{ν_α} that of the flavour α at its temperature T_{ν_α}. Unit: GeV^4. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\rho_{\rm tot}$, $p_{\rm tot}$: The total background energy density and pressure of four-dimensional cosmology (all species). Unit: GeV^4. Status: STANDARD.
- $M_{\rm Pl}$: The Planck mass, $M_{\rm Pl} = G_{\rm N}^{-1/2}$. Unit: GeV. Value: 1.22091×10^{19} GeV. Status: STANDARD.

The terms. The momentum-averaged form of FE-110 for the temperatures of the photons and of the neutrino flavours: the photon temperature falls by the expansion (the dilution of the photons, of the electron-positron plasma and of the QED interaction pressure $p_{\rm QED}$) and rises by the energy that the collisions C_{ν_α} take from the neutrinos; each neutrino temperature falls with the expansion and changes by its collisions. The Hubble rate follows from the total energy density. Status: STANDARD (integrated from 20 MeV to 5 keV with LSODA, relative tolerance 10^{-11}; the method reproduces the published values of Escudero (2020) to 5×10^{-4}; ν_τ follows ν_μ). Residual 2.7×10^{-12}.

$$
\begin{aligned}
C_{\nu_e} &= \frac{G_{\rm F}^2}{\pi^5}\Big[4\Big(\big(\tilde g_L^{\nu_e e}\big)^2 + \big(\tilde g_R^{\nu_e e}\big)^2\Big)\mathcal{F}_{\rm tr}(T_\gamma, T_{\nu_e}) + 2\mathcal{F}_{\rm tr}(T_{\nu_\mu}, T_{\nu_e})\Big],\\
C_{\nu_\mu} &= \frac{G_{\rm F}^2}{\pi^5}\Big[4\Big(\big(\tilde g_L^{\nu_\mu e}\big)^2 + \big(\tilde g_R^{\nu_\mu e}\big)^2\Big)\mathcal{F}_{\rm tr}(T_\gamma, T_{\nu_\mu}) - \mathcal{F}_{\rm tr}(T_{\nu_\mu}, T_{\nu_e})\Big],\\
\mathcal{F}_{\rm tr}(T_{\rm A}, T_{\rm B}) &= 32 f_{\rm a}\big(T_{\rm A}^9 - T_{\rm B}^9\big) + 56 f_{\rm s} T_{\rm A}^4 T_{\rm B}^4\big(T_{\rm A} - T_{\rm B}\big),\\
\tilde g_L^{\nu_e e} &= \frac{1}{2} + \sin^2\theta_{\rm W}, \qquad \tilde g_R^{\nu_e e} = \tilde g_R^{\nu_\mu e} = \sin^2\theta_{\rm W}, \qquad \tilde g_L^{\nu_\mu e} = -\frac{1}{2} + \sin^2\theta_{\rm W}
\end{aligned}
\tag{FE-112}
$$

Symbols in (FE-112).

- C_{ν_α}, C_{ν_e}, C_{ν_μ}: The energy transfer rate per volume into the neutrinos of the flavour α by collisions. Unit: GeV^5. Status: STANDARD.
- $G_{\rm F}$: The Fermi constant. Unit: GeV^{-2}. Value: 1.1663788×10^{-5} GeV^{-2} (measured). Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\tilde g_L^{\nu_e e}$, $\tilde g_R^{\nu_e e}$: The effective left- and right-handed couplings of electron neutrinos to electrons (charged and neutral current), $\tilde g_L^{\nu_e e} = \frac{1}{2} + \sin^2\theta_{\rm W}$, $\tilde g_R^{\nu_e e} = \sin^2\theta_{\rm W}$. Unit: dimensionless. Value: 0.727 and 0.233 (radiatively corrected). Status: STANDARD.
- L: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1 + \Gamma_4)/2$. Status: STANDARD.
- R: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1 - \Gamma_4)/2$. Status: STANDARD.
- $\mathcal{F}_{\rm tr}$: The energy-transfer function of two species with the temperatures $T_{\rm A}$ and $T_{\rm B}$ (Maxwell-Boltzmann with the corrections $f_{\rm a}$, $f_{\rm s}$). Unit: GeV^9. Status: STANDARD.
- T_γ: The temperature of the photons (equal to the plasma temperature T; the subscript is written where the neutrinos have their own temperature). Unit: GeV. Value: 2.7255 K today. Status: STANDARD.
- T_{ν_α}, T_{ν_e}, T_{ν_μ}: The temperature of the neutrinos of the flavour α (of ν_e, of ν_μ and ν_τ) during decoupling. Unit: GeV. Status: STANDARD.
- $\tilde g_L^{\nu_\mu e}$, $\tilde g_R^{\nu_\mu e}$: The effective couplings of muon (and tau) neutrinos to electrons (neutral current only), $\tilde g_L^{\nu_\mu e} = -\frac{1}{2} + \sin^2\theta_{\rm W}$, $\tilde g_R^{\nu_\mu e} = \sin^2\theta_{\rm W}$. Unit: dimensionless. Value: -0.273 and 0.233 (radiatively corrected). Status: STANDARD.
- $T_{\rm A}$, $T_{\rm B}$: The temperatures of the two species that exchange energy in $\mathcal{F}_{\rm tr}$. Unit: GeV. Status: STANDARD.
- $f_{\rm a}$, $f_{\rm s}$: The Fermi-Dirac corrections of the annihilation and scattering parts of the energy transfer. Unit: dimensionless. Value: 0.884 and 0.829. Status: STANDARD.
- $\theta_{\rm W}$: The weak mixing angle of the couplings, $\cos\theta_{\rm W} = \hat g_2/\hat g_Z$, $\sin\theta_{\rm W} = \hat g_1/\hat g_Z$; fixed by the measured W and Z masses with $\rho_{\rm EW,0}$. Unit: dimensionless. Value: $\cos^2\theta_{\rm W} = 0.776368$. Status: DERIVED.

The terms. The rates of energy transfer between the electron-positron plasma and the neutrinos (annihilation and scattering, with the couplings $\tilde g^{\nu_e e}$ and $\tilde g^{\nu_\mu e}$; the electron neutrinos also have the charged current) and between the neutrino flavours; $\mathcal{F}_{\rm tr}$ is the transfer function between two thermal baths of temperatures $T_{\rm A}$ and $T_{\rm B}$, with the

factors $f_\mathrm{a} = 0.884$ and $f_\mathrm{s} = 0.829$ of the Fermi-Dirac statistics. Status: STANDARD (the radiatively corrected couplings 0.727, 0.233, −0.273, 0.233 are used; the tree-level ones change N_eff by 4×10^{-4}).

$$
\begin{aligned}
&p_\mathrm{QED} = p_\mathrm{QED}^{(2)} + p_\mathrm{QED}^{(3)}, \qquad p_\mathrm{QED}^{(2)} = -\frac{e^2 T^2}{12\pi^2}\,\mathcal{Q}_1 - \frac{e^2}{8\pi^4}\,\mathcal{Q}_1^2 + \frac{e^2 m_\mathrm{e}^2}{16\pi^4}\,\mathcal{Q}_2, \\
&p_\mathrm{QED}^{(3)} = \frac{e^3 T}{12\pi^4}\,\mathcal{Q}_3^{3/2}, \qquad \rho_\mathrm{QED} = -p_\mathrm{QED} + T\,\frac{\mathrm{d}p_\mathrm{QED}}{\mathrm{d}T}
\end{aligned}
\tag{FE-113}
$$

Symbols in (FE-113).

- p_QED, $p_\mathrm{QED}^{(2)}$, $p_\mathrm{QED}^{(3)}$: The finite-temperature QED correction to the pressure of the electron-photon plasma, and its parts of order e^2 and e^3. Unit: GeV^4. Status: STANDARD.
- e: The elementary charge (the charge of the positron), $e^2 = 4\pi\alpha_\mathrm{em}$; W^+ has the charge $+e$ and W^- the charge $-e$. Unit: dimensionless. Value: $e^2 = 4\pi/137.036$ at low energy. Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\mathcal{Q}_1$, $\mathcal{Q}_2$, $\mathcal{Q}_3$: The momentum integrals of the QED pressure. Unit: GeV^2. Status: STANDARD.
- m_e: The electron mass. Unit: GeV. Value: 0.51099895 MeV. Status: STANDARD.
- ρ_QED: The QED correction to the energy density of the plasma. Unit: GeV^4. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.

The terms. The finite-temperature QED correction to the pressure of the electron-photon plasma at order e^2 and e^3, from the integrals of FE-114; ρ_QED follows by the thermodynamic relation. It raises N_eff by 0.009 (the decoupling computed with and without it). Status: STANDARD (the massless limits $-5e^2T^4/288$ and $e^3T^4/(36\sqrt{3}\,\pi)$ reproduced to 10^{-16}).

$$
\begin{aligned}
&\mathcal{Q}_1 = \int_0^\infty \mathrm{d}k\,\frac{k^2}{E}\,n_\mathrm{D}(k), \qquad \mathcal{Q}_2 = \int_0^\infty\!\!\int_0^\infty \mathrm{d}k\,\mathrm{d}k_2\,\frac{k k_2}{E(k)E(k_2)}\,\ln\left|\frac{k+k_2}{k-k_2}\right|\,n_\mathrm{D}(k)\,n_\mathrm{D}(k_2), \\
&\mathcal{Q}_3 = \int_0^\infty \mathrm{d}k\,\frac{k^2+E^2}{E}\,n_\mathrm{D}(k), \qquad n_\mathrm{D}(k) = \frac{2}{\mathrm{e}^{E/T}+1}, \qquad E = \sqrt{k^2+m_\mathrm{e}^2}
\end{aligned}
\tag{FE-114}
$$

Symbols in (FE-114).

- $\mathcal{Q}_1$, $\mathcal{Q}_2$, $\mathcal{Q}_3$: The momentum integrals of the QED pressure. Unit: GeV^2. Status: STANDARD.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no upper bound. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- E: The energy of a particle: of the neutrino in the flavour equations, $E = (k^2 + m^2)^{1/2}$ of a particle of momentum k in the thermal integrals. Unit: GeV. Status: STANDARD.
- $n_\mathrm{D}(k)$: The Fermi-Dirac occupation of the electrons and positrons summed over the two spin states. Unit: dimensionless. Status: STANDARD.
- k_2, $\vec{k}_2$: A second momentum (variable of integration). Unit: GeV. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- m_e: The electron mass. Unit: GeV. Value: 0.51099895 MeV. Status: STANDARD.

The terms. The momentum integrals over the electron and positron distribution n_{D} (two spin states, energy E with the electron mass m_{e}). Status: STANDARD (Gauss-Legendre panels; the logarithmic singularity at $k_2 = k$ removed by a substitution).

For instantaneous decoupling the entropy of the electron-positron plasma goes to the photons alone, and

$$T_{\nu_e} = T_{\nu_\mu} = \Big(\frac{4}{11}\Big)^{1/3} T_\gamma \quad \Rightarrow \quad N_{\mathrm{eff}} = \Big(\frac{11}{4}\Big)^{4/3} \cdot 3 \Big(\frac{4}{11}\Big)^{4/3} = 3.$$

Symbols in this equation.

- $T_{\nu_\alpha},\ T_{\nu_e},\ T_{\nu_\mu}$: The temperature of the neutrinos of the flavour α (of ν_e, of ν_μ and ν_τ) during decoupling. Unit: GeV. Status: STANDARD.
- T_γ: The temperature of the photons (equal to the plasma temperature T; the subscript is written where the neutrinos have their own temperature). Unit: GeV. Value: 2.7255 K today. Status: STANDARD.
- N_{eff}: The effective number of neutrino species (the radiation beyond the photons in units of one neutrino species with $T_\nu/T_\gamma = (4/11)^{1/3}$); ΔN_{eff} an addition to it. Unit: dimensionless. Value: 3.0432 (neutrinos) + 0.0010 (fold bands). Status: STANDARD.

$$N_{\mathrm{eff}} = \Big(\frac{11}{4}\Big)^{4/3} \Big[\Big(\frac{T_{\nu_e}}{T_\gamma}\Big)^4 + 2 \Big(\frac{T_{\nu_\mu}}{T_\gamma}\Big)^4 \Big], \qquad \frac{\mathrm{d}T_\nu}{\mathrm{d}x_4} = -\mathcal{H} T_\nu + \frac{C_{\nu_e} + 2C_{\nu_\mu}}{3\,\mathrm{d}\rho_\nu/\mathrm{d}T_\nu} \tag{FE-115}$$

Symbols in (FE-115).

- N_{eff}: The effective number of neutrino species (the radiation beyond the photons in units of one neutrino species with $T_\nu/T_\gamma = (4/11)^{1/3}$); ΔN_{eff} an addition to it. Unit: dimensionless. Value: 3.0432 (neutrinos) + 0.0010 (fold bands). Status: STANDARD.
- $T_{\nu_\alpha},\ T_{\nu_e},\ T_{\nu_\mu}$: The temperature of the neutrinos of the flavour α (of ν_e, of ν_μ and ν_τ) during decoupling. Unit: GeV. Status: STANDARD.
- T_γ: The temperature of the photons (equal to the plasma temperature T; the subscript is written where the neutrinos have their own temperature). Unit: GeV. Value: 2.7255 K today. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- T_ν: The temperature of the relic neutrinos (one common temperature). Unit: GeV. Value: 1.9454 K today. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- $C_{\nu_\alpha},\ C_{\nu_e},\ C_{\nu_\mu}$: The energy transfer rate per volume into the neutrinos of the flavour α by collisions. Unit: GeV^5. Status: STANDARD.
- $\rho_\nu,\ \rho_{\nu_\alpha}$: The energy density of one neutrino flavour (neutrinos and antineutrinos) at the common temperature T_ν, $\frac{7}{8}\frac{\pi^2}{15} T_\nu^4$ when massless; ρ_{ν_α} that of the flavour α at its temperature T_{ν_α}. Unit: GeV^4. Status: STANDARD.

The terms. The effective number of neutrino species measures the energy density of the neutrinos relative to that of three species decoupled instantaneously; the second relation is the common-temperature form (efficient oscillations). Status: STANDARD. UM-2 computes 3.0476 (separate temperatures) and 3.0485 (common temperature) and adopts the precision value 3.0432 (Cielo et al. 2023); the difference, 4×10^{-3}, is the electron-mass and momentum dependence that the temperature method omits. The fold bands add 0.0010 (FE-129, FE-130); the sum agrees with the value in the CMB computation to 5×10^{-16}.

9.4 Dense neutrino gases

$$G_{\mathrm{ELN}}(u) = \sqrt{2}\, G_{\mathrm{F}} \big[n_{\nu_e}\, g_{\nu_e}(u) - n_{\bar{\nu}_e}\, g_{\bar{\nu}_e}(u) \big], \qquad \int_{-1}^{1} g_{\nu_e} \frac{\mathrm{d}u}{2} = \int_{-1}^{1} g_{\bar{\nu}_e} \frac{\mathrm{d}u}{2} = 1, \qquad \mathcal{I}_l = \int_{-1}^{1} \frac{\mathrm{d}u}{2}\, G_{\mathrm{ELN}}(u)\, u^l \tag{FE-116}$$

Symbols in (FE-116).

- $G_{\mathrm{ELN}}(u)$: The angular distribution of the electron lepton number of a dense neutrino gas (in units of a potential). Unit: GeV. Status: STANDARD.
- u: The cosine of the angle between a momentum and the symmetry axis x_3 of the neutrino gas. Unit: dimensionless. Status: STANDARD.
- G_{F}: The Fermi constant. Unit: GeV^{-2}. Value: 1.1663788×10^{-5} GeV^{-2} (measured). Status: STANDARD.
- $n_{\nu_e},\ n_{\bar\nu_e}$: The number densities of electron neutrinos and electron antineutrinos. Unit: GeV^3. Status: STANDARD.
- $g_{\nu_e}(u),\ g_{\bar\nu_e}(u)$: The normalised angular distributions of electron neutrinos and antineutrinos. Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $\mathcal{I}_l$: The l-th angular moment of the electron-lepton-number distribution. Unit: GeV. Status: STANDARD.
- l: The multipole (angular moment) number $l = 0, 1, 2, \ldots$ of a Boltzmann hierarchy or of an angular moment $\int u^l \cdots$; as a coefficient it is the same number. Status: STANDARD.

The terms. In a dense, homogeneous and axially symmetric gas of electron neutrinos and antineutrinos (near the neutrinosphere of a supernova), G_{ELN} is the electron-lepton-number distribution over the cosine u of the angle to the symmetry axis x_3, with the normalised angular distributions g_{ν_e}, $g_{\bar\nu_e}$ and the densities n_{ν_e}, $n_{\bar\nu_e}$; $\mathcal{I}_l$ are its moments. Status: STANDARD.

$$\mathcal{C}(u) \propto \mathrm{e}^{-\mathrm{i}(\omega_{\mathrm{f}} x_4 - k_{\mathrm{f}} x_3)}, \qquad \bar\omega = \omega_{\mathrm{f}} - \mathcal{I}_0, \qquad \bar k = k_{\mathrm{f}} - \mathcal{I}_1 : \qquad \det \Pi(\bar\omega, \bar k) = 0,$$
$$\Pi = \mathrm{diag}(1,\ -1) + \int_{-1}^{1} \frac{\mathrm{d}u}{2}\, \frac{G_{\mathrm{ELN}}(u)}{\bar\omega - \bar k u} \begin{pmatrix} 1 & u \\ u & u^2 \end{pmatrix} \tag{FE-117}$$

Symbols in (FE-117).

- $\mathcal{C}(u)$: The flavour coherence (off-diagonal element of the density matrix) of the neutrinos moving in the direction u. Unit: dimensionless. Status: STANDARD.
- u: The cosine of the angle between a momentum and the symmetry axis x_3 of the neutrino gas. Unit: dimensionless. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\omega_{\mathrm{f}},\ k_{\mathrm{f}}$: The frequency and the wavenumber (along x_3) of a flavour wave. Unit: GeV. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space (also the symmetry axis of the fast-flavour analysis). Unit: length. Status: AUTHOR.
- $\bar\omega,\ \bar k$: The frequency and wavenumber of a flavour wave shifted by the moments, $\bar\omega = \omega_{\mathrm{f}} - \mathcal{I}_0$, $\bar k = k_{\mathrm{f}} - \mathcal{I}_1$; a mode with $\mathrm{Im}\,\bar\omega > 0$ grows. Unit: GeV. Status: STANDARD.
- $\mathcal{I}_l$: The l-th angular moment of the electron-lepton-number distribution. Unit: GeV. Status: STANDARD.
- Π: The 2×2 dispersion matrix of the fast flavour instability. Unit: dimensionless. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $G_{\mathrm{ELN}}(u)$: The angular distribution of the electron lepton number of a dense neutrino gas (in units of a potential). Unit: GeV. Status: STANDARD.

The terms. The linearised FE-110 for a small flavour coherence $\mathcal{C}(u)$ with the frequency ω_{f} and the wave number k_{f} along the axis (shifted by the moments, $\bar\omega$, $\bar k$): non-trivial solutions need $\det \Pi = 0$, the dispersion relation of fast flavour waves. Status: STANDARD (roots for real k_{f} by vectorised Newton iterations; three independent evaluations agree to 1.2×10^{-14}).

$$\big(1 + \mathcal{J}_0\big)\big(-1 + \mathcal{J}_2\big) - \mathcal{J}_1^2 = 0, \qquad \mathcal{J}_l = \int_{-1}^{1} \frac{\mathrm{d}u}{2}\, \frac{G_{\mathrm{ELN}}(u)\, u^l}{\bar\omega - \bar k u}; \qquad -1 + \int_{-1}^{1} \frac{\mathrm{d}u}{2}\, \frac{G_{\mathrm{ELN}}(u)\,(1 - u^2)}{2(\bar\omega - \bar k u)} = 0 \tag{FE-118}$$

Symbols in (FE-118).

- $\mathcal{J}_l$: The l-th moment of the dispersion integral of the fast flavour instability. Unit: dimensionless. Status: STANDARD.
- l: The multipole (angular moment) number $l = 0, 1, 2, \ldots$ of a Boltzmann hierarchy or of an angular moment $\int u^l \cdots$; as a coefficient it is the same number. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- u: The cosine of the angle between a momentum and the symmetry axis x_3 of the neutrino gas. Unit: dimensionless. Status: STANDARD.
- $G_{\mathrm{ELN}}(u)$: The angular distribution of the electron lepton number of a dense neutrino gas (in units of a potential). Unit: GeV. Status: STANDARD.
- $\bar{\omega}$, $\bar{k}$: The frequency and wavenumber of a flavour wave shifted by the moments, $\bar{\omega} = \omega_{\mathrm{f}} - \mathcal{I}_0$, $\bar{k} = k_{\mathrm{f}} - \mathcal{I}_1$; a mode with $\mathrm{Im}\,\bar{\omega} > 0$ grows. Unit: GeV. Status: STANDARD.

The terms. The dispersion relation splits into the axisymmetric branch (the first equation, with the moments $\mathcal{J}_l$) and the azimuthal branch (the second). A root with $\mathrm{Im}\,\bar{\omega} > 0$ for real $\bar{k}$ grows exponentially: the fast flavour instability. Status: STANDARD (with a crossing of the electron-lepton-number distribution the largest growth rate is $0.088\sqrt{2}\,G_{\mathrm{F}} n_{\nu_e}$; without one no mode grows).

$$\begin{aligned} &\mathrm{i}\,\frac{\mathrm{d}\mathcal{C}(u)}{\mathrm{d}x_4} = \big(\mathcal{I}_0 - u\,\mathcal{I}_1\big)\mathcal{C}(u) - \int_{-1}^{1} \frac{\mathrm{d}\tilde{u}}{2}\, G_{\mathrm{ELN}}(\tilde{u})\,(1 - u\tilde{u})\,\mathcal{C}(\tilde{u}); \\ &\bar{\omega}^2 - \big(\mathcal{I}_2 - \mathcal{I}_0\big)\bar{\omega} - \big(\mathcal{I}_0\mathcal{I}_2 - \mathcal{I}_1^2\big) = 0, \qquad \mathrm{Im}\,\bar{\omega} = \frac{1}{2}\sqrt{4\mathcal{I}_1^2 - \big(\mathcal{I}_0 + \mathcal{I}_2\big)^2} \end{aligned} \tag{FE-119}$$

Symbols in (FE-119).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $\mathcal{C}(u)$: The flavour coherence (off-diagonal element of the density matrix) of the neutrinos moving in the direction u. Unit: dimensionless. Status: STANDARD.
- u: The cosine of the angle between a momentum and the symmetry axis x_3 of the neutrino gas. Unit: dimensionless. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{I}_l$: The l-th angular moment of the electron-lepton-number distribution. Unit: GeV. Status: STANDARD.
- $\tilde{u}$: A second direction cosine (variable of integration). Unit: dimensionless. Status: STANDARD.
- $G_{\mathrm{ELN}}(u)$: The angular distribution of the electron lepton number of a dense neutrino gas (in units of a potential). Unit: GeV. Status: STANDARD.
- $\bar{\omega}$, $\bar{k}$: The frequency and wavenumber of a flavour wave shifted by the moments, $\bar{\omega} = \omega_{\mathrm{f}} - \mathcal{I}_0$, $\bar{k} = k_{\mathrm{f}} - \mathcal{I}_1$; a mode with $\mathrm{Im}\,\bar{\omega} > 0$ grows. Unit: GeV. Status: STANDARD.
- Im: The imaginary part of a complex number. Status: STANDARD.

The terms. The homogeneous equation for $\mathcal{C}(u)$ and the quadratic equation of the mode $k_{\mathrm{f}} = \mathcal{I}_1$, whose imaginary part gives the growth rate in closed form. Status: STANDARD (the eigenvalues of the discretised equation, 300 nodes, agree with the dispersion relation; near the neutrinosphere the coherence grows by a factor e in 6×10^{-10} to 6×10^{-11} s). Residual of the stored moments, growth rates and eigenvalues 3.3×10^{-13}.

9.5 The cosmic neutrino background

$$\begin{aligned} &f_{\mathrm{FD}}(k) = \frac{1}{\mathrm{e}^{k/T_\nu} + 1}, \qquad n_i = 2\int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\, f_{\mathrm{FD}} = \frac{3\zeta(3)}{2\pi^2}\, T_\nu^3, \qquad \rho_i = 2\int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\, \sqrt{k^2 + m_i^2}\, f_{\mathrm{FD}}, \\ &\bar{v}_i = \frac{\int_0^\infty k^2\, \frac{k}{\sqrt{k^2 + m_i^2}}\, f_{\mathrm{FD}}\, \mathrm{d}k}{\int_0^\infty k^2\, f_{\mathrm{FD}}\, \mathrm{d}k}, \qquad \text{flavour content of } \nu_i : \ |U_{\alpha i}|^2 \end{aligned} \tag{FE-120}$$

Symbols in (FE-120).

- f_{FD}: The Fermi-Dirac distribution of the relic neutrinos, $f_{\mathrm{FD}} = 1/(\mathrm{e}^{k/T_\nu} + 1)$. Unit: dimensionless. Status: STANDARD.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- T_ν: The temperature of the relic neutrinos (one common temperature). Unit: GeV. Value: 1.9454 K today. Status: STANDARD.
- n, n_o, $n_{\psi i}$, n_f, n_i: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and n_o the contribution of the orbital o, $n_{\psi i}(y)$ the density of the Standard-Model fermion ψ_i (per proper 7-volume of eight dimensions); n_f the density of the fermion component f and n_i that of the neutrino mass state i (per 3-volume of four dimensions). Unit: GeV^7 (eight dimensions), GeV^3 (four dimensions). Status: STANDARD.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $\vec{k}$, $\vec{k}_o$: The wave vector (momentum) in ordinary 3-space; $\vec{k}_o$ that of the orbital o, a point of the lattice $k_{\mathrm{lat}}\vec{n}_o$. Unit: GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- ζ: The Riemann zeta function; only its value $\zeta(3)$ occurs, the sum of $1/n^3$ over the positive integers n. Unit: dimensionless. Value: $\zeta(3) = 1.2020569$. Status: STANDARD.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- m, m_s, m_i: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.
- $\bar{v}_i$: The mean velocity of the relic neutrinos of the mass state i. Unit: dimensionless ($\mathrm{c} = 1$). Value: 0.435, 0.0604, 0.0105 today. Status: STANDARD.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no upper bound. Status: STANDARD.
- ν_α, ν_i: A left-handed neutrino: ν_α of the flavour $\alpha = e, \mu, \tau$ (ν_e, ν_μ, ν_τ), ν_i of the mass state i (mass m_i). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- U: The PMNS matrix, the unitary 3×3 mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states i); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212^\circ$ (NuFIT 6.0). Status: STANDARD.
- α: A flavour index running over e, μ, τ (the flavour states of the leptons and neutrinos). Status: STANDARD.

The terms. Today the relic neutrinos are three mass states (decoherent), each with the Fermi-Dirac distribution of the temperature T_ν (the momenta redshift, so the massless form is kept), the number density n_i, the energy density ρ_i with the mass m_i, and the mean velocity $\bar{v}_i$; the flavour content of ν_i is $|U_{\alpha i}|^2$. Status: STANDARD (exact integrals): $T_\nu = 1.9454$ K, 112.0 per cm^3 per mass state, mean velocities 0.435, 0.0604 and 0.0105 (in units of the speed of light), and $\Omega_\nu h^2_{100} = 6.382 \times 10^{-4}$ (the value in the CMB computation agrees to 1.9×10^{-5}). Residual 2.9×10^{-12}.

10 The thermal history

The hot Universe of UM-2 is the plasma of the Standard-Model zero modes with their x_0 masses (no Higgs particle), in which the dark matter freezes out through the portals I12 and I13, the fold bands of dirac16complex are produced out of equilibrium through I11, and the right-handed neutrinos are never produced. The Hubble rate is that of four-dimensional gravity on the stabilised block (FE-47, FE-49).

10.1 The degrees of freedom

$$\rho_s = \frac{d_s}{2\pi^2}\int_0^\infty \frac{k^2 E\,\mathrm{d}k}{\mathrm{e}^{E/T}\pm 1}, \qquad p_s = \frac{d_s}{6\pi^2}\int_0^\infty \frac{k^4\,\mathrm{d}k}{E\,(\mathrm{e}^{E/T}\pm 1)}, \qquad s_s = \frac{\rho_s + p_s}{T}, \qquad E = \sqrt{k^2+m_s^2} \qquad \text{(FE-121)}$$

Symbols in (FE-121).

- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- s: The label of a particle species of the plasma or of matter (photon, gluons, $W^\pm$, Z, quarks, leptons, nucleons, dark matter); a sum over s runs over the species present. Status: STANDARD.
- d_o, d_s: A degeneracy: $d_o = 4r_3(|\vec{n}_o|^2)$ the number of states of the momentum shell of the orbital o; d_s the number of internal states (spin, colour, particle and antiparticle) of the species s. Unit: dimensionless. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no upper bound. Status: STANDARD.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- E: The energy of a particle: of the neutrino in the flavour equations, $E = (k^2+m^2)^{1/2}$ of a particle of momentum k in the thermal integrals. Unit: GeV. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- p: A pressure: $p_M = \mathcal{T}^M{}_M$ (no sum) for the direction $M \neq x_4$, p_s the pressure of the species s; Δp the pressure perturbation in the cosmological perturbation equations; in the Feynman-Hellmann relation the pressure of a gauge plasma. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- s, s_s: The entropy density of the plasma, $s = 2\pi^2 g_s T^3/45$; s_s that of the species s. Unit: GeV^3. Status: STANDARD.
- m, m_s, m_i: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.

The terms. The energy density, pressure and entropy density of a species s in equilibrium at the temperature T: d_s internal states, the energy E with the x_0 mass m_s, Fermi-Dirac (upper sign) or Bose-Einstein statistics. The species are the photon, the gluons, W^+, W^- and Z (three polarisations each), the six quarks, the three charged leptons and the three neutrinos; there is no Higgs particle, and the right-handed neutrinos, the fold bands, the radion and the Kaluza-Klein modes are not in equilibrium. Status: STANDARD (exact integrals).

$$g_\rho = \frac{30}{\pi^2 T^4}\sum_s \rho_s, \qquad g_s = \frac{45}{2\pi^2 T^3}\sum_s s_s, \qquad \mathcal{H} = \Big(\frac{8\pi^3 g_\rho}{90}\Big)^{1/2}\frac{T^2}{M_{\mathrm{Pl}}} \qquad \text{(FE-122)}$$

Symbols in (FE-122).

- g_ρ, g_s: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s = 43/11$ today. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.

- s: The label of a particle species of the plasma or of matter (photon, gluons, $W^\pm$, Z, quarks, leptons, nucleons, dark matter); a sum over s runs over the species present. Status: STANDARD.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- s, s_s: The entropy density of the plasma, $s = 2\pi^2 g_s T^3/45$; s_s that of the species s. Unit: GeV^3. Status: STANDARD.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- M_{Pl}: The Planck mass, $M_{\mathrm{Pl}} = G_{\mathrm{N}}^{-1/2}$. Unit: GeV. Value: 1.22091×10^{19} GeV. Status: STANDARD.

The terms. The effective numbers of relativistic degrees of freedom for the energy (g_ρ) and the entropy (g_s), and the Hubble rate of the radiation era (the four-dimensional Friedmann equation with $\rho = \frac{\pi^2}{30} g_\rho T^4$). Status: STANDARD (checked against the exact limits 105.75 above the electroweak scale, which is the Standard-Model value 106.75 without the Higgs particle, 86.25, 10.75 and 43/11). Residual (adaptive quadrature against the tabulated values) 5.0×10^{-9}; the neglected order-α interaction pressure of the plasma is -5.4 % of the ideal-gas pressure at the dark-matter freeze-out (QCD -4.7 %, SU(2) -0.7 %, U(1) -0.1 %).

$$w_{\mathrm{QGP}}(T) = \frac{1}{2}\Big[1 + \tanh\frac{T - T_{\mathrm{c}}}{T_{\mathrm{w}}}\Big], \qquad \frac{T_\nu}{T} = \Big(\frac{g_{s,e\gamma}(T)}{11/2}\Big)^{1/3}, \qquad \frac{\mathrm{d}\ln T}{\mathrm{d}x_4} = -\frac{\mathcal{H}}{1 + \frac{1}{3}\,\mathrm{d}\ln g_s/\mathrm{d}\ln T} \tag{FE-123}$$

Symbols in (FE-123).

- $w_{\mathrm{QGP}}(T)$: The weight of the quark-gluon plasma in the model of the QCD crossover (1 above, 0 below). Unit: dimensionless. Status: ASSUMED.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- T_{c}, T_{w}: The temperature and the width of the QCD crossover. Unit: GeV. Value: 155 MeV and 15 MeV. Status: ASSUMED.
- T_ν: The temperature of the relic neutrinos (one common temperature). Unit: GeV. Value: 1.9454 K today. Status: STANDARD.
- $g_{s,e\gamma}$: The entropy degrees of freedom of the electrons, positrons and photons only (after neutrino decoupling). Unit: dimensionless. Value: 11/2 before and 2 after the annihilation of $e^\pm$. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- g_ρ, g_s: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s = 43/11$ today. Status: STANDARD.

The terms. The QCD crossover as a smooth switch w_{QGP} between the quark-gluon plasma and a gas of the twelve lightest hadron multiplets at $T_{\mathrm{c}} = 155$ MeV with the width $T_{\mathrm{w}} = 15$ MeV; the neutrino temperature after an instantaneous decoupling at 2 MeV from the entropy $g_{s,e\gamma}$ of the electron-photon plasma; and the cooling law of the plasma from the conservation of entropy. Status: ASSUMED (the model of the crossover and the instantaneous decoupling; the radiation density after decoupling differs from that of the CMB computation by 5.9×10^{-3}, which enters only the entropy today).

10.2 The dark-matter relic

The dark matter is two Dirac species (the lowest odd levels of Ψ_A and Ψ_B), each with its antiparticle. For one species with $n_{\mathrm{DM}} = \bar{n}_{\mathrm{DM}}$, the abundance per entropy and the variable $x = M_{\mathrm{DM}}/T$ obey (entropy conservation, FE-123)

$$\frac{\mathrm{d}Y_{\mathrm{DM}}}{\mathrm{d}x_4} = -\frac{s\,\langle\sigma v_{\mathrm{rel}}\rangle}{2}\big(Y_{\mathrm{DM}}^2 - Y_{\mathrm{eq}}^2\big), \qquad \frac{\mathrm{d}x}{\mathrm{d}x_4} = \frac{x\,\mathcal{H}}{1 + \frac{1}{3}\,\mathrm{d}\ln g_s/\mathrm{d}\ln T},$$

Symbols in this equation.

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- Y_{DM}, Y_{eq}, Y_∞: The abundance (number density per entropy density) of one dark-matter species (particles and antiparticles), its equilibrium value and its final value. Unit: dimensionless. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- s, s_s: The entropy density of the plasma, $s = 2\pi^2 g_s T^3/45$; s_s that of the species s. Unit: GeV^3. Status: STANDARD.
- σ: A cross-section (of the dark-matter annihilation or production process named in the equation; summed over the spins where marked). Unit: GeV^{-2}. Status: STANDARD.
- v_{rel}: The relative velocity of the two annihilating particles; $\langle\sigma v_{\mathrm{rel}}\rangle$ the thermal average. Unit: dimensionless (c = 1). Status: STANDARD.
- x: The ratio $x = M_{\mathrm{DM}}/T$ of the dark-matter mass to the temperature. Unit: dimensionless. Value: $x_{\mathrm{f}} = 31.23$ at the freeze-out. Status: STANDARD.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- g_ρ, g_s: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s = 43/11$ today. Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.

which combine to the second form of:

$$\begin{gathered}\frac{\mathrm{d}n_{\mathrm{DM}}}{\mathrm{d}x_4} + 3\mathcal{H} n_{\mathrm{DM}} = -\langle\sigma v_{\mathrm{rel}}\rangle\big(n_{\mathrm{DM}}\bar{n}_{\mathrm{DM}} - n_{\mathrm{DM,eq}}^2\big) \;\Rightarrow\; \frac{\mathrm{d}Y_{\mathrm{DM}}}{\mathrm{d}x} = -\Big(1 + \frac{1}{3}\frac{\mathrm{d}\ln g_s}{\mathrm{d}\ln T}\Big)\frac{s\,\langle\sigma v_{\mathrm{rel}}\rangle}{2x\mathcal{H}}\big(Y_{\mathrm{DM}}^2 - Y_{\mathrm{eq}}^2\big), \\ Y_{\mathrm{DM}} = \frac{n_{\mathrm{DM}} + \bar{n}_{\mathrm{DM}}}{s}, \qquad x = \frac{M_{\mathrm{DM}}}{T}, \qquad s = \frac{2\pi^2}{45}\,g_s T^3, \qquad Y_{\mathrm{eq}} = \frac{4M_{\mathrm{DM}}^2 T K_2(x)}{2\pi^2 s}\end{gathered} \tag{FE-124}$$

Symbols in (FE-124).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- n_{DM}, $\bar{n}_{\mathrm{DM}}$: The number densities of dark-matter particles and antiparticles of one species. Unit: GeV^3. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- σ: A cross-section (of the dark-matter annihilation or production process named in the equation; summed over the spins where marked). Unit: GeV^{-2}. Status: STANDARD.
- v_{rel}: The relative velocity of the two annihilating particles; $\langle\sigma v_{\mathrm{rel}}\rangle$ the thermal average. Unit: dimensionless (c = 1). Status: STANDARD.
- $n_{\mathrm{DM,eq}}$, $\bar{n}_{\mathrm{DM,eq}}$: The equilibrium number densities of dark-matter particles and antiparticles of one species. Unit: GeV^3. Status: STANDARD.
- Y_{DM}, Y_{eq}, Y_∞: The abundance (number density per entropy density) of one dark-matter species (particles and antiparticles), its equilibrium value and its final value. Unit: dimensionless. Status: STANDARD.
- x: The ratio $x = M_{\mathrm{DM}}/T$ of the dark-matter mass to the temperature. Unit: dimensionless. Value: $x_{\mathrm{f}} = 31.23$ at the freeze-out. Status: STANDARD.
- g_ρ, g_s: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s = 43/11$ today. Status: STANDARD.

- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- s, s_s: The entropy density of the plasma, $s = 2\pi^2 g_s T^3/45$; s_s that of the species s. Unit: GeV^3. Status: STANDARD.
- M_{DM}: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- K_1, K_2: The modified Bessel functions of the second kind of orders 1 and 2 (not related to the block variable K). Unit: dimensionless. Status: STANDARD.

The terms. The number density of the dark matter changes by the expansion ($3\mathcal{H}n_{\mathrm{DM}}$) and by annihilation and inverse annihilation with the thermally averaged cross-section times relative velocity $\langle\sigma v_{\mathrm{rel}}\rangle$; Y_{DM} is the abundance per entropy density s (particles and antiparticles), Y_{eq} its equilibrium value (Bessel function K_2). Status: STANDARD (integrated implicitly, Radau, $\ln Y_{\mathrm{DM}}$ from $x = 5$ to 2000 with relative tolerance 10^{-10}, with the degrees of freedom of UM-2; each of the two species freezes out separately). Residual 6.1×10^{-9} (the relic condition re-solved and the late-time integral identity); the leading neglected effect is the missing order-α pressure of the plasma, which would shift Ωh^2 by about +2.7 % (ESTIMATE).

$$\langle\sigma v_{\mathrm{rel}}\rangle = \frac{1}{8M_{\mathrm{DM}}^4 T K_2(x)^2} \int_{4M_{\mathrm{DM}}^2}^{\infty} \sigma\left(E_{\mathrm{cm}}^2 - 4M_{\mathrm{DM}}^2\right) E_{\mathrm{cm}}\, K_1\!\left(\frac{E_{\mathrm{cm}}}{T}\right) \mathrm{d}\left(E_{\mathrm{cm}}^2\right) \tag{FE-125}$$

Symbols in (FE-125).

- σ: A cross-section (of the dark-matter annihilation or production process named in the equation; summed over the spins where marked). Unit: GeV^{-2}. Status: STANDARD.
- v_{rel}: The relative velocity of the two annihilating particles; $\langle\sigma v_{\mathrm{rel}}\rangle$ the thermal average. Unit: dimensionless ($c = 1$). Status: STANDARD.
- M_{DM}: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- K_1, K_2: The modified Bessel functions of the second kind of orders 1 and 2 (not related to the block variable K). Unit: dimensionless. Status: STANDARD.
- x: The ratio $x = M_{\mathrm{DM}}/T$ of the dark-matter mass to the temperature. Unit: dimensionless. Value: $x_{\mathrm{f}} = 31.23$ at the freeze-out. Status: STANDARD.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no upper bound. Status: STANDARD.
- E_{cm}: The total energy of the colliding pair in its centre-of-mass frame. Unit: GeV. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.

The terms. The relativistic thermal average over the centre-of-mass energy E_{cm} of the pair, with the Bessel functions K_1, K_2 of the Maxwell-Boltzmann distributions. Status: STANDARD (Gondolo and Gelmini; adaptive quadrature in the kinetic energy of the pair with exponentially scaled Bessel functions).

$$\Omega_{\mathrm{DM}} h_{100}^2 = \frac{M_{\mathrm{DM}}\, Y_\infty\, s_0}{\rho_{\mathrm{crit}}/h_{100}^2}, \qquad \frac{\rho_{\mathrm{crit}}}{h_{100}^2} = 1.05368 \times 10^{-5}\ \mathrm{GeV/cm}^3 \tag{FE-126}$$

Symbols in (FE-126).

- Ω_{DM}: The dark-matter density today in units of the critical density. Unit: dimensionless. Value: $\Omega_{\mathrm{DM}} h_{100}^2 = 0.120$ (both species). Status: DERIVED.
- h_{100}: The reduced Hubble constant, $\mathcal{H}_0/(100\ \mathrm{km\ s^{-1}\ Mpc^{-1}})$. Unit: dimensionless. Value: 0.6738. Status: STANDARD.
- M_{DM}: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.

- $Y_{\rm DM}$, $Y_{\rm eq}$, Y_∞: The abundance (number density per entropy density) of one dark-matter species (particles and antiparticles), its equilibrium value and its final value. Unit: dimensionless. Status: STANDARD.
- s_0: The entropy density today. Unit: GeV^3 (quoted per cm^3). Value: 2891.3 cm^{-3}. Status: STANDARD.
- $\rho_{\rm crit}$: The critical density, $3\mathcal{H}_0^2 M_{\rm Pl}^2/(8\pi)$. Unit: GeV^4 (quoted in GeV/cm^3). Value: $\rho_{\rm crit}/h_{100}^2 = 1.05368 \times 10^{-5}$ GeV/cm^3. Status: STANDARD.

The terms. The relic density from the final abundance Y_∞, the entropy density today s_0 and the critical density. Status: STANDARD. Each species gives 0.060 (0.120 together, the measured cold dark matter, which is the input of the CMB computation; the two agree to 3.1×10^{-9}) for $G_{\rm A} = 1.278 \times 10^{-9}$ GeV^{-2}, with the freeze-out at $x = 31.23$ ($T = 1.42$ TeV).

$$\begin{aligned}
\sigma_{\rm I12} &= \frac{G_{\rm I12}^2\, E_{\rm cm}^3 \big(E_{\rm cm}^2 - 4M_{\rm DM}^2\big)^{1/2}}{8\pi}, \qquad G_{\rm I12}^2 = 8G_{\rm g}^2 + 3G_{\rm W}^2 + G_{\rm B}^2, \\
\sigma_{\rm I13} &= \sum_f \frac{N_{\rm c}\, G_f^2\, E_{\rm cm}^2 \big(1 + 2M_{\rm DM}^2/E_{\rm cm}^2\big)}{24\pi\beta_{\rm cm}}, \qquad G_f = G_{\rm A}\, q_f^B\, \tilde{I}_{of}^{(b)}, \\
\beta_{\rm cm} &= \sqrt{1 - \frac{4M_{\rm DM}^2}{E_{\rm cm}^2}}, \qquad k_{\rm cm} = \sqrt{\frac{E_{\rm cm}^2}{4} - M_{\rm DM}^2}
\end{aligned} \tag{FE-127}$$

Symbols in (FE-127).

- $\sigma_{\rm I12}$, $\sigma_{\rm I13}$: The spin-averaged annihilation cross-sections of a dark-matter pair through I12 (into gauge-boson pairs) and through I13 (into quark pairs). Unit: GeV^{-2}. Status: DERIVED.
- $G_{\rm I12}$: The combined coupling of I12 to all gauge-boson pairs, $G_{\rm I12}^2 = 8G_{\rm g}^2 + 3G_{\rm W}^2 + G_{\rm B}^2$. Unit: GeV^{-3}. Value: benchmark $\pi/(2M_{\rm DM}^3) = 1.801 \times 10^{-14}$ GeV^{-3}. Status: DERIVED.
- $E_{\rm cm}$: The total energy of the colliding pair in its centre-of-mass frame. Unit: GeV. Status: STANDARD.
- $M_{\rm DM}$: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $G_{\rm g}$: The coupling of a dark-matter pair to canonically normalised gluon pairs through I12. Unit: GeV^{-3}. Value: 6.09×10^{-15} GeV^{-3} (benchmark). Status: DERIVED.
- $G_{\rm W}$: The coupling of a dark-matter pair to canonically normalised SU(2) gauge pairs through I12. Unit: GeV^{-3}. Value: 2.96×10^{-15} GeV^{-3}. Status: DERIVED.
- $G_{\rm B}$: The coupling of a dark-matter pair to canonically normalised hypercharge pairs through I12. Unit: GeV^{-3}. Value: 1.08×10^{-15} GeV^{-3}. Status: DERIVED.
- f: The label of a fermion species: a Standard-Model multiplet of one generation (its $N_{\rm c}$ Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over f runs over these species. Status: STANDARD.
- $N_{\rm c}$: The number of Weyl components of a fermion multiplet that share its zero-mode profile and overlap (colour times isospin): 6 for a quark doublet, 3 for a quark singlet, 2 for a lepton doublet, 1 for a charged-lepton singlet. Unit: dimensionless. Status: STANDARD.
- G_f: The I13 coupling of the dark matter to the quark component f, $G_f = G_{\rm A} q_f^B \tilde{I}_{of}^{(b)}$. Unit: GeV^{-2}. Status: DERIVED.
- $\beta_{\rm cm}$: The velocity of each particle of the pair in the centre-of-mass frame. Unit: dimensionless. Status: STANDARD.
- $G_{\rm A}$: The four-dimensional strength of the portal I13 on the zero modes, $G_{\rm A} = c_{\rm A} H/(2f^4 v_{\rm t} \Lambda_\star^6)$. Unit: GeV^{-2}. Value: 1.278×10^{-9} GeV^{-2} (dark-matter relic density). Status: DERIVED.
- q_ψ^B, q_f^B: The baryon number of the multiplet ψ or of the fermion component f ($\frac{1}{3}$ for quarks, 0 for leptons). Unit: dimensionless. Status: STANDARD.
- $\tilde{I}_{of}^{(b)}$: The overlap of the lower component of the state o with the zero mode of f, $\tilde{I}_{of}^{(b)} = H^{-1} \int_{-L}^0 b_o^2 \varrho_f {\rm e}^{-3Hy} {\rm d}y$ (the overlap of I13); $\tilde{I}_Q^{(b)}$, $\tilde{I}_{u_R}^{(b)}$, $\tilde{I}_{d_R}^{(b)}$, $\tilde{I}_{\nu_R}^{(b)}$ those of the dark matter. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q^{(b)} = 2.6135$. Status: DERIVED.
- o: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over o runs over the orbitals. Status: RECORD.

- k_{cm}: The momentum of each particle of the pair in the centre-of-mass frame. Unit: GeV. Status: STANDARD.

The terms. The two annihilation channels: through I12 into pairs of gauge bosons (eight gluons, three W, one B; a p-wave from the state ${}^3P_0$, so $\sigma v_{\mathrm{rel}} \propto v_{\mathrm{rel}}^2$ at low velocity), and through I13 into quark-antiquark pairs of the multiplets f, each with N_{c} Weyl components (colour times isospin; an s-wave from ${}^3S_1$), with the couplings G_f of FE-70 (o the dark-matter orbital; the sum of $N_{\mathrm{c}}(q_f^B \tilde{I}_{of}^{(b)})^2$ over the multiplets is $\mathcal{S}_B = 16.351$); β_{cm} and k_{cm} are the velocity and the momentum of the dark matter in the centre-of-mass frame. Status: DERIVED (spin averaged; the polarisation sum and the spin trace checked numerically); I12 at the unitarity-safe benchmark $G_{\mathrm{I12}} = \pi/(2M_{\mathrm{DM}}^3) = 1.801 \times 10^{-14}$ GeV^{-3}, which alone would leave far too much dark matter, so I13 sets the relic.

$$\sigma_{\mathrm{I12}} \leq \frac{\pi}{4k_{\mathrm{cm}}^2} \ \text{(total angular momentum 0)}, \qquad \sigma_{\mathrm{I13}} \leq \frac{3\pi}{4k_{\mathrm{cm}}^2} \ \text{(total angular momentum 1)} \tag{FE-128}$$

Symbols in (FE-128).

- σ_{I12}, σ_{I13}: The spin-averaged annihilation cross-sections of a dark-matter pair through I12 (into gauge-boson pairs) and through I13 (into quark pairs). Unit: GeV^{-2}. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- k_{cm}: The momentum of each particle of the pair in the centre-of-mass frame. Unit: GeV. Status: STANDARD.

The terms. The partial-wave unitarity bounds of the two channels (total angular momentum 0 for I12, 1 for I13). Where an effective cross-section exceeds its bound it is replaced by the bound. Status: DERIVED (I13 reaches its bound only above $v_{\mathrm{rel}} = 0.51$, which gives 22 % of the freeze-out rate; I12 alone would need a strongly coupled ${}^3P_0$ channel).

10.3 The fold bands

$$\Gamma_{\mathrm{fi}} = \frac{T}{32\pi^4} \int_0^\infty \mathrm{d}\big(E_{\mathrm{cm}}^2\big) \Big(\sum_{\text{spins}} \sigma \Big) E_{\mathrm{cm}}^3 K_1\Big(\frac{E_{\mathrm{cm}}}{T}\Big) = \frac{4}{\pi^5} N_{\mathrm{fold}} G_{\mathrm{V}}^2 T^8 \sum_f N_{\mathrm{c}} \big(q_f \tilde{I}_{of}\big)^2, \qquad \frac{\mathrm{d}\big(\rho_{\mathrm{fold}}\, \mathrm{e}^{4a_4}\big)}{\mathrm{d}x_4} = 8T\, \Gamma_{\mathrm{fi}}\, \mathrm{e}^{4a_4} \tag{FE-129}$$

Symbols in (FE-129).

- Γ_{fi}: The rate per volume of the production of fold-band pairs from the plasma through I11 (freeze-in). Unit: GeV^4. Status: DERIVED.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no upper bound. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- E_{cm}: The total energy of the colliding pair in its centre-of-mass frame. Unit: GeV. Status: STANDARD.
- σ: A cross-section (of the dark-matter annihilation or production process named in the equation; summed over the spins where marked). Unit: GeV^{-2}. Status: STANDARD.
- K_1, K_2: The modified Bessel functions of the second kind of orders 1 and 2 (not related to the block variable K). Unit: dimensionless. Status: STANDARD.
- N_{fold}: The number of fold bands: the massless fold zero modes of Ψ_A and of Ψ_B (two Dirac fermions). Unit: dimensionless. Value: 2. Status: DERIVED.
- G_{V}: The four-dimensional strength of the vector portal I11 on the zero modes, $G_{\mathrm{V}} = c_{\mathrm{V}} H/(2f^4 v_{\mathrm{t}} \Lambda_\star^6)$. Unit: GeV^{-2}. Value: benchmark 4.38×10^{-17} GeV^{-2} ($\Delta N_{\mathrm{eff}} = 0.001$). Status: DERIVED.
- f: The label of a fermion species: a Standard-Model multiplet of one generation (its N_{c} Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over f runs over these species. Status: STANDARD.

- N_{c}: The number of Weyl components of a fermion multiplet that share its zero-mode profile and overlap (colour times isospin): 6 for a quark doublet, 3 for a quark singlet, 2 for a lepton doublet, 1 for a charged-lepton singlet. Unit: dimensionless. Status: STANDARD.
- q_ψ, q_f: The $B-L$ charge: q_ψ of the multiplet ψ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), q_f of the fermion component f. Unit: dimensionless. Status: STANDARD.
- $\tilde{I}_{of}$: The overlap of the state o of dirac16complex with the zero mode of the fermion component f, $\tilde{I}_{of} = H^{-1}\int_{-L}^{0}(a_o^2+b_o^2)\varrho_f \mathrm{e}^{-3Hy}\mathrm{d}y$ (the integral in units of H); $\tilde{I}_Q$, $\tilde{I}_{u_R}$, $\tilde{I}_{d_R}$, $\tilde{I}_\ell$, $\tilde{I}_{e_R}$ those of the dark matter with Q, u_R, d_R, ℓ, e_R, $\tilde{I}_{\nu_R,i}$ with the right-handed neutrino of the generation i. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q = 8.2722$, $\tilde{I}_\ell = 1.2524$, $\tilde{I}_{u_R} = 0.3854$, $\tilde{I}_{d_R} = 0.3961$, $\tilde{I}_{e_R} = 0.3972$. Status: DERIVED.
- o: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over o runs over the orbitals. Status: RECORD.
- ρ_{fold}: The energy density of the fold bands. Unit: GeV^4. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.

The terms. The massless fold bands of dirac16complex (N_{fold} species, two Dirac fermions) are produced from pairs of Standard-Model fermions through I11: Γ_{fi} is the production rate per volume (Maxwell-Boltzmann statistics, massless particles; σ the cross-section summed over spins, K_1 the Bessel function), proportional to $G_{\mathrm{V}}^2 T^8$ and to the sum over the fifteen multiplets of three generations of $N_{\mathrm{c}}(q_f \tilde{I}_{of})^2$ (o the fold band; 126.85); each production puts the energy $8T$ on average into the fold bands, whose comoving energy density $\rho_{\mathrm{fold}}\mathrm{e}^{4a_4}$ grows accordingly. Status: DERIVED (integrated from the highest temperature 10 TeV, ASSUMED): $\Delta N_{\mathrm{eff}} = 5.21\times10^{29}\,G_{\mathrm{V}}^2$ GeV^4, so the benchmark $G_{\mathrm{V}} = 4.38\times10^{-17}$ GeV^{-2} gives 0.001. Residual (re-integration with Gauss-Legendre panels) 3.9×10^{-5}.

$$\Delta N_{\mathrm{eff}} = \frac{\rho_{\mathrm{fold}}}{\rho_\nu}, \qquad \rho_\nu = \frac{7}{8}\,\frac{\pi^2}{15}\,T_\nu^4, \qquad \frac{2\Gamma_{\mathrm{fi}}}{n_{\mathrm{fold,eq}}\,\mathcal{H}} \ll 1, \qquad n_{\mathrm{fold,eq}} = N_{\mathrm{fold}}\cdot 4\cdot\frac{3}{4}\,\frac{\zeta(3)}{\pi^2}\,T^3 \tag{FE-130}$$

Symbols in (FE-130).

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- N_{eff}: The effective number of neutrino species (the radiation beyond the photons in units of one neutrino species with $T_\nu/T_\gamma = (4/11)^{1/3}$); ΔN_{eff} an addition to it. Unit: dimensionless. Value: 3.0432 (neutrinos) + 0.0010 (fold bands). Status: STANDARD.
- ρ_{fold}: The energy density of the fold bands. Unit: GeV^4. Status: DERIVED.
- ρ_ν, ρ_{ν_α}: The energy density of one neutrino flavour (neutrinos and antineutrinos) at the common temperature T_ν, $\frac{7}{8}\frac{\pi^2}{15}T_\nu^4$ when massless; ρ_{ν_α} that of the flavour α at its temperature T_{ν_α}. Unit: GeV^4. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- T_ν: The temperature of the relic neutrinos (one common temperature). Unit: GeV. Value: 1.9454 K today. Status: STANDARD.
- Γ_{fi}: The rate per volume of the production of fold-band pairs from the plasma through I11 (freeze-in). Unit: GeV^4. Status: DERIVED.
- $n_{\mathrm{fold,eq}}$: The number density the fold bands would have in thermal equilibrium. Unit: GeV^3. Status: DERIVED.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- N_{fold}: The number of fold bands: the massless fold zero modes of Ψ_A and of Ψ_B (two Dirac fermions). Unit: dimensionless. Value: 2. Status: DERIVED.
- ζ: The Riemann zeta function; only its value $\zeta(3)$ occurs, the sum of $1/n^3$ over the positive integers n. Unit: dimensionless. Value: $\zeta(3) = 1.2020569$. Status: STANDARD.

- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.

The terms. The extra radiation ΔN_{eff} is the energy density of the fold bands in units of one neutrino species, and the condition for the production to stay far from equilibrium: the production rate per fold quantum, $2\Gamma_{\mathrm{fi}}/n_{\mathrm{fold,eq}}$, compared with the Hubble rate ($n_{\mathrm{fold,eq}}$ the equilibrium density of the fold bands). Status: DERIVED (the ratio is 0.012 at the highest temperature 10 TeV and falls afterwards: the fold bands are never in equilibrium).

10.4 The right-handed neutrinos

The right-handed neutrinos are gauge singlets and carry $B-L$ charge -1; the portals could produce them from the dark matter. The baryon-number current of I13 is the choice that prevents it; the following equations show the counterfactual and the actual rates.

$$
\begin{aligned}
&G_{\mathrm{A}}^{B-L} = G_{\mathrm{A}}\Big[\frac{\mathcal{S}_B}{\mathcal{S}_{B-L}}\Big]^{1/2}, \qquad \mathcal{S}_B = \sum_f N_{\mathrm{c}}\big(q_f^B \tilde{I}_{of}^{(b)}\big)^2, \qquad \mathcal{S}_{B-L} = \sum_f N_{\mathrm{c}}\big(q_f \tilde{I}_{of}^{(b)}\big)^2, \\
&\frac{\mathrm{d}n_{\nu_R}}{\mathrm{d}x_4} + 3\mathcal{H} n_{\nu_R} = \Gamma_\nu\Big(1 - \frac{n_{\nu_R}^2}{(n_{\nu_R}^{\mathrm{eq}})^2}\Big), \qquad \Gamma_\nu = 2\, n_{\mathrm{DM,eq}}\, \bar{n}_{\mathrm{DM,eq}}\, \frac{G_\nu^2 M_{\mathrm{DM}}^2}{2\pi}, \qquad G_\nu = G_{\mathrm{A}}^{B-L}\, \tilde{I}_{\nu_R}^{(b)}, \qquad \text{(FE-131)} \\
&n_{\nu_R}^{\mathrm{eq}} = \frac{3}{4}\,\frac{\zeta(3)}{\pi^2}\, T^3, \qquad n_{\mathrm{DM,eq}} = \bar{n}_{\mathrm{DM,eq}} = 2\Big(\frac{M_{\mathrm{DM}} T}{2\pi}\Big)^{3/2} \mathrm{e}^{-x}
\end{aligned}
$$

Symbols in (FE-131).

- G_{A}^{B-L}: The strength I13 would need if it coupled to the $B-L$ current (the same relic density). Unit: GeV^{-2}. Status: DERIVED.
- G_{A}: The four-dimensional strength of the portal I13 on the zero modes, $G_{\mathrm{A}} = c_{\mathrm{A}} H/(2 f^4 v_{\mathrm{t}} \Lambda_\star^6)$. Unit: GeV^{-2}. Value: 1.278×10^{-9} GeV^{-2} (dark-matter relic density). Status: DERIVED.
- $\mathcal{S}_B$, $\mathcal{S}_{B-L}$: The colour-weighted sums of the squared I13 overlaps of the dark matter with the baryon-number and with the $B-L$ charges. Unit: dimensionless. Value: $\mathcal{S}_B = 16.351$, $\mathcal{S}_{B-L} = 22.92$. Status: DERIVED.
- f: The label of a fermion species: a Standard-Model multiplet of one generation (its N_{c} Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over f runs over these species. Status: STANDARD.
- N_{c}: The number of Weyl components of a fermion multiplet that share its zero-mode profile and overlap (colour times isospin): 6 for a quark doublet, 3 for a quark singlet, 2 for a lepton doublet, 1 for a charged-lepton singlet. Unit: dimensionless. Status: STANDARD.
- q_ψ^B, q_f^B: The baryon number of the multiplet ψ or of the fermion component f ($\frac{1}{3}$ for quarks, 0 for leptons). Unit: dimensionless. Status: STANDARD.
- $\tilde{I}_{of}^{(b)}$: The overlap of the lower component of the state o with the zero mode of f, $\tilde{I}_{of}^{(b)} = H^{-1}\int_{-L}^{0} b_o^2 \varrho_f \mathrm{e}^{-3Hy}\mathrm{d}y$ (the overlap of I13); $\tilde{I}_Q^{(b)}$, $\tilde{I}_{u_R}^{(b)}$, $\tilde{I}_{d_R}^{(b)}$, $\tilde{I}_{\nu_R}^{(b)}$ those of the dark matter. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q^{(b)} = 2.6135$. Status: DERIVED.
- o: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over o runs over the orbitals. Status: RECORD.
- q_ψ, q_f: The $B-L$ charge: q_ψ of the multiplet ψ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), q_f of the fermion component f. Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- n_{ν_R}, $n_{\nu_R}^{\mathrm{eq}}$: The number density of the right-handed neutrinos (three generations) and its equilibrium value. Unit: GeV^3. Status: DERIVED.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- Γ_ν: The production rate per volume of right-handed neutrinos in the counterfactual $B-L$ variant of I13. Unit: GeV^4. Status: DERIVED.
- $n_{\mathrm{DM,eq}}$, $\bar{n}_{\mathrm{DM,eq}}$: The equilibrium number densities of dark-matter particles and antiparticles of one species. Unit: GeV^3. Status: STANDARD.

- G_ν: The four-dimensional coupling of the dark matter to a right-handed neutrino in the $B-L$ variant of I13. Unit: GeV^{-2}. Status: DERIVED.
- M_{DM}: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- ζ: The Riemann zeta function; only its value $\zeta(3)$ occurs, the sum of $1/n^3$ over the positive integers n. Unit: dimensionless. Value: $\zeta(3) = 1.2020569$. Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- x: The ratio $x = M_{\mathrm{DM}}/T$ of the dark-matter mass to the temperature. Unit: dimensionless. Value: $x_{\mathrm{f}} = 31.23$ at the freeze-out. Status: STANDARD.

The terms. If I13 coupled to the $B-L$ current, its coupling G_{A}^{B-L} would be fixed by the same relic density through the sums $\mathcal{S}_B$ and $\mathcal{S}_{B-L} = 22.92$ (the latter including the leptons), and the dark matter in equilibrium would produce right-handed neutrinos at the rate Γ_ν (both species, G_ν the coupling with the overlap of the right-handed neutrino) towards their equilibrium density $n_{\nu_R}^{\mathrm{eq}}$. Status: DERIVED (the three ν_R would reach 0.9997 of equilibrium by $x = 40$ and add $\Delta N_{\mathrm{eff}} = 0.144$, excluded by the CMB: this is why I13 couples to the baryon number). Residual 2.5×10^{-12}.

$$\Gamma_{\mathrm{el}} = 2\big(n_{\mathrm{DM}} + \bar{n}_{\mathrm{DM}}\big)\,\sigma_{\mathrm{el}} > \mathcal{H}, \qquad \sigma_{\mathrm{el}} = \frac{G_\nu^2 E^2}{2\pi}, \qquad E \simeq 3T \tag{FE-132}$$

Symbols in (FE-132).

- Γ_{el}, σ_{el}: The rate of elastic scattering of a right-handed neutrino on the dark matter, and its cross-section. Unit: GeV; GeV^{-2}. Status: ESTIMATE.
- n_{DM}, $\bar{n}_{\mathrm{DM}}$: The number densities of dark-matter particles and antiparticles of one species. Unit: GeV^3. Status: STANDARD.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- G_ν: The four-dimensional coupling of the dark matter to a right-handed neutrino in the $B-L$ variant of I13. Unit: GeV^{-2}. Status: DERIVED.
- E: The energy of a particle: of the neutrino in the flavour equations, $E = (k^2 + m^2)^{1/2}$ of a particle of momentum k in the thermal integrals. Unit: GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.

The terms. The elastic scattering of the right-handed neutrinos on the dark matter, which keeps them in kinetic equilibrium while its rate exceeds the Hubble rate ($E \simeq 3T$ the neutrino energy). Status: ESTIMATE (for a Weyl ν_R on a heavy Dirac fermion; kinetic decoupling at $x = 24.4$; a factor 2 in the rate moves it to 23.8 or 25.1).

$$\Delta N_{\mathrm{eff}} = N_{\mathrm{W}} \Big(\frac{10.75}{g_s}\Big)^{4/3} F_{\mathrm{h}}, \qquad F_{\mathrm{h}} = \Big[\frac{T_\nu}{T_\gamma}\Big(\frac{11}{4}\Big)^{1/3} \Big/ \big(1 + h_\nu\big)\Big]^4 = 1.0095 \tag{FE-133}$$

Symbols in (FE-133).

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- N_{eff}: The effective number of neutrino species (the radiation beyond the photons in units of one neutrino species with $T_\nu/T_\gamma = (4/11)^{1/3}$); ΔN_{eff} an addition to it. Unit: dimensionless. Value: 3.0432 (neutrinos) + 0.0010 (fold bands). Status: STANDARD.
- N_{W}: The number of Weyl species that decouple early. Unit: dimensionless. Status: DERIVED.

- $g_\rho,\ g_s$: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s = 43/11$ today. Status: STANDARD.
- $F_{\rm h}$: The correction factor of the early-decoupled radiation for the heating of the neutrinos. Unit: dimensionless. Value: 1.0095. Status: DERIVED.
- T_ν: The temperature of the relic neutrinos (one common temperature). Unit: GeV. Value: 1.9454 K today. Status: STANDARD.
- T_γ: The temperature of the photons (equal to the plasma temperature T; the subscript is written where the neutrinos have their own temperature). Unit: GeV. Value: 2.7255 K today. Status: STANDARD.
- h_ν: The relative heating of the neutrinos by the electron-positron annihilation (beyond instantaneous decoupling). Unit: dimensionless. Value: 0.00165. Status: DERIVED.

The terms. The extra radiation of $N_{\rm W}$ Weyl species that decouple when the entropy degrees of freedom are g_s, diluted by the later annihilations; $F_{\rm h}$ corrects for the non-instantaneous neutrino decoupling ($T_\nu/T_\gamma = 0.71664$ and the heating $h_\nu = 0.00165$ of the common-temperature decoupling, FE-115). Status: DERIVED ($F_{\rm h} = 1.0095$; used for the $B-L$ variant, 0.144, and for the self-interaction channel).

$$r_{\nu_R} = \frac{G_{\rm V}^2 \sum_{i=1}^{3} \tilde{I}_{\nu_R,i}^2}{G_{\rm A}^2\, \mathcal{S}_B} \tag{FE-134}$$

Symbols in (FE-134).

- r_{ν_R}: The ratio of the I11 production of right-handed neutrinos to the I13 annihilation rate of the dark matter. Unit: dimensionless. Value: 2.0×10^{-17}. Status: DERIVED.
- $G_{\rm V}$: The four-dimensional strength of the vector portal I11 on the zero modes, $G_{\rm V} = c_{\rm V} H/(2 f^4 v_{\rm t} \Lambda_\star^6)$. Unit: GeV^{-2}. Value: benchmark 4.38×10^{-17} GeV^{-2} ($\Delta N_{\rm eff} = 0.001$). Status: DERIVED.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $\tilde{I}_{of}$: The overlap of the state o of dirac16complex with the zero mode of the fermion component f, $\tilde{I}_{of} = H^{-1} \int_{-L}^{0} (a_o^2 + b_o^2) \varrho_f \mathrm{e}^{-3Hy} \mathrm{d}y$ (the integral in units of H); $\tilde{I}_Q$, $\tilde{I}_{u_R}$, $\tilde{I}_{d_R}$, $\tilde{I}_\ell$, $\tilde{I}_{e_R}$ those of the dark matter with Q, u_R, d_R, ℓ, e_R, $\tilde{I}_{\nu_R,i}$ with the right-handed neutrino of the generation i. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q = 8.2722$, $\tilde{I}_\ell = 1.2524$, $\tilde{I}_{u_R} = 0.3854$, $\tilde{I}_{d_R} = 0.3961$, $\tilde{I}_{e_R} = 0.3972$. Status: DERIVED.
- $G_{\rm A}$: The four-dimensional strength of the portal I13 on the zero modes, $G_{\rm A} = c_{\rm A} H/(2 f^4 v_{\rm t} \Lambda_\star^6)$. Unit: GeV^{-2}. Value: 1.278×10^{-9} GeV^{-2} (dark-matter relic density). Status: DERIVED.
- $\mathcal{S}_B,\ \mathcal{S}_{B-L}$: The colour-weighted sums of the squared I13 overlaps of the dark matter with the baryon-number and with the $B-L$ charges. Unit: dimensionless. Value: $\mathcal{S}_B = 16.351$, $\mathcal{S}_{B-L} = 22.92$. Status: DERIVED.

The terms. With the baryon current in I13 only I11 can produce right-handed neutrinos; r_{ν_R} is the ratio of that production to the annihilation rate that sets the relic, with the overlaps $\tilde{I}_{\nu_R,i}$ of the three right-handed neutrinos. Status: DERIVED ($r_{\nu_R} = 2.0 \times 10^{-17}$: the right-handed neutrinos are never thermalised).

10.5 The self-interaction channel

$$\begin{aligned} &\mathcal{L}_4^\lambda = -\lambda \left(\bar{\Psi}_{\rm DM}\Psi_{\rm fold}\right)\left(\bar{\Psi}_{\rm fold}\Psi_{\rm DM}\right)\left(n_{\rm DF}^{\rm sc}\right)^2, \qquad n_{\rm DF}^{\rm sc} = a_{\rm DM} b_{\rm fold} + b_{\rm DM} a_{\rm fold}, \qquad b_{\rm fold} = 0, \\ &G_\lambda = \frac{\lambda H \tilde{I}_\lambda}{2 f^4 v_{\rm t}}, \qquad \tilde{I}_\lambda = \frac{1}{H}\int_{-L}^{0} \left(n_{\rm DF}^{\rm sc}\right)^2 \mathrm{e}^{-3Hy}\, \mathrm{d}y, \qquad \sigma_\lambda v_{\rm rel} = \frac{G_\lambda^2 M_{\rm DM}^2}{2\pi} \quad \text{(per final fold species)} \end{aligned} \tag{FE-135}$$

Symbols in (FE-135).

- $\mathcal{L}_4^\lambda$: The four-dimensional term of the contact self-interaction that couples a dark-matter pair to a fold-band pair. Unit: GeV^4. Status: ESTIMATE.
- λ: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar{\Psi}\Psi)^2$ of the record). Unit: GeV^{-6}. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm 7 \times 10^{-5} H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4 \times 10^{-10} H^{-6}$ (for $\Delta N_{\rm eff} \le 0.001$). Status: RECORD.
- $\Psi_{\rm DM},\ \bar{\Psi}_{\rm DM}$: The four-dimensional field of a dark-matter quantum (the lowest odd level of dirac16complex on the static block, in either half) and its Dirac adjoint. Unit: GeV$^{3/2}$. Status: DERIVED.

- $\Psi_{\rm fold}$, $\bar\Psi_{\rm fold}$: The four-dimensional field of a fold-band quantum (an even zero mode of dirac16complex, massless) and its Dirac adjoint. Unit: GeV$^{3/2}$. Status: DERIVED.
- $n_{\rm DF}^{\rm sc}$: The transition scalar density between the dark-matter orbital and a fold-band orbital. Unit: GeV (normalised orbitals). Status: DERIVED.
- $a_{\rm DM}$, $b_{\rm DM}$: The upper and lower components of the dark-matter orbital (block spinor $(a, {\rm i}b)$, normalised on one half). Unit: GeV$^{1/2}$. Status: DERIVED.
- $a_{\rm fold}$, $b_{\rm fold}$: The upper and lower components of a fold-band orbital ($b_{\rm fold}=0$ for the even zero mode). Unit: GeV$^{1/2}$. Status: DERIVED.
- G_λ: The four-dimensional strength of the self-interaction channel of the dark matter into fold-band pairs. Unit: GeV^{-2}. Value: $\le 2.3\times10^{-15}$ GeV^{-2} (for $\Delta N_{\rm eff}\le 0.001$). Status: ESTIMATE.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda=-10H^2$ and the mass parameter $m_\Psi=H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\rm W}$ and $\rho_{\rm EW,0}$). Status: AUTHOR.
- $\tilde I_\lambda$: The overlap of the squared transition scalar density, $\tilde I_\lambda = H^{-1}\int_{-L}^{0}(n_{\rm DF}^{\rm sc})^2{\rm e}^{-3Hy}{\rm d}y$. Unit: dimensionless. Value: 14.38. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f={\rm e}^{-a_4}b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $v_{\rm t}$: The coordinate volume of the compact extra times x_5,x_6,x_7. Unit: GeV^{-3}. Value: not fixed; $v_{\rm t}=H^{-3}$ in the quoted conversions. Status: ASSUMED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y=-L$; $HL=3$. Unit: length. Value: $L=3/H$ ($fL=8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- e: Euler's number, the base of the natural logarithm; ${\rm e}^X$ is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- y: The hidden coordinate of the y chart, $y=\ln|\sin z|/(6H)$, with $y=0$ at the fold ($z=\pi/2$) and $y=-L$ at the regular tip; each half of a period of x_0 is the interval $-L\le y\le 0$. Unit: length. Status: DERIVED.
- d: The differential: ${\rm d}X$ is an infinitesimal change of X, ${\rm d}X/{\rm d}Y$ the derivative of X with respect to Y, ${\rm d}^nX$ the volume element of n variables. Status: STANDARD.
- σ_λ: The cross-section of the self-interaction channel (per final fold species). Unit: GeV^{-2}. Status: ESTIMATE.
- $v_{\rm rel}$: The relative velocity of the two annihilating particles; $\langle\sigma v_{\rm rel}\rangle$ the thermal average. Unit: dimensionless (c = 1). Status: STANDARD.
- $M_{\rm DM}$: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

The terms. The contact self-interaction $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ of IN-5, projected on the dark matter and the fold bands, contains the transition $(\bar\Psi_{\rm DM}\Psi_{\rm fold})(\bar\Psi_{\rm fold}\Psi_{\rm DM})$ with the scalar transition density $n_{\rm DF}^{\rm sc}$ of the two orbitals ($b_{\rm fold}=0$, so only $b_{\rm DM}a_{\rm fold}$ remains; it is odd under the fold reflection, its square even): the dark matter annihilates into pairs of fold quanta with the four-dimensional coupling G_λ and the overlap $\tilde I_\lambda=14.38$ (with its factor $1/H$). Status: ESTIMATE (the cross-section; the Fierz coefficients are not computed), $\tilde I_\lambda$ DERIVED.

$$\frac{{\rm d}\big(\rho_{\rm fold}\,{\rm e}^{4a_4}\big)}{{\rm d}x_4}=2M_{\rm DM}\,\Gamma_\lambda\,{\rm e}^{4a_4},\qquad \Gamma_\lambda=2\,n_{\rm DM,eq}\,\bar n_{\rm DM,eq}\,\sigma_\lambda v_{\rm rel}\,N_{\rm fold},\qquad \frac{2\Gamma_\lambda}{n_{\rm fold,eq}\,\mathcal{H}}\ll 1 \tag{FE-136}$$

Symbols in (FE-136).

- d: The differential: ${\rm d}X$ is an infinitesimal change of X, ${\rm d}X/{\rm d}Y$ the derivative of X with respect to Y, ${\rm d}^nX$ the volume element of n variables. Status: STANDARD.
- $\rho_{\rm fold}$: The energy density of the fold bands. Unit: GeV^4. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; ${\rm e}^X$ is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is ${\rm e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- x_4: The time; dots denote ${\rm d}/{\rm d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.

- M_{DM}: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- Γ_λ: The rate per volume of the self-interaction channel. Unit: GeV^4. Status: ESTIMATE.
- $n_{\mathrm{DM,eq}}$, $\bar{n}_{\mathrm{DM,eq}}$: The equilibrium number densities of dark-matter particles and antiparticles of one species. Unit: GeV^3. Status: STANDARD.
- σ_λ: The cross-section of the self-interaction channel (per final fold species). Unit: GeV^{-2}. Status: ESTIMATE.
- v_{rel}: The relative velocity of the two annihilating particles; $\langle \sigma v_{\mathrm{rel}} \rangle$ the thermal average. Unit: dimensionless (c = 1). Status: STANDARD.
- N_{fold}: The number of fold bands: the massless fold zero modes of Ψ_A and of Ψ_B (two Dirac fermions). Unit: dimensionless. Value: 2. Status: DERIVED.
- $n_{\mathrm{fold,eq}}$: The number density the fold bands would have in thermal equilibrium. Unit: GeV^3. Status: DERIVED.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.

The terms. While the dark matter is in equilibrium each annihilation puts the energy $2M_{\mathrm{DM}}$ into the fold bands (Γ_λ the annihilation rate per volume into all fold species); the condition on the right keeps the fold bands out of equilibrium. Status: DERIVED (integrated from the highest temperature to the freeze-out): $\Delta N_{\mathrm{eff}} = 1.83 \times 10^{26}\, G_\lambda^2$ GeV^4, so $\Delta N_{\mathrm{eff}} \leq 0.001$ requires $G_\lambda \leq 2.3 \times 10^{-15}$ GeV^{-2}, i.e. $\lambda \leq 1.4 \times 10^{-10}\, H^{-6}$ (ASSUMED bound; otherwise the fold bands thermalise and decouple as four Weyl species, FE-133). Residual 1.1×10^{-5}.

11 The coupled system

11.1 Which fields appear in which equations

Every field of UM-2 obeys its own field equation and enters the equations of the other fields as a source, a mean field, a current or a parameter. The table lists, for each field, its own equations and the other equations in which it appears.

field	its own equations	it appears in
Einstein-Lovelock gravity g_{MN} (with the block scale f)	FE-1, FE-11; the background FE-31 to FE-49; the graviton tower FE-53, FE-54; the block response FE-80; the perturbations FE-87 to FE-90, FE-97	every equation: the metric in every kinetic term, the spin connection FE-13, the block scale in every x_0 problem (FE-50 to FE-79), the expansion rate in FE-110 to FE-136
dirac16complex Ψ	FE-12, FE-13, FE-28; the reduction FE-56; the spectrum and the Kohn-Sham equations FE-62 to FE-69; its levels in matter FE-70 to FE-73; as dark matter FE-91, FE-124 to FE-128; the fold bands FE-129, FE-130, FE-135, FE-136	the sources of gravity FE-2, FE-3, FE-5, FE-9, FE-10, FE-45 to FE-47, FE-67, FE-80, FE-81, FE-86; the fermion equations FE-14 to FE-19 (I11, I13); the gauge couplings FE-21 to FE-24 and FE-79 (I12); the neutrino hierarchy FE-96 (fold bands); FE-131, FE-132
neutrinos: ν_e, ν_μ, ν_τ in ℓ_i, and ν_{Ri}	FE-17, FE-19, FE-29, FE-30; the masses FE-57 to FE-60, FE-100; FE-96; the flavour evolution FE-101 to FE-120; FE-131 to FE-134	FE-6; the I11 current in FE-12; the currents FE-20, FE-23 to FE-25; FE-26; the mean fields FE-64, FE-70; the plasma FE-121 to FE-123; N_{eff} in FE-97
charged leptons: ℓ_i (lower members) and e_{Ri}	FE-17, FE-18, FE-29, FE-30, FE-57 to FE-60	FE-6; FE-12 (I11); FE-20, FE-23 to FE-25; FE-26; FE-64, FE-70, FE-73; the baryon fluid FE-92; the neutrino interactions FE-101 to FE-103, FE-109, FE-111 to FE-114; FE-121
quarks: Q_i, u_{Ri}, d_{Ri}	FE-14 to FE-16, FE-29, FE-30, FE-57 to FE-61	FE-6; FE-12 (I11, I13); FE-20, FE-22 to FE-25; FE-26; FE-54; FE-64, FE-70, FE-73; the QCD plasma and the nucleon FE-75, FE-83 to FE-85; FE-92; FE-103; FE-121; FE-127

field	its own equations	it appears in
gluons G^A_M	FE-22, FE-27; the massless zero mode FE-50 to FE-52; the condensate FE-74, FE-75	FE-7, FE-10; FE-12 (I12); FE-14 to FE-16; FE-21; FE-64, FE-71, FE-72, FE-79; the QCD scale FE-81 to FE-84; FE-121; FE-127
photon A_M	FE-25; the massless zero mode FE-50 to FE-52, FE-79; the CMB photons FE-93 to FE-95	FE-7; FE-12 (I12, through W^3 and B); FE-20; FE-92; the photon temperature FE-111 to FE-115; FE-121, FE-122
W^+, W^-, Z	FE-23 to FE-25, FE-51, FE-52, FE-55; the condensates FE-76 to FE-78; FE-79	FE-7, FE-8; FE-12 (I12); FE-14 to FE-19; FE-26; FE-71, FE-72; the Fermi interactions FE-101, FE-102, FE-109; FE-121; FE-127
tip sigma field Σ	FE-26	the tip energy-momentum tensor FE-8; the tip Yukawa terms of FE-14 to FE-19; the tip currents of FE-23, FE-24; the tip mixing FE-30, FE-60; the Robin parameter of FE-51
block stabiliser ρ_{stab}	FE-4, FE-41, FE-43	FE-3; the static block FE-42; the radion and the block response FE-44 to FE-47; the reduction FE-48

No Higgs field and no Higgs particle appear in any equation: the masses of $W^\pm$ and Z come from the tip sigma field (FE-51), those of the fermions from their localisation in x_0 and the tip mixing (FE-59, FE-60).

11.2 How the system is solved

The coupled system is solved in eight steps. Each step solves a group of equations exactly or with a controlled approximation, uses the results of the earlier steps, and is checked by the residual of its solution in the equations it uses (section "What was verified").

Step 1: the background. The background equations FE-38 to FE-40 (with the tensors FE-35 to FE-37 and the conservation law FE-41) are solved in closed form for a static block with the stabiliser FE-4: FE-42 and FE-43 give $K_* = 12 + \frac{4}{3}\sqrt{51}$, $f_* = 0.215556$ and $\mathcal{G}(K_*) = 13.8081$. Linearised (FE-44 to FE-46) they give the stable radion ($\omega_{\text{r}} = 3.38705\,H$) and the adiabatic response of the block to matter (FE-47), and integrated numerically from a deflating primordial block they show how the block reaches the static point within $x_4 < 1/H$ (deflation rates up to $|\mathcal{H}_f| = 2.63\,H$, the largest deflation of the block scale $f/f_* = 2.917$, then radion oscillations at $3.3868\,H$ and a dust-like expansion). The reduction FE-48, FE-49 gives four-dimensional general relativity with $8\pi G_{\text{N}} = \kappa/(\mathcal{G}(K_*)V_\perp)$ and $\Lambda_4 = 0$ at the static point. The block-averaged sources of matter are included at first order only, with the hidden pressure $\bar{p}$; the block-shape source $p_0 - p_5$ is not (OPEN).

Step 2: the masses. On the static block every mass is an x_0 eigenvalue. For the bosons FE-50 to FE-52 give the massless graviton, photon and gluons, the tower FE-53 (with the potential FE-54) and the $W^\pm$, Z eigenvalues from the tip condition FE-51; FE-55 fixes β_{W} by the measured electroweak parameter, reproduces m_{Z} from m_{W}, and fixes the Kaluza-Klein scale $m_{\text{KK}} = 121.32$ TeV and with it $H = 14.296$ TeV. For the fermions the exact reduction FE-56 gives the zero modes FE-57 and the tip values FE-58; the mass conditions FE-59 (one generation) and FE-60 (three generations, with the tip matrices of FE-61) reproduce the twelve measured masses with the right-handed localisations and the measured mixing matrices U and V_{CKM}; the neutrino masses FE-100 are among them. FE-62 gives the spectrum of dirac16complex: the massless fold bands and the dark matter at 44.35 TeV.

Step 3: dirac16complex in Kohn-Sham theory and its back-reaction on the block. The Kohn-Sham equations FE-63 with the mean fields FE-64 (densities FE-65, energy FE-66, boundary conditions FE-69) are solved to self-consistency for many-body states of dirac16complex on the block. Their energy-momentum tensor FE-67 (checked by the identities FE-68) is the source of the background equations: its block averages ρ, p_3 and $\bar{p} = \frac{1}{4}(p_0 + 3p_5)$ enter the block response FE-45 to FE-47, which changes f and with it every mass of step 2. The block-shape source $p_0 - p_5$ of these states is up to 4 times $\bar{p}$ (OPEN).

Step 4: the portals and the condensates. On the zero modes the portals become the contact interactions FE-70 (I11, I13) and FE-71 (I12), with overlaps computed from the Kohn-Sham orbitals. In an environment the levels of dirac16complex shift by the mean fields of the Standard-Model densities and condensates (FE-72, FE-73),

with the condensates of the plasma from FE-74 to FE-78 (QCD trace anomaly, massive vectors, order-α pressure, running couplings without a Higgs particle). Conversely the dark matter weakens the gauge couplings and changes the x_0 problem of $W^{\pm}$ and Z (FE-79), while the photon and the gluons stay massless.

Step 5: gravity closes the loop. In ten environments, from the cosmic mean today to a neutron-star core and the dark-matter freeze-out (densities FE-86), the block responds to every species through FE-80, with the hidden pressure FE-81 of matter and of the hot plasma, the block dependence of the gauge couplings FE-82, of the QCD scale and the nucleon mass FE-83, FE-84, and of the fermion masses FE-85. The loop mass, source, block, mass is iterated to self-consistency; its gain is at most 1.05×10^{-33}, so the block moves by at most $\Delta \ln f = 1.1 \times 10^{-34}$ (at the freeze-out) and by 7×10^{-94} today.

Step 6: the neutrino flavours. With the masses and the tip matrix of step 2, the four-fermion interactions FE-101, FE-102 and the matter potentials FE-103, the flavour evolution FE-104 is solved exactly in vacuum and in matter (FE-105 to FE-108: accelerator baselines, the Sun, the Earth, supernova envelopes). In the early Universe the thermal potential FE-109 delays the onset of oscillations, and the decoupling equations FE-110 to FE-114 give N_{eff} (FE-115). In dense neutrino gases the linearised kinetic equations give the fast flavour instability (FE-116 to FE-119). Today the neutrinos form the cosmic neutrino background FE-120.

Step 7: the thermal history. With the x_0 masses of step 2 the degrees of freedom FE-121 to FE-123 give the expansion of the hot Universe. The Boltzmann equation FE-124 with the thermal average FE-125, the cross-sections FE-127 and the unitarity bounds FE-128 fixes the I13 coupling $G_{\mathrm{A}} = 1.278 \times 10^{-9}$ GeV^{-2} by the measured relic density FE-126. The freeze-in FE-129, FE-130 bounds the I11 coupling by the extra radiation (benchmark $G_{\mathrm{V}} = 4.38 \times 10^{-17}$ GeV^{-2} for $\Delta N_{\mathrm{eff}} = 0.001$), the self-interaction channel FE-135, FE-136 bounds λ, and FE-131 to FE-134 show that the right-handed neutrinos are never thermalised with the baryon current in I13 (with a $B - L$ current they would add $\Delta N_{\mathrm{eff}} = 0.144$).

Step 8: the linear perturbations. The metric series FE-87 is evaluated term by term: order 1 with $n = 0$ is the four-dimensional cosmological perturbation theory FE-88 to FE-90, with the species of UM-2 (the dark matter of step 7, FE-91; baryons FE-92; photons FE-93 to FE-95; the three neutrino mass states of step 2 and the fold bands of step 7, FE-96), the background and primordial spectrum FE-97; it gives the CMB, CNB and BAO, the potentials FE-98 and σ_8 (FE-99). The terms $n \geq 1$ are the graviton tower of step 2, the radion term is the block response of step 5, and order 2 is estimated.

11.3 The solution in concert

The steps share quantities: the same mass, coupling or density enters equations of different steps. UM-2 checks 26 such links, each the difference between the values of one quantity in two or more equations. 21 of them agree to between 5×10^{-16} and 1.9×10^{-5}, the precision of the stored numbers and of the numerical methods. Five differ by design, each by an amount that the results tolerate: the weak angle of the x_0 masses and of the Z widths (relative change of the total width 8.3×10^{-3}, the size of the electroweak loop corrections); the Hubble rate of the thermal history and of four-dimensional gravity at the freeze-out (1.7×10^{-4}: the thermal history leaves out the fold bands); the radiation density after neutrino decoupling in the thermal history and in the CMB computation (5.9×10^{-3}: instantaneous decoupling in the thermal history); the plasma energy density at the freeze-out in the thermal history and in the block sources (1.3×10^{-3}: the trace anomaly of the interacting plasma is in the block sources only); and the radion frequency of the linear and of the nonlinear evolution (7.9×10^{-5}: measured from the zero crossings of a finite oscillation).

12 What was verified

12.1 The checks

The computations of UM-2 are done by sixteen Python scripts (folder `universe_model_2/code`), each of which tests its own results against independent computations, closed forms or limits, with a stated tolerance. There are 287 checks in all, and all pass: 235 of the computations of version 3 and 52 of the residual analysis of the solution (`um2_residuals.py`). The scripts are deterministic (the Kohn-Sham computations run with one numerical thread), so every number of this document can be reproduced by running them.

script	checks	what is verified
`um2_algebra.py`	31	the author's 16×16 Dirac matrices rebuilt from his formulas: the Clifford relations of signature $(4, 4)$, C, the chirality, $\mathcal{B}$, the irreducibility of the representations of Pin(4,4) and Spin(4,4), the block basis and the block forms of FE-56
`um2_geometry.py`	25	the curvature of the order-0 metric in both charts (FE-33), the Lovelock tensors by four routes (FE-34 to FE-37), the Bianchi identities ($p = 1, 2, 3$), the minisuperspace variation, the equations of the record, the static block, the vielbein and the spin connection (FE-13)
`um2_background.py`	22	the equal block components (x_0 and x_5), the static block (FE-42), the sourceless vacuum (repulsive), the stabiliser and its roots (FE-43), $\mathcal{G}(K_*) > 0$, the stability of the block for the stabiliser family (FE-44), the first-order response (FE-45 to FE-47), the dimensional reduction (FE-48)
`um2_primordial.py`	5	the nonlinear primordial evolution: the constraint preserved to 6×10^{-10}, the radion oscillation, the dust-like expansion, the basin of the stable block, the largest deflation
`um2_fields.py`	16	the exact Dirac reduction in all eight blocks (FE-56), the selection operator and the boundary conditions (FE-28, FE-29, self-adjoint), the zero modes of dirac16complex, of the doublets and of the singlets (FE-57), their helicity and dispersion, the tip mixing for one and for three generations (FE-30), the gauge x_0 operator and its boundary conditions (FE-50, FE-51), the gauge invariance of the tip terms
`um2_spectra.py`	14	the x_0 spectra of the bosons and the fermions (FE-52, FE-59): exact roots, finite elements and shooting, the massless zero modes, the gap, the Dirichlet bound, the custodial relation, the fit of β_{W}, m_{Z}, the twelve fermion masses, the first-order formula
`um2_electroweak.py`	19	the charges of three generations, the charges of W^+ and W^-, the photon and Z couplings, the cancellation of the gauge anomalies, the chiral projector, the halves Ψ_A and Ψ_B, the Z charges, the invisible width, the Z widths of three generations
`um2_flavour.py`	7	the PMNS and CKM matrices (unitarity, Jarlskog invariant, FE-61), the exact three-generation tip condition and a general tip matrix (FE-60), no tree-level flavour-changing couplings of the Kaluza-Klein modes, $\mu \to e\gamma$, lepton universality
`um2_ks_jobs.py`	22	seven Kohn-Sham jobs: the exact spectrum and the massless fold band (FE-62), basis convergence, the y-conservation law and the energy identity of the interacting states (FE-68), radiation of the fold-band gas, the record's reference states reproduced, the dark matter (energy bookkeeping, cold, hidden pressure), the block response (FE-47), the Hellmann-Feynman responses to the portal mean fields (FE-72)
`um2_coupled.py`	16	the overlaps and their screening (FE-70), the fold band massless, the freeze-in benchmark, direct detection, the I13 algebra, overlaps and relic, the $B - L$ alternative, the self-interaction channel, the proton sensitivity (FE-83), the mean fields in ten environments, the gravity loop (FE-80), W, Z and photon with the dark matter (FE-79), the domain of validity
`um2_neutrino.py`	16	the masses and the cosmological bound, vacuum oscillations, the MSW resonances, long baselines with CP violation, the solar survival probability, the onset of oscillations in the early Universe, the QED integrals, the decoupling method against published values and N_{eff}, the CNB, supernova resonances, the fast flavour instability (three checks), Dirac or Majorana neutrinos
`um2_cosmology.py`	8	the CMB with the angular scale matched, three massive neutrino states, the TT and EE spectra against Planck ΛCDM, BAO against DESI DR2, dark-radiation variants, N_{eff} and helium, the freeze-in radiation, σ_8 and the first-order potential

script	checks	what is verified
`um2_thermal.py`	7	the degrees of freedom, the thermal average, the Boltzmann solver, the relic, the unitarity of the effective operator and the unitarity-limited relic, the I12 benchmark
`um2_series.py`	4	the terms of the metric series: order 1 with $n = 0$, the Kaluza-Klein gravitons, the radion, the decreasing magnitudes
`um2_analysis.py`	23	the x_0 mass mechanism and the uncertainty relation, dirac16complex as dark matter (mass, stability, neutrality, cross-sections, direct detection) and as dark energy (it cannot be)
`um2_residuals.py`	52	the residuals of the solution in 26 field equations and 26 links between the equations (below)

12.2 The residuals of the solution

For 26 of the field equations the stored solution is substituted afresh into the equation as printed in this document (or into its closed form). The residual is the relative amount by which the equation fails; the neglected terms are those that the approximation of the solution leaves out. Every residual is below its tolerance.

field equations	residual	the largest neglected term
background, FE-38 to FE-43	3.9×10^{-13}	matter against the stabiliser: 3.0×10^{-32} at the freeze-out, 2.4×10^{-93} today
block response, FE-44 to FE-47	1.4×10^{-7} (second order, for sources of $10^{-5} H^2$)	second order at the freeze-out 2.3×10^{-32}; inertia 3.4×10^{-33}
primordial era, FE-38 to FE-40	1.3×10^{-11}	radiation 1.0×10^{-6} of the final block; the block-shape mode (OPEN)
four-dimensional gravity, FE-48, FE-49	6.9×10^{-13}	$\Lambda_4 / H^2 \sim 10^{-92}$ (tuned, OPEN)
boson x_0 eigenvalues, FE-50 to FE-52	5.9×10^{-12}	the dark-matter kinetic function 1.9×10^{-17}; the Lovelock corrections to the graviton operator (OPEN)
fermion tip conditions, FE-58 to FE-60	5.4×10^{-11}	mixing with the Kaluza-Klein modes, $(m/m_{\mathrm{KK}})^2 \leq 2.0 \times 10^{-6}$
spectrum of dirac16complex, FE-62	8.5×10^{-13}	the Standard-Model mean fields, 1.5×10^{-6} of the dark-matter mass
Kohn-Sham equations, FE-63 to FE-69	$7.4 \times 10^{-14}\, H$	the self-interaction mean field at the bound on λ, $2.7 \times 10^{-9}\, m_\Psi$; correlation beyond the local exchange (OPEN)
Kohn-Sham source of gravity, FE-67 with FE-47	1.4×10^{-12}	the block-shape source, $\lvert p_0 - p_5 \rvert / \bar{p}$ up to 4 (OPEN)
portal overlaps, FE-70, FE-71	3.8×10^{-12}	screening by the dirac16complex gas, 7.2×10^{-10} at the bound on λ
coupled environments, FE-72 to FE-86	2.0×10^{-11}	the loop gain, 1.05×10^{-33}
W, Z with the dark matter, FE-79	3.3×10^{-12}	the physical shift, 1.9×10^{-17}
$W^{\pm}$, Z masses and Z widths, FE-25, FE-55	2.7×10^{-12}	electroweak loop corrections, 4.1×10^{-3} of the total width
CMB and BAO perturbations, FE-90 to FE-98	1.5×10^{-6}	order ϵ^2, 2.4×10^{-5}
graviton tower, FE-53, FE-54	1.2×10^{-12}	the Kaluza-Klein correction to Newton's law at 50 μm: zero in double precision

field equations	residual	the largest neglected term
metric series, FE-87	2.4×10^{-5}	order 2 at a neutron star, 0.03
neutrino flavour evolution, FE-103 to FE-108	7.1×10^{-12}	non-adiabatic solar effects, 1.3×10^{-3}
thermal potential, FE-109	8.7×10^{-12}	the CP-asymmetric potential, 1.1×10^{-2} (ESTIMATE)
neutrino decoupling, FE-111 to FE-115	2.7×10^{-12}	the electron mass and the momentum dependence, 4.4×10^{-3} in N_{eff}
fast flavour instability, FE-116 to FE-119	3.3×10^{-13}	the vacuum term, 9.0×10^{-6} of the self-interaction
cosmic neutrino background, FE-120	2.9×10^{-12}	non-instantaneous heating, 1.1×10^{-2} in $\Omega_\nu h^2_{100}$ (included in the CMB computation)
degrees of freedom, FE-121 to FE-123	5.0×10^{-9}	the order-α pressure, -5.4 % at the freeze-out
dark-matter relic, FE-124 to FE-128	6.1×10^{-9}	the order-α pressure, about $+2.7$ % in Ωh^2 (ESTIMATE)
freeze-in, FE-129, FE-130	3.9×10^{-5}	inverse processes, 1.2×10^{-2} at 10 TeV
right-handed neutrinos, FE-131 to FE-134	2.5×10^{-12}	the elastic rate (ESTIMATE), 9.5×10^{-6} in ΔN_{eff}
self-interaction channel, FE-135, FE-136	1.1×10^{-5}	inverse annihilation at the bound, 2.7×10^{-2}; the Fierz coefficients (not computed)

12.3 The links between the equations

quantity	field equations	difference
the Kohn-Sham energy density and pressures in the block response	FE-63 to FE-69, FE-45 to FE-47	3.4×10^{-12}
the hidden pressure of the dark matter, $p_0/\rho = -\mathrm{d}\ln\varepsilon_{\mathrm{DM}}/\mathrm{d}\ln f$	FE-62, FE-67, FE-81	6.1×10^{-12}
m_{Z} from the x_0 eigenvalues and in the Z widths	FE-55, FE-25	2.0×10^{-12}
the weak angle of the x_0 problem and of the Z widths (change of the total width)	FE-51, FE-55, FE-102	8.3×10^{-3} (by design)
the fermion masses in the thermal history	FE-59, FE-121	3.1×10^{-12}
the neutrino masses in the flavour evolution, the CNB and the CMB	FE-100, FE-104, FE-120, FE-96	5.7×10^{-12}
the PMNS matrix in the flavour evolution and the CNB	FE-61, FE-104, FE-120	8.2×10^{-13}
the neutrino density today in the CNB and the CMB computation	FE-120, FE-96	1.9×10^{-5}
N_{eff} of the decoupling and the freeze-in in the CMB computation	FE-115, FE-130, FE-97	5.4×10^{-16}
the relic density as the cold dark matter of the CMB computation	FE-126, FE-91	3.1×10^{-9}
the Hubble rate at the freeze-out: thermal history and four-dimensional gravity	FE-122, FE-49	1.7×10^{-4} (by design)
the radiation after neutrino decoupling: thermal history and CMB computation	FE-123, FE-97	5.9×10^{-3} (by design)
the plasma energy density at the freeze-out: thermal history and block sources	FE-121, FE-80, FE-81	1.3×10^{-3} (by design)
the portal responses of the Kohn-Sham jobs and the overlaps of the coupled solution	FE-70, FE-72	3.0×10^{-12}

quantity	field equations	difference
the I12 response $\mathcal{O}_{\mathrm{DM}}$ across the computations	FE-71, FE-72, FE-79	5.7×10^{-6}
the I12 benchmark in the relic and in the mean fields	FE-127, FE-72	1.6×10^{-12}
the dark-matter mass of the spectrum in the Boltzmann equation	FE-62, FE-124	8.1×10^{-13}
the relic and freeze-in couplings in the mean fields	FE-126, FE-129, FE-72	6.7×10^{-13}
the W eigenvalue in the problem with the dark-matter kinetic function	FE-52, FE-79	2.3×10^{-6}
the block-response coefficients across the computations	FE-46, FE-47, FE-80	2.0×10^{-13}
the radion frequency of the linear and of the nonlinear evolution	FE-44, FE-38 to FE-40	7.9×10^{-5} (by design)
the block scale f_{*} across the computations	FE-43, FE-52, FE-59, FE-62, FE-63	2.2×10^{-12}
the heating factor F_{h} of the neutrino decoupling	FE-115, FE-133	5.8×10^{-7}
the terms of the metric series in the cosmology, the coupled solution and the masses	FE-87, FE-53, FE-98, FE-80	4.7×10^{-12}
the freeze-out temperature of the relic and of the freeze-out environment	FE-124, FE-86	2.1×10^{-12}
the block sensitivities of the masses in the hidden pressure	FE-85, FE-81	4.5×10^{-11}

12.4 What is not verified

The following are not settled and are labelled OPEN where they occur: the block-shape mode driven by $p_0 - p_5$ (the order-0 metric has one block scale; its response to a block-anisotropic source is not computed, and the source is not small compared with the hidden pressure that is kept); the Gauss-Bonnet and cubic corrections to the graviton operator of the Kaluza-Klein tower; the tuned value of Λ_4 that the observed dark energy needs; the normalisation that relates the tip mixing matrices $\mathcal{K}$ to the Yukawa matrices $\mathcal{Y}$ of the action; correlation beyond the local exchange in the Kohn-Sham treatment of dirac16complex; and the nonlinear evolution of the fast flavour instability. The cross-section of the self-interaction channel (FE-135), the electroweak condensates below m_{W} (FE-78), the QCD interaction measure below 1 GeV (FE-75) and the block dependence of the QCD scale and the nucleon mass (FE-83, FE-84) are estimates.